



PTT PHENOL COMPANY
LIMITED

PROCESS SPECIFICATION

CONTROL VALVE PROCESS DATA SHEET

Docu. No. : 14780-8120-PS-0010
Project No. : 120117
Rev. No. : Z1
Date : MAY 10, 2016
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TITLE :

CONTROL VALVE PROCESS DATA SHEET

(CDN UNIT)

DISCIPLINE : PROCESS
DOCUMENT NO. : 14780-8120-PS-0010
PROJECT NAME : PTT Phenol Train II Project
PROJECT NO. : 120117
OWNER : PTT Phenol Company Limited (PPCL)

Z1	MAY 10, 2016	AS BUILT	S.W.Park	J.S.Chung	-	S.W.Lee
F3	MAR 04, 2015	FINAL ISSUE	S.W.Park	J.S.Chung	-	S.W.Lee
F2	FEB 06, 2015	FINAL ISSUE	S.W.Park	J.S.Chung	-	S.W.Lee
F1	JUN 12, 2014	FINAL ISSUE	S.W.Park	J.S.Chung	-	S.W.Lee
A1	DEC 13, 2013	APPROVED FOR DESIGN	S.W.Park	J.S.Chung	-	S.W.Lee
O1	AUG 14, 2013	COMMENT ISSUE	S.W.Park	J.S.Chung	-	S.W.Lee
REV. NO.	DATE	DESCRIPTION	PREP'N	REVIEW	QA REVIEW	APPROVAL
REV. NO.	PREPARATION DEPT.	PREPARATION	REVIEW	QA REVIEW	APPROVAL	
Z1	PROCESS	S.W.Park	J.S.Chung	-	S.W.Lee	
		MAY 10, 2016	MAY 10, 2016	-	MAY 10, 2016	

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		PROCESS SPECIFICATION			
PTT PHENOL TRAIN II PROJECT JOB NO. 120117		Doc. No. : 14780-8120-PS-0010		Rev. Z1	Sheet 2 / 136
Record of Revision					
NO.	REV.	PAGE NO.	DESCRIPTION		
1	O1	10	Pressure rating of UV-23-0301 has been changed from class 150 to class 300 according to technical clarification meeting (31-JUL-13 ~ 09-AUG-13).		
2	O1	12	Pressure rating of LV-23-0801 has been changed from class 150 to class 300 according to technical clarification meeting (31-JUL-13 ~ 09-AUG-13).		
3	O1	14	Pressure rating of UV-23-0401 has been changed from class 150 to class 300 according to technical clarification meeting (31-JUL-13 ~ 09-AUG-13).		
4	O1	16	Pressure rating of HV-23-0401 has been changed from class 150 to class 300 according to technical clarification meeting (31-JUL-13 ~ 09-AUG-13).		
5	O1	20	Pressure rating of FV-23-0501 has been changed from class 300 to class 150 according to technical clarification meeting (31-JUL-13 ~ 09-AUG-13).		
6	O1	24	Pressure rating of UV-23-0501 has been changed from class 150 to class 300 according to technical clarification meeting (31-JUL-13 ~ 09-AUG-13).		
7	O1	36	Pressure rating of UV-23-0702 has been changed from class 150 to class 300 according to technical clarification meeting (31-JUL-13 ~ 09-AUG-13).		
8	O1	46	Pressure rating of UV-23-0705 has been changed from class 150 to class 300 according to technical clarification meeting (31-JUL-13 ~ 09-AUG-13).		
9	O1	56	Pressure rating of UV-23-0802 has been changed from class 150 to class 300 according to technical clarification meeting (31-JUL-13 ~ 09-AUG-13).		
10	O1	64	Pressure rating of UV-23-0804 has been changed from class 150 to class 300 according to technical clarification meeting (31-JUL-13 ~ 09-AUG-13).		
11	O1	66	Pressure rating of UV-23-0805 has been changed from class 150 to class 300 according to technical clarification meeting (31-JUL-13 ~ 09-AUG-13).		
12	O1	84	Pressure rating of UV-23-1203 has been changed from class 150 to class 300 according to technical clarification meeting (31-JUL-13 ~ 09-AUG-13).		
13	O1	86	Pressure rating of UV-23-1202 has been changed from class 150 to class 300 according to technical clarification meeting (31-JUL-13 ~ 09-AUG-13).		
14	O1	88	Pressure rating of UV-23-1204 has been changed from class 150 to class 300 according to technical clarification meeting (31-JUL-13 ~ 09-AUG-13).		
15	O1	96	Pressure rating of UV-23-1205 has been changed from class 150 to class 300 according to technical clarification meeting (31-JUL-13 ~ 09-AUG-13).		
16	O1	98	Pressure rating of UV-23-1206 has been changed from class 150 to class 300 according to technical clarification meeting (31-JUL-13 ~ 09-AUG-13).		
17	O1	122	Pressure rating of TV-23-1302B has been changed from class 300 to class 150 according to technical clarification meeting (31-JUL-13 ~ 09-AUG-13).		
18	A1	General	Tag number of control valves and On-off valves which are controlled by SIS system have been revised as per P&ID.		
19	A1	General	Design pressure and design temperature have been added to inform design condition to instrument team.		
20	A1	4 - 7	Control valves and on-off valves index has been updated as per new numbering sequence.		
21	A1	134	Data sheet for UXV-23-0803 has been added as per P&ID.		
22	F1	GENERAL	Design pressure for control valve has been revised to meet line design pressure.		
23	F1	134 - 136	PCV DATA FOR PUMP SEAL HAS BEEN ADDED.		
24	F2	134	Leackage class of UXV-23-0803 has been revised from type 4 to class 6 as per final vendor data.		
25	F3	72	Fail position has been reivsded as per UOP-PET-E-016.		
26	F3	116	Body size of TV-23-1401 has been revised as per the latest vendor data for consistency between documents and drawing.		
27	F3	118, 120	Body size of UXV-23-1403, 1402 have been revised as per revision of line size by UOP.		

Revision
Indication



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CONTROL VALVES

PROJECT SPECIFICATION

963766 — **616-G**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

GENERAL REQUIREMENTS AND INFORMATION

A GENERAL REQUIREMENTS

UOP Standard Specification 6-14 is part of this specification.

B PROJECT INFORMATION

- Instrument Air Supply pressure range is:
Minimum: ~ Normal: 7.00 kg/cm2(g) Maximum: ~
- Liquid volume flow rates are at **flowing** temperature. Gas volume flow rate base temperature is 0°C and base pressure is 1.033 kg/cm2 absolute.

Other specifications referenced in this specification:	UOP Standard Specification: 6-14
	UOP Standard Drawing:
	UOP Project Specification: 601, this project

**CONTROL VALVES****PROJECT SPECIFICATION**963766 — **616-I**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

INDEX

TAG	SERVICE	P&ID DRAWING	SHEETS
FV-23-0403	Preflash Column Reflux	04	18 - 19
FV-23-0501	S1.5 Steam to Preflash Column Steam Heater	05	20 - 21
FV-23-0801	Flash Column Reflux	08	60 - 61
FV-23-1001	Nitrogen to Concentration Vacuum Producing Equipment	10	72 - 73
FV-23-1201	Process Water to Decomposer Feed	12A	106 - 107
FV-23-1202	Decomposer Feed Startup bypass	12	82 - 83
FV-23-1205	Decomposer Feed	12	92 - 83
FV-23-1301	Decomposer Product to Dehydrators	13	112 - 113
FV-23-1801, 1802	Circulating Decomposer Liquid to Calorimeter	18	126 - 127
FV-23-2001	Acid Aromatics to Direct Neutralization Static Mixers	20A	130 - 131
HV-23-0401	Cumene Quench to Preflash Column Bottom (Manual)	04	16 - 17
HV-23-0601	Cumene Quench to Preflash Column (Manual)	06	30 - 31
HV-23-0801	Cumene Quench to Flash Column Collector Tray (Manual)	08	54 - 55
HV-23-0802	Cumene Quench to Flash Column Collector Tray (Manual)	08	58 - 59
HV-23-1001	Chilled Water to Flash Column Overhead Vapor Chiller	10	68 - 69
HV-23-1201	Makeup Cumene to Decomposer Feed Flush Drum	12A	104 - 105
LV-23-0501	Preflash Column Steam Heater Condensate to Condensate Header	05A	26 - 27
LV-23-0601	Cumene to Concentration Cumene Quench Drum	06	28 - 29
LV-23-0701	Flash Column Vaporizer Condensate to SC3 Condensate Header	07A	50 - 51
LV-23-0801	Oxidate Feed to Preflash Column	04	12 - 13

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INDEX

TAG	SERVICE	P&ID DRAWING	SHEETS
LV-23-0803	Preflash / Flash Column Net Overhead Liquid	08B	62 - 63
LV-23-1101	Cumene to Cumene Flush Drum	11	76 - 77
PCV-13-2001	Nitrogen Purge to Acid Aromatics Sump	20A	132 - 133
PCV-23-1003	Nitrogen Purge to Concentration Vacuum Producing Equipment	10	70 - 71
PV-23-1101A	Nitrogen to Cumene Flush Drum	11	78 - 79
PV-23-1101B	Vent Gas from Cumene Flush Drum	11	80 - 81
PV-23-1202A	Nitrogen to Decomposer Feed Flush Drum	12A	100 - 101
PV-23-1202B	Vent Gas from Decomposer Feed Flush Drum	12A	102 - 103
PV-23-1301A	Nitrogen to Decomposer Drum	13	108 - 109
PV-23-1301B	Vent Gas from Decomposer Drum	13	110 - 111
PV-23-1401	Crude Product to Fractionation Feed Tanks at Fractionation Section	19	128 - 129
TV-23-0703	S3 Steam to Flash Column Vaporizer	07	40 - 41
TV-23-1302A	Reliable Cooling Water to Decomposer Cooler	17	124 - 125
TV-23-1302B	Reliable Cooling Water bypassing Decomposer Cooler	17	122 - 123
TV-23-1401	S4 Steam to Dehydrators	14	116 - 117
UXV-23-0301	Oxidate Feed to Preflash Column Shutoff	03	10 - 11
UXV-23-0302	Oxidate to CHP Closed Drain	03	8 - 9
UXV-23-0401	Cumene Quench to Oxidate Feed Line	04	14 - 15
UXV-23-0501	Preflash Column Steam Heater Steam Line Vent to Atmosphere	05	24 - 25
UXV-23-0502	S1.5 Steam to Preflash Column Steam Heater Shutoff	05	22 - 23

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**CONTROL VALVES****PROJECT SPECIFICATION**963766 — **616-I**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
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TAG	SERVICE	P&ID DRAWING	SHEETS
UXV-23-0601	Cumene Quench to Preflash Column	06	32 - 33
UXV-23-0701	S3 Steam Condensate from Flash Column Vaporizer Shutoff	07	34 - 35
UXV-23-0702	Flash Column Vaporizer Steam Line Vent to Atmosphere	07	36 - 37
UXV-23-0703	S3 Steam to Flash Column Vaporizer Shutoff	07	38 - 39
UXV-23-0704	Reliable Cooling Water to Flash Column Vaporizer Shutoff	07	42 - 43
UXV-23-0705	Cumene Quench to Flash Column Vaporizer	07	46 - 47
UXV-23-0706	Reliable Cooling Water from Flash Column Vaporizer	07	44 - 45
UXV-23-0707	Steam to / from Flash Column Vaporizer Condensate Drum	07	48 - 49
UXV-23-0801	Cumene Quench to Flash Column Collector Tray	08	52 - 53
UXV-23-0802	Cumene Quench to Flash Column Bottom	08	56 - 57
UXV-23-0804	Recycle Cumene to Feed Wash Column Shutoff	08B	64 - 65
UXV-23-0805	Oxidate Long Circulation to Oxidizer No. 2 Shutoff	08B	66 - 67
UXV-23-1001	Concentration Vacuum Producing Equipment Vent	10	74 - 75
UXV-23-1201A	Cumene Flush from Decomposer Feed Line	12	90 - 91
UXV-23-1201B	Cumene Flush to Decomposer Feed Line	12	94 - 95
UXV-23-1202	Cumene Flush to Oxidate Circulation Line	12	86 - 87
UXV-23-1203	Decomposer Feed Startup Bypass to Oxidate Circulation Line	12	84 - 85
UXV-23-1204	Decomposer Feed Shutoff	12	88 - 89
UXV-23-1205	Decomposer Feed Shutoff	12	96 - 97
UXV-23-1206	Decomposer Feed Flush Drum to Decomposer Feed Line Shutoff	12A	98 - 99

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**CONTROL VALVES****PROJECT SPECIFICATION**963766 — **616-I**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
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TAG	SERVICE	P&ID DRAWING	SHEETS
UXV-23-1401	S4 Steam to Dehydrators Shutoff	14	114 - 115
UXV-23-1402	Decomposer Product to Crude Product Cooler Shutoff	14A	120 - 121
UXV-23-1403	Decomposer Product Return to Decomposer	14A	118 - 119
UXV-23-0803	Reliable Cooling Water to Preflash and Flash Column Condenser Shutoff	08A	134
PCV-23-0901	N2 Control Valve For P-2301A	01J	134
PCV-23-0902	N2 Control Valve For P-2301B	01J	134
PCV-23-1702	N2 Control Valve For P-2302A	01K	134
PCV-23-1704	N2 Control Valve For P-2302B	01K	135
PCV-23-1302	N2 Control Valve For P-2303A	01K	135
PCV-23-1303	N2 Control Valve For P-2303B	01K	135
PCV-23-0801	N2 Control Valve For P-2307A	01K	135
PCV-23-0802	N2 Control Valve For P-2307B	01K	136

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**posco**
ENGINEERING**CONTROL VALVES****PROJECT SPECIFICATION**963766 — **616**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

GENERAL	Item Number: 1 Quantity: 1 Valve Tag Number: UXV-23-0302 Data Page: 1 of 2									
	Service: Oxidate to CHP Closed Drain Process Material: Hydrocarbon with aromatics P&ID Drawing: 03									
SIZING CONDITIONS	Process Data Case	General		General						
	Sizing Case	Open		Closed						
	Calculated C _v									
		Inlet	Outlet	Inlet	Outlet	Inlet	Outlet	Inlet	Outlet	Units
	Liquid Flow at Flowing Cond									m3/h
	Vapor Flow at Base Cond									normal m3/h
	Steam Flow									kg/h
	Temperature	82		82						°C
	Liquid Specific Gravity									----
	Vapor Spec. Gravity (Air=1)									----
	Pressure	0.10	0.10	7.04	0.10					kg/cm2(g)
	Liq Critical Press (Pseudo)									kg/cm2(a)
	Liquid Vapor Pressure									kg/cm2(a)
	Liquid Viscosity									cP
	Compressibility Z									
	Specific Heats Cp/Cv (Real)									
Estimate Sound Press Level									dBA	
VALVE BODY & BONNET	Body Type: Globe §	TRIM		Approximate Valve C _v : 225		F _L :		x _T :		
	Body Size: 4" # Port Type: Single			Trim Style: Man. Standard		Trim Size:				
	Inlet Conn Size: 4" # Type: Flanged Class: 300			Inherent Characteristic: Equal Percentage						
	Outlet Conn Size: 4" # Type: Flanged Class: 300			Balance Style: Balanced		Full Open Position: 100 %				
	Flange Facing: Raised Face Flange Face Finish:			Guiding Style: Cage						
	Body & Bonnet Material: 316 Stainless Steel ‡			Closure Member Material: 316 Stainless Steel						
	Bonnet Style: Integral or Bolted			Seat Ring Material: 316 SS with Stellite sheath						
	Packing Style: Manufacturer's Recommendation			Seat Ring Retainer Material:						
	Packing Material: Manufacturer's Recommendation			Bearing Material:						
	Body/Bonnet Gasket Material: Manufacturer's Recommendation			Bushing/Cage Material: 316 SS hard chrome plated						
	Lub/Iso Valve Material: Lub Material:			Stem/Shaft Material: 316 Stainless Steel						
	Manufacturer: Masoneilan (or equal)			Flow Direction: Flow to close						
Model Number:	Seat Leakage Class: ANSI Class V									
ACTUATOR	Type: Spring Diaphragm, Pneumatic	INSTALLATION		Inlet Pipe Nominal Size:		Est Wall Thick:				
	Action Style: Fail Open			Outlet Pipe Nominal Size:		Est Wall Thick:				
	Service Type: On-Off			Est Insulation Type:						
	Max Shutoff Press: In: 11.40 Out: 0 kg/cm2(g)			Est Insulation Thickness:						
	Maximum Stroke Time to Open: 5 Seconds			Installation:						
	Maximum Stroke Time to Close:			Design Pressure : 11.5 kg/cm2g (F.V)						
	Stroke Cycle Service:			Design Temperature : 120 C						
	DATA PAGE 1 REMARKS			§ The information is based on Masoneilan 41000 series.						
# Line size valve.										
‡ Preferred material is L grade stainless steel. Standard grade stainless steel that is not welded may be used for process wetted parts including the body. Stainless steel with weld repair is acceptable if low carbon (<= 0.03 %) filler material is used.										

ALL ITEMS SHALL COMPLY WITH THE GENERAL REQUIREMENTS SHOWN ON THE GENERAL REQUIREMENTS SECTION OF THIS SPECIFICATION.



posco
ENGINEERING

CONTROL VALVES

PROJECT SPECIFICATION

963766 — 616

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Item Number: 1 Quantity: 1 Valve Tag Number: UXV-23-0302 Data Page: 2 of 2																																																																																									
<table border="1"> <tr> <td rowspan="4">TRANSDUCER</td> <td>Transducer Tag Number:</td> <td rowspan="4">HANDWHEEL</td> <td>Handwheel Position:</td> </tr> <tr> <td>Transducer Input Signal Range:</td> <td>Usage:</td> </tr> <tr> <td>Transducer Output Signal Range:</td> <td></td> </tr> <tr> <td>Manufacturer:</td> <td></td> </tr> </table>	TRANSDUCER	Transducer Tag Number:	HANDWHEEL	Handwheel Position:	Transducer Input Signal Range:	Usage:	Transducer Output Signal Range:		Manufacturer:		<table border="1"> <tr> <td rowspan="12">SOLENOID VALVE</td> <td>Solenoid Valve Tag Number: UXY-23-0302</td> <td rowspan="12">POSITION SWITCH</td> <td>Open Position Switch Tag Number:</td> </tr> <tr> <td>Body Type: 3-Way Body Material: Brass</td> <td>Close Position Switch Tag Number:</td> </tr> <tr> <td>Seat Material: Stainless Steel Disc Material: Stainless Steel</td> <td>Sensor Type:</td> </tr> <tr> <td>Connection Size/Type: †</td> <td>Contact Form:</td> </tr> <tr> <td>Packing Style: Packless</td> <td>Open Position Switch Contact Action:</td> </tr> <tr> <td>Fluid: Instrument Air</td> <td>Close Position Switch Contact Action:</td> </tr> <tr> <td>Pressure: Contractor to determine</td> <td>Contact Load: Contact Rating:</td> </tr> <tr> <td>Minimum Pressure Port Orifice C_v: MAX Size: MAX</td> <td>Mounting Location:</td> </tr> <tr> <td>Minimum Exhaust Port Orifice C_v: MAX Size: MAX</td> <td>Manufacturer:</td> </tr> <tr> <td>Operation Style: †</td> <td rowspan="3">POSITION TRANSMITTER</td> <td>Position Transmitter Tag Number:</td> </tr> <tr> <td>Coil Configuration: Single</td> <td>Sensor Type:</td> </tr> <tr> <td>Operating Voltage: §§ Frequency:</td> <td>Output Signal Range:</td> </tr> <tr> <td>Action Coil 1 Deenergized: Vent air from valve</td> <td rowspan="3">LOCK-UP VALVE</td> <td>Travel Measurement:</td> </tr> <tr> <td>Action Coil 1 Energized: Supply air to valve</td> <td>Mounting Location:</td> </tr> <tr> <td>Action Coil 2 Deenergized:</td> <td>Manufacturer:</td> </tr> <tr> <td>Action Coil 2 Energized:</td> <td rowspan="3">QUICK EXHAUST</td> <td>Action Style:</td> </tr> <tr> <td>Manual Reset:</td> <td>Mounting Location:</td> </tr> <tr> <td>Manual Operator:</td> <td>Model Number:</td> </tr> <tr> <td>Mounting Location: Nearest item to actuator in air line. †</td> <td rowspan="3">VOLUME BOOSTER</td> <td>Manufacturer:</td> </tr> <tr> <td>Manufacturer: ASCO (or equal)</td> <td>Connection Size/Type:</td> </tr> <tr> <td></td> <td>Body Material:</td> </tr> <tr> <td>POSITIONER</td> <td></td> <td>Diaphragm Material:</td> </tr> <tr> <td rowspan="3">AIR FILTER</td> <td>Positioner Tag Number:</td> <td>Exhaust Action:</td> </tr> <tr> <td>Positioner Type:</td> <td>Mounting Location:</td> </tr> <tr> <td>Positioner Input Signal Range:</td> <td>Manufacturer:</td> </tr> <tr> <td rowspan="3">TUBING</td> <td>Positioner Output Signal Range:</td> <td rowspan="3">VOLUME BOOSTER</td> <td>Connection Size/Type:</td> </tr> <tr> <td>Manufacturer:</td> <td>Body Material:</td> </tr> <tr> <td>Quantity: one</td> <td>Seat Ring Material:</td> </tr> <tr> <td></td> <td>Type: Filter/Regulator</td> <td>Diaphragm Material:</td> </tr> <tr> <td></td> <td>Manufacturer:</td> <td>Mounting Location:</td> </tr> <tr> <td></td> <td>Tubing Size:</td> <td>Manufacturer:</td> </tr> <tr> <td></td> <td>Tubing Material:</td> <td></td> </tr> </table>	SOLENOID VALVE	Solenoid Valve Tag Number: UXY-23-0302	POSITION SWITCH	Open Position Switch Tag Number:	Body Type: 3-Way Body Material: Brass	Close Position Switch Tag Number:	Seat Material: Stainless Steel Disc Material: Stainless Steel	Sensor Type:	Connection Size/Type: †	Contact Form:	Packing Style: Packless	Open Position Switch Contact Action:	Fluid: Instrument Air	Close Position Switch Contact Action:	Pressure: Contractor to determine	Contact Load: Contact Rating:	Minimum Pressure Port Orifice C_v : MAX Size: MAX	Mounting Location:	Minimum Exhaust Port Orifice C_v : MAX Size: MAX	Manufacturer:	Operation Style: †	POSITION TRANSMITTER	Position Transmitter Tag Number:	Coil Configuration: Single	Sensor Type:	Operating Voltage: §§ Frequency:	Output Signal Range:	Action Coil 1 Deenergized: Vent air from valve	LOCK-UP VALVE	Travel Measurement:	Action Coil 1 Energized: Supply air to valve	Mounting Location:	Action Coil 2 Deenergized:	Manufacturer:	Action Coil 2 Energized:	QUICK EXHAUST	Action Style:	Manual Reset:	Mounting Location:	Manual Operator:	Model Number:	Mounting Location: Nearest item to actuator in air line. †	VOLUME BOOSTER	Manufacturer:	Manufacturer: ASCO (or equal)	Connection Size/Type:		Body Material:	POSITIONER		Diaphragm Material:	AIR FILTER	Positioner Tag Number:	Exhaust Action:	Positioner Type:	Mounting Location:	Positioner Input Signal Range:	Manufacturer:	TUBING	Positioner Output Signal Range:	VOLUME BOOSTER	Connection Size/Type:	Manufacturer:	Body Material:	Quantity: one	Seat Ring Material:		Type: Filter/Regulator	Diaphragm Material:		Manufacturer:	Mounting Location:		Tubing Size:	Manufacturer:		Tubing Material:	
TRANSDUCER		Transducer Tag Number:		HANDWHEEL	Handwheel Position:																																																																																				
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Action Coil 1 Energized: Supply air to valve		Mounting Location:																																																																																							
Action Coil 2 Deenergized:		Manufacturer:																																																																																							
Action Coil 2 Energized:	QUICK EXHAUST	Action Style:																																																																																							
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AIR FILTER	Positioner Tag Number:	Exhaust Action:																																																																																							
	Positioner Type:	Mounting Location:																																																																																							
	Positioner Input Signal Range:	Manufacturer:																																																																																							
TUBING	Positioner Output Signal Range:	VOLUME BOOSTER	Connection Size/Type:																																																																																						
	Manufacturer:		Body Material:																																																																																						
	Quantity: one		Seat Ring Material:																																																																																						
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**posco**
ENGINEERING**CONTROL VALVES****PROJECT SPECIFICATION**963766 — **616**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

GENERAL	Item Number: 2 Quantity: 1 Valve Tag Number: UXV-23-0301 Data Page: 1 of 2									
	Service: Oxidate Feed to Preflash Column Shutoff Process Material: Hydrocarbon with aromatics P&ID Drawing: 03									
SIZING CONDITIONS	Process Data Case	General		General						
	Sizing Case	Open		Closed						
	Calculated C _v									
		Inlet	Outlet	Inlet	Outlet	Inlet	Outlet	Inlet	Outlet	Units
	Liquid Flow at Flowing Cond									m3/h
	Vapor Flow at Base Cond									normal m3/h
	Steam Flow									kg/h
	Temperature	82		82						°C
	Liquid Specific Gravity									----
	Vapor Spec. Gravity (Air=1)									----
	Pressure	7.04	7.04	7.04	0.10					kg/cm2(g)
	Liq Critical Press (Pseudo)									kg/cm2(a)
	Liquid Vapor Pressure									kg/cm2(a)
	Liquid Viscosity									cP
	Compressibility Z									
	Specific Heats Cp/Cv (Real)									
Estimate Sound Press Level									dBA	
VALVE BODY & BONNET	Body Type: Eccentric Disc §	TRIM		Approximate Valve C _v : 1350 F _L : x _T :						
	Body Size: 8" ¶ Port Type: Single			Trim Style: Man. Standard Trim Size:						
	Inlet Conn Size: 8" ¶ Type: Flanged Class: 300			Inherent Characteristic: Equal Percentage						
	Outlet Conn Size: 8" ¶ Type: Flanged Class: 300			Balance Style: Full Open Position: 90°						
	Flange Facing: Flange Face Finish:			Guiding Style: Shaft						
	Body & Bonnet Material: 316 Stainless Steel ‡			Closure Member Material: 316 Stainless Steel						
	Bonnet Style:			Seat Ring Material: Incoloy 825 with Hard Chrome						
	Packing Style: Manufacturer's Recommendation			Seat Ring Retainer Material: 316 Stainless Steel						
	Packing Material: Manufacturer's Recommendation			Bearing Material: Stellite						
	Body/Bonnet Gasket Material:			Bushing/Cage Material:						
	Lub/Iso Valve Material: Lub Material:			Stem/Shaft Material: 17-4PH						
	Manufacturer: Metso Automation (or equal)			Flow Direction:						
	Model Number:			Seat Leakage Class: ANSI Class V						
	ACTUATOR			Type: Spring Return Piston, Pneumatic	INSTALLATION		Inlet Pipe Nominal Size: Est Wall Thick:			
Action Style: Fail Close		Outlet Pipe Nominal Size: Est Wall Thick:								
Service Type: On-Off		Est Insulation Type:								
Max Shutoff Press: In: 11.40 Out: 0 kg/cm2(g)		Est Insulation Thickness:								
Maximum Stroke Time to Open:		Installation:								
Maximum Stroke Time to Close: 5 Seconds		Design Pressure : 11.5 kg/cm2g (F.V)								
Stroke Cycle Service:		Design Temperature : 120 C								
DATA PAGE 1 REMARKS		§ The information is based on Neldisc L6 series. Valve selected shall be of double flanged construction with minimal flange-to-flange dimension and no pockets within the valve body.								
	¶ Line size valve.									
	‡ Preferred material is L grade stainless steel. Standard grade stainless steel that is not welded may be used for process wetted parts including the body. Stainless steel with weld repair is acceptable if low carbon (<= 0.03 %) filler material is used.									

ALL ITEMS SHALL COMPLY WITH THE GENERAL REQUIREMENTS SHOWN ON THE GENERAL REQUIREMENTS SECTION OF THIS SPECIFICATION.



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ENGINEERING

CONTROL VALVES

PROJECT SPECIFICATION

963766 — 616

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Item Number: 2 Quantity: 1 Valve Tag Number: UXV-23-0301 Data Page: 2 of 2																																																																																				
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DATA PAGE 2 REMARKS

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ENGINEERING**CONTROL VALVES****PROJECT SPECIFICATION**963766 — **616**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				
1	28AUG12	MVD	MW				

GENERAL	Item Number: 3 Quantity: 1 Valve Tag Number: LV-23-0801 Data Page: 1 of 2									
	Service: Oxidate Feed to Preflash Column Process Material: Hydrocarbon with aromatics P&ID Drawing: 04									
SIZING CONDITIONS	Process Data Case	Single Case		Single Case		Single Case		Single Case		
	Sizing Case	Normal ‡		Maximum ‡		Minimum ‡		Start-up ‡		
	Calculated C _v	298		354		121		58		
		Inlet	Outlet	Inlet	Outlet	Inlet	Outlet	Inlet	Outlet	Units
	Liquid Flow at Flowing Cond	241.9	206.4	266.1	227.1	121.0	103.2	58.1	49.5	m3/h
	Vapor Flow at Base Cond		4628		5090		2314		1111	normal m3/h
	Steam Flow									kg/h
	Temperature	82	58	82	58	82	58	82	58	°C
	Liquid Specific Gravity	0.846	0.874	0.846	0.874	0.846	0.874	0.846	0.874	----
	Vapor Spec. Gravity (Air=1)		4.071		4.071		4.071		4.071	----
	Pressure	5.17	-1.01	4.63	-1.00	6.91	-1.01	6.91	-1.01	kg/cm2(g)
	Liq Critical Press (Pseudo)	33.43		33.43		33.43		33.43		kg/cm2(a)
	Liquid Vapor Pressure	0.31		0.31		0.31		0.31		kg/cm2(a)
	Liquid Viscosity	0.607	0.900	0.607	0.900	0.607	0.900	0.607	0.900	cP
	Compressibility Z		1.000		1.000		1.000		1.000	
Specific Heats Cp/Cv (Real)		1.054		1.054		1.054		1.054		
Estimate Sound Press Level									dBA	
VALVE BODY & BONNET	Body Type: Eccentric Disc §									
	Body Size: 8" Port Type: Single									
	Inlet Conn Size: 8" Type: Flanged Class: 300									
	Outlet Conn Size: 8" Type: Flanged Class: 300									
	Flange Facing: Flange Face Finish:									
	Body & Bonnet Material: 316 Stainless Steel ‡‡									
	Bonnet Style:									
	Packing Style: Manufacturer's Recommendation †									
	Packing Material: Manufacturer's Recommendation									
	Body/Bonnet Gasket Material:									
	Lub/Iso Valve Material: Lub Material:									
	Manufacturer: Metso Automation (or equal)									
	Model Number:									
ACTUATOR	Type: Spring Return Piston, Pneumatic									
	Action Style: Fail Close									
	Service Type: Modulating									
	Max Shutoff Press: In: 11.40 Out: -1.01 kg/cm2(g)									
	Maximum Stroke Time to Open:									
	Maximum Stroke Time to Close: 5 Seconds									
	Stroke Cycle Service:									
TRIM	1	Approximate Valve C _v : 880		F _L : 0.6		x _T : 0.35				
		Trim Style: Man. Standard		Trim Size:						
		Inherent Characteristic: Equal Percentage								
		Balance Style:		Full Open Position: 90°						
		Guiding Style: Shaft								
		Closure Member Material: 316 Stainless Steel								
		Seat Ring Material: Incoloy 825 with Hard Chrome								
		Seat Ring Retainer Material: 316 Stainless Steel								
		Bearing Material: Stellite								
		Bushing/Cage Material:								
INSTALLATION		Stem/Shaft Material: 17-4PH								
		Flow Direction:								
		Seat Leakage Class: ANSI Class V								
		Inlet Pipe Nominal Size:		Est Wall Thick:						
		Outlet Pipe Nominal Size:		Est Wall Thick:						
		Est Insulation Type:								
		Est Insulation Thickness:								
		Installation:								
		Design Pressure : 11.5 kg/cm2g (F.V)								
		Design Temperature : 120 C								
DATA PAGE 1 REMARKS	§ The information is based on Neldisc L6 series. Valve selected shall be of double flanged construction with minimal flange-to-flange dimension and no pockets within the valve body.									
	‡‡ Preferred material is L grade stainless steel. Standard grade stainless steel that is not welded may be used for process wetted parts including the body. Stainless steel with weld repair is acceptable if low carbon (<= 0.03 %) filler material is used.									
	† Provide packing suitable for vacuum service to minimize air leakage into the process line.									
	‡ The calculated C _v for flashing service is determined by adding the Calculated C _v for the liquid and the vapor (or steam) at the outlet conditions of the valve.									
ALL ITEMS SHALL COMPLY WITH THE GENERAL REQUIREMENTS SHOWN ON THE GENERAL REQUIREMENTS SECTION OF THIS SPECIFICATION.										



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ENGINEERING

CONTROL VALVES

PROJECT SPECIFICATION

963766 — 616

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Item Number: 3 Quantity: 1 Valve Tag Number: LV-23-0801 Data Page: 2 of 2																																																																																										
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	Operation Style: †		POSITION TRANSMITTER	Position Transmitter Tag Number:																																																																																						
	Coil Configuration: Single			Sensor Type:																																																																																						
	Operating Voltage: \$\$ Frequency:			Output Signal Range:																																																																																						
Action Coil 1 Deenergized: Vent air from valve	Travel Measurement:	Mounting Location:																																																																																								
Action Coil 1 Energized: Supply air to valve	Manufacturer:																																																																																									
Action Coil 2 Deenergized:	LOCK-UP VALVE	Action Style:																																																																																								
Action Coil 2 Energized:		Mounting Location:																																																																																								
Manual Reset:		Model Number:																																																																																								
Manual Operator:	Manufacturer:																																																																																									
Mounting Location: Nearest item to actuator in air line. †	QUICK EXHAUST	Connection Size/Type:																																																																																								
Manufacturer: ASCO (or equal)		Body Material:																																																																																								
		Diaphragm Material:																																																																																								
POSITIONER	Positioner Tag Number:	Exhaust Action:																																																																																								
	Positioner Type: Digital	Mounting Location:																																																																																								
	Positioner Input Signal Range: 4 - 20 mA	Manufacturer:																																																																																								
AIR FILTER	Positioner Output Signal Range: 0 - 100 %	VOLUME BOOSTER	Connection Size/Type:																																																																																							
	Manufacturer:		Body Material:																																																																																							
	Quantity: one		Seat Ring Material:																																																																																							
TUBING	Type: Filter/Regulator	Diaphragm Material:																																																																																								
	Manufacturer:	Mounting Location:																																																																																								
	Tubing Size:	Manufacturer:																																																																																								
Tubing Material:																																																																																										
† The solenoid valve shall be mounted and tubed by the control valve vendor. The solenoid valve size and type, the tubing size, the actuator air connection size, and the air/filter regulator capacity shall be determined by the control valve vendor to meet the maximum stroking time requirements specified for the control valve. Pilot operated solenoid valves shall be suitable for operation in low pressure control signal air. \$\$ The Contractor shall coordinate the solenoid valve operating voltage and coil watt rating with the supplier of the Concentration Section Shutdown System (UC-2301). See UOP Project Specification 660, this project.																																																																																										

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ENGINEERING**CONTROL VALVES****PROJECT SPECIFICATION**963766 — **616**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

GENERAL	Item Number: 4 Quantity: 1 Valve Tag Number: UXV-23-0401 Data Page: 1 of 2									
	Service: Cumene Quench to Oxidate Feed Line Process Material: Hydrocarbon with aromatics P&ID Drawing: 04									
SIZING CONDITIONS	Process Data Case	General		General						
	Sizing Case	Open		Closed						
	Calculated C _v									
		Inlet	Outlet	Inlet	Outlet	Inlet	Outlet	Inlet	Outlet	Units
	Liquid Flow at Flowing Cond									m3/h
	Vapor Flow at Base Cond									normal m3/h
	Steam Flow									kg/h
	Temperature	46		46						°C
	Liquid Specific Gravity									----
	Vapor Spec. Gravity (Air=1)									----
	Pressure	-0.64	-0.64	-0.64	6.93					kg/cm2(g)
	Liq Critical Press (Pseudo)									kg/cm2(a)
	Liquid Vapor Pressure									kg/cm2(a)
	Liquid Viscosity									cP
	Compressibility Z									
Specific Heats Cp/Cv (Real)										
Estimate Sound Press Level									dBA	
VALVE BODY & BONNET	Body Type: Eccentric Disc §	TRIM		Approximate Valve C _v : 180 F _L : x _T :						
	Body Size: 4" ¶¶ Port Type: Single			Trim Style: Man. Standard Trim Size:						
	Inlet Conn Size: 4" ¶¶ Type: Flanged Class: 300			Inherent Characteristic: Equal Percentage						
	Outlet Conn Size: 4" ¶¶ Type: Flanged Class: 300			Balance Style: Full Open Position: 90°						
	Flange Facing: Flange Face Finish:			Guiding Style: Shaft						
	Body & Bonnet Material: 316 Stainless Steel ‡			Closure Member Material: 316 Stainless Steel						
	Bonnet Style:			Seat Ring Material: Incoloy 825 with Hard Chrome						
	Packing Style: Manufacturer's Recommendation †			Seat Ring Retainer Material: 316 Stainless Steel						
	Packing Material: Manufacturer's Recommendation			Bearing Material: Stellite						
	Body/Bonnet Gasket Material:			Bushing/Cage Material:						
	Lub/Iso Valve Material: Lub Material:			Stem/Shaft Material: 17-4PH						
	Manufacturer: Metso Automation (or equal)			Flow Direction:						
	Model Number:			Seat Leakage Class: ANSI Class V						
ACTUATOR	Type: Spring Return Piston, Pneumatic	INSTALLATION		Inlet Pipe Nominal Size: Est Wall Thick:						
	Action Style: Fail Open			Outlet Pipe Nominal Size: Est Wall Thick:						
	Service Type: On-Off			Est Insulation Type:						
	Max Shutoff Press: In: -0.64 Out: 11.40 kg/cm2(g)			Est Insulation Thickness:						
	Maximum Stroke Time to Open: 5 Seconds			Installation:						
	Maximum Stroke Time to Close:			Design Pressure : 11.5 kg/cm2g (F.V)						
	Stroke Cycle Service:			Design Temperature : 120 C						
	DATA PAGE 1 REMARKS			§ The information is based on Neldisc L6 series. Valve selected shall be of double flanged construction with minimal flange-to-flange dimension and no pockets within the valve body.						
‡ Preferred material is L grade stainless steel. Standard grade stainless steel that is not welded may be used for process wetted parts including the body. Stainless steel with weld repair is acceptable if low carbon (<= 0.03 %) filler material is used.										
¶¶ Line size valve.										
† Provide packing suitable for vacuum service to minimize air leakage into the process line.										

ALL ITEMS SHALL COMPLY WITH THE GENERAL REQUIREMENTS SHOWN ON THE GENERAL REQUIREMENTS SECTION OF THIS SPECIFICATION.



CONTROL VALVES

PROJECT SPECIFICATION

963766 — 616

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Note: The information in this document is confidential and the property of UOP LLC and must not be disclosed to others or reproduced in any manner or used for any purpose whatsoever without its written permission.

		Item Number: 4 Quantity: 1	Valve Tag Number: UXV-23-0401		Data Page: 2 of 2		
TRANSDUCER		Transducer Tag Number:	HANDWHEEL		Handwheel Position:		
		Transducer Input Signal Range:			Usage:		
		Transducer Output Signal Range:					
		Manufacturer:					
SOLENOID VALVE		Solenoid Valve Tag Number: UXY-23-0401	POSITION SWITCH		Open Position Switch Tag Number:		
		Body Type: 3-Way Body Material: Brass			Close Position Switch Tag Number:		
		Seat Material: Stainless Steel Disc Material: Stainless Steel			Sensor Type:		
		Connection Size/Type: †			Contact Form:		
		Packing Style: Packless			Open Position Switch Contact Action:		
		Fluid: Instrument Air			Close Position Switch Contact Action:		
		Pressure: Contractor to determine			Contact Load: Contact Rating:		
		Minimum Pressure Port Orifice C _v : MAX Size: MAX			Mounting Location:		
		Minimum Exhaust Port Orifice C _v : MAX Size: MAX			Manufacturer:		
		Operation Style: †		POSITION TRANSMITTER		Position Transmitter Tag Number:	
		Coil Configuration: Single				Sensor Type:	
		Operating Voltage: §§ Frequency:				Output Signal Range:	
		Action Coil 1 Deenergized: Vent air from valve				Travel Measurement:	
		Action Coil 1 Energized: Supply air to valve				Mounting Location:	
		Action Coil 2 Deenergized:				Manufacturer:	
		Action Coil 2 Energized:		LOCK-UP VALVE		Action Style:	
		Manual Reset:				Mounting Location:	
		Manual Operator:				Model Number:	
	Mounting Location: Nearest item to actuator in air line. †		Manufacturer:				
	Manufacturer: ASCO (or equal)	QUICK EXHAUST		Connection Size/Type:			
POSITIONER			Positioner Tag Number:		Body Material:		
			Positioner Type:		Diaphragm Material:		
			Positioner Input Signal Range:		Exhaust Action:		
		Positioner Output Signal Range:		Mounting Location:			
	Manufacturer:		Manufacturer:				
AIR FILTER		Quantity: one	VOLUME BOOSTER		Connection Size/Type:		
		Type: Filter/Regulator			Body Material:		
		Manufacturer:			Seat Ring Material:		
					Diaphragm Material:		
TUBING		Tubing Size:		Mounting Location:			
		Tubing Material:		Manufacturer:			
DATA PAGE 2 REMARKS		† The solenoid valve shall be mounted and tubed by the control valve vendor. The solenoid valve size and type, the tubing size, the actuator air connection size, and the air/filter regulator capacity shall be determined by the control valve vendor to meet the maximum stroking time requirements specified for the control valve.					
		§§ The Contractor shall coordinate the solenoid valve operating voltage and coil watt rating with the supplier of the Concentration Shutdown System (UC-2301). See UOP Project Specification 660, this project.					



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CONTROL VALVES

PROJECT SPECIFICATION

963766 — 616

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

GENERAL	Item Number: 5 Quantity: 1 Valve Tag Number: HV-23-0401 Data Page: 1 of 2																																																																																																																																																																										
	Service: Cumene Quench to Preflash Column Bottom (Manual) Process Material: Hydrocarbon with aromatics P&ID Drawing: 04																																																																																																																																																																										
SIZING CONDITIONS	<table border="1"> <tr> <th>Process Data Case</th> <th colspan="2">General</th> <th colspan="2">General</th> <th colspan="2"></th> <th colspan="2"></th> <th></th> </tr> <tr> <th>Sizing Case</th> <th colspan="2">Open</th> <th colspan="2">Closed</th> <th colspan="2"></th> <th colspan="2"></th> <th></th> </tr> <tr> <td>Calculated C_v</td> <td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td> </tr> <tr> <td></td> <td>Inlet</td><td>Outlet</td><td>Inlet</td><td>Outlet</td><td>Inlet</td><td>Outlet</td><td>Inlet</td><td>Outlet</td><td>Units</td> </tr> <tr> <td>Liquid Flow at Flowing Cond</td> <td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>m3/h</td> </tr> <tr> <td>Vapor Flow at Base Cond</td> <td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>normal m3/h</td> </tr> <tr> <td>Steam Flow</td> <td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>kg/h</td> </tr> <tr> <td>Temperature</td> <td>46</td><td></td><td>46</td><td></td><td></td><td></td><td></td><td></td><td>°C</td> </tr> <tr> <td>Liquid Specific Gravity</td> <td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>----</td> </tr> <tr> <td>Vapor Spec. Gravity (Air=1)</td> <td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>----</td> </tr> <tr> <td>Pressure</td> <td>-0.46</td><td>-0.46</td><td>-0.46</td><td>-1.01</td><td></td><td></td><td></td><td></td><td>kg/cm2(g)</td> </tr> <tr> <td>Liq Critical Press (Pseudo)</td> <td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>kg/cm2(a)</td> </tr> <tr> <td>Liquid Vapor Pressure</td> <td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>kg/cm2(a)</td> </tr> <tr> <td>Liquid Viscosity</td> <td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>cP</td> </tr> <tr> <td>Compressibility Z</td> <td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td> </tr> <tr> <td>Specific Heats Cp/Cv (Real)</td> <td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td> </tr> <tr> <td>Estimate Sound Press Level</td> <td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>dBA</td> </tr> </table>	Process Data Case	General		General							Sizing Case	Open		Closed							Calculated C _v											Inlet	Outlet	Inlet	Outlet	Inlet	Outlet	Inlet	Outlet	Units	Liquid Flow at Flowing Cond									m3/h	Vapor Flow at Base Cond									normal m3/h	Steam Flow									kg/h	Temperature	46		46						°C	Liquid Specific Gravity									----	Vapor Spec. Gravity (Air=1)									----	Pressure	-0.46	-0.46	-0.46	-1.01					kg/cm2(g)	Liq Critical Press (Pseudo)									kg/cm2(a)	Liquid Vapor Pressure									kg/cm2(a)	Liquid Viscosity									cP	Compressibility Z										Specific Heats Cp/Cv (Real)										Estimate Sound Press Level									dBA
	Process Data Case	General		General																																																																																																																																																																							
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Estimate Sound Press Level									dBA																																																																																																																																																																		
VALVE BODY & BONNET	Body Type: Eccentric Disc § Body Size: 3" ¶¶ Port Type: Single Inlet Conn Size: 3" ¶¶ Type: Flanged Class: 300 Outlet Conn Size: 3" ¶¶ Type: Flanged Class: 300 Flange Facing: Flange Face Finish: Body & Bonnet Material: 316 Stainless Steel ‡ Bonnet Style: Packing Style: Manufacturer's Recommendation † Packing Material: Manufacturer's Recommendation Body/Bonnet Gasket Material: Lub/Iso Valve Material: Lub Material: Manufacturer: Metso Automation (or equal) Model Number:	TRIM	Approximate Valve C _v : 245 F _L : x _T : Trim Style: Man. Standard Trim Size: Inherent Characteristic: Equal Percentage Balance Style: Full Open Position: 90° Guiding Style: Shaft Closure Member Material: 316 Stainless Steel Seat Ring Material: Incoloy 825 with Hard Chrome Seat Ring Retainer Material: 316 Stainless Steel Bearing Material: Stellite Bushing/Cage Material: Stem/Shaft Material: 17-4PH Flow Direction: Seat Leakage Class: ANSI Class V																																																																																																																																																																								
	ACTUATOR		INSTALLATION	Type: Spring Diaphragm, Pneumatic Action Style: Fail Close Service Type: Modulating Max Shutoff Press: In: 3.85 Out: -1.02 kg/cm2(g) Maximum Stroke Time to Open: Maximum Stroke Time to Close: Stroke Cycle Service:	Inlet Pipe Nominal Size: Est Wall Thick: Outlet Pipe Nominal Size: Est Wall Thick: Est Insulation Type: Est Insulation Thickness: Installation: Design Pressure : 3.5 kg/cm2g (F.V) Design Temperature : 120 C																																																																																																																																																																						
				§ The information is based on Neldisc LG series. Valve selected shall be of lugged construction and no pockets within the valve body.																																																																																																																																																																							
				‡ Preferred material is L grade stainless steel. Standard grade stainless steel that is not welded may be used for process wetted parts including the body. Stainless steel with weld repair is acceptable if low carbon (<= 0.03 %) filler material is used.																																																																																																																																																																							
				¶¶ Line size valve.																																																																																																																																																																							
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ALL ITEMS SHALL COMPLY WITH THE GENERAL REQUIREMENTS SHOWN ON THE GENERAL REQUIREMENTS SECTION OF THIS SPECIFICATION.



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CONTROL VALVES

PROJECT SPECIFICATION

963766 — **616**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Item Number: 5 Quantity: 1		Valve Tag Number: HV-23-0401		Data Page: 2 of 2	
TRANSducer	Transducer Tag Number:		HANDWHEEL	Handwheel Position:	
	Transducer Input Signal Range:			Usage:	
	Transducer Output Signal Range:				
	Manufacturer:				
SOLENOID VALVE	Solenoid Valve Tag Number:		POSITION SWITCH	Open Position Switch Tag Number:	
	Body Type: Body Material:			Close Position Switch Tag Number:	
	Seat Material: Disc Material:			Sensor Type:	
	Connection Size/Type:			Contact Form:	
	Packing Style:			Open Position Switch Contact Action:	
	Fluid:			Close Position Switch Contact Action:	
	Pressure:			Contact Load: Contact Rating:	
	Minimum Pressure Port Orifice C _v : Size:			Mounting Location:	
	Minimum Exhaust Port Orifice C _v : Size:			Manufacturer:	
	Operation Style:			POSITION TRANSMITTER	Position Transmitter Tag Number:
	Coil Configuration:		Sensor Type:		
	Operating Voltage: Frequency:		Output Signal Range:		
	Action Coil 1 Deenergized:		Travel Measurement:		
	Action Coil 1 Energized:		Mounting Location:		
	Action Coil 2 Deenergized:		LOCK-UP VALVE	Manufacturer:	
	Action Coil 2 Energized:			Action Style:	
	Manual Reset:			Mounting Location:	
	Manual Operator:		QUICK EXHAUST	Model Number:	
	Mounting Location:			Manufacturer:	
	Manufacturer:			Connection Size/Type:	
POSITIONER	Positioner Tag Number:		VOLUME BOOSTER	Body Material:	
	Positioner Type: Digital			Diaphragm Material:	
	Positioner Input Signal Range: 4 - 20 mA			Exhaust Action:	
AIR FILTER	Positioner Output Signal Range: 0 - 100 %		TUBING	Mounting Location:	
	Manufacturer:			Manufacturer:	
	Quantity: one			Connection Size/Type:	
TUBING	Type: Filter/Regulator		TUBING	Body Material:	
	Manufacturer:			Seat Ring Material:	
	Tubing Size:			Diaphragm Material:	
DATA PAGE 2 REMARKS	Tubing Material:		TUBING	Mounting Location:	
				Manufacturer:	

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CONTROL VALVES

PROJECT SPECIFICATION

963766 — **616**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

GENERAL	Item Number: 6 Quantity: 1		Valve Tag Number: FV-23-0403		Data Page: 1 of 2			
	Service: Preflash Column Reflux		P&ID Drawing: 04					
Process Material: Hydrocarbon with aromatics								
SIZING CONDITIONS	Process Data Case	Single Case		Single Case		Single Case		
	Sizing Case	Normal		Maximum		Minimum		
	Calculated C _v	10.3		11.8		4.67		
		Inlet	Outlet	Inlet	Outlet	Inlet	Outlet	Units
	Liquid Flow at Flowing Cond	22.9		25.2		11.5		m ³ /h
	Vapor Flow at Base Cond							normal m ³ /h
	Steam Flow							kg/h
	Temperature	39		39		39		°C
	Liquid Specific Gravity	0.848		0.848		0.848		----
	Vapor Spec. Gravity (Air=1)							----
	Pressure	7.05	1.37	6.66	1.41	8.18	1.24	kg/cm ² (g)
	Liq Critical Press (Pseudo)	32.74		32.74		32.74		kg/cm ² (a)
	Liquid Vapor Pressure	0.02		0.02		0.02		kg/cm ² (a)
	Liquid Viscosity	0.622		0.622		0.622		cP
	Compressibility Z							
Specific Heats Cp/Cv (Real)								
Estimate Sound Press Level							dBA	
VALVE BODY & BONNET	Body Type: Globe §		Approximate Valve C _v : 18		F _L : 0.9 x _T : 0.72			
	Body Size: 1-1/2" Port Type: Single		Trim Style: Man. Standard Trim Size:					
	Inlet Conn Size: 1-1/2" Type: Flanged Class: 300		Inherent Characteristic: Linear					
	Outlet Conn Size: 1-1/2" Type: Flanged Class: 300		Balance Style: Unbalanced Full Open Position: 100 %					
	Flange Facing: Raised Face Flange Face Finish:		Guiding Style: Top					
	Body & Bonnet Material: 316 Stainless Steel ‡		Closure Member Material: 316 Stainless Steel					
	Bonnet Style: Integral or Bolted		Seat Ring Material: 316 Stainless Steel					
	Packing Style: Manufacturer's Recommendation †		Seat Ring Retainer Material: 304 Stainless Steel					
	Packing Material: Manufacturer's Recommendation		Bearing Material:					
	Body/Bonnet Gasket Material: Manufacturer's Recommend		Bushing/Cage Material: Stellite					
	Lub/Iso Valve Material: Lub Material:		Stem/Shaft Material: 316 Stainless Steel					
	ACTUATOR	Type: Spring Diaphragm, Pneumatic		Flow Direction:				
Action Style: Fail Open		Seat Leakage Class: ANSI Class IV						
Service Type: Modulating		Inlet Pipe Nominal Size: Est Wall Thick:						
Max Shutoff Press: In: 13.55 Out: -1.02 kg/cm ² (g)		Outlet Pipe Nominal Size: Est Wall Thick:						
Maximum Stroke Time to Open:		Est Insulation Type:						
Maximum Stroke Time to Close:		Est Insulation Thickness:						
Stroke Cycle Service:		Installation:						
DATA PAGE 1 REMARKS	Design Pressure : 13.55 kg/cm ² g (F.V)							
	Design Temperature : 120 C							
	§ The information is based on Masoneilan 21000 series. ‡ Preferred material is L grade stainless steel. Standard grade stainless steel that is not welded may be used for process wetted parts including the body. Stainless steel with weld repair is acceptable if low carbon (<= 0.03 %) filler material is used. † Provide packing suitable for vacuum service to minimize air leakage into the process line.							

ALL ITEMS SHALL COMPLY WITH THE GENERAL REQUIREMENTS SHOWN ON THE GENERAL REQUIREMENTS SECTION OF THIS SPECIFICATION.

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CONTROL VALVES

PROJECT SPECIFICATION

963766 — **616**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Item Number: 6 Quantity: 1		Valve Tag Number: FV-23-0403		Data Page: 2 of 2	
TRANS-DUCER	Transducer Tag Number:		HANDWHEEL	Handwheel Position:	
	Transducer Input Signal Range:			Usage:	
	Transducer Output Signal Range:				
	Manufacturer:				
SOLENOID VALVE	Solenoid Valve Tag Number:		POSITION SWITCH	Open Position Switch Tag Number:	
	Body Type: Body Material:			Close Position Switch Tag Number:	
	Seat Material: Disc Material:			Sensor Type:	
	Connection Size/Type:			Contact Form:	
	Packing Style:			Open Position Switch Contact Action:	
	Fluid:			Close Position Switch Contact Action:	
	Pressure:			Contact Load: Contact Rating:	
	Minimum Pressure Port Orifice C _v : Size:			Mounting Location:	
	Minimum Exhaust Port Orifice C _v : Size:			Manufacturer:	
	Operation Style:			POSITION TRANSMITTER	Position Transmitter Tag Number:
	Coil Configuration:		Sensor Type:		
	Operating Voltage: Frequency:		Output Signal Range:		
	Action Coil 1 Deenergized:		Travel Measurement:		
	Action Coil 1 Energized:		Mounting Location:		
	Action Coil 2 Deenergized:		Manufacturer:		
	Action Coil 2 Energized:		LOCK-UP VALVE	Action Style:	
	Manual Reset:			Mounting Location:	
	Manual Operator:			Model Number:	
Mounting Location:		Manufacturer:			
Manufacturer:		QUICK EXHAUST	Connection Size/Type:		
			Body Material:		
			Diaphragm Material:		
			Exhaust Action:		
POSITIONER	Positioner Tag Number:		Mounting Location:		
	Positioner Type: Digital		Manufacturer:		
	Positioner Input Signal Range: 4 - 20 mA				
AIR FILTER	Positioner Output Signal Range: 0 - 100 %		VOLUME BOOSTER	Connection Size/Type:	
	Manufacturer:			Body Material:	
	Quantity: one			Seat Ring Material:	
Type: Filter/Regulator		Diaphragm Material:			
TUBING	Manufacturer:		Mounting Location:		
	Tubing Size:		Manufacturer:		
Tubing Material:					
DATA PAGE 2 REMARKS					

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CONTROL VALVES

PROJECT SPECIFICATION

963766 — **616**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Note: The information in this document is confidential and the property of UOP LLC and must not be disclosed to others or reproduced in any manner or used for any purpose whatsoever without its written permission.

GENERAL	Item Number: 7 Quantity: 1		Valve Tag Number: FV-23-0501				Data Page: 1 of 2			
	Service: S1.5 Steam to Preflash Column Steam Heater									
	Process Material: Steam P&ID Drawing: 05									
SIZING CONDITIONS	Process Data Case	Design		Design						
	Sizing Case	Normal		Maximum						
	Calculated C _v	472		733						
		Inlet	Outlet	Inlet	Outlet	Inlet	Outlet	Inlet	Outlet	Units
	Liquid Flow at Flowing Cond									m3/h
	Vapor Flow at Base Cond									normal m3/h
	Steam Flow	11388		14235						kg/h
	Temperature	135		123						°C
	Liquid Specific Gravity									----
	Vapor Spec. Gravity (Air=1)	0.622		0.622						----
	Pressure	1.50	0.66	1.20	0.66					kg/cm2(g)
	Liq Critical Press (Pseudo)									kg/cm2(a)
	Liquid Vapor Pressure									kg/cm2(a)
	Liquid Viscosity									cP
	Compressibility Z	0.974		0.974						
	Specific Heats Cp/Cv (Real)	1.320		1.321						
	Estimate Sound Press Level									dBA
	VALVE BODY & BONNET	Body Type: Globe §				Approximate Valve C _v : 1280 F _L : 0.9 x _T : 0.72				
Body Size: 16" Port Type: Single				Trim Style: Man. Standard Trim Size:						
Inlet Conn Size: 16" Type: Flanged Class: 150				Inherent Characteristic: Linear						
Outlet Conn Size: 16" Type: Flanged Class: 150				Balance Style: Balanced Full Open Position: 100 %						
Flange Facing: Raised Face Flange Face Finish:				Guiding Style: Cage						
Body & Bonnet Material: Carbon Steel				Closure Member Material: 17-4 PH Stainless Steel						
Bonnet Style: Integral or Bolted				Seat Ring Material: 316 SS with Stellite sheath						
Packing Style: Manufacturer's Recommendation				Seat Ring Retainer Material:						
Packing Material: Manufacturer's Recommendation				Bearing Material:						
Body/Bonnet Gasket Material: Manufacturer's Recommend				Bushing/Cage Material: CA6NM hard chrome plated						
Lub/Iso Valve Material: Lub Material:				Stem/Shaft Material: 17-4 PH Stainless Steel						
Manufacturer: Masoneilan (or equal)				Flow Direction:						
Model Number:				Seat Leakage Class: ANSI Class V						
ACTUATOR		Type: Spring Diaphragm, Pneumatic				INSTALLATION				
	Action Style: Fail Close									
	Service Type: Modulating									
	Max Shutoff Press: In: 2.00 Out: 0 kg/cm2(g)									
	Maximum Stroke Time to Open:									
	Maximum Stroke Time to Close: 5 Seconds									
	Stroke Cycle Service:									
DATA PAGE 1 REMARKS	§ The information is based on Masoneilan 41000 series.									
	UOP calculations indicate that the noise level of this valve may exceed 85 dBA. The estimated noise level shall be verified based on final design criteria and Vendor/Contractor's detailed calculations. It is the responsibility of the detailed design Contractor to select a valve size, style and/or trim style that complies with all the specified process requirements, including noise limitation.									

ALL ITEMS SHALL COMPLY WITH THE GENERAL REQUIREMENTS SHOWN ON THE GENERAL REQUIREMENTS SECTION OF THIS SPECIFICATION.



CONTROL VALVES

PROJECT SPECIFICATION

963766 — **616**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Item Number: 7 Quantity: 1 Valve Tag Number: FV-23-0501 Data Page: 2 of 2	
TRANS-DUCER Transducer Tag Number: Transducer Input Signal Range: Transducer Output Signal Range: Manufacturer:	HANDWHEEL Handwheel Position: Usage:
SOLENOID VALVE Solenoid Valve Tag Number: FXY-23-0501 Body Type: 3-Way Body Material: Brass Seat Material: Stainless Steel Disc Material: Stainless Steel Connection Size/Type: † Packing Style: Packless Fluid: Control Signal Air Pressure: Contractor to determine Minimum Pressure Port Orifice C_v: MAX Size: MAX Minimum Exhaust Port Orifice C_v: MAX Size: MAX Operation Style: † Coil Configuration: Single Operating Voltage: §§ Frequency: Action Coil 1 Deenergized: Vent air from valve Action Coil 1 Energized: Supply air to valve Action Coil 2 Deenergized: Action Coil 2 Energized: Manual Reset: Manual Operator: Mounting Location: Nearest item to actuator in air line. † Manufacturer: ASCO (or equal)	POSITION SWITCH Open Position Switch Tag Number: Close Position Switch Tag Number: Sensor Type: Contact Form: Open Position Switch Contact Action: Close Position Switch Contact Action: Contact Load: Contact Rating: Mounting Location: Manufacturer: POSITION TRANSMITTER Position Transmitter Tag Number: Sensor Type: Output Signal Range: Travel Measurement: Mounting Location: Manufacturer: LOCK-UP VALVE Action Style: Mounting Location: Model Number: Manufacturer: QUICK EXHAUST Connection Size/Type: Body Material: Diaphragm Material: Exhaust Action: Mounting Location: Manufacturer: VOLUME BOOSTER Connection Size/Type: Body Material: Seat Ring Material: Diaphragm Material: Mounting Location: Manufacturer:
POSITIONER Positioner Tag Number: Positioner Type: Digital Positioner Input Signal Range: 4 - 20 mA Positioner Output Signal Range: 0 - 100 % Manufacturer:	
AIR FILTER Quantity: one Type: Filter/Regulator Manufacturer:	
TUBING Tubing Size: Tubing Material:	
DATA PAGE 2 REMARKS † The solenoid valve shall be mounted and tubed by the control valve vendor. The solenoid valve size and type, the tubing size, the actuator air connection size, and the air/filter regulator capacity shall be determined by the control valve vendor to meet the maximum stroking time requirements specified for the control valve. Pilot operated solenoid valves shall be suitable for operation in low pressure control signal air. §§ The Contractor shall coordinate the solenoid valve operating voltage and coil watt rating with the supplier of the Concentration Section Shutdown System (UC-2301). See UOP Project Specification 660, this project.	

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CONTROL VALVES

PROJECT SPECIFICATION

963766 — **616**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

GENERAL	Item Number: 8 Quantity: 1 Valve Tag Number: UXV-23-0502 Data Page: 1 of 2																																																																																																																																																																
	Service: S1.5 Steam to Preflash Column Steam Heater Shutoff Process Material: Steam P&ID Drawing: 05																																																																																																																																																																
SIZING CONDITIONS	<table border="1"> <thead> <tr> <th>Process Data Case</th> <th colspan="2">General</th> <th colspan="2">General</th> <th colspan="2"></th> <th colspan="2"></th> <th></th> </tr> <tr> <th>Sizing Case</th> <th colspan="2">Open</th> <th colspan="2">Closed</th> <th colspan="2"></th> <th colspan="2"></th> <th></th> </tr> <tr> <th>Calculated C_v</th> <th>Inlet</th> <th>Outlet</th> <th>Inlet</th> <th>Outlet</th> <th>Inlet</th> <th>Outlet</th> <th>Inlet</th> <th>Outlet</th> <th>Units</th> </tr> </thead> <tbody> <tr> <td>Liquid Flow at Flowing Cond</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>m3/h</td> </tr> <tr> <td>Vapor Flow at Base Cond</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>normal m3/h</td> </tr> <tr> <td>Steam Flow</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>kg/h</td> </tr> <tr> <td>Temperature</td> <td>135</td> <td></td> <td>135</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>°C</td> </tr> <tr> <td>Liquid Specific Gravity</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>----</td> </tr> <tr> <td>Vapor Spec. Gravity (Air=1)</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>----</td> </tr> <tr> <td>Pressure</td> <td>1.50</td> <td>1.50</td> <td>1.50</td> <td>0</td> <td></td> <td></td> <td></td> <td></td> <td>kg/cm2(g)</td> </tr> <tr> <td>Liq Critical Press (Pseudo)</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>kg/cm2(a)</td> </tr> <tr> <td>Liquid Vapor Pressure</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>kg/cm2(a)</td> </tr> <tr> <td>Liquid Viscosity</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>cP</td> </tr> <tr> <td>Compressibility Z</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> <tr> <td>Specific Heats Cp/Cv (Real)</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> <tr> <td>Estimate Sound Press Level</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>dBA</td> </tr> </tbody> </table>	Process Data Case	General		General							Sizing Case	Open		Closed							Calculated C _v	Inlet	Outlet	Inlet	Outlet	Inlet	Outlet	Inlet	Outlet	Units	Liquid Flow at Flowing Cond									m3/h	Vapor Flow at Base Cond									normal m3/h	Steam Flow									kg/h	Temperature	135		135						°C	Liquid Specific Gravity									----	Vapor Spec. Gravity (Air=1)									----	Pressure	1.50	1.50	1.50	0					kg/cm2(g)	Liq Critical Press (Pseudo)									kg/cm2(a)	Liquid Vapor Pressure									kg/cm2(a)	Liquid Viscosity									cP	Compressibility Z										Specific Heats Cp/Cv (Real)										Estimate Sound Press Level									dBA
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VALVE BODY & BONNET	Body Type: Eccentric Disc § Body Size: 18" # Port Type: Single Inlet Conn Size: 18" # Type: Flanged Class: 150 Outlet Conn Size: 18" # Type: Flanged Class: 150 Flange Facing: Flange Face Finish: Body & Bonnet Material: Carbon Steel Bonnet Style: Packing Style: Manufacturer's Recommendation Packing Material: Manufacturer's Recommendation Body/Bonnet Gasket Material: Lub/Iso Valve Material: Lub Material: Manufacturer: Masoneilan (or equal) Model Number:	TRIM	Approximate Valve C _v : 11400 F _L : x _T : Trim Style: Man. Standard Trim Size: Inherent Characteristic: Equal Percentage Balance Style: Full Open Position: 90° Guiding Style: Shaft Closure Member Material: 316 Stainless Steel Seat Ring Material: RTFE Seat Ring Retainer Material: Carbon Steel Bearing Material: Nitrided Stainless Steel Bushing/Cage Material: Stem/Shaft Material: 17-4PH Stainless Steel Flow Direction: Flow to close Seat Leakage Class: ANSI Class VI																																																																																																																																																														
	ACTUATOR		INSTALLATION	Type: Spring Diaph, Piston/Double Act Piston, Pneu ## Action Style: Fail Close Service Type: On-Off Max Shutoff Press: In: 2.00 Out: 0 kg/cm2(g) Maximum Stroke Time to Open: Maximum Stroke Time to Close: 5 Seconds Stroke Cycle Service:	Inlet Pipe Nominal Size: Est Wall Thick: Outlet Pipe Nominal Size: Est Wall Thick: Est Insulation Type: Est Insulation Thickness: Installation: Design Pressure : 7.0 kg/cm2g (F.V) Design Temperature : 195 C																																																																																																																																																												
				§ The information is based on Masoneilan 39003 series.																																																																																																																																																													
				# Line size valve.																																																																																																																																																													
				## If a double acting piston actuator is provided for this valve, then a volume tank with the required accessories shall be supplied to move the control valve to the minimum signal position as specified, upon loss of instrument air pressure.																																																																																																																																																													
	DATA PAGE 1 REMARKS																																																																																																																																																																

ALL ITEMS SHALL COMPLY WITH THE GENERAL REQUIREMENTS SHOWN ON THE GENERAL REQUIREMENTS SECTION OF THIS SPECIFICATION.

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ENGINEERING

CONTROL VALVES

PROJECT SPECIFICATION

963766 — **616**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Item Number: 8 Quantity: 1 Valve Tag Number: UXV-23-0502 Data Page: 2 of 2	
TRANS-DUCER Transducer Tag Number: Transducer Input Signal Range: Transducer Output Signal Range: Manufacturer:	HANDWHEEL Handwheel Position: Usage:
SOLENOID VALVE Solenoid Valve Tag Number: UXY-23-0502 Body Type: 3-Way Body Material: Brass Seat Material: Stainless Steel Disc Material: Stainless Steel Connection Size/Type: † Packing Style: Packless Fluid: Instrument Air Pressure: Contractor to determine Minimum Pressure Port Orifice C_v: MAX Size: MAX Minimum Exhaust Port Orifice C_v: MAX Size: MAX Operation Style: † Coil Configuration: Single Operating Voltage: §§ Frequency: Action Coil 1 Deenergized: Vent air from valve Action Coil 1 Energized: Supply air to valve Action Coil 2 Deenergized: Action Coil 2 Energized: Manual Reset: Manual Operator: Mounting Location: Nearest item to actuator in air line. † Manufacturer: ASCO (or equal)	POSITION SWITCH Open Position Switch Tag Number: Close Position Switch Tag Number: Sensor Type: Contact Form: Open Position Switch Contact Action: Close Position Switch Contact Action: Contact Load: Contact Rating: Mounting Location: Manufacturer: POSITION TRANSMITTER Position Transmitter Tag Number: Sensor Type: Output Signal Range: Travel Measurement: Mounting Location: Manufacturer: LOCK-UP VALVE Action Style: Mounting Location: Model Number: Manufacturer: QUICK EXHAUST Connection Size/Type: Body Material: Diaphragm Material: Exhaust Action: Mounting Location: Manufacturer: VOLUME BOOSTER Connection Size/Type: Body Material: Seat Ring Material: Diaphragm Material: Mounting Location: Manufacturer:
POSITIONER Positioner Tag Number: Positioner Type: Positioner Input Signal Range: Positioner Output Signal Range: Manufacturer:	
AIR FILTER Quantity: one Type: Filter/Regulator Manufacturer:	
TUBING Tubing Size: Tubing Material:	
DATA PAGE 2 REMARKS † The solenoid valve shall be mounted and tubed by the control valve vendor. The solenoid valve size and type, the tubing size, the actuator air connection size, and the air/filter regulator capacity shall be determined by the control valve vendor to meet the maximum stroking time requirements specified for the control valve. §§ The Contractor shall coordinate the solenoid valve operating voltage and coil watt rating with the supplier of the Concentration Section Shutdown System (UC-2301). See UOP Project Specification 660, this project.	

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CONTROL VALVES

PROJECT SPECIFICATION

963766 — **616**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Note: The information in this document is confidential and the property of UOP LLC and must not be disclosed to others or reproduced in any manner or used for any purpose whatsoever without its written permission.

GENERAL	Item Number: 9 Quantity: 1 Valve Tag Number: UXV-23-0501 Data Page: 1 of 2							
	Service: Preflash Column Steam Heater Steam Line Vent to Atmosphere Process Material: Steam P&ID Drawing: 05							
SIZING CONDITIONS	Process Data Case	General	General					
	Sizing Case	Open	Closed					
	Calculated C _v							
		Inlet	Outlet	Inlet	Outlet	Inlet	Outlet	Units
	Liquid Flow at Flowing Cond							m3/h
	Vapor Flow at Base Cond							normal m3/h
	Steam Flow							kg/h
	Temperature	135		135				°C
	Liquid Specific Gravity							----
	Vapor Spec. Gravity (Air=1)							----
	Pressure	1.50	1.50	1.50	0			kg/cm2(g)
	Liq Critical Press (Pseudo)							kg/cm2(a)
	Liquid Vapor Pressure							kg/cm2(a)
	Liquid Viscosity							cP
	Compressibility Z							
Specific Heats Cp/Cv (Real)								
Estimate Sound Press Level							dBA	
VALVE BODY & BONNET	Body Type: Eccentric Disc §	Approximate Valve C _v : 451 F _L : x _T :						
	Body Size: 4" ± Port Type: Single	Trim Style: Man. Standard Trim Size:						
	Inlet Conn Size: 4" ± Type: Flanged Class: 300	Inherent Characteristic: Equal Percentage						
	Outlet Conn Size: 4" ± Type: Flanged Class: 300	Balance Style: Full Open Position: 90°						
	Flange Facing: Flange Face Finish:	Guiding Style: Shaft						
	Body & Bonnet Material: Carbon Steel	Closure Member Material: 316 Stainless Steel						
	Bonnet Style:	Seat Ring Material: RTFE						
	Packing Style: Manufacturer's Recommendation	Seat Ring Retainer Material: Carbon Steel						
	Packing Material: Manufacturer's Recommendation	Bearing Material: Nitrided Stainless Steel						
	Body/Bonnet Gasket Material:	Bushing/Cage Material:						
	Lub/Iso Valve Material: Lub Material:	Stem/Shaft Material: 17-4PH Stainless Steel						
	Manufacturer: Masoneilan (or equal)	Flow Direction: Flow to close						
	Model Number:	Seat Leakage Class: ANSI Class VI						
	ACTUATOR	Type: Spring Diaphragm, Pneumatic	Inlet Pipe Nominal Size: Est Wall Thick:					
Action Style: Fail Open		Outlet Pipe Nominal Size: Est Wall Thick:						
Service Type: On-Off		Est Insulation Type:						
Max Shutoff Press: In: 2.00 Out: 0 kg/cm2(g)		Est Insulation Thickness:						
Maximum Stroke Time to Open: 5 Seconds		Installation:						
Maximum Stroke Time to Close:		Design Pressure : 7.0 kg/cm2g (F.V)						
Stroke Cycle Service:		Design Temperature : 195 C						
DATA PAGE 1 REMARKS		§ The information is based on Masoneilan 39003 series.						
	± Line size valve.							
	Valve will normally be in the closed position. Valve opening will be intermittent and generally short-term.							

ALL ITEMS SHALL COMPLY WITH THE GENERAL REQUIREMENTS SHOWN ON THE GENERAL REQUIREMENTS SECTION OF THIS SPECIFICATION.



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CONTROL VALVES

PROJECT SPECIFICATION

963766 — **616**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Item Number: 9 Quantity: 1		Valve Tag Number: UXV-23-0501		Data Page: 2 of 2		
TRANSDUCER	Transducer Tag Number:	HANDWHEEL	Handwheel Position:			
	Transducer Input Signal Range:		Usage:			
	Transducer Output Signal Range:					
	Manufacturer:					
SOLENOID VALVE	Solenoid Valve Tag Number: UXY-23-0501	POSITION SWITCH	Open Position Switch Tag Number:			
	Body Type: 3-Way Body Material: Brass		Close Position Switch Tag Number:			
	Seat Material: Stainless Steel Disc Material: Stainless Steel		Sensor Type:			
	Connection Size/Type: †		Contact Form:			
	Packing Style: Packless		Open Position Switch Contact Action:			
	Fluid: Instrument Air		Close Position Switch Contact Action:			
	Pressure: Contractor to determine		Contact Load: Contact Rating:			
	Minimum Pressure Port Orifice C _v : MAX Size: MAX		Mounting Location:			
	Minimum Exhaust Port Orifice C _v : MAX Size: MAX		Manufacturer:			
	Operation Style: †					
	Coil Configuration: Single		POSITION TRANSMITTER	Position Transmitter Tag Number:		
	Operating Voltage: \$\$ Frequency:			Sensor Type:		
	Action Coil 1 Deenergized: Vent air from valve			Output Signal Range:		
	Action Coil 1 Energized: Supply air to valve			Travel Measurement:		
	Action Coil 2 Deenergized:		Mounting Location:			
	Action Coil 2 Energized:		Manufacturer:			
	Manual Reset:		LOCK-UP VALVE	Action Style:		
	Manual Operator:			Mounting Location:		
Mounting Location: Nearest item to actuator in air line. †	Model Number:					
Manufacturer: ASCO (or equal)	Manufacturer:					
POSITIONER	Positioner Tag Number:	QUICK EXHAUST	Connection Size/Type:			
	Positioner Type:		Body Material:			
	Positioner Input Signal Range:		Diaphragm Material:			
	Positioner Output Signal Range:		Exhaust Action:			
Manufacturer:	Mounting Location:					
AIR FILTER	Quantity: one	VOLUME BOOSTER	Connection Size/Type:			
	Type: Filter/Regulator		Body Material:			
	Manufacturer:		Seat Ring Material:			
			Diaphragm Material:			
TUBING	Tubing Size:		Mounting Location:			
	Tubing Material:		Manufacturer:			
DATA PAGE 2 REMARKS	† The solenoid valve shall be mounted and tubed by the control valve vendor. The solenoid valve size and type, the tubing size, the actuator air connection size, and the air/filter regulator capacity shall be determined by the control valve vendor to meet the maximum stroking time requirements specified for the control valve.					
	\$\$ The Contractor shall coordinate the solenoid valve operating voltage and coil watt rating with the supplier of the Concentration Section Shutdown System (UC-2301). See UOP Project Specification 660, this project.					

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CONTROL VALVES

PROJECT SPECIFICATION

963766 — **616**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

GENERAL	Item Number: 10 Quantity: 1		Valve Tag Number: LV-23-0501		Data Page: 1 of 2			
	Service: Preflash Column Steam Heater Condensate to Condensate Header							
	Process Material: Water and Steam		P&ID Drawing: 05A					
SIZING CONDITIONS	Process Data Case	Single Case		Single Case		Single Case		
	Sizing Case	Normal		Maximum		Minimum ‡		
	Calculated C _v	10.9		12.5		4.66		
		Inlet	Outlet	Inlet	Outlet	Inlet	Outlet	Units
	Liquid Flow at Flowing Cond	12.2		13.4		6.1	6.1	m3/h
	Vapor Flow at Base Cond							normal m3/h
	Steam Flow						5	kg/h
	Temperature	117		117		117	116	°C
	Liquid Specific Gravity	0.947		0.947		0.947	0.947	----
	Vapor Spec. Gravity (Air=1)						0.622	----
	Pressure	2.67	0.92	2.51	0.96	3.05	0.76	kg/cm2(g)
	Liq Critical Press (Pseudo)	225.55		225.55		225.55		kg/cm2(a)
	Liquid Vapor Pressure	1.81		1.81		1.81		kg/cm2(a)
	Liquid Viscosity	0.248		0.248		0.248	0.243	cP
	Compressibility Z						0.976	
	Specific Heats Cp/Cv (Real)						1.321	
	Estimate Sound Press Level							dBA
VALVE BODY & BONNET	Body Type: Globe ‡		Approximate Valve C _v : 19		F _L : 0.9 x _T : 0.72			
	Body Size: 2" † Port Type: Single		Trim Style: Man. Standard		Trim Size:			
	Inlet Conn Size: 2" † Type: Flanged Class: 300		Inherent Characteristic: Linear		Balance Style: Unbalanced Full Open Position: 100 %			
	Outlet Conn Size: 2" † Type: Flanged Class: 300		Guiding Style: Top		Closure Member Material: 316 SS, Stellite sheath			
	Flange Facing: Raised Face Flange Face Finish:		Seat Ring Material: 316 SS with Stellite sheath		Seat Ring Retainer Material: 304 Stainless Steel			
	Body & Bonnet Material: Carbon Steel		Bearing Material:		Bushing/Cage Material: Stellite			
	Bonnet Style: Integral or Bolted		Stem/Shaft Material: 316 Stainless Steel		Flow Direction:			
	Packing Style: Manufacturer's Recommendation		Seat Leakage Class: ANSI Class IV					
	Packing Material: Manufacturer's Recommendation							
	Body/Bonnet Gasket Material: Manufacturer's Recommend							
	Lub/Iso Valve Material: Lub Material:							
	Manufacturer: Masoneilan (or equal)							
	Model Number:							
	ACTUATOR	Type: Spring Diaphragm, Pneumatic		Inlet Pipe Nominal Size:		Est Wall Thick:		
Action Style: Fail Close		Outlet Pipe Nominal Size:		Est Wall Thick:				
Service Type: Modulating		Est Insulation Type:						
Max Shutoff Press: In: 5.85 Out: 0.70 kg/cm2(g)		Est Insulation Thickness:						
Maximum Stroke Time to Open:		Installation:						
Maximum Stroke Time to Close:		Design Pressure :		9.3 kg/cm2g (F.V)				
Stroke Cycle Service:		Design Temperature :		195 C				
DATA PAGE 1 REMARKS	§ The information is based on Masoneilan 21000 series.							
	‡ The calculated Cv for flashing service is determined by adding the Calculated Cv for the liquid and the vapor (or steam) at the outlet conditions of the valve.							
	† Oversize body selected to reduce velocity damage to body.							

ALL ITEMS SHALL COMPLY WITH THE GENERAL REQUIREMENTS SHOWN ON THE GENERAL REQUIREMENTS SECTION OF THIS SPECIFICATION.

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CONTROL VALVES

PROJECT SPECIFICATION

963766 — **616**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Item Number: 10 Quantity: 1		Valve Tag Number: LV-23-0501		Data Page: 2 of 2	
TRANS-DUCER	Transducer Tag Number:		HANDWHEEL	Handwheel Position:	
	Transducer Input Signal Range:			Usage:	
	Transducer Output Signal Range:				
	Manufacturer:				
SOLENOID VALVE	Solenoid Valve Tag Number:		POSITION SWITCH	Open Position Switch Tag Number:	
	Body Type: Body Material:			Close Position Switch Tag Number:	
	Seat Material: Disc Material:			Sensor Type:	
	Connection Size/Type:			Contact Form:	
	Packing Style:			Open Position Switch Contact Action:	
	Fluid:			Close Position Switch Contact Action:	
	Pressure:			Contact Load: Contact Rating:	
	Minimum Pressure Port Orifice C _v : Size:			Mounting Location:	
	Minimum Exhaust Port Orifice C _v : Size:			Manufacturer:	
	Operation Style:			POSITION TRANSMITTER	Position Transmitter Tag Number:
	Coil Configuration:		Sensor Type:		
	Operating Voltage: Frequency:		Output Signal Range:		
	Action Coil 1 Deenergized:		Travel Measurement:		
	Action Coil 1 Energized:		Mounting Location:		
	Action Coil 2 Deenergized:		LOCK-UP VALVE	Manufacturer:	
	Action Coil 2 Energized:			Action Style:	
	Manual Reset:			Mounting Location:	
	Manual Operator:		QUICK EXHAUST	Model Number:	
	Mounting Location:			Manufacturer:	
	Manufacturer:			Connection Size/Type:	
POSITIONER	Positioner Tag Number:		VOLUME BOOSTER	Body Material:	
	Positioner Type: Digital			Diaphragm Material:	
	Positioner Input Signal Range: 4 - 20 mA			Exhaust Action:	
	Positioner Output Signal Range: 0 - 100 %			Mounting Location:	
AIR FILTER	Manufacturer:		Manufacturer:		
	Quantity: one		Connection Size/Type:		
	Type: Filter/Regulator		Body Material:		
TUBING	Manufacturer:		Seat Ring Material:		
	Tubing Size:		Diaphragm Material:		
Tubing Material:		Mounting Location:			
		Manufacturer:			
DATA PAGE 2 REMARKS					

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PROJECT SPECIFICATION

963766 — **616**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Note: The information in this document is confidential and the property of UOP LLC and must not be disclosed to others or reproduced in any manner or used for any purpose whatsoever without its written permission.

GENERAL	Item Number: 11 Quantity: 1 Valve Tag Number: LV-23-0601 Data Page: 1 of 2 Service: Cumene to Concentration Cumene Quench Drum Process Material: Hydrocarbon with aromatics P&ID Drawing: 06																																																																																																																																																																																																				
SIZING CONDITIONS	<table border="1" style="width: 100%; border-collapse: collapse;"> <tr> <td style="width: 30%;">Process Data Case</td> <td colspan="2">Single Case</td> <td colspan="2"></td> <td colspan="2"></td> <td colspan="2"></td> <td colspan="2"></td> </tr> <tr> <td>Sizing Case</td> <td colspan="2">Normal</td> <td colspan="2"></td> <td colspan="2"></td> <td colspan="2"></td> <td colspan="2"></td> </tr> <tr> <td>Calculated C_v</td> <td colspan="2">42.6</td> <td colspan="2"></td> <td colspan="2"></td> <td colspan="2"></td> <td colspan="2"></td> </tr> <tr> <td></td> <td>Inlet</td> <td>Outlet</td> <td>Inlet</td> <td>Outlet</td> <td>Inlet</td> <td>Outlet</td> <td>Inlet</td> <td>Outlet</td> <td colspan="2">Units</td> </tr> <tr> <td>Liquid Flow at Flowing Cond</td> <td>52.5</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td colspan="2">m3/h</td> </tr> <tr> <td>Vapor Flow at Base Cond</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td colspan="2">normal m3/h</td> </tr> <tr> <td>Steam Flow</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td colspan="2">kg/h</td> </tr> <tr> <td>Temperature</td> <td>46</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td colspan="2">°C</td> </tr> <tr> <td>Liquid Specific Gravity</td> <td>0.841</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td colspan="2">----</td> </tr> <tr> <td>Vapor Spec. Gravity (Air=1)</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td colspan="2">----</td> </tr> <tr> <td>Pressure</td> <td>4.20</td> <td>2.46</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td colspan="2">kg/cm2(g)</td> </tr> <tr> <td>Liq Critical Press (Pseudo)</td> <td>32.72</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td colspan="2">kg/cm2(a)</td> </tr> <tr> <td>Liquid Vapor Pressure</td> <td>0.90</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td colspan="2">kg/cm2(a)</td> </tr> <tr> <td>Liquid Viscosity</td> <td>0.425</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td colspan="2">cP</td> </tr> <tr> <td>Compressibility Z</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td colspan="2"></td> </tr> <tr> <td>Specific Heats Cp/Cv (Real)</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td colspan="2"></td> </tr> <tr> <td>Estimate Sound Press Level</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td colspan="2">dBA</td> </tr> </table>										Process Data Case	Single Case										Sizing Case	Normal										Calculated C _v	42.6											Inlet	Outlet	Inlet	Outlet	Inlet	Outlet	Inlet	Outlet	Units		Liquid Flow at Flowing Cond	52.5								m3/h		Vapor Flow at Base Cond									normal m3/h		Steam Flow									kg/h		Temperature	46								°C		Liquid Specific Gravity	0.841								----		Vapor Spec. Gravity (Air=1)									----		Pressure	4.20	2.46							kg/cm2(g)		Liq Critical Press (Pseudo)	32.72								kg/cm2(a)		Liquid Vapor Pressure	0.90								kg/cm2(a)		Liquid Viscosity	0.425								cP		Compressibility Z											Specific Heats Cp/Cv (Real)											Estimate Sound Press Level									dBA	
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VALVE BODY & BONNET	Body Type: Eccentric Rotary Plug § Body Size: 3" Port Type: Single Inlet Conn Size: 3" Type: Flanged Class: 300 Outlet Conn Size: 3" Type: Flanged Class: 300 Flange Facing: Raised Face Flange Face Finish: Body & Bonnet Material: Carbon Steel Bonnet Style: Integral or Bolted Packing Style: Manufacturer's Recommendation Packing Material: Manufacturer's Recommendation Body/Bonnet Gasket Material: Lub/Iso Valve Material: Lub Material: Manufacturer: Masoneilan (or equal) Model Number:					TRIM	Approximate Valve C _v : 81 F _L : 0.68 x _T : 0.4 Trim Style: Man. Standard Trim Size: Inherent Characteristic: Linear Balance Style: Full Open Position: 100 % Guiding Style: Shaft Closure Member Material: 316 SS, Stellite sheath Seat Ring Material: 316 SS with Stellite sheath Seat Ring Retainer Material: 316 Stainless Steel Bearing Material: Stellite Bushing/Cage Material: Stem/Shaft Material: 17-4 PH Stainless Steel Flow Direction: Seat Leakage Class: ANSI Class IV																																																																																																																																																																																														
ACTUATOR	Type: Spring Diaphragm, Pneumatic Action Style: Fail Open Service Type: Modulating Max Shutoff Press: In: 6.42 Out: 0.07 kg/cm2(g) Maximum Stroke Time to Open: Maximum Stroke Time to Close: Stroke Cycle Service:					INSTALLATION	Inlet Pipe Nominal Size: Est Wall Thick: Outlet Pipe Nominal Size: Est Wall Thick: Est Insulation Type: Est Insulation Thickness: Installation: Design Pressure : 10.6 kg/cm2g (F.V) Design Temperature : 195 C																																																																																																																																																																																														
DATA PAGE 1 REMARKS	§ The information is based on Masoneilan 35002 series. Valve will normally be in the closed position. Valve opening will be intermittent and generally short-term.																																																																																																																																																																																																				

ALL ITEMS SHALL COMPLY WITH THE GENERAL REQUIREMENTS SHOWN ON THE GENERAL REQUIREMENTS SECTION OF THIS SPECIFICATION.



CONTROL VALVES

PROJECT SPECIFICATION

963766 — **616**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Item Number: 11 Quantity: 1		Valve Tag Number: LV-23-0601		Data Page: 2 of 2	
TRANS-DUCER	Transducer Tag Number:		HANDWHEEL	Handwheel Position:	
	Transducer Input Signal Range:			Usage:	
	Transducer Output Signal Range:				
	Manufacturer:				
SOLENOID VALVE	Solenoid Valve Tag Number:		POSITION SWITCH	Open Position Switch Tag Number:	
	Body Type: Body Material:			Close Position Switch Tag Number:	
	Seat Material: Disc Material:			Sensor Type:	
	Connection Size/Type:			Contact Form:	
	Packing Style:			Open Position Switch Contact Action:	
	Fluid:			Close Position Switch Contact Action:	
	Pressure:			Contact Load: Contact Rating:	
	Minimum Pressure Port Orifice C _v : Size:			Mounting Location:	
	Minimum Exhaust Port Orifice C _v : Size:			Manufacturer:	
	Operation Style:			POSITION TRANSMITTER	Position Transmitter Tag Number:
	Coil Configuration:		Sensor Type:		
	Operating Voltage: Frequency:		Output Signal Range:		
	Action Coil 1 Deenergized:		Travel Measurement:		
	Action Coil 1 Energized:		Mounting Location:		
	Action Coil 2 Deenergized:		LOCK-UP VALVE	Manufacturer:	
	Action Coil 2 Energized:			Action Style:	
	Manual Reset:			Mounting Location:	
	Manual Operator:		QUICK EXHAUST	Model Number:	
	Mounting Location:			Manufacturer:	
	Manufacturer:			Connection Size/Type:	
POSITIONER	Positioner Tag Number:		VOLUME BOOSTER	Body Material:	
	Positioner Type: Digital			Diaphragm Material:	
	Positioner Input Signal Range: 4 - 20 mA			Exhaust Action:	
AIR FILTER	Positioner Output Signal Range: 0 - 100 %		TUBING	Mounting Location:	
	Manufacturer:			Manufacturer:	
	Quantity: one				
TUBING	Type: Filter/Regulator		TUBING	Connection Size/Type:	
	Manufacturer:			Body Material:	
	Tubing Size:			Seat Ring Material:	
DATA PAGE 2 REMARKS	Tubing Material:		TUBING	Diaphragm Material:	
				Mounting Location:	
				Manufacturer:	

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PROJECT SPECIFICATION

963766 — **616**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

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GENERAL	Item Number: 12 Quantity: 1 Valve Tag Number: HV-23-0601 Data Page: 1 of 2																																																																																																																																																																
	Service: Cumene Quench to Preflash Column (Manual) Process Material: Hydrocarbon with aromatics P&ID Drawing: 06																																																																																																																																																																
SIZING CONDITIONS	<table border="1"> <thead> <tr> <th>Process Data Case</th> <th colspan="2">General</th> <th colspan="2">General</th> <th colspan="2"></th> <th colspan="2"></th> <th></th> </tr> <tr> <th>Sizing Case</th> <th colspan="2">Open</th> <th colspan="2">Closed</th> <th colspan="2"></th> <th colspan="2"></th> <th></th> </tr> <tr> <th>Calculated C_v</th> <th>Inlet</th> <th>Outlet</th> <th>Inlet</th> <th>Outlet</th> <th>Inlet</th> <th>Outlet</th> <th>Inlet</th> <th>Outlet</th> <th>Units</th> </tr> </thead> <tbody> <tr> <td>Liquid Flow at Flowing Cond</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>m³/h</td> </tr> <tr> <td>Vapor Flow at Base Cond</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>normal m³/h</td> </tr> <tr> <td>Steam Flow</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>kg/h</td> </tr> <tr> <td>Temperature</td> <td>46</td> <td></td> <td>46</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>°C</td> </tr> <tr> <td>Liquid Specific Gravity</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>----</td> </tr> <tr> <td>Vapor Spec. Gravity (Air=1)</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>----</td> </tr> <tr> <td>Pressure</td> <td>-0.39</td> <td>-0.39</td> <td>-0.39</td> <td>-0.95</td> <td></td> <td></td> <td></td> <td></td> <td>kg/cm²(g)</td> </tr> <tr> <td>Liq Critical Press (Pseudo)</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>kg/cm²(a)</td> </tr> <tr> <td>Liquid Vapor Pressure</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>kg/cm²(a)</td> </tr> <tr> <td>Liquid Viscosity</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>cP</td> </tr> <tr> <td>Compressibility Z</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> <tr> <td>Specific Heats Cp/Cv (Real)</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> <tr> <td>Estimate Sound Press Level</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>dBA</td> </tr> </tbody> </table>	Process Data Case	General		General							Sizing Case	Open		Closed							Calculated C _v	Inlet	Outlet	Inlet	Outlet	Inlet	Outlet	Inlet	Outlet	Units	Liquid Flow at Flowing Cond									m ³ /h	Vapor Flow at Base Cond									normal m ³ /h	Steam Flow									kg/h	Temperature	46		46						°C	Liquid Specific Gravity									----	Vapor Spec. Gravity (Air=1)									----	Pressure	-0.39	-0.39	-0.39	-0.95					kg/cm ² (g)	Liq Critical Press (Pseudo)									kg/cm ² (a)	Liquid Vapor Pressure									kg/cm ² (a)	Liquid Viscosity									cP	Compressibility Z										Specific Heats Cp/Cv (Real)										Estimate Sound Press Level									dBA
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VALVE BODY & BONNET	Body Type: Globe § Body Size: 1-1/2" ¶¶ Port Type: Single Inlet Conn Size: 1-1/2" ¶¶ Type: Flanged Class: 300 Outlet Conn Size: 1-1/2" ¶¶ Type: Flanged Class: 300 Flange Facing: Raised Face Flange Face Finish: Body & Bonnet Material: 316 Stainless Steel ‡ Bonnet Style: Integral or Bolted Packing Style: Manufacturer's Recommendation † Packing Material: Manufacturer's Recommendation Body/Bonnet Gasket Material: Manufacturer's Recommend Lub/Iso Valve Material: Lub Material: Manufacturer: Masoneilan (or equal) Model Number:	TRIM	Approximate Valve C _v : 25 F _L : x _T : Trim Style: Man. Standard Trim Size: Inherent Characteristic: Linear Balance Style: Unbalanced Full Open Position: 100 % Guiding Style: Top Closure Member Material: 316 Stainless Steel Seat Ring Material: 316 Stainless Steel Seat Ring Retainer Material: 304 Stainless Steel Bearing Material: Bushing/Cage Material: Stellite Stem/Shaft Material: 316 Stainless Steel Flow Direction: Seat Leakage Class: ANSI Class V																																																																																																																																																														
	ACTUATOR		INSTALLATION	Type: Spring Diaphragm, Pneumatic Action Style: Fail Close Service Type: Modulating Max Shutoff Press: In: 3.85 Out: -1.02 kg/cm²(g) Maximum Stroke Time to Open: Maximum Stroke Time to Close: Stroke Cycle Service:	Inlet Pipe Nominal Size: Est Wall Thick: Outlet Pipe Nominal Size: Est Wall Thick: Est Insulation Type: Est Insulation Thickness: Installation: Design Pressure : 4.2 kg/cm²g (F.V) Design Temperature : 120 C																																																																																																																																																												
				§ The information is based on Masoneilan 21000 series.																																																																																																																																																													
				¶ Preferred material is L grade stainless steel. Standard grade stainless steel that is not welded may be used for process wetted parts including the body. Stainless steel with weld repair is acceptable if low carbon (<= 0.03 %) filler material is used.																																																																																																																																																													
				¶¶ Line size valve.																																																																																																																																																													
				† Provide packing suitable for vacuum service to minimize air leakage into the process line.																																																																																																																																																													
	ALL ITEMS SHALL COMPLY WITH THE GENERAL REQUIREMENTS SHOWN ON THE GENERAL REQUIREMENTS SECTION OF THIS SPECIFICATION.																																																																																																																																																																



PROJECT SPECIFICATION

963766 — 616

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

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PROJECT SPECIFICATION

963766 — **616**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

GENERAL	Item Number: 13 Quantity: 1 Valve Tag Number: UXV-23-0601 Data Page: 1 of 2																																																																																																																																																																								
	Service: Cumene Quench to Preflash Column Process Material: Hydrocarbon with aromatics P&ID Drawing: 06																																																																																																																																																																								
SIZING CONDITIONS	<table border="1"> <thead> <tr> <th rowspan="2">Process Data Case</th> <th colspan="2">General</th> <th colspan="2">General</th> <th colspan="2"></th> <th colspan="2"></th> <th rowspan="2">Units</th> </tr> <tr> <th colspan="2">Open</th> <th colspan="2">Closed</th> <th colspan="2"></th> <th colspan="2"></th> </tr> </thead> <tbody> <tr> <td>Sizing Case</td> <td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td> </tr> <tr> <td>Calculated C_v</td> <td>Inlet</td><td>Outlet</td><td>Inlet</td><td>Outlet</td><td>Inlet</td><td>Outlet</td><td>Inlet</td><td>Outlet</td><td></td> </tr> <tr> <td>Liquid Flow at Flowing Cond</td> <td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>m3/h</td> </tr> <tr> <td>Vapor Flow at Base Cond</td> <td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>normal m3/h</td> </tr> <tr> <td>Steam Flow</td> <td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>kg/h</td> </tr> <tr> <td>Temperature</td> <td>46</td><td></td><td>46</td><td></td><td></td><td></td><td></td><td></td><td>°C</td> </tr> <tr> <td>Liquid Specific Gravity</td> <td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>----</td> </tr> <tr> <td>Vapor Spec. Gravity (Air=1)</td> <td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>----</td> </tr> <tr> <td>Pressure</td> <td>-0.39</td><td>-0.39</td><td>-0.39</td><td>-0.95</td><td></td><td></td><td></td><td></td><td>kg/cm2(g)</td> </tr> <tr> <td>Liq Critical Press (Pseudo)</td> <td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>kg/cm2(a)</td> </tr> <tr> <td>Liquid Vapor Pressure</td> <td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>kg/cm2(a)</td> </tr> <tr> <td>Liquid Viscosity</td> <td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>cP</td> </tr> <tr> <td>Compressibility Z</td> <td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td> </tr> <tr> <td>Specific Heats Cp/Cv (Real)</td> <td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td> </tr> <tr> <td>Estimate Sound Press Level</td> <td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>dBA</td> </tr> </tbody> </table>	Process Data Case	General		General						Units	Open		Closed						Sizing Case										Calculated C _v	Inlet	Outlet	Inlet	Outlet	Inlet	Outlet	Inlet	Outlet		Liquid Flow at Flowing Cond									m3/h	Vapor Flow at Base Cond									normal m3/h	Steam Flow									kg/h	Temperature	46		46						°C	Liquid Specific Gravity									----	Vapor Spec. Gravity (Air=1)									----	Pressure	-0.39	-0.39	-0.39	-0.95					kg/cm2(g)	Liq Critical Press (Pseudo)									kg/cm2(a)	Liquid Vapor Pressure									kg/cm2(a)	Liquid Viscosity									cP	Compressibility Z										Specific Heats Cp/Cv (Real)										Estimate Sound Press Level									dBA
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VALVE BODY & BONNET	Body Type: Eccentric Disc § Body Size: 10" #1 Port Type: Single Inlet Conn Size: 10" #1 Type: Flanged Class: 150 Outlet Conn Size: 10" #1 Type: Flanged Class: 150 Flange Facing: Flange Face Finish: Body & Bonnet Material: 316 Stainless Steel ‡ Bonnet Style: Packing Style: Manufacturer's Recommendation † Packing Material: Manufacturer's Recommendation Body/Bonnet Gasket Material: Lub/Iso Valve Material: Lub Material: Manufacturer: Metso Automation (or equal) Model Number:	TRIM	Approximate Valve C _v : 2730 F _L : x _T : Trim Style: Man. Standard Trim Size: Inherent Characteristic: Equal Percentage Balance Style: Full Open Position: 90° Guiding Style: Shaft Closure Member Material: 316 Stainless Steel Seat Ring Material: Incoloy 825 with Hard Chrome Seat Ring Retainer Material: 316 Stainless Steel Bearing Material: Stellite Bushing/Cage Material: Stem/Shaft Material: 17-4PH Flow Direction: Seat Leakage Class: ANSI Class V																																																																																																																																																																						
	ACTUATOR		INSTALLATION	Type: Spring Return Piston, Pneumatic Action Style: Fail Open Service Type: On-Off Max Shutoff Press: In: 3.85 Out: -1.02 kg/cm2(g) Maximum Stroke Time to Open: 5 Seconds Maximum Stroke Time to Close: Stroke Cycle Service:	Inlet Pipe Nominal Size: Est Wall Thick: Outlet Pipe Nominal Size: Est Wall Thick: Est Insulation Type: Est Insulation Thickness: Installation: Design Pressure : 4.2 kg/cm2g (F.V) Design Temperature : 120 C																																																																																																																																																																				
				§ The information is based on Neldisc L6 series. Valve selected shall be of double flanged construction with minimal flange-to-flange dimension and no pockets within the valve body.																																																																																																																																																																					
				‡ Preferred material is L grade stainless steel. Standard grade stainless steel that is not welded may be used for process wetted parts including the body. Stainless steel with weld repair is acceptable if low carbon (<= 0.03 %) filler material is used.																																																																																																																																																																					
				#1 Line size valve.																																																																																																																																																																					
				† Provide packing suitable for vacuum service to minimize air leakage into the process line.																																																																																																																																																																					

ALL ITEMS SHALL COMPLY WITH THE GENERAL REQUIREMENTS SHOWN ON THE GENERAL REQUIREMENTS SECTION OF THIS SPECIFICATION.

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CONTROL VALVES

PROJECT SPECIFICATION

963766 — **616**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Item Number: 13 Quantity: 1 Valve Tag Number: UXV-23-0601 Data Page: 2 of 2	
TRANS-DUCER Transducer Tag Number: Transducer Input Signal Range: Transducer Output Signal Range: Manufacturer:	HANDWHEEL Handwheel Position: Usage:
SOLENOID VALVE Solenoid Valve Tag Number: UXY-23-0601 Body Type: 3-Way Body Material: Brass Seat Material: Stainless Steel Disc Material: Stainless Steel Connection Size/Type: † Packing Style: Packless Fluid: Instrument Air Pressure: Contractor to determine Minimum Pressure Port Orifice C_v: MAX Size: MAX Minimum Exhaust Port Orifice C_v: MAX Size: MAX Operation Style: † Coil Configuration: Single Operating Voltage: §§ Frequency: Action Coil 1 Deenergized: Vent air from valve Action Coil 1 Energized: Supply air to valve Action Coil 2 Deenergized: Action Coil 2 Energized: Manual Reset: Manual Operator: Mounting Location: Nearest item to actuator in air line. † Manufacturer: ASCO (or equal)	POSITION SWITCH Open Position Switch Tag Number: Close Position Switch Tag Number: Sensor Type: Contact Form: Open Position Switch Contact Action: Close Position Switch Contact Action: Contact Load: Contact Rating: Mounting Location: Manufacturer:
POSITION TRANSMITTER Position Transmitter Tag Number: Sensor Type: Output Signal Range: Travel Measurement: Mounting Location: Manufacturer:	LOCK-UP VALVE Action Style: Mounting Location: Model Number: Manufacturer:
QUICK EXHAUST Connection Size/Type: Body Material: Diaphragm Material: Exhaust Action: Mounting Location: Manufacturer:	VOLUME BOOSTER Connection Size/Type: Body Material: Seat Ring Material: Diaphragm Material: Mounting Location: Manufacturer:
POSITIONER Positioner Tag Number: Positioner Type: Positioner Input Signal Range: Positioner Output Signal Range: Manufacturer:	
AIR FILTER Quantity: one Type: Filter/Regulator Manufacturer:	
TUBING Tubing Size: Tubing Material:	
DATA PAGE 2 REMARKS † The solenoid valve shall be mounted and tubed by the control valve vendor. The solenoid valve size and type, the tubing size, the actuator air connection size, and the air/filter regulator capacity shall be determined by the control valve vendor to meet the maximum stroking time requirements specified for the control valve. §§ The Contractor shall coordinate the solenoid valve operating voltage and coil watt rating with the supplier of the Concentration Section Shutdown System (UC-2301). See UOP Project Specification 660, this project.	

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GENERAL	Item Number: 14 Quantity: 1 Valve Tag Number: UXV-23-0701 Data Page: 1 of 2							
	Service: S3 Steam Condensate from Flash Column Vaporizer Shutoff Process Material: Water P&ID Drawing: 07							
SIZING CONDITIONS	Process Data Case	General	General					
	Sizing Case	Open	Closed					
	Calculated C _v							
		Inlet	Outlet	Inlet	Outlet	Inlet	Outlet	Units
	Liquid Flow at Flowing Cond							m3/h
	Vapor Flow at Base Cond							normal m3/h
	Steam Flow							kg/h
	Temperature	131		131				°C
	Liquid Specific Gravity							----
	Vapor Spec. Gravity (Air=1)							----
	Pressure	1.78	1.78	3.90	2.32			kg/cm2(g)
	Liq Critical Press (Pseudo)							kg/cm2(a)
	Liquid Vapor Pressure							kg/cm2(a)
	Liquid Viscosity							cP
	Compressibility Z							
Specific Heats Cp/Cv (Real)								
Estimate Sound Press Level							dBA	
VALVE BODY & BONNET	Body Type: Globe §	Approximate Valve C _v : 225 F _L : x _T :						
	Body Size: 4" ± Port Type: Single	Trim Style: Man. Standard Trim Size:						
	Inlet Conn Size: 4" ± Type: Flanged Class: 300	Inherent Characteristic: Equal Percentage						
	Outlet Conn Size: 4" ± Type: Flanged Class: 300	Balance Style: Balanced Full Open Position: 100 %						
	Flange Facing: Raised Face Flange Face Finish:	Guiding Style: Cage						
	Body & Bonnet Material: Carbon Steel	Closure Member Material: 316 SS with Stellite sheath						
	Bonnet Style: Integral or Bolted	Seat Ring Material: 316 SS with Stellite sheath						
	Packing Style: Manufacturer's Recommendation	Seat Ring Retainer Material:						
	Packing Material: Manufacturer's Recommendation	Bearing Material:						
	Body/Bonnet Gasket Material: Manufacturer's Recommend	Bushing/Cage Material: 316 SS hard chrome plated						
	Lub/Iso Valve Material: Lub Material:	Stem/Shaft Material: 316 Stainless Steel						
	Manufacturer: Masoneilan (or equal)	Flow Direction: Flow to close						
	Model Number:	Seat Leakage Class: ANSI Class V						
	ACTUATOR	Type: Spring Diaphragm, Pneumatic	Inlet Pipe Nominal Size: Est Wall Thick:					
Action Style: Fail Close		Outlet Pipe Nominal Size: Est Wall Thick:						
Service Type: On-Off		Est Insulation Type:						
Max Shutoff Press: In: 5.66 Out: 2.32 kg/cm2(g)		Est Insulation Thickness:						
Maximum Stroke Time to Open:		Installation:						
Maximum Stroke Time to Close: 5 Seconds		Design Pressure : 10.5 kg/cm2g (F.V)						
Stroke Cycle Service:		Design Temperature : 195 C						
DATA PAGE 1 REMARKS	§ The information is based on Masoneilan 41000 series.							
	± Line size valve.							

ALL ITEMS SHALL COMPLY WITH THE GENERAL REQUIREMENTS SHOWN ON THE GENERAL REQUIREMENTS SECTION OF THIS SPECIFICATION.



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CONTROL VALVES

PROJECT SPECIFICATION

963766 — **616**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Item Number: 14 Quantity: 1		Valve Tag Number: UXV-23-0701		Data Page: 2 of 2			
TRANS-DUCER	Transducer Tag Number:		HANDWHEEL	Handwheel Position:			
	Transducer Input Signal Range:			Usage:			
Transducer Output Signal Range:							
Manufacturer:							
SOLENOID VALVE	Solenoid Valve Tag Number: UXY-23-0701		POSITION SWITCH	Open Position Switch Tag Number:			
	Body Type: 3-Way Body Material: Brass			Close Position Switch Tag Number:			
	Seat Material: Stainless Steel Disc Material: Stainless Steel			Sensor Type:			
	Connection Size/Type: †			Contact Form:			
	Packing Style: Packless			Open Position Switch Contact Action:			
	Fluid: Instrument Air			Close Position Switch Contact Action:			
	Pressure: Contractor to determine			Contact Load: Contact Rating:			
	Minimum Pressure Port Orifice C _v : MAX Size: MAX			Mounting Location:			
	Minimum Exhaust Port Orifice C _v : MAX Size: MAX			Manufacturer:			
	Operation Style: †						
	Coil Configuration: Single			Position Transmitter Tag Number:			
	Operating Voltage: \$\$ Frequency:			Sensor Type:			
	Action Coil 1 Deenergized: Vent air from valve			Output Signal Range:			
	Action Coil 1 Energized: Supply air to valve			Travel Measurement:			
	Action Coil 2 Deenergized:			Mounting Location:			
	Action Coil 2 Energized:			Manufacturer:			
	POSITIONER	Manual Reset:		LOCK-UP VALVE	Action Style:		
		Manual Operator:			Mounting Location:		
Mounting Location: Nearest item to actuator in air line. †		Model Number:					
Manufacturer: ASCO (or equal)		Manufacturer:					
Positioner Tag Number:		Connection Size/Type:					
Positioner Type:		Body Material:					
Positioner Input Signal Range:		Diaphragm Material:					
Positioner Output Signal Range:		Exhaust Action:					
Manufacturer:		Mounting Location:					
		Manufacturer:					
AIR FILTER	Quantity: one		VOLUME BOOSTER	Connection Size/Type:			
	Type: Filter/Regulator			Body Material:			
	Manufacturer:			Seat Ring Material:			
TUBING	Tubing Size:			Diaphragm Material:			
	Tubing Material:			Mounting Location:			
				Manufacturer:			
DATA PAGE 2 REMARKS	† The solenoid valve shall be mounted and tubed by the control valve vendor. The solenoid valve size and type, the tubing size, the actuator air connection size, and the air/filter regulator capacity shall be determined by the control valve vendor to meet the maximum stroking time requirements specified for the control valve.						
	\$\$ The Contractor shall coordinate the solenoid valve operating voltage and coil watt rating with the supplier of the Concentration Section Shutdown System (UC-2301). See UOP Project Specification 660, this project.						

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CONTROL VALVES

PROJECT SPECIFICATION

963766 — **616**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

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GENERAL	Item Number: 15 Quantity: 1 Valve Tag Number: UXV-23-0702 Data Page: 1 of 2																																																																																																																																																																
	Service: Flash Column Vaporizer Steam Line Vent to Atmosphere Process Material: Steam P&ID Drawing: 07																																																																																																																																																																
SIZING CONDITIONS	<table border="1"> <thead> <tr> <th>Process Data Case</th> <th colspan="2">General</th> <th colspan="2">General</th> <th colspan="2"></th> <th colspan="2"></th> <th></th> </tr> <tr> <th>Sizing Case</th> <th colspan="2">Open</th> <th colspan="2">Closed</th> <th colspan="2"></th> <th colspan="2"></th> <th></th> </tr> <tr> <th>Calculated C_v</th> <th>Inlet</th> <th>Outlet</th> <th>Inlet</th> <th>Outlet</th> <th>Inlet</th> <th>Outlet</th> <th>Inlet</th> <th>Outlet</th> <th>Units</th> </tr> </thead> <tbody> <tr> <td>Liquid Flow at Flowing Cond</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>m3/h</td> </tr> <tr> <td>Vapor Flow at Base Cond</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>normal m3/h</td> </tr> <tr> <td>Steam Flow</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>kg/h</td> </tr> <tr> <td>Temperature</td> <td>148</td> <td></td> <td>148</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>°C</td> </tr> <tr> <td>Liquid Specific Gravity</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>----</td> </tr> <tr> <td>Vapor Spec. Gravity (Air=1)</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>----</td> </tr> <tr> <td>Pressure</td> <td>2.70</td> <td>2.70</td> <td>2.70</td> <td>0</td> <td></td> <td></td> <td></td> <td></td> <td>kg/cm2(g)</td> </tr> <tr> <td>Liq Critical Press (Pseudo)</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>kg/cm2(a)</td> </tr> <tr> <td>Liquid Vapor Pressure</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>kg/cm2(a)</td> </tr> <tr> <td>Liquid Viscosity</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>cP</td> </tr> <tr> <td>Compressibility Z</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> <tr> <td>Specific Heats Cp/Cv (Real)</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> <tr> <td>Estimate Sound Press Level</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>dBA</td> </tr> </tbody> </table>	Process Data Case	General		General							Sizing Case	Open		Closed							Calculated C _v	Inlet	Outlet	Inlet	Outlet	Inlet	Outlet	Inlet	Outlet	Units	Liquid Flow at Flowing Cond									m3/h	Vapor Flow at Base Cond									normal m3/h	Steam Flow									kg/h	Temperature	148		148						°C	Liquid Specific Gravity									----	Vapor Spec. Gravity (Air=1)									----	Pressure	2.70	2.70	2.70	0					kg/cm2(g)	Liq Critical Press (Pseudo)									kg/cm2(a)	Liquid Vapor Pressure									kg/cm2(a)	Liquid Viscosity									cP	Compressibility Z										Specific Heats Cp/Cv (Real)										Estimate Sound Press Level									dBA
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REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Item Number: 15 Quantity: 1		Valve Tag Number: UXV-23-0702		Data Page: 2 of 2		
TRANS-DUCER	Transducer Tag Number:		HANDWHEEL	Handwheel Position:		
	Transducer Input Signal Range:			Usage:		
	Transducer Output Signal Range:					
	Manufacturer:					
SOLENOID VALVE	Solenoid Valve Tag Number: UXY-23-0702		POSITION SWITCH	Open Position Switch Tag Number:		
	Body Type: 3-Way Body Material: Brass			Close Position Switch Tag Number:		
	Seat Material: Stainless Steel Disc Material: Stainless Steel			Sensor Type:		
	Connection Size/Type: †			Contact Form:		
	Packing Style: Packless			Open Position Switch Contact Action:		
	Fluid: Instrument Air			Close Position Switch Contact Action:		
	Pressure: Contractor to determine			Contact Load: Contact Rating:		
	Minimum Pressure Port Orifice C _v : MAX Size: MAX			Mounting Location:		
	Minimum Exhaust Port Orifice C _v : MAX Size: MAX			Manufacturer:		
	Operation Style: †			POSITION TRANSMITTER	Position Transmitter Tag Number:	
	Coil Configuration: Single				Sensor Type:	
	Operating Voltage: \$\$ Frequency:				Output Signal Range:	
	Action Coil 1 Deenergized: Vent air from valve				Travel Measurement:	
	Action Coil 1 Energized: Supply air to valve				Mounting Location:	
	Action Coil 2 Deenergized:				Manufacturer:	
	Action Coil 2 Energized:			LOCK-UP VALVE	Action Style:	
	Manual Reset:				Mounting Location:	
	Manual Operator:				Model Number:	
Mounting Location: Nearest item to actuator in air line. †		QUICK EXHAUST	Manufacturer:			
Manufacturer: ASCO (or equal)			Connection Size/Type:			
			Body Material:			
POSITIONER	Positioner Tag Number:			Diaphragm Material:		
	Positioner Type:			Exhaust Action:		
	Positioner Input Signal Range:			Mounting Location:		
AIR FILTER	Positioner Output Signal Range:		VOLUME BOOSTER	Manufacturer:		
	Manufacturer:			Connection Size/Type:		
	Quantity: one			Body Material:		
TUBING	Type: Filter/Regulator			Seat Ring Material:		
	Manufacturer:			Diaphragm Material:		
	Tubing Size:			Mounting Location:		
DATA PAGE 2 REMARKS	Tubing Material:			Manufacturer:		
† The solenoid valve shall be mounted and tubed by the control valve vendor. The solenoid valve size and type, the tubing size, the actuator air connection size, and the air/filter regulator capacity shall be determined by the control valve vendor to meet the maximum stroking time requirements specified for the control valve. \$\$ The Contractor shall coordinate the solenoid valve operating voltage and coil watt rating with the supplier of the Concentration Section Shutdown System (UC-2301). See UOP Project Specification 660, this project.						

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CONTROL VALVES

PROJECT SPECIFICATION

963766 — **616**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

GENERAL	Item Number: 16 Quantity: 1 Valve Tag Number: UXV-23-0703 Data Page: 1 of 2																																																																																																																																																																								
	Service: S3 Steam to Flash Column Vaporizer Shutoff Process Material: Steam P&ID Drawing: 07																																																																																																																																																																								
SIZING CONDITIONS	<table border="1"> <thead> <tr> <th rowspan="2">Process Data Case</th> <th colspan="2">General</th> <th colspan="2">General</th> <th colspan="2"></th> <th colspan="2"></th> <th rowspan="2">Units</th> </tr> <tr> <th colspan="2">Open</th> <th colspan="2">Closed</th> <th colspan="2"></th> <th colspan="2"></th> </tr> </thead> <tbody> <tr> <td>Sizing Case</td> <td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td> </tr> <tr> <td>Calculated C_v</td> <td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td> </tr> <tr> <td>Liquid Flow at Flowing Cond</td> <td>Inlet</td><td>Outlet</td><td>Inlet</td><td>Outlet</td><td>Inlet</td><td>Outlet</td><td>Inlet</td><td>Outlet</td><td>m3/h</td> </tr> <tr> <td>Vapor Flow at Base Cond</td> <td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>normal m3/h</td> </tr> <tr> <td>Steam Flow</td> <td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>kg/h</td> </tr> <tr> <td>Temperature</td> <td>148</td><td></td><td>148</td><td></td><td></td><td></td><td></td><td></td><td>°C</td> </tr> <tr> <td>Liquid Specific Gravity</td> <td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>----</td> </tr> <tr> <td>Vapor Spec. Gravity (Air=1)</td> <td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>----</td> </tr> <tr> <td>Pressure</td> <td>2.70</td><td>2.70</td><td>2.70</td><td>0</td><td></td><td></td><td></td><td></td><td>kg/cm2(g)</td> </tr> <tr> <td>Liq Critical Press (Pseudo)</td> <td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>kg/cm2(a)</td> </tr> <tr> <td>Liquid Vapor Pressure</td> <td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>kg/cm2(a)</td> </tr> <tr> <td>Liquid Viscosity</td> <td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>cP</td> </tr> <tr> <td>Compressibility Z</td> <td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td> </tr> <tr> <td>Specific Heats Cp/Cv (Real)</td> <td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td> </tr> <tr> <td>Estimate Sound Press Level</td> <td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>dBA</td> </tr> </tbody> </table>	Process Data Case	General		General						Units	Open		Closed						Sizing Case										Calculated C _v										Liquid Flow at Flowing Cond	Inlet	Outlet	Inlet	Outlet	Inlet	Outlet	Inlet	Outlet	m3/h	Vapor Flow at Base Cond									normal m3/h	Steam Flow									kg/h	Temperature	148		148						°C	Liquid Specific Gravity									----	Vapor Spec. Gravity (Air=1)									----	Pressure	2.70	2.70	2.70	0					kg/cm2(g)	Liq Critical Press (Pseudo)									kg/cm2(a)	Liquid Vapor Pressure									kg/cm2(a)	Liquid Viscosity									cP	Compressibility Z										Specific Heats Cp/Cv (Real)										Estimate Sound Press Level									dBA
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VALVE BODY & BONNET	Body Type: Eccentric Disc § Body Size: 12" # Port Type: Single Inlet Conn Size: 12" # Type: Flanged Class: 150 Outlet Conn Size: 12" # Type: Flanged Class: 150 Flange Facing: Flange Face Finish: Body & Bonnet Material: Carbon Steel Bonnet Style: Packing Style: Manufacturer's Recommendation Packing Material: Manufacturer's Recommendation Body/Bonnet Gasket Material: Lub/Iso Valve Material: Lub Material: Manufacturer: Masoneilan (or equal) Model Number:	TRIM	Approximate Valve C _v : 4837 F _L : x _T : Trim Style: Man. Standard Trim Size: Inherent Characteristic: Equal Percentage Balance Style: Full Open Position: 90° Guiding Style: Shaft Closure Member Material: 316 Stainless Steel Seat Ring Material: RTFE Seat Ring Retainer Material: Carbon Steel Bearing Material: Nitrided Stainless Steel Bushing/Cage Material: Stem/Shaft Material: 17-4PH Stainless Steel Flow Direction: Flow to close Seat Leakage Class: ANSI Class VI																																																																																																																																																																						
	ACTUATOR		INSTALLATION	Type: Spring Diaph, Piston/Double Act Piston, Pneu ## Action Style: Fail Close Service Type: On-Off Max Shutoff Press: In: 2.80 Out: 0 kg/cm2(g) Maximum Stroke Time to Open: Maximum Stroke Time to Close: 5 Seconds Stroke Cycle Service:	Inlet Pipe Nominal Size: Est Wall Thick: Outlet Pipe Nominal Size: Est Wall Thick: Est Insulation Type: Est Insulation Thickness: Installation: Design Pressure : 10.5 kg/cm2g (F.V) Design Temperature : 195 C																																																																																																																																																																				
				§ The information is based on Masoneilan 39003 series.																																																																																																																																																																					
				# Line size valve.																																																																																																																																																																					
				## If a double acting piston actuator is provided for this valve, then a volume tank with the required accessories shall be supplied to move the control valve to the minimum signal position as specified, upon loss of instrument air pressure.																																																																																																																																																																					
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ALL ITEMS SHALL COMPLY WITH THE GENERAL REQUIREMENTS SHOWN ON THE GENERAL REQUIREMENTS SECTION OF THIS SPECIFICATION.

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CONTROL VALVES

PROJECT SPECIFICATION

963766 — **616**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Item Number: 16 Quantity: 1		Valve Tag Number: UXV-23-0703		Data Page: 2 of 2			
TRANS-DUCER	Transducer Tag Number:		HANDWHEEL	Handwheel Position:			
	Transducer Input Signal Range:			Usage:			
	Transducer Output Signal Range:						
	Manufacturer:						
SOLENOID VALVE	Solenoid Valve Tag Number: UXY-23-0703		POSITION SWITCH	Open Position Switch Tag Number:			
	Body Type: 3-Way Body Material: Brass			Close Position Switch Tag Number:			
	Seat Material: Stainless Steel Disc Material: Stainless Steel			Sensor Type:			
	Connection Size/Type: †			Contact Form:			
	Packing Style: Packless			Open Position Switch Contact Action:			
	Fluid: Instrument Air			Close Position Switch Contact Action:			
	Pressure: Contractor to determine			Contact Load: Contact Rating:			
	Minimum Pressure Port Orifice C _v : MAX Size: MAX			Mounting Location:			
	Minimum Exhaust Port Orifice C _v : MAX Size: MAX			Manufacturer:			
	Operation Style: †						
	Coil Configuration: Single			Position Transmitter Tag Number:			
	Operating Voltage: \$\$ Frequency:			Sensor Type:			
	Action Coil 1 Deenergized: Vent air from valve			Output Signal Range:			
	Action Coil 1 Energized: Supply air to valve			Travel Measurement:			
	Action Coil 2 Deenergized:			Mounting Location:			
	Action Coil 2 Energized:			Manufacturer:			
	POSITIONER	Manual Reset:		LOCK-UP VALVE	Action Style:		
		Manual Operator:			Mounting Location:		
Mounting Location: Nearest item to actuator in air line. †		Model Number:					
Manufacturer: ASCO (or equal)		Manufacturer:					
Positioner Tag Number:		Connection Size/Type:					
Positioner Type:		Body Material:					
Positioner Input Signal Range:		Diaphragm Material:					
Positioner Output Signal Range:		Exhaust Action:					
Manufacturer:		Mounting Location:					
		Manufacturer:					
AIR FILTER	Quantity: one		VOLUME BOOSTER	Connection Size/Type:			
	Type: Filter/Regulator			Body Material:			
	Manufacturer:			Seat Ring Material:			
TUBING	Tubing Size:			Diaphragm Material:			
	Tubing Material:			Mounting Location:			
				Manufacturer:			
DATA PAGE 2 REMARKS	† The solenoid valve shall be mounted and tubed by the control valve vendor. The solenoid valve size and type, the tubing size, the actuator air connection size, and the air/filter regulator capacity shall be determined by the control valve vendor to meet the maximum stroking time requirements specified for the control valve.						
	\$\$ The Contractor shall coordinate the solenoid valve operating voltage and coil watt rating with the supplier of the Concentration Section Shutdown System (UC-2301). See UOP Project Specification 660, this project.						

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CONTROL VALVES

PROJECT SPECIFICATION

963766 — **616**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

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GENERAL	Item Number: 17 Quantity: 1		Valve Tag Number: TV-23-0703		Data Page: 1 of 2	
	Service: S3 Steam to Flash Column Vaporizer		P&ID Drawing: 07			
Process Material: Steam						
SIZING CONDITIONS	Process Data Case	Design		Design		Design
	Sizing Case	Normal		Maximum		Minimum
	Calculated C _v	250		309		119
		Inlet	Outlet	Inlet	Outlet	Inlet
	Liquid Flow at Flowing Cond					Units
	Vapor Flow at Base Cond					m3/h
	Steam Flow	7947		8742		kg/h
	Temperature	148		138		°C
	Liquid Specific Gravity					----
	Vapor Spec. Gravity (Air=1)	0.622		0.622		----
	Pressure	2.70	1.78	2.50	1.78	kg/cm2(g)
	Liq Critical Press (Pseudo)					kg/cm2(a)
	Liquid Vapor Pressure					kg/cm2(a)
	Liquid Viscosity					cP
	Compressibility Z	0.967		0.964		
	Specific Heats Cp/Cv (Real)	1.318		1.319		
Estimate Sound Press Level					dBA	
VALVE BODY & BONNET	Body Type: Globe	Approximate Valve C _v : 415 F _L : 0.9 x _T : 0.72				
	Body Size: 8" Port Type: Single	Trim Style: Man. Standard Trim Size:				
	Inlet Conn Size: 8" Type: Flanged Class: 300	Inherent Characteristic: Linear				
	Outlet Conn Size: 8" Type: Flanged Class: 300	Balance Style: Balanced Full Open Position: 100 %				
	Flange Facing: Raised Face Flange Face Finish:	Guiding Style: Cage				
	Body & Bonnet Material: Carbon Steel	Closure Member Material: 17-4 PH Stainless Steel				
	Bonnet Style: Integral or Bolted	Seat Ring Material: 316 SS with Stellite sheath				
	Packing Style: Manufacturer's Recommendation	Seat Ring Retainer Material:				
	Packing Material: Manufacturer's Recommendation	Bearing Material:				
	Body/Bonnet Gasket Material: Manufacturer's Recommend	Bushing/Cage Material: CA6NM hard chrome plated				
	Lub/Iso Valve Material: Lub Material:	Stem/Shaft Material: 17-4 PH Stainless Steel				
	Manufacturer: Masoneilan (or equal)	Flow Direction:				
	Model Number:	Seat Leakage Class: ANSI Class V				
	ACTUATOR	Type: Spring Diaphragm, Pneumatic	Inlet Pipe Nominal Size: Est Wall Thick:			
Action Style: Fail Close		Outlet Pipe Nominal Size: Est Wall Thick:				
Service Type: Modulating		Est Insulation Type:				
Max Shutoff Press: In: 2.80 Out: 0 kg/cm2(g)		Est Insulation Thickness:				
Maximum Stroke Time to Open:		Installation:				
Maximum Stroke Time to Close: 5 Seconds		Design Pressure : 10.5 kg/cm2g (F.V)				
Stroke Cycle Service:		Design Temperature : 195 C				
DATA PAGE 1 REMARKS		§ The information is based on Masoneilan 41000 series.				
	UOP calculations indicate that the noise level of this valve may exceed 85 dBA. The estimated noise level shall be verified based on final design criteria and Vendor/Contractor's detailed calculations. It is the responsibility of the detailed design Contractor to select a valve size, style and/or trim style that complies with all the specified process requirements, including noise limitation.					
	Alternate Cooling Water Operation Maximum Shutoff Pressure: In: 3.90 kg/cm2(g) Out: 0 kg/cm2(g) .					
ALL ITEMS SHALL COMPLY WITH THE GENERAL REQUIREMENTS SHOWN ON THE GENERAL REQUIREMENTS SECTION OF THIS SPECIFICATION.						



CONTROL VALVES

PROJECT SPECIFICATION

963766 — 616

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Item Number: 17 Quantity: 1		Valve Tag Number: TV-23-0703		Data Page: 2 of 2		
TRANSDUCER	Transducer Tag Number:		HANDWHEEL	Handwheel Position:		
	Transducer Input Signal Range:			Usage:		
	Transducer Output Signal Range:					
	Manufacturer:					
SOLENOID VALVE	Solenoid Valve Tag Number: TXY-23-0703		POSITION SWITCH	Open Position Switch Tag Number:		
	Body Type: 3-Way Body Material: Brass			Close Position Switch Tag Number:		
	Seat Material: Stainless Steel Disc Material: Stainless Steel			Sensor Type:		
	Connection Size/Type: †			Contact Form:		
	Packing Style: Packless		POSITION TRANSMITTER	Open Position Switch Contact Action:		
	Fluid: Control Signal Air			Close Position Switch Contact Action:		
	Pressure: Contractor to determine			Contact Load: Contact Rating:		
	Minimum Pressure Port Orifice C _v : MAX Size: MAX			Mounting Location:		
	Minimum Exhaust Port Orifice C _v : MAX Size: MAX			Manufacturer:		
	Operation Style: †			POSITION TRANSMITTER	Position Transmitter Tag Number:	
	Coil Configuration: Single				Sensor Type:	
	Operating Voltage: \$\$ Frequency:				Output Signal Range:	
	Action Coil 1 Deenergized: Vent air from valve				Travel Measurement:	
	Action Coil 1 Energized: Supply air to valve			LOCK-UP VALVE	Mounting Location:	
	Action Coil 2 Deenergized:		Manufacturer:			
	Action Coil 2 Energized:		Action Style:			
	Manual Reset:		LOCK-UP VALVE	Mounting Location:		
	Manual Operator:			Model Number:		
	Mounting Location: Nearest item to actuator in air line. †			Manufacturer:		
	Manufacturer: ASCO (or equal)		QUICK EXHAUST	Connection Size/Type:		
POSITIONER	Positioner Tag Number:			Body Material:		
	Positioner Type: Digital			Diaphragm Material:		
	Positioner Input Signal Range: 4 - 20 mA			Exhaust Action:		
	Positioner Output Signal Range: 0 - 100 %		Mounting Location:			
	Manufacturer:		Manufacturer:			
AIR FILTER	Quantity: one		VOLUME BOOSTER	Connection Size/Type:		
	Type: Filter/Regulator			Body Material:		
	Manufacturer:			Seat Ring Material:		
TUBING	Tubing Size:			VOLUME BOOSTER	Diaphragm Material:	
	Tubing Material:		Mounting Location:			
			Manufacturer:			
DATA PAGE 2 REMARKS	† The solenoid valve shall be mounted and tubed by the control valve vendor. The solenoid valve size and type, the tubing size, the actuator air connection size, and the air/filter regulator capacity shall be determined by the control valve vendor to meet the maximum stroking time requirements specified for the control valve. Pilot operated solenoid valves shall be suitable for operation in low pressure control signal air.					
	\$\$ The Contractor shall coordinate the solenoid valve operating voltage and coil watt rating with the supplier of the Concetration Section Shutdown System (UC-2301). See UOP Project Specification 660, this project.					

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CONTROL VALVES

PROJECT SPECIFICATION

963766 — **616**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

GENERAL	Item Number: 18 Quantity: 1 Valve Tag Number: UXV-23-0704 Data Page: 1 of 2																																																																																																																																																																
	Service: Reliable Cooling Water to Flash Column Vaporizer Shutoff Process Material: Water P&ID Drawing: 07																																																																																																																																																																
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CONTROL VALVES

PROJECT SPECIFICATION

963766 — **616**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Item Number: 18 Quantity: 1		Valve Tag Number: UXV-23-0704		Data Page: 2 of 2	
TRANS-DUCER	Transducer Tag Number:		HANDWHEEL	Handwheel Position:	
	Transducer Input Signal Range:			Usage:	
	Transducer Output Signal Range:				
	Manufacturer:				
SOLENOID VALVE	Solenoid Valve Tag Number: UXY-23-0704		POSITION SWITCH	Open Position Switch Tag Number:	
	Body Type: 3-Way Body Material: Brass			Close Position Switch Tag Number:	
	Seat Material: Stainless Steel Disc Material: Stainless Steel			Sensor Type:	
	Connection Size/Type: †			Contact Form:	
	Packing Style: Packless		Open Position Switch Contact Action:		
	Fluid: Instrument Air		Close Position Switch Contact Action:		
	Pressure: Contractor to determine		Contact Load: Contact Rating:		
	Minimum Pressure Port Orifice C _v : MAX Size: MAX		Mounting Location:		
	Minimum Exhaust Port Orifice C _v : MAX Size: MAX		Manufacturer:		
	Operation Style: †		POSITION TRANSMITTER	Position Transmitter Tag Number:	
	Coil Configuration: Single			Sensor Type:	
	Operating Voltage: \$\$ Frequency:			Output Signal Range:	
	Action Coil 1 Deenergized: Vent air from valve			Travel Measurement:	
	Action Coil 1 Energized: Supply air to valve		Mounting Location:		
	Action Coil 2 Deenergized:		Manufacturer:		
	Action Coil 2 Energized:		LOCK-UP VALVE	Action Style:	
	Manual Reset:			Mounting Location:	
	Manual Operator:			Model Number:	
Mounting Location: Nearest item to actuator in air line. †		Manufacturer:			
Manufacturer: ASCO (or equal)		QUICK EXHAUST	Connection Size/Type:		
Positioner Tag Number:			Body Material:		
Positioner Type:			Diaphragm Material:		
Positioner Input Signal Range:			Exhaust Action:		
Positioner Output Signal Range:		Mounting Location:			
Manufacturer:		Manufacturer:			
AIR FILTER	Quantity: one		VOLUME BOOSTER	Connection Size/Type:	
	Type: Filter/Regulator			Body Material:	
	Manufacturer:			Seat Ring Material:	
Tubing Size:		Diaphragm Material:			
Tubing Material:		Mounting Location:			
		Manufacturer:			
DATA PAGE 2 REMARKS	† The solenoid valve shall be mounted and tubed by the control valve vendor. The solenoid valve size and type, the tubing size, the actuator air connection size, and the air/filter regulator capacity shall be determined by the control valve vendor to meet the maximum stroking time requirements specified for the control valve.				
	\$\$ The Contractor shall coordinate the solenoid valve operating voltage and coil watt rating with the supplier of the Concentration Section Shutdown System (UC-2301). See UOP Project Specification 660, this project.				

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CONTROL VALVES

PROJECT SPECIFICATION

963766 — **616**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

GENERAL	Item Number: 19 Quantity: 1 Valve Tag Number: UXV-23-0706 Data Page: 1 of 2																																																																																																																																																																
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CONTROL VALVES

PROJECT SPECIFICATION

963766 — **616**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Item Number: 19 Quantity: 1		Valve Tag Number: UXV-23-0706		Data Page: 2 of 2		
TRANSDUCER	Transducer Tag Number:		HANDWHEEL	Handwheel Position:		
	Transducer Input Signal Range:			Usage:		
	Transducer Output Signal Range:					
	Manufacturer:					
SOLENOID VALVE	Solenoid Valve Tag Number: UXY-23-0706		POSITION SWITCH	Open Position Switch Tag Number:		
	Body Type: 3-Way Body Material: Brass			Close Position Switch Tag Number:		
	Seat Material: Stainless Steel Disc Material: Stainless Steel			Sensor Type:		
	Connection Size/Type: †			Contact Form:		
	Packing Style: Packless			Open Position Switch Contact Action:		
	Fluid: Instrument Air			Close Position Switch Contact Action:		
	Pressure: Contractor to determine			Contact Load: Contact Rating:		
	Minimum Pressure Port Orifice C _v : MAX Size: MAX			Mounting Location:		
	Minimum Exhaust Port Orifice C _v : MAX Size: MAX			Manufacturer:		
	Operation Style: †			POSITION TRANSMITTER	Position Transmitter Tag Number:	
	Coil Configuration: Single				Sensor Type:	
	Operating Voltage: \$\$ Frequency:				Output Signal Range:	
	Action Coil 1 Deenergized: Vent air from valve				Travel Measurement:	
	Action Coil 1 Energized: Supply air to valve				Mounting Location:	
	Action Coil 2 Deenergized:				Manufacturer:	
	Action Coil 2 Energized:			LOCK-UP VALVE	Action Style:	
	Manual Reset:				Mounting Location:	
	Manual Operator:				Model Number:	
Mounting Location: Nearest item to actuator in air line. †		QUICK EXHAUST	Manufacturer:			
Manufacturer: ASCO (or equal)			Connection Size/Type:			
			Body Material:			
POSITIONER	Positioner Tag Number:			Diaphragm Material:		
	Positioner Type:			Exhaust Action:		
	Positioner Input Signal Range:			Mounting Location:		
AIR FILTER	Positioner Output Signal Range:		VOLUME BOOSTER	Manufacturer:		
	Manufacturer:			Connection Size/Type:		
	Quantity: one			Body Material:		
TUBING	Type: Filter/Regulator			Seat Ring Material:		
	Manufacturer:			Diaphragm Material:		
	Tubing Size:			Mounting Location:		
DATA PAGE 2 REMARKS	Tubing Material:			Manufacturer:		
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CONTROL VALVES

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REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

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GENERAL	Item Number: 20 Quantity: 1 Valve Tag Number: UXV-23-0705 Data Page: 1 of 2																																																																																																																																																																
	Service: Cumene Quench to Flash Column Vaporizer Process Material: Hydrocarbon with aromatics P&ID Drawing: 07																																																																																																																																																																
SIZING CONDITIONS	<table border="1"> <thead> <tr> <th>Process Data Case</th> <th colspan="2">General</th> <th colspan="2">General</th> <th colspan="2"></th> <th colspan="2"></th> <th></th> </tr> <tr> <th>Sizing Case</th> <th colspan="2">Open</th> <th colspan="2">Closed</th> <th colspan="2"></th> <th colspan="2"></th> <th></th> </tr> <tr> <th>Calculated C_v</th> <th>Inlet</th> <th>Outlet</th> <th>Inlet</th> <th>Outlet</th> <th>Inlet</th> <th>Outlet</th> <th>Inlet</th> <th>Outlet</th> <th>Units</th> </tr> </thead> <tbody> <tr> <td>Liquid Flow at Flowing Cond</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>m3/h</td> </tr> <tr> <td>Vapor Flow at Base Cond</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>normal m3/h</td> </tr> <tr> <td>Steam Flow</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>kg/h</td> </tr> <tr> <td>Temperature</td> <td>46</td> <td></td> <td>46</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>°C</td> </tr> <tr> <td>Liquid Specific Gravity</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>----</td> </tr> <tr> <td>Vapor Spec. Gravity (Air=1)</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>----</td> </tr> <tr> <td>Pressure</td> <td>0.07</td> <td>0.07</td> <td>0.07</td> <td>-0.74</td> <td></td> <td></td> <td></td> <td></td> <td>kg/cm2(g)</td> </tr> <tr> <td>Liq Critical Press (Pseudo)</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>kg/cm2(a)</td> </tr> <tr> <td>Liquid Vapor Pressure</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>kg/cm2(a)</td> </tr> <tr> <td>Liquid Viscosity</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>cP</td> </tr> <tr> <td>Compressibility Z</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> <tr> <td>Specific Heats Cp/Cv (Real)</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> <tr> <td>Estimate Sound Press Level</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>dBA</td> </tr> </tbody> </table>	Process Data Case	General		General							Sizing Case	Open		Closed							Calculated C _v	Inlet	Outlet	Inlet	Outlet	Inlet	Outlet	Inlet	Outlet	Units	Liquid Flow at Flowing Cond									m3/h	Vapor Flow at Base Cond									normal m3/h	Steam Flow									kg/h	Temperature	46		46						°C	Liquid Specific Gravity									----	Vapor Spec. Gravity (Air=1)									----	Pressure	0.07	0.07	0.07	-0.74					kg/cm2(g)	Liq Critical Press (Pseudo)									kg/cm2(a)	Liquid Vapor Pressure									kg/cm2(a)	Liquid Viscosity									cP	Compressibility Z										Specific Heats Cp/Cv (Real)										Estimate Sound Press Level									dBA
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VALVE BODY & BONNET	Body Type: Eccentric Disc § Body Size: 6" #2 Port Type: Single Inlet Conn Size: 6" #2 Type: Flanged Class: 300 Outlet Conn Size: 6" #2 Type: Flanged Class: 300 Flange Facing: Flange Face Finish: Body & Bonnet Material: 316 Stainless Steel ‡ Bonnet Style: Packing Style: Manufacturer's Recommendation † Packing Material: Manufacturer's Recommendation Body/Bonnet Gasket Material: Lub/Iso Valve Material: Lub Material: Manufacturer: Metso Automation (or equal) Model Number:	TRIM	Approximate Valve C _v : 510 F _L : x _T : Trim Style: Man. Standard Trim Size: Inherent Characteristic: Equal Percentage Balance Style: Full Open Position: 90° Guiding Style: Shaft Closure Member Material: 316 Stainless Steel Seat Ring Material: Incoloy 825 with Hard Chrome Seat Ring Retainer Material: 316 Stainless Steel Bearing Material: Stellite Bushing/Cage Material: Stem/Shaft Material: 17-4PH Flow Direction: Seat Leakage Class: ANSI Class V																																																																																																																																																														
	ACTUATOR		INSTALLATION	Type: Spring Return Piston, Pneumatic Action Style: Fail Open Service Type: On-Off Max Shutoff Press: In: 3.85 Out: -1.02 kg/cm2(g) Maximum Stroke Time to Open: 5 Seconds Maximum Stroke Time to Close: Stroke Cycle Service:	Inlet Pipe Nominal Size: Est Wall Thick: Outlet Pipe Nominal Size: Est Wall Thick: Est Insulation Type: Est Insulation Thickness: Installation: Design Pressure : 4.2 kg/cm2g (F.V) Design Temperature : 120 C																																																																																																																																																												
				§ The information is based on Neldisc L6 series. Valve selected shall be of double flanged construction with minimal flange-to-flange dimension and no pockets within the valve body.																																																																																																																																																													
				‡ Preferred material is L grade stainless steel. Standard grade stainless steel that is not welded may be used for process wetted parts including the body. Stainless steel with weld repair is acceptable if low carbon (<= 0.03 %) filler material is used.																																																																																																																																																													
				#2 Line size valve.																																																																																																																																																													
				† Provide packing suitable for vacuum service to minimize air leakage into the process line.																																																																																																																																																													

ALL ITEMS SHALL COMPLY WITH THE GENERAL REQUIREMENTS SHOWN ON THE GENERAL REQUIREMENTS SECTION OF THIS SPECIFICATION.



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CONTROL VALVES

PROJECT SPECIFICATION

963766 — **616**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Item Number: 20 Quantity: 1		Valve Tag Number: UXV-23-0705		Data Page: 2 of 2			
TRANS-DUCER	Transducer Tag Number:		HANDWHEEL	Handwheel Position:			
	Transducer Input Signal Range:			Usage:			
Transducer Output Signal Range:							
Manufacturer:							
SOLENOID VALVE	Solenoid Valve Tag Number: UXY-23-0705		POSITION SWITCH	Open Position Switch Tag Number:			
	Body Type: 3-Way Body Material: Brass			Close Position Switch Tag Number:			
	Seat Material: Stainless Steel Disc Material: Stainless Steel			Sensor Type:			
	Connection Size/Type: †			Contact Form:			
	Packing Style: Packless			Open Position Switch Contact Action:			
	Fluid: Instrument Air			Close Position Switch Contact Action:			
	Pressure: Contractor to determine			Contact Load: Contact Rating:			
	Minimum Pressure Port Orifice C _v : MAX Size: MAX			Mounting Location:			
	Minimum Exhaust Port Orifice C _v : MAX Size: MAX			Manufacturer:			
	Operation Style: †						
	Coil Configuration: Single			Position Transmitter Tag Number:			
	Operating Voltage: \$\$ Frequency:			Sensor Type:			
	Action Coil 1 Deenergized: Vent air from valve			Output Signal Range:			
	Action Coil 1 Energized: Supply air to valve			Travel Measurement:			
	Action Coil 2 Deenergized:			Mounting Location:			
	Action Coil 2 Energized:			Manufacturer:			
	POSITIONER	Manual Reset:		LOCK-UP VALVE	Action Style:		
		Manual Operator:			Mounting Location:		
Mounting Location: Nearest item to actuator in air line. †		Model Number:					
Manufacturer: ASCO (or equal)		Manufacturer:					
Positioner Tag Number:		Connection Size/Type:					
Positioner Type:		Body Material:					
Positioner Input Signal Range:		Diaphragm Material:					
Positioner Output Signal Range:		Exhaust Action:					
Manufacturer:		Mounting Location:					
		Manufacturer:					
AIR FILTER	Quantity: one		VOLUME BOOSTER	Connection Size/Type:			
	Type: Filter/Regulator			Body Material:			
	Manufacturer:			Seat Ring Material:			
TUBING	Tubing Size:			Diaphragm Material:			
	Tubing Material:			Mounting Location:			
		Manufacturer:					
DATA PAGE 2 REMARKS	† The solenoid valve shall be mounted and tubed by the control valve vendor. The solenoid valve size and type, the tubing size, the actuator air connection size, and the air/filter regulator capacity shall be determined by the control valve vendor to meet the maximum stroking time requirements specified for the control valve.						
	\$\$ The Contractor shall coordinate the solenoid valve operating voltage and coil watt rating with the supplier of the Concentration Section Shutdown System (UC-2301). See UOP Project Specification 660, this project.						

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CONTROL VALVES

PROJECT SPECIFICATION

963766 — **616**

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0	25JUN12	MVD	MW				

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GENERAL	Item Number: 21 Quantity: 1 Valve Tag Number: UXV-23-0707 Data Page: 1 of 2																																																																																																																																																																
	Service: Steam to / from Flash Column Vaporizer Condensate Drum Process Material: Steam P&ID Drawing: 07																																																																																																																																																																
SIZING CONDITIONS	<table border="1"> <thead> <tr> <th>Process Data Case</th> <th colspan="2">General</th> <th colspan="2">General</th> <th colspan="2"></th> <th colspan="2"></th> <th></th> </tr> <tr> <th>Sizing Case</th> <th colspan="2">Open</th> <th colspan="2">Closed</th> <th colspan="2"></th> <th colspan="2"></th> <th></th> </tr> <tr> <th>Calculated C_v</th> <th>Inlet</th> <th>Outlet</th> <th>Inlet</th> <th>Outlet</th> <th>Inlet</th> <th>Outlet</th> <th>Inlet</th> <th>Outlet</th> <th>Units</th> </tr> </thead> <tbody> <tr> <td>Liquid Flow at Flowing Cond</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>m3/h</td> </tr> <tr> <td>Vapor Flow at Base Cond</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>normal m3/h</td> </tr> <tr> <td>Steam Flow</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>kg/h</td> </tr> <tr> <td>Temperature</td> <td>135</td> <td></td> <td>135</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>°C</td> </tr> <tr> <td>Liquid Specific Gravity</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>----</td> </tr> <tr> <td>Vapor Spec. Gravity (Air=1)</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>----</td> </tr> <tr> <td>Pressure</td> <td>1.78</td> <td>1.78</td> <td>3.90</td> <td>2.32</td> <td></td> <td></td> <td></td> <td></td> <td>kg/cm2(g)</td> </tr> <tr> <td>Liq Critical Press (Pseudo)</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>kg/cm2(a)</td> </tr> <tr> <td>Liquid Vapor Pressure</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>kg/cm2(a)</td> </tr> <tr> <td>Liquid Viscosity</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>cP</td> </tr> <tr> <td>Compressibility Z</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> <tr> <td>Specific Heats Cp/Cv (Real)</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> <tr> <td>Estimate Sound Press Level</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>dBA</td> </tr> </tbody> </table>	Process Data Case	General		General							Sizing Case	Open		Closed							Calculated C _v	Inlet	Outlet	Inlet	Outlet	Inlet	Outlet	Inlet	Outlet	Units	Liquid Flow at Flowing Cond									m3/h	Vapor Flow at Base Cond									normal m3/h	Steam Flow									kg/h	Temperature	135		135						°C	Liquid Specific Gravity									----	Vapor Spec. Gravity (Air=1)									----	Pressure	1.78	1.78	3.90	2.32					kg/cm2(g)	Liq Critical Press (Pseudo)									kg/cm2(a)	Liquid Vapor Pressure									kg/cm2(a)	Liquid Viscosity									cP	Compressibility Z										Specific Heats Cp/Cv (Real)										Estimate Sound Press Level									dBA
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VALVE BODY & BONNET	Body Type: Globe § Body Size: 1-1/2" ± Port Type: Single Inlet Conn Size: 1-1/2" ± Type: Flanged Class: 300 Outlet Conn Size: 1-1/2" ± Type: Flanged Class: 300 Flange Facing: Raised Face Flange Face Finish: Body & Bonnet Material: Carbon Steel Bonnet Style: Integral or Bolted Packing Style: Manufacturer's Recommendation Packing Material: Manufacturer's Recommendation Body/Bonnet Gasket Material: Manufacturer's Recommend Lub/Iso Valve Material: Lub Material: Manufacturer: Masoneilan (or equal) Model Number:	TRIM	Approximate Valve C _v : 25 F _L : x _T : Trim Style: Man. Standard Trim Size: Inherent Characteristic: Equal Percentage Balance Style: Unbalanced Full Open Position: 100 % Guiding Style: Top Closure Member Material: 316 Stainless Steel + soft seat Seat Ring Material: 316 Stainless Steel Seat Ring Retainer Material: 304 Stainless Steel Bearing Material: Bushing/Cage Material: Stellite Stem/Shaft Material: 316 Stainless Steel Flow Direction: Flow to close Seat Leakage Class: ANSI Class VI																																																																																																																																																														
	ACTUATOR		INSTALLATION	Type: Spring Diaphragm, Pneumatic Action Style: Fail Close Service Type: On-Off Max Shutoff Press: In: 7.70 Out: 0 kg/cm2(g) Maximum Stroke Time to Open: Maximum Stroke Time to Close: 5 Seconds Stroke Cycle Service:	Inlet Pipe Nominal Size: Est Wall Thick: Outlet Pipe Nominal Size: Est Wall Thick: Est Insulation Type: Est Insulation Thickness: Installation: Design Pressure : 10.5 kg/cm2g (F.V) Design Temperature : 195 C																																																																																																																																																												
				§ The information is based on Masoneilan 21000 series.																																																																																																																																																													
				± Line size valve.																																																																																																																																																													

ALL ITEMS SHALL COMPLY WITH THE GENERAL REQUIREMENTS SHOWN ON THE GENERAL REQUIREMENTS SECTION OF THIS SPECIFICATION.



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CONTROL VALVES

PROJECT SPECIFICATION

963766 — **616**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Item Number: 21 Quantity: 1		Valve Tag Number: UXV-23-0707		Data Page: 2 of 2			
TRANS-DUCER	Transducer Tag Number:		HANDWHEEL	Handwheel Position:			
	Transducer Input Signal Range:			Usage:			
Transducer Output Signal Range:							
Manufacturer:							
SOLENOID VALVE	Solenoid Valve Tag Number: UXY-23-0707		POSITION SWITCH	Open Position Switch Tag Number:			
	Body Type: 3-Way Body Material: Brass			Close Position Switch Tag Number:			
	Seat Material: Stainless Steel Disc Material: Stainless Steel			Sensor Type:			
	Connection Size/Type: †			Contact Form:			
	Packing Style: Packless			Open Position Switch Contact Action:			
	Fluid: Instrument Air			Close Position Switch Contact Action:			
	Pressure: Contractor to determine			Contact Load: Contact Rating:			
	Minimum Pressure Port Orifice C _v : MAX Size: MAX			Mounting Location:			
	Minimum Exhaust Port Orifice C _v : MAX Size: MAX			Manufacturer:			
	Operation Style: †						
	Coil Configuration: Single			Position Transmitter Tag Number:			
	Operating Voltage: \$\$ Frequency:			Sensor Type:			
	Action Coil 1 Deenergized: Vent air from valve			Output Signal Range:			
	Action Coil 1 Energized: Supply air to valve			Travel Measurement:			
	Action Coil 2 Deenergized:			Mounting Location:			
	Action Coil 2 Energized:			Manufacturer:			
	POSITIONER	Manual Reset:		LOCK-UP VALVE	Action Style:		
		Manual Operator:			Mounting Location:		
Mounting Location: Nearest item to actuator in air line. †		Model Number:					
Manufacturer: ASCO (or equal)		Manufacturer:					
Positioner Tag Number:		Connection Size/Type:					
Positioner Type:		Body Material:					
Positioner Input Signal Range:		Diaphragm Material:					
Positioner Output Signal Range:		Exhaust Action:					
Manufacturer:		Mounting Location:					
		Manufacturer:					
AIR FILTER	Quantity: one		VOLUME BOOSTER	Connection Size/Type:			
	Type: Filter/Regulator			Body Material:			
	Manufacturer:			Seat Ring Material:			
TUBING	Tubing Size:			Diaphragm Material:			
	Tubing Material:			Mounting Location:			
		Manufacturer:					
DATA PAGE 2 REMARKS	† The solenoid valve shall be mounted and tubed by the control valve vendor. The solenoid valve size and type, the tubing size, the actuator air connection size, and the air/filter regulator capacity shall be determined by the control valve vendor to meet the maximum stroking time requirements specified for the control valve.						
	\$\$ The Contractor shall coordinate the solenoid valve operating voltage and coil watt rating with the supplier of the Concentration Section Shutdown System (UC-2301). See UOP Project Specification 660, this project.						

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PROJECT SPECIFICATION

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REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

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GENERAL	Item Number: 22 Quantity: 1		Valve Tag Number: LV-23-0701		Data Page: 1 of 2			
	Service: Flash Column Vaporizer Condensate to SC3 Condensate Header							
	Process Material: Water and Steam			P&ID Drawing: 07A				
SIZING CONDITIONS	Process Data Case	Single Case		Single Case		Single Case		
	Sizing Case	Normal		Maximum		Minimum		
	Calculated C _v	7.6		8.7		3.50		
		Inlet	Outlet	Inlet	Outlet	Inlet	Outlet	Units
	Liquid Flow at Flowing Cond	8.6		9.5		4.3		m ³ /h
	Vapor Flow at Base Cond							normal m ³ /h
	Steam Flow							kg/h
	Temperature	135		135		135		°C
	Liquid Specific Gravity	0.931		0.931		0.931		----
	Vapor Spec. Gravity (Air=1)							----
	Pressure	3.99	2.23	3.83	2.24	4.35	2.21	kg/cm ² (g)
	Liq Critical Press (Pseudo)	224.90		224.90		224.90		kg/cm ² (a)
	Liquid Vapor Pressure	3.23		3.23		3.23		kg/cm ² (a)
	Liquid Viscosity	0.204		0.204		0.204		cP
	Compressibility Z							
Specific Heats Cp/Cv (Real)								
Estimate Sound Press Level							dBA	
VALVE BODY & BONNET	Body Type: Globe §			Approximate Valve C _v : 13 F _L : 0.9 x _T : 0.72				
	Body Size: 1-1/2" Port Type: Single			Trim Style: Man. Standard Trim Size:				
	Inlet Conn Size: 1-1/2" Type: Flanged Class: 300			Inherent Characteristic: Linear				
	Outlet Conn Size: 1-1/2" Type: Flanged Class: 300			Balance Style: Unbalanced Full Open Position: 100 %				
	Flange Facing: Raised Face Flange Face Finish:			Guiding Style: Top				
	Body & Bonnet Material: Carbon Steel			Closure Member Material: 316 SS, Stellite sheath				
	Bonnet Style: Integral or Bolted			Seat Ring Material: 316 SS with Stellite sheath				
	Packing Style: Manufacturer's Recommendation			Seat Ring Retainer Material: 304 Stainless Steel				
	Packing Material: Manufacturer's Recommendation			Bearing Material:				
	Body/Bonnet Gasket Material: Manufacturer's Recommend			Bushing/Cage Material: Stellite				
	Lub/Iso Valve Material: Lub Material:			Stem/Shaft Material: 316 Stainless Steel				
	Manufacturer: Masoneilan (or equal)			Flow Direction:				
	Model Number:			Seat Leakage Class: ANSI Class IV				
	ACTUATOR	Type: Spring Diaphragm, Pneumatic			Inlet Pipe Nominal Size: Est Wall Thick:			
Action Style: Fail Close			Outlet Pipe Nominal Size: Est Wall Thick:					
Service Type: Modulating			Est Insulation Type:					
Max Shutoff Press: In: 6.38 Out: 2.20 kg/cm ² (g)			Est Insulation Thickness:					
Maximum Stroke Time to Open:			Installation:					
Maximum Stroke Time to Close:			Design Pressure : 9.3 kg/cm ² g (F.V)					
Stroke Cycle Service:			Design Temperature : 195 C					
DATA PAGE 1 REMARKS	§ The information is based on Masoneilan 21000 series.							
	Hardened trim is recommended for this service based upon the possibility of cavitation occurring.							

ALL ITEMS SHALL COMPLY WITH THE GENERAL REQUIREMENTS SHOWN ON THE GENERAL REQUIREMENTS SECTION OF THIS SPECIFICATION.



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CONTROL VALVES

PROJECT SPECIFICATION

963766 — **616**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Item Number: 22 Quantity: 1		Valve Tag Number: LV-23-0701		Data Page: 2 of 2	
TRANS-DUCER	Transducer Tag Number:		HANDWHEEL	Handwheel Position:	
	Transducer Input Signal Range:			Usage:	
	Transducer Output Signal Range:				
	Manufacturer:				
SOLENOID VALVE	Solenoid Valve Tag Number:		POSITION SWITCH	Open Position Switch Tag Number:	
	Body Type: Body Material:			Close Position Switch Tag Number:	
	Seat Material: Disc Material:			Sensor Type:	
	Connection Size/Type:			Contact Form:	
	Packing Style:			Open Position Switch Contact Action:	
	Fluid:			Close Position Switch Contact Action:	
	Pressure:			Contact Load: Contact Rating:	
	Minimum Pressure Port Orifice C _v : Size:			Mounting Location:	
	Minimum Exhaust Port Orifice C _v : Size:			Manufacturer:	
	Operation Style:			POSITION TRANSMITTER	Position Transmitter Tag Number:
	Coil Configuration:		Sensor Type:		
	Operating Voltage: Frequency:		Output Signal Range:		
	Action Coil 1 Deenergized:		Travel Measurement:		
	Action Coil 1 Energized:		Mounting Location:		
	Action Coil 2 Deenergized:		LOCK-UP VALVE	Manufacturer:	
	Action Coil 2 Energized:			Action Style:	
	Manual Reset:			Mounting Location:	
	Manual Operator:		QUICK EXHAUST	Model Number:	
	Mounting Location:			Manufacturer:	
	Manufacturer:			Connection Size/Type:	
POSITIONER	Positioner Tag Number:		VOLUME BOOSTER	Body Material:	
	Positioner Type: Digital			Seat Ring Material:	
	Positioner Input Signal Range: 4 - 20 mA			Diaphragm Material:	
AIR FILTER	Positioner Output Signal Range: 0 - 100 %		TUBING	Mounting Location:	
	Manufacturer:			Manufacturer:	
	Quantity: one				
TUBING	Type: Filter/Regulator				
	Manufacturer:				
	Tubing Size:				
DATA PAGE 2 REMARKS	Tubing Material:				

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PROJECT SPECIFICATION

963766 — **616**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

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GENERAL	Item Number: 23 Quantity: 1 Valve Tag Number: UXV-23-0801 Data Page: 1 of 2																																																																																																																																																																
	Service: Cumene Quench to Flash Column Collector Tray Process Material: Hydrocarbon with aromatics P&ID Drawing: 08																																																																																																																																																																
SIZING CONDITIONS	<table border="1"> <thead> <tr> <th>Process Data Case</th> <th colspan="2">General</th> <th colspan="2">General</th> <th colspan="2"></th> <th colspan="2"></th> <th></th> </tr> <tr> <th>Sizing Case</th> <th colspan="2">Open</th> <th colspan="2">Closed</th> <th colspan="2"></th> <th colspan="2"></th> <th></th> </tr> <tr> <th>Calculated C_v</th> <th>Inlet</th> <th>Outlet</th> <th>Inlet</th> <th>Outlet</th> <th>Inlet</th> <th>Outlet</th> <th>Inlet</th> <th>Outlet</th> <th>Units</th> </tr> </thead> <tbody> <tr> <td>Liquid Flow at Flowing Cond</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>m3/h</td> </tr> <tr> <td>Vapor Flow at Base Cond</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>normal m3/h</td> </tr> <tr> <td>Steam Flow</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>kg/h</td> </tr> <tr> <td>Temperature</td> <td>46</td> <td></td> <td>46</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>°C</td> </tr> <tr> <td>Liquid Specific Gravity</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>----</td> </tr> <tr> <td>Vapor Spec. Gravity (Air=1)</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>----</td> </tr> <tr> <td>Pressure</td> <td>-0.72</td> <td>-0.72</td> <td>-0.72</td> <td>-1.01</td> <td></td> <td></td> <td></td> <td></td> <td>kg/cm2(g)</td> </tr> <tr> <td>Liq Critical Press (Pseudo)</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>kg/cm2(a)</td> </tr> <tr> <td>Liquid Vapor Pressure</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>kg/cm2(a)</td> </tr> <tr> <td>Liquid Viscosity</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>cP</td> </tr> <tr> <td>Compressibility Z</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> <tr> <td>Specific Heats Cp/Cv (Real)</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> <tr> <td>Estimate Sound Press Level</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>dBA</td> </tr> </tbody> </table>	Process Data Case	General		General							Sizing Case	Open		Closed							Calculated C _v	Inlet	Outlet	Inlet	Outlet	Inlet	Outlet	Inlet	Outlet	Units	Liquid Flow at Flowing Cond									m3/h	Vapor Flow at Base Cond									normal m3/h	Steam Flow									kg/h	Temperature	46		46						°C	Liquid Specific Gravity									----	Vapor Spec. Gravity (Air=1)									----	Pressure	-0.72	-0.72	-0.72	-1.01					kg/cm2(g)	Liq Critical Press (Pseudo)									kg/cm2(a)	Liquid Vapor Pressure									kg/cm2(a)	Liquid Viscosity									cP	Compressibility Z										Specific Heats Cp/Cv (Real)										Estimate Sound Press Level									dBA
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VALVE BODY & BONNET	Body Type: Eccentric Disc § Body Size: 6" #2 Port Type: Single Inlet Conn Size: 6" #2 Type: Flanged Class: 300 Outlet Conn Size: 6" #2 Type: Flanged Class: 300 Flange Facing: Flange Face Finish: Body & Bonnet Material: 316 Stainless Steel ‡ Bonnet Style: Packing Style: Manufacturer's Recommendation † Packing Material: Manufacturer's Recommendation Body/Bonnet Gasket Material: Lub/Iso Valve Material: Lub Material: Manufacturer: Metso Automation (or equal) Model Number:	TRIM	Approximate Valve C _v : 510 F _L : x _T : Trim Style: Man. Standard Trim Size: Inherent Characteristic: Equal Percentage Balance Style: Full Open Position: 90° Guiding Style: Shaft Closure Member Material: 316 Stainless Steel Seat Ring Material: Incoloy 825 with Hard Chrome Seat Ring Retainer Material: 316 Stainless Steel Bearing Material: Stellite Bushing/Cage Material: Stem/Shaft Material: 17-4PH Flow Direction: Seat Leakage Class: ANSI Class V																																																																																																																																																														
	ACTUATOR		INSTALLATION	Type: Spring Return Piston, Pneumatic Action Style: Fail Open Service Type: On-Off Max Shutoff Press: In: 3.85 Out: -1.02 kg/cm2(g) Maximum Stroke Time to Open: 5 Seconds Maximum Stroke Time to Close: Stroke Cycle Service:	Inlet Pipe Nominal Size: Est Wall Thick: Outlet Pipe Nominal Size: Est Wall Thick: Est Insulation Type: Est Insulation Thickness: Installation: Design Pressure : 4.2 kg/cm2g (F.V) Design Temperature : 120 C																																																																																																																																																												
				§ The information is based on Neldisc L6 series. Valve selected shall be of double flanged construction with minimal flange-to-flange dimension and no pockets within the valve body.																																																																																																																																																													
				‡ Preferred material is L grade stainless steel. Standard grade stainless steel that is not welded may be used for process wetted parts including the body. Stainless steel with weld repair is acceptable if low carbon (<= 0.03 %) filler material is used.																																																																																																																																																													
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CONTROL VALVES

PROJECT SPECIFICATION

963766 — **616**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Item Number: 23 Quantity: 1		Valve Tag Number: UXV-23-0801		Data Page: 2 of 2		
TRANS-DUCER	Transducer Tag Number:		HANDWHEEL	Handwheel Position:		
	Transducer Input Signal Range:			Usage:		
	Transducer Output Signal Range:					
	Manufacturer:					
SOLENOID VALVE	Solenoid Valve Tag Number: UXY-23-0801		POSITION SWITCH	Open Position Switch Tag Number:		
	Body Type: 3-Way Body Material: Brass			Close Position Switch Tag Number:		
	Seat Material: Stainless Steel Disc Material: Stainless Steel			Sensor Type:		
	Connection Size/Type: †			Contact Form:		
	Packing Style: Packless			Open Position Switch Contact Action:		
	Fluid: Instrument Air			Close Position Switch Contact Action:		
	Pressure: Contractor to determine			Contact Load: Contact Rating:		
	Minimum Pressure Port Orifice C _v : MAX Size: MAX			Mounting Location:		
	Minimum Exhaust Port Orifice C _v : MAX Size: MAX			Manufacturer:		
	Operation Style: †			POSITION TRANSMITTER	Position Transmitter Tag Number:	
	Coil Configuration: Single				Sensor Type:	
	Operating Voltage: \$\$ Frequency:				Output Signal Range:	
	Action Coil 1 Deenergized: Vent air from valve				Travel Measurement:	
	Action Coil 1 Energized: Supply air to valve				Mounting Location:	
	Action Coil 2 Deenergized:				Manufacturer:	
	Action Coil 2 Energized:			LOCK-UP VALVE	Action Style:	
	Manual Reset:				Mounting Location:	
	Manual Operator:				Model Number:	
Mounting Location: Nearest item to actuator in air line. †		QUICK EXHAUST	Manufacturer:			
Manufacturer: ASCO (or equal)			Connection Size/Type:			
			Body Material:			
POSITIONER	Positioner Tag Number:			Diaphragm Material:		
	Positioner Type:			Exhaust Action:		
	Positioner Input Signal Range:			Mounting Location:		
AIR FILTER	Positioner Output Signal Range:		VOLUME BOOSTER	Manufacturer:		
	Manufacturer:			Connection Size/Type:		
	Quantity: one			Body Material:		
TUBING	Type: Filter/Regulator			Seat Ring Material:		
	Manufacturer:			Diaphragm Material:		
	Tubing Size:			Mounting Location:		
DATA PAGE 2 REMARKS	Tubing Material:			Manufacturer:		
† The solenoid valve shall be mounted and tubed by the control valve vendor. The solenoid valve size and type, the tubing size, the actuator air connection size, and the air/filter regulator capacity shall be determined by the control valve vendor to meet the maximum stroking time requirements specified for the control valve. \$\$ The Contractor shall coordinate the solenoid valve operating voltage and coil watt rating with the supplier of the Concentration Section Shutdown System (UC-2301). See UOP Project Specification 660, this project.						

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CONTROL VALVES

PROJECT SPECIFICATION

963766 — **616**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

GENERAL	Item Number: 24 Quantity: 1 Valve Tag Number: HV-23-0801 Data Page: 1 of 2																																																																																																																																																																										
	Service: Cumene Quench to Flash Column Collector Tray (Manual) Process Material: Hydrocarbon with aromatics P&ID Drawing: 08																																																																																																																																																																										
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VALVE BODY & BONNET	Body Type: Globe § Body Size: 1" ## Port Type: Single Inlet Conn Size: 1" ## Type: Flanged Class: 300 Outlet Conn Size: 1" ## Type: Flanged Class: 300 Flange Facing: Raised Face Flange Face Finish: Body & Bonnet Material: 316 Stainless Steel ‡ Bonnet Style: Integral or Bolted Packing Style: Manufacturer's Recommendation † Packing Material: Manufacturer's Recommendation Body/Bonnet Gasket Material: Manufacturer's Recommend Lub/Iso Valve Material: Lub Material: Manufacturer: Masoneilan (or equal) Model Number:	TRIM	Approximate Valve C _v : 12 F _L : x _T : Trim Style: Man. Standard Trim Size: Inherent Characteristic: Linear Balance Style: Unbalanced Full Open Position: 100 % Guiding Style: Top Closure Member Material: 316 Stainless Steel Seat Ring Material: 316 Stainless Steel Seat Ring Retainer Material: 304 Stainless Steel Bearing Material: Bushing/Cage Material: Stellite Stem/Shaft Material: 316 Stainless Steel Flow Direction: Seat Leakage Class: ANSI Class V																																																																																																																																																																								
	ACTUATOR		INSTALLATION	Type: Spring Diaphragm, Pneumatic Action Style: Fail Close Service Type: Modulating Max Shutoff Press: In: 3.85 Out: -1.02 kg/cm2(g) Maximum Stroke Time to Open: Maximum Stroke Time to Close: Stroke Cycle Service:	Inlet Pipe Nominal Size: Est Wall Thick: Outlet Pipe Nominal Size: Est Wall Thick: Est Insulation Type: Est Insulation Thickness: Installation: Design Pressure : 4.2 kg/cm2g (F.V) Design Temperature : 120 C																																																																																																																																																																						
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ALL ITEMS SHALL COMPLY WITH THE GENERAL REQUIREMENTS SHOWN ON THE GENERAL REQUIREMENTS SECTION OF THIS SPECIFICATION.

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CONTROL VALVES

PROJECT SPECIFICATION

963766 — 616

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

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PROJECT SPECIFICATION

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0	25JUN12	MVD	MW				

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GENERAL	Item Number: 25 Quantity: 1 Valve Tag Number: UXV-23-0802 Data Page: 1 of 2																																																																																																																																																																
	Service: Cumene Quench to Flash Column Bottom Process Material: Hydrocarbon with aromatics P&ID Drawing: 08																																																																																																																																																																
SIZING CONDITIONS	<table border="1"> <thead> <tr> <th>Process Data Case</th> <th colspan="2">General</th> <th colspan="2">General</th> <th colspan="2"></th> <th colspan="2"></th> <th></th> </tr> <tr> <th>Sizing Case</th> <th colspan="2">Open</th> <th colspan="2">Closed</th> <th colspan="2"></th> <th colspan="2"></th> <th></th> </tr> <tr> <th>Calculated C_v</th> <th>Inlet</th> <th>Outlet</th> <th>Inlet</th> <th>Outlet</th> <th>Inlet</th> <th>Outlet</th> <th>Inlet</th> <th>Outlet</th> <th>Units</th> </tr> </thead> <tbody> <tr> <td>Liquid Flow at Flowing Cond</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>m3/h</td> </tr> <tr> <td>Vapor Flow at Base Cond</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>normal m3/h</td> </tr> <tr> <td>Steam Flow</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>kg/h</td> </tr> <tr> <td>Temperature</td> <td>46</td> <td></td> <td>46</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>°C</td> </tr> <tr> <td>Liquid Specific Gravity</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>----</td> </tr> <tr> <td>Vapor Spec. Gravity (Air=1)</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>----</td> </tr> <tr> <td>Pressure</td> <td>-0.12</td> <td>-0.12</td> <td>-0.12</td> <td>-0.24</td> <td></td> <td></td> <td></td> <td></td> <td>kg/cm2(g)</td> </tr> <tr> <td>Liq Critical Press (Pseudo)</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>kg/cm2(a)</td> </tr> <tr> <td>Liquid Vapor Pressure</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>kg/cm2(a)</td> </tr> <tr> <td>Liquid Viscosity</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>cP</td> </tr> <tr> <td>Compressibility Z</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> <tr> <td>Specific Heats Cp/Cv (Real)</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> <tr> <td>Estimate Sound Press Level</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>dBA</td> </tr> </tbody> </table>	Process Data Case	General		General							Sizing Case	Open		Closed							Calculated C _v	Inlet	Outlet	Inlet	Outlet	Inlet	Outlet	Inlet	Outlet	Units	Liquid Flow at Flowing Cond									m3/h	Vapor Flow at Base Cond									normal m3/h	Steam Flow									kg/h	Temperature	46		46						°C	Liquid Specific Gravity									----	Vapor Spec. Gravity (Air=1)									----	Pressure	-0.12	-0.12	-0.12	-0.24					kg/cm2(g)	Liq Critical Press (Pseudo)									kg/cm2(a)	Liquid Vapor Pressure									kg/cm2(a)	Liquid Viscosity									cP	Compressibility Z										Specific Heats Cp/Cv (Real)										Estimate Sound Press Level									dBA
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VALVE BODY & BONNET	Body Type: Eccentric Disc § Body Size: 6" #2 Port Type: Single Inlet Conn Size: 6" #2 Type: Flanged Class: 300 Outlet Conn Size: 6" #2 Type: Flanged Class: 300 Flange Facing: Flange Face Finish: Body & Bonnet Material: 316 Stainless Steel ‡ Bonnet Style: Packing Style: Manufacturer's Recommendation † Packing Material: Manufacturer's Recommendation Body/Bonnet Gasket Material: Lub/Iso Valve Material: Lub Material: Manufacturer: Metso Automation (or equal) Model Number:	TRIM	Approximate Valve C _v : 510 F _L : x _T : Trim Style: Man. Standard Trim Size: Inherent Characteristic: Equal Percentage Balance Style: Full Open Position: 90° Guiding Style: Shaft Closure Member Material: 316 Stainless Steel Seat Ring Material: Incoloy 825 with Hard Chrome Seat Ring Retainer Material: 316 Stainless Steel Bearing Material: Stellite Bushing/Cage Material: Stem/Shaft Material: 17-4PH Flow Direction: Seat Leakage Class: ANSI Class V																																																																																																																																																														
	ACTUATOR		INSTALLATION	Type: Spring Return Piston, Pneumatic Action Style: Fail Open Service Type: On-Off Max Shutoff Press: In: 3.85 Out: -1.02 kg/cm2(g) Maximum Stroke Time to Open: 5 Seconds Maximum Stroke Time to Close: Stroke Cycle Service:	Inlet Pipe Nominal Size: Est Wall Thick: Outlet Pipe Nominal Size: Est Wall Thick: Est Insulation Type: Est Insulation Thickness: Installation: Design Pressure : 4.2 kg/cm2g (F.V) Design Temperature : 120 C																																																																																																																																																												
				§ The information is based on Neldisc L6 series. Valve selected shall be of double flanged construction with minimal flange-to-flange dimension and no pockets within the valve body.																																																																																																																																																													
				‡ Preferred material is L grade stainless steel. Standard grade stainless steel that is not welded may be used for process wetted parts including the body. Stainless steel with weld repair is acceptable if low carbon (<= 0.03 %) filler material is used.																																																																																																																																																													
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CONTROL VALVES

PROJECT SPECIFICATION

963766 — **616**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Item Number: 25 Quantity: 1		Valve Tag Number: UXV-23-0802		Data Page: 2 of 2			
TRANS-DUCER	Transducer Tag Number:		HANDWHEEL	Handwheel Position:			
	Transducer Input Signal Range:			Usage:			
TRANS-DUCER	Transducer Output Signal Range:		HANDWHEEL				
	Manufacturer:						
SOLENOID VALVE	Solenoid Valve Tag Number: UXY-23-0802		POSITION SWITCH	Open Position Switch Tag Number:			
	Body Type: 3-Way Body Material: Brass			Close Position Switch Tag Number:			
	Seat Material: Stainless Steel Disc Material: Stainless Steel			Sensor Type:			
	Connection Size/Type: †			Contact Form:			
	Packing Style: Packless			Open Position Switch Contact Action:			
	Fluid: Instrument Air			Close Position Switch Contact Action:			
	Pressure: Contractor to determine			Contact Load: Contact Rating:			
	Minimum Pressure Port Orifice C _v : MAX Size: MAX			Mounting Location:			
	Minimum Exhaust Port Orifice C _v : MAX Size: MAX			Manufacturer:			
	Operation Style: †						
	Coil Configuration: Single			Position Transmitter Tag Number:			
	Operating Voltage: \$\$ Frequency:			Sensor Type:			
	Action Coil 1 Deenergized: Vent air from valve			Output Signal Range:			
	Action Coil 1 Energized: Supply air to valve			Travel Measurement:			
	Action Coil 2 Deenergized:			Mounting Location:			
	Action Coil 2 Energized:			Manufacturer:			
	POSITIONER	Manual Reset:		LOCK-UP VALVE	Action Style:		
		Manual Operator:			Mounting Location:		
Mounting Location: Nearest item to actuator in air line. †		Model Number:					
Manufacturer: ASCO (or equal)		Manufacturer:					
Positioner Tag Number:		Connection Size/Type:					
Positioner Type:		Body Material:					
Positioner Input Signal Range:		Diaphragm Material:					
Positioner Output Signal Range:		Exhaust Action:					
Manufacturer:		Mounting Location:					
		Manufacturer:					
		Connection Size/Type:					
		Body Material:					
		Seat Ring Material:					
		Diaphragm Material:					
		Mounting Location:					
		Manufacturer:					
AIR FILTER		Quantity: one			VOLUME BOOSTER	Connection Size/Type:	
		Type: Filter/Regulator				Body Material:	
TUBING	Manufacturer:		VOLUME BOOSTER	Seat Ring Material:			
	Tubing Size:			Diaphragm Material:			
TUBING	Tubing Material:		VOLUME BOOSTER	Mounting Location:			
				Manufacturer:			
DATA PAGE 2 REMARKS	† The solenoid valve shall be mounted and tubed by the control valve vendor. The solenoid valve size and type, the tubing size, the actuator air connection size, and the air/filter regulator capacity shall be determined by the control valve vendor to meet the maximum stroking time requirements specified for the control valve.						
	\$\$ The Contractor shall coordinate the solenoid valve operating voltage and coil watt rating with the supplier of the Concentration Section Shutdown System (UC-2301). See UOP Project Specification 660, this project.						

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CONTROL VALVES

PROJECT SPECIFICATION

963766 — **616**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

GENERAL	Item Number: 26 Quantity: 1 Valve Tag Number: HV-23-0802 Data Page: 1 of 2																																																																																																																																																																										
	Service: Cumene Quench to Flash Column Collector Tray (Manual) Process Material: Hydrocarbon with aromatics P&ID Drawing: 08																																																																																																																																																																										
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VALVE BODY & BONNET	Body Type: Globe § Body Size: 1" ## Port Type: Single Inlet Conn Size: 1" ## Type: Flanged Class: 300 Outlet Conn Size: 1" ## Type: Flanged Class: 300 Flange Facing: Raised Face Flange Face Finish: Body & Bonnet Material: 316 Stainless Steel ‡ Bonnet Style: Integral or Bolted Packing Style: Manufacturer's Recommendation † Packing Material: Manufacturer's Recommendation Body/Bonnet Gasket Material: Manufacturer's Recommend Lub/Iso Valve Material: Lub Material: Manufacturer: Masoneilan (or equal) Model Number:	TRIM	Approximate Valve C _v : 12 F _L : x _T : Trim Style: Man. Standard Trim Size: Inherent Characteristic: Linear Balance Style: Unbalanced Full Open Position: 100 % Guiding Style: Top Closure Member Material: 316 Stainless Steel Seat Ring Material: 316 Stainless Steel Seat Ring Retainer Material: 304 Stainless Steel Bearing Material: Bushing/Cage Material: Stellite Stem/Shaft Material: 316 Stainless Steel Flow Direction: Seat Leakage Class: ANSI Class V																																																																																																																																																																								
	ACTUATOR		INSTALLATION	Type: Spring Diaphragm, Pneumatic Action Style: Fail Close Service Type: Modulating Max Shutoff Press: In: 3.85 Out: -1.02 kg/cm²(g) Maximum Stroke Time to Open: Maximum Stroke Time to Close: Stroke Cycle Service:	Inlet Pipe Nominal Size: Est Wall Thick: Outlet Pipe Nominal Size: Est Wall Thick: Est Insulation Type: Est Insulation Thickness: Installation: Design Pressure : 4.2 kg/cm²g (F.V) Design Temperature : 120 C																																																																																																																																																																						
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ALL ITEMS SHALL COMPLY WITH THE GENERAL REQUIREMENTS SHOWN ON THE GENERAL REQUIREMENTS SECTION OF THIS SPECIFICATION.

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CONTROL VALVES

PROJECT SPECIFICATION

963766 — **616**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Item Number: 26 Quantity: 1		Valve Tag Number: HV-23-0802		Data Page: 2 of 2	
TRANS-DUCER	Transducer Tag Number:		HANDWHEEL	Handwheel Position:	
	Transducer Input Signal Range:			Usage:	
	Transducer Output Signal Range:				
	Manufacturer:				
SOLENOID VALVE	Solenoid Valve Tag Number:		POSITION SWITCH	Open Position Switch Tag Number:	
	Body Type: Body Material:			Close Position Switch Tag Number:	
	Seat Material: Disc Material:			Sensor Type:	
	Connection Size/Type:			Contact Form:	
	Packing Style:			Open Position Switch Contact Action:	
	Fluid:			Close Position Switch Contact Action:	
	Pressure:			Contact Load: Contact Rating:	
	Minimum Pressure Port Orifice C_v : Size:			Mounting Location:	
	Minimum Exhaust Port Orifice C_v : Size:			Manufacturer:	
	Operation Style:			POSITION TRANSMITTER	Position Transmitter Tag Number:
	Coil Configuration:		Sensor Type:		
	Operating Voltage: Frequency:		Output Signal Range:		
	Action Coil 1 Deenergized:		Travel Measurement:		
	Action Coil 1 Energized:		Mounting Location:		
	Action Coil 2 Deenergized:		LOCK-UP VALVE	Manufacturer:	
	Action Coil 2 Energized:			Action Style:	
	Manual Reset:			Mounting Location:	
	Manual Operator:		QUICK EXHAUST	Model Number:	
	Mounting Location:			Manufacturer:	
	Manufacturer:			Connection Size/Type:	
POSITIONER	Positioner Tag Number:		VOLUME BOOSTER	Body Material:	
	Positioner Type: Digital			Seat Ring Material:	
	Positioner Input Signal Range: 4 - 20 mA			Diaphragm Material:	
	Positioner Output Signal Range: 0 - 100 %			Mounting Location:	
AIR FILTER	Quantity: one		TUBING	Manufacturer:	
	Type: Filter/Regulator			Connection Size/Type:	
	Manufacturer:			Body Material:	
TUBING	Tubing Size:		TUBING	Seat Ring Material:	
	Tubing Material:			Diaphragm Material:	
TUBING			TUBING	Mounting Location:	
				Manufacturer:	
DATA PAGE 2 REMARKS					

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CONTROL VALVES

PROJECT SPECIFICATION

963766 — **616**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

GENERAL	Item Number: 27 Quantity: 1		Valve Tag Number: FV-23-0801		Data Page: 1 of 2	
	Service: Flash Column Reflux		P&ID Drawing: 08			
Process Material: Hydrocarbon with aromatics						
SIZING CONDITIONS	Process Data Case	Single Case		Single Case		Single Case
	Sizing Case	Normal		Maximum		Minimum
	Calculated C _v	9.4		10.7		4.31
		Inlet	Outlet	Inlet	Outlet	Inlet
	Liquid Flow at Flowing Cond	21.9		24.1		10.9
	Vapor Flow at Base Cond					
	Steam Flow					
	Temperature	39		39		39
	Liquid Specific Gravity	0.848		0.848		0.848
	Vapor Spec. Gravity (Air=1)					
	Pressure	7.03	0.76	6.64	0.77	8.17
	Liq Critical Press (Pseudo)	32.74		32.74		32.74
	Liquid Vapor Pressure	0.02		0.02		0.02
	Liquid Viscosity	0.622		0.622		0.622
	Compressibility Z					
Specific Heats Cp/Cv (Real)						
Estimate Sound Press Level						
VALVE BODY & BONNET	Body Type: Globe §					Approximate Valve C _v : 18
	Body Size: 1-1/2" Port Type: Single					F _L : 0.9 x _T : 0.72
	Inlet Conn Size: 1-1/2" Type: Flanged Class: 300					Trim Style: Man. Standard Trim Size:
	Outlet Conn Size: 1-1/2" Type: Flanged Class: 300					Inherent Characteristic: Linear
	Flange Facing: Raised Face Flange Face Finish:					Balance Style: Unbalanced Full Open Position: 100 %
	Body & Bonnet Material: 316 Stainless Steel ‡					Guiding Style: Top
	Bonnet Style: Integral or Bolted					Closure Member Material: 316 SS, Stellite sheath
	Packing Style: Manufacturer's Recommendation †					Seat Ring Material: 316 SS with Stellite sheath
	Packing Material: Manufacturer's Recommendation					Seat Ring Retainer Material: 304 Stainless Steel
	Body/Bonnet Gasket Material: Manufacturer's Recommend					Bearing Material:
	Lub/Iso Valve Material: Lub Material:					Bushing/Cage Material: Stellite
	Manufacturer: Masoneilan (or equal)					Stem/Shaft Material: 316 Stainless Steel
	Model Number:					Flow Direction:
						Seat Leakage Class: ANSI Class IV
	ACTUATOR	Type: Spring Diaphragm, Pneumatic				
Action Style: Fail Open						Outlet Pipe Nominal Size: Est Wall Thick:
Service Type: Modulating						Est Insulation Type:
Max Shutoff Press: In: 13.55 Out: -1.02 kg/cm2(g)						Est Insulation Thickness:
Maximum Stroke Time to Open:						Installation:
Maximum Stroke Time to Close:						Design Pressure : 13.35 kg/cm2g (F.V)
Stroke Cycle Service:						Design Temperature : 120 C
DATA PAGE 1 REMARKS	§ The information is based on Masoneilan 21000 series.					
	Hardened trim is recommended for this service based upon the possibility of cavitation occurring.					
	‡ Preferred material is L grade stainless steel. Standard grade stainless steel that is not welded may be used for process wetted parts including the body. Stainless steel with weld repair is acceptable if low carbon (<= 0.03 %) filler material is used.					
	† Provide packing suitable for vacuum service to minimize air leakage into the process line.					

ALL ITEMS SHALL COMPLY WITH THE GENERAL REQUIREMENTS SHOWN ON THE GENERAL REQUIREMENTS SECTION OF THIS SPECIFICATION.

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CONTROL VALVES

PROJECT SPECIFICATION

963766 — **616**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Item Number: 27 Quantity: 1		Valve Tag Number: FV-23-0801		Data Page: 2 of 2	
TRANS-DUCER	Transducer Tag Number:		HANDWHEEL	Handwheel Position:	
	Transducer Input Signal Range:			Usage:	
	Transducer Output Signal Range:				
	Manufacturer:				
SOLENOID VALVE	Solenoid Valve Tag Number:		POSITION SWITCH	Open Position Switch Tag Number:	
	Body Type: Body Material:			Close Position Switch Tag Number:	
	Seat Material: Disc Material:			Sensor Type:	
	Connection Size/Type:			Contact Form:	
	Packing Style:			Open Position Switch Contact Action:	
	Fluid:			Close Position Switch Contact Action:	
	Pressure:			Contact Load: Contact Rating:	
	Minimum Pressure Port Orifice C_v : Size:			Mounting Location:	
	Minimum Exhaust Port Orifice C_v : Size:			Manufacturer:	
	Operation Style:			POSITION TRANSMITTER	Position Transmitter Tag Number:
	Coil Configuration:		Sensor Type:		
	Operating Voltage: Frequency:		Output Signal Range:		
	Action Coil 1 Deenergized:		Travel Measurement:		
	Action Coil 1 Energized:		Mounting Location:		
	Action Coil 2 Deenergized:		LOCK-UP VALVE	Manufacturer:	
	Action Coil 2 Energized:			Action Style:	
	Manual Reset:			Mounting Location:	
	Manual Operator:		QUICK EXHAUST	Model Number:	
	Mounting Location:			Manufacturer:	
	Manufacturer:			Connection Size/Type:	
POSITIONER	Positioner Tag Number:		VOLUME BOOSTER	Body Material:	
	Positioner Type: Digital			Diaphragm Material:	
	Positioner Input Signal Range: 4 - 20 mA			Exhaust Action:	
	Positioner Output Signal Range: 0 - 100 %			Mounting Location:	
AIR FILTER	Manufacturer:		Manufacturer:		
	Quantity: one		Connection Size/Type:		
	Type: Filter/Regulator		Body Material:		
TUBING	Manufacturer:		Seat Ring Material:		
	Tubing Size:		Diaphragm Material:		
Tubing Material:		Mounting Location:			
		Manufacturer:			
DATA PAGE 2 REMARKS					

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CONTROL VALVES

PROJECT SPECIFICATION

963766 — **616**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Note: The information in this document is confidential and the property of UOP LLC and must not be disclosed to others or reproduced in any manner or used for any purpose whatsoever without its written permission.

GENERAL	Item Number: 28 Quantity: 1		Valve Tag Number: LV-23-0803		Data Page: 1 of 2			
	Service: Preflash / Flash Column Net Overhead Liquid		P&ID Drawing: 08B					
Process Material: Hydrocarbon with aromatics								
SIZING CONDITIONS	Process Data Case	Single Case		Single Case		Long Circulation		
	Sizing Case	Normal		Maximum		Minimum		
	Calculated C _v	135		173		43.3		
		Inlet	Outlet	Inlet	Outlet	Inlet	Outlet	Units
	Liquid Flow at Flowing Cond	166.3		182.9		83.1		m ³ /h
	Vapor Flow at Base Cond							normal m ³ /h
	Steam Flow							kg/h
	Temperature	39		39		39		°C
	Liquid Specific Gravity	0.848		0.848		0.848		----
	Vapor Spec. Gravity (Air=1)							----
	Pressure	7.12	5.36	6.74	5.46	8.19	1.61	kg/cm ² (g)
	Liq Critical Press (Pseudo)	32.74		32.74		32.74		kg/cm ² (a)
	Liquid Vapor Pressure	0.02		0.02		0.02		kg/cm ² (a)
	Liquid Viscosity	0.622		0.622		0.622		cP
	Compressibility Z							
Specific Heats Cp/Cv (Real)								
Estimate Sound Press Level							dBA	
VALVE BODY & BONNET	Body Type: Eccentric Rotary Plug §		Approximate Valve C _v : 230		F _L : 0.68 x _T : 0.4			
	Body Size: 4" Port Type: Single		Trim Style: Man. Standard		Trim Size:			
	Inlet Conn Size: 4" Type: Flanged Class: 300		Inherent Characteristic: Linear		Balance Style:			
	Outlet Conn Size: 4" Type: Flanged Class: 300		Full Open Position: 100 %		Guiding Style: Shaft			
	Flange Facing: Raised Face Flange Face Finish:		Closure Member Material: 316 SS, Stellite sheath		Seat Ring Material: 316 SS with Stellite sheath			
	Body & Bonnet Material: 316 Stainless Steel ‡		Seat Ring Retainer Material: Stellite		Bearing Material: Stellite			
	Bonnet Style: Integral or Bolted		Bushing/Cage Material:		Stem/Shaft Material: 316 Stainless Steel			
	Packing Style: Manufacturer's Recommendation		Flow Direction: Flow to close		Seat Leakage Class: ANSI Class IV			
	Packing Material: Manufacturer's Recommendation		Inlet Pipe Nominal Size:		Est Wall Thick:			
	Body/Bonnet Gasket Material:		Outlet Pipe Nominal Size:		Est Wall Thick:			
	Lub/Iso Valve Material: Lub Material:		Est Insulation Type:		Est Insulation Thickness:			
	Manufacturer: Masoneilan (or equal)		Installation:		Design Pressure : 13.35 kg/cm ² g (F.V)			
Model Number:		Design Temperature : 120 C						
ACTUATOR	Type: Spring Diaphragm, Pneumatic							
	Action Style: Fail Close							
	Service Type: Modulating							
	Max Shutoff Press: In: 13.55 Out: 0.81 kg/cm ² (g)							
DATA PAGE 1 REMARKS	Maximum Stroke Time to Open:							
	Maximum Stroke Time to Close:							
	Stroke Cycle Service:							
	§ The information is based on Masoneilan 35002 series.							
‡ Preferred material is L grade stainless steel. Standard grade stainless steel that is not welded may be used for process wetted parts including the body. Stainless steel with weld repair is acceptable if low carbon (<= 0.03 %) filler material is used.								
Hardened trim is recommended for this service based upon the possibility of cavitation occurring.								

ALL ITEMS SHALL COMPLY WITH THE GENERAL REQUIREMENTS SHOWN ON THE GENERAL REQUIREMENTS SECTION OF THIS SPECIFICATION.



CONTROL VALVES

PROJECT SPECIFICATION

963766 — **616**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Item Number: 28 Quantity: 1		Valve Tag Number: LV-23-0803		Data Page: 2 of 2	
TRANS-DUCER	Transducer Tag Number:		HANDWHEEL	Handwheel Position:	
	Transducer Input Signal Range:			Usage:	
	Transducer Output Signal Range:				
	Manufacturer:				
SOLENOID VALVE	Solenoid Valve Tag Number:		POSITION SWITCH	Open Position Switch Tag Number:	
	Body Type: Body Material:			Close Position Switch Tag Number:	
	Seat Material: Disc Material:			Sensor Type:	
	Connection Size/Type:			Contact Form:	
	Packing Style:			Open Position Switch Contact Action:	
	Fluid:			Close Position Switch Contact Action:	
	Pressure:			Contact Load: Contact Rating:	
	Minimum Pressure Port Orifice C _v : Size:			Mounting Location:	
	Minimum Exhaust Port Orifice C _v : Size:			Manufacturer:	
	Operation Style:			POSITION TRANSMITTER	Position Transmitter Tag Number:
	Coil Configuration:		Sensor Type:		
	Operating Voltage: Frequency:		Output Signal Range:		
	Action Coil 1 Deenergized:		Travel Measurement:		
	Action Coil 1 Energized:		Mounting Location:		
	Action Coil 2 Deenergized:		LOCK-UP VALVE	Manufacturer:	
	Action Coil 2 Energized:			Action Style:	
	Manual Reset:			Mounting Location:	
	Manual Operator:		QUICK EXHAUST	Model Number:	
	Mounting Location:			Manufacturer:	
	Manufacturer:			Connection Size/Type:	
POSITIONER	Positioner Tag Number:		VOLUME BOOSTER	Body Material:	
	Positioner Type: Digital			Diaphragm Material:	
	Positioner Input Signal Range: 4 - 20 mA			Exhaust Action:	
	Positioner Output Signal Range: 0 - 100 %			Mounting Location:	
AIR FILTER	Quantity: one		Manufacturer:		
	Type: Filter/Regulator		Connection Size/Type:		
	Manufacturer:		Body Material:		
TUBING	Tubing Size:		Seat Ring Material:		
	Tubing Material:		Diaphragm Material:		
		Mounting Location:			
		Manufacturer:			
DATA PAGE 2 REMARKS					

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CONTROL VALVES

PROJECT SPECIFICATION

963766 — **616**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

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GENERAL	Item Number: 29 Quantity: 1 Valve Tag Number: UXV-23-0804 Data Page: 1 of 2																																																																																																																																																																						
	Service: Recycle Cumene to Feed Wash Column Shutoff Process Material: Hydrocarbon with aromatics P&ID Drawing: 08B																																																																																																																																																																						
SIZING CONDITIONS	<table border="1"> <thead> <tr> <th rowspan="2">Process Data Case</th> <th colspan="2">General</th> <th colspan="2">General</th> <th colspan="2"></th> <th colspan="2"></th> <th rowspan="2">Units</th> </tr> <tr> <th colspan="2">Open</th> <th colspan="2">Closed</th> <th colspan="2"></th> <th colspan="2"></th> </tr> </thead> <tbody> <tr> <td>Sizing Case</td> <td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr> <tr> <td>Calculated C_v</td> <td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr> <tr> <td>Liquid Flow at Flowing Cond</td> <td>Inlet</td><td>Outlet</td><td>Inlet</td><td>Outlet</td><td>Inlet</td><td>Outlet</td><td>Inlet</td><td>Outlet</td><td>m3/h</td></tr> <tr> <td>Vapor Flow at Base Cond</td> <td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>normal m3/h</td></tr> <tr> <td>Steam Flow</td> <td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>kg/h</td></tr> <tr> <td>Temperature</td> <td>39</td><td></td><td>39</td><td></td><td></td><td></td><td></td><td></td><td>°C</td></tr> <tr> <td>Liquid Specific Gravity</td> <td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>----</td></tr> <tr> <td>Vapor Spec. Gravity (Air=1)</td> <td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>----</td></tr> <tr> <td>Pressure</td> <td>5.36</td><td>5.36</td><td>2.99</td><td>5.17</td><td></td><td></td><td></td><td></td><td>kg/cm2(g)</td></tr> <tr> <td>Liq Critical Press (Pseudo)</td> <td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>kg/cm2(a)</td></tr> <tr> <td>Liquid Vapor Pressure</td> <td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>kg/cm2(a)</td></tr> <tr> <td>Liquid Viscosity</td> <td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>cP</td></tr> <tr> <td>Compressibility Z</td> <td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr> <tr> <td>Specific Heats Cp/Cv (Real)</td> <td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr> <tr> <td>Estimate Sound Press Level</td> <td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>dBA</td></tr> </tbody> </table>	Process Data Case	General		General						Units	Open		Closed						Sizing Case									Calculated C _v									Liquid Flow at Flowing Cond	Inlet	Outlet	Inlet	Outlet	Inlet	Outlet	Inlet	Outlet	m3/h	Vapor Flow at Base Cond									normal m3/h	Steam Flow									kg/h	Temperature	39		39						°C	Liquid Specific Gravity									----	Vapor Spec. Gravity (Air=1)									----	Pressure	5.36	5.36	2.99	5.17					kg/cm2(g)	Liq Critical Press (Pseudo)									kg/cm2(a)	Liquid Vapor Pressure									kg/cm2(a)	Liquid Viscosity									cP	Compressibility Z										Specific Heats Cp/Cv (Real)										Estimate Sound Press Level									dBA
	Process Data Case		General		General							Units																																																																																																																																																											
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Estimate Sound Press Level									dBA																																																																																																																																																														
VALVE BODY & BONNET	Body Type: Eccentric Disc § Body Size: 8" #1 Port Type: Single Inlet Conn Size: 8" #1 Type: Flanged Class: 300 Outlet Conn Size: 8" #1 Type: Flanged Class: 300 Flange Facing: Flange Face Finish: Body & Bonnet Material: 316 Stainless Steel # Bonnet Style: Packing Style: Manufacturer's Recommendation Packing Material: Manufacturer's Recommendation Body/Bonnet Gasket Material: Lub/Iso Valve Material: Lub Material: Manufacturer: Metso Automation (or equal) Model Number:	TRIM	Approximate Valve C _v : 1030 F _L : x _T : Trim Style: Man. Standard Trim Size: Inherent Characteristic: Equal Percentage Balance Style: Full Open Position: 90° Guiding Style: Shaft Closure Member Material: 316 Stainless Steel Seat Ring Material: Incoloy 825 with Hard Chrome Seat Ring Retainer Material: 316 Stainless Steel Bearing Material: Stellite Bushing/Cage Material: Stem/Shaft Material: 17-4PH Flow Direction: Flow to close Seat Leakage Class: ANSI Class V																																																																																																																																																																				
	ACTUATOR		INSTALLATION	Type: Spring Return Piston, Pneumatic Action Style: Fail Close Service Type: On-Off Max Shutoff Press: In: 13.55 Out: 1.76 kg/cm2(g) Maximum Stroke Time to Open: 5 Seconds Maximum Stroke Time to Close: 5 Seconds Stroke Cycle Service:	Inlet Pipe Nominal Size: Est Wall Thick: Outlet Pipe Nominal Size: Est Wall Thick: Est Insulation Type: Est Insulation Thickness: Installation:																																																																																																																																																																		
				Design Pressure : 13.55 kg/cm2g (F.V) Design Temperature : 120 C																																																																																																																																																																			
				§ The information is based on Neldisc L6 series.																																																																																																																																																																			
				# Preferred material is L grade stainless steel. Standard grade stainless steel that is not welded may be used for process wetted parts including the body. Stainless steel with weld repair is acceptable if low carbon (<= 0.03 %) filler material is used.																																																																																																																																																																			
				# Line size valve.																																																																																																																																																																			
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CONTROL VALVES

PROJECT SPECIFICATION

963766 — **616**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Item Number: 29 Quantity: 1		Valve Tag Number: UXV-23-0804		Data Page: 2 of 2			
TRANSDUCER	Transducer Tag Number:		HANDWHEEL	Handwheel Position:			
	Transducer Input Signal Range:			Usage:			
Transducer Output Signal Range:							
Manufacturer:							
SOLENOID VALVE	Solenoid Valve Tag Number: UXY-23-0804		POSITION SWITCH	Open Position Switch Tag Number:			
	Body Type: 3-Way Body Material: Brass			Close Position Switch Tag Number:			
	Seat Material: Stainless Steel Disc Material: Stainless Steel			Sensor Type:			
	Connection Size/Type: †			Contact Form:			
	Packing Style: Packless			Open Position Switch Contact Action:			
	Fluid: Instrument Air			Close Position Switch Contact Action:			
	Pressure: Contractor to determine			Contact Load: Contact Rating:			
	Minimum Pressure Port Orifice C_v : MAX Size: MAX			Mounting Location:			
	Minimum Exhaust Port Orifice C_v : MAX Size: MAX			Manufacturer:			
	Operation Style: †						
	Coil Configuration: Single						
	Operating Voltage: \$\$ Frequency:						
	Action Coil 1 Deenergized: Vent air from valve						
	Action Coil 1 Energized: Supply air to valve						
	Action Coil 2 Deenergized:						
	Action Coil 2 Energized:						
	POSITIONER	Manual Reset:		LOCK-UP VALVE	Action Style:		
		Manual Operator:			Mounting Location:		
Mounting Location: Nearest item to actuator in air line. †		Model Number:					
Manufacturer: ASCO (or equal)		Manufacturer:					
AIR FILTER	Positioner Tag Number:		QUICK EXHAUST	Connection Size/Type:			
	Positioner Type:			Body Material:			
	Positioner Input Signal Range:			Diaphragm Material:			
	Positioner Output Signal Range:			Exhaust Action:			
TUBING	Manufacturer:		VOLUME BOOSTER	Mounting Location:			
	Quantity: one			Manufacturer:			
	Type: Filter/Regulator			Connection Size/Type:			
	Manufacturer:			Body Material:			
DATA PAGE 2 REMARKS	Tubing Size:			Seat Ring Material:			
	Tubing Material:			Diaphragm Material:			
				Mounting Location:			
				Manufacturer:			
† The solenoid valve shall be mounted and tubed by the control valve vendor. The solenoid valve size and type, the tubing size, the actuator air connection size, and the air/filter regulator capacity shall be determined by the control valve vendor to meet the maximum stroking time requirements specified for the control valve. \$\$ The Contractor shall coordinate the solenoid valve operating voltage and coil watt rating with the supplier of the Decomposition Section Shutdown System (UC-2302). See UOP Project Specification 660, this project.							

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CONTROL VALVES

PROJECT SPECIFICATION

963766 — **616**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Note: The information in this document is confidential and the property of UOP LLC and must not be disclosed to others or reproduced in any manner or used for any purpose whatsoever without its written permission.

GENERAL	Item Number: 30 Quantity: 1		Valve Tag Number: UXV-23-0805		Data Page: 1 of 2	
	Service: Oxidate Long Circulation to Oxidizer No. 2 Shutoff		P&ID Drawing: 08B			
SIZING CONDITIONS	Process Data Case		General		General	
	Sizing Case		Open		Closed	
	Calculated C _v					
		Inlet	Outlet	Inlet	Outlet	Units
	Liquid Flow at Flowing Cond					m3/h
	Vapor Flow at Base Cond					normal m3/h
	Steam Flow					kg/h
	Temperature	39		39		°C
	Liquid Specific Gravity					----
	Vapor Spec. Gravity (Air=1)					----
	Pressure	2.99	2.99	5.36	2.99	kg/cm2(g)
	Liq Critical Press (Pseudo)					kg/cm2(a)
	Liquid Vapor Pressure					kg/cm2(a)
	Liquid Viscosity					cP
	Compressibility Z					
	Specific Heats Cp/Cv (Real)					
	Estimate Sound Press Level					dBA
	VALVE BODY & BONNET	Body Type: Eccentric Disc §		Approximate Valve C _v : 1030		F _L : x _T :
Body Size: 8" #1 Port Type: Single		Trim Style: Man. Standard Trim Size:				
Inlet Conn Size: 8" #1 Type: Flanged Class: 300		Inherent Characteristic: Equal Percentage				
Outlet Conn Size: 8" #1 Type: Flanged Class: 300		Balance Style: Full Open Position: 90°				
Flange Facing: Flange Face Finish:		Guiding Style: Shaft				
Body & Bonnet Material: 316 Stainless Steel #		Closure Member Material: 316 Stainless Steel				
Bonnet Style:		Seat Ring Material: Incoloy 825 with Hard Chrome				
Packing Style: Manufacturer's Recommendation		Seat Ring Retainer Material: 316 Stainless Steel				
Packing Material: Manufacturer's Recommendation		Bearing Material: Stellite				
Body/Bonnet Gasket Material:		Bushing/Cage Material:				
Lub/Iso Valve Material: Lub Material:		Stem/Shaft Material: 17-4PH				
Manufacturer: Metso Automation (or equal)		Flow Direction: Flow to close				
Model Number:		Seat Leakage Class: ANSI Class V				
ACTUATOR		Type: Spring Return Piston, Pneumatic		Inlet Pipe Nominal Size: Est Wall Thick:		
	Action Style: Fail Close		Outlet Pipe Nominal Size: Est Wall Thick:			
	Service Type: On-Off		Est Insulation Type:			
	Max Shutoff Press: In: 13.55 Out: 0.81 kg/cm2(g)		Est Insulation Thickness:			
	Maximum Stroke Time to Open: 5 Seconds		Installation:			
	Maximum Stroke Time to Close: 5 Seconds		Design Pressure : 13.55 kg/cm2g (F.V)			
	Stroke Cycle Service:		Design Temperature : 250 C			
DATA PAGE 1 REMARKS	§ The information is based on Neldisc L6 series.					
	# Preferred material is L grade stainless steel. Standard grade stainless steel that is not welded may be used for process wetted parts including the body. Stainless steel with weld repair is acceptable if low carbon (<= 0.03 %) filler material is used.					
	# Line size valve.					
	Valve will normally be in the closed position. Valve opening will be intermittent and generally short-term.					

ALL ITEMS SHALL COMPLY WITH THE GENERAL REQUIREMENTS SHOWN ON THE GENERAL REQUIREMENTS SECTION OF THIS SPECIFICATION.



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CONTROL VALVES

PROJECT SPECIFICATION

963766 — **616**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Item Number: 30 Quantity: 1		Valve Tag Number: UXV-23-0805		Data Page: 2 of 2		
TRANSDUCER	Transducer Tag Number:		HANDWHEEL	Handwheel Position:		
	Transducer Input Signal Range:			Usage:		
	Transducer Output Signal Range:					
	Manufacturer:					
SOLENOID VALVE	Solenoid Valve Tag Number: UXY-23-0805		POSITION SWITCH	Open Position Switch Tag Number:		
	Body Type: 3-Way Body Material: Brass			Close Position Switch Tag Number:		
	Seat Material: Stainless Steel Disc Material: Stainless Steel			Sensor Type:		
	Connection Size/Type: †			Contact Form:		
	Packing Style: Packless			Open Position Switch Contact Action:		
	Fluid: Instrument Air			Close Position Switch Contact Action:		
	Pressure: Contractor to determine			Contact Load: Contact Rating:		
	Minimum Pressure Port Orifice C _v : MAX Size: MAX			Mounting Location:		
	Minimum Exhaust Port Orifice C _v : MAX Size: MAX			Manufacturer:		
	Operation Style: †			POSITION TRANSMITTER	Position Transmitter Tag Number:	
	Coil Configuration: Single				Sensor Type:	
	Operating Voltage: \$\$ Frequency:				Output Signal Range:	
	Action Coil 1 Deenergized: Vent air from valve				Travel Measurement:	
	Action Coil 1 Energized: Supply air to valve				Mounting Location:	
	Action Coil 2 Deenergized:				Manufacturer:	
	Action Coil 2 Energized:			LOCK-UP VALVE	Action Style:	
	Manual Reset:				Mounting Location:	
	Manual Operator:				Model Number:	
Mounting Location: Nearest item to actuator in air line. †		QUICK EXHAUST	Manufacturer:			
Manufacturer: ASCO (or equal)			Connection Size/Type:			
			Body Material:			
POSITIONER	Positioner Tag Number:			Diaphragm Material:		
	Positioner Type:			Exhaust Action:		
	Positioner Input Signal Range:			Mounting Location:		
AIR FILTER	Positioner Output Signal Range:		VOLUME BOOSTER	Manufacturer:		
	Manufacturer:			Connection Size/Type:		
	Quantity: one			Body Material:		
TUBING	Type: Filter/Regulator			Seat Ring Material:		
	Manufacturer:			Diaphragm Material:		
	Tubing Size:			Mounting Location:		
DATA PAGE 2 REMARKS	Tubing Material:			Manufacturer:		
† The solenoid valve shall be mounted and tubed by the control valve vendor. The solenoid valve size and type, the tubing size, the actuator air connection size, and the air/filter regulator capacity shall be determined by the control valve vendor to meet the maximum stroking time requirements specified for the control valve. \$\$ The Contractor shall coordinate the solenoid valve operating voltage and coil watt rating with the supplier of the Decomposition Section Shutdown System (UC-2302). See UOP Project Specification 660, this project.						

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posco
ENGINEERING

CONTROL VALVES

PROJECT SPECIFICATION

963766 — **616**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				
1	28AUG12	MVD	MW				

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GENERAL	Item Number: 31 Quantity: 1 Valve Tag Number: HV-23-1001 Data Page: 1 of 2																																																																																																																																																																
	Service: Chilled Water to Flash Column Overhead Vapor Chiller Process Material: Water P&ID Drawing: 10																																																																																																																																																																
SIZING CONDITIONS	<table border="1"> <thead> <tr> <th>Process Data Case</th> <th colspan="2">General</th> <th colspan="2">General</th> <th colspan="2"></th> <th colspan="2"></th> <th></th> </tr> <tr> <th>Sizing Case</th> <th colspan="2">Open</th> <th colspan="2">Closed</th> <th colspan="2"></th> <th colspan="2"></th> <th></th> </tr> <tr> <th>Calculated C_v</th> <th>Inlet</th> <th>Outlet</th> <th>Inlet</th> <th>Outlet</th> <th>Inlet</th> <th>Outlet</th> <th>Inlet</th> <th>Outlet</th> <th>Units</th> </tr> </thead> <tbody> <tr> <td>Liquid Flow at Flowing Cond</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>m3/h</td> </tr> <tr> <td>Vapor Flow at Base Cond</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>normal m3/h</td> </tr> <tr> <td>Steam Flow</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>kg/h</td> </tr> <tr> <td>Temperature</td> <td>0</td> <td></td> <td>0</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>°C</td> </tr> <tr> <td>Liquid Specific Gravity</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>----</td> </tr> <tr> <td>Vapor Spec. Gravity (Air=1)</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>----</td> </tr> <tr> <td>Pressure</td> <td>6.75</td> <td>6.75</td> <td>6.75</td> <td>2.50</td> <td></td> <td></td> <td></td> <td></td> <td>kg/cm2(g)</td> </tr> <tr> <td>Liq Critical Press (Pseudo)</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>kg/cm2(a)</td> </tr> <tr> <td>Liquid Vapor Pressure</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>kg/cm2(a)</td> </tr> <tr> <td>Liquid Viscosity</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>cP</td> </tr> <tr> <td>Compressibility Z</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> <tr> <td>Specific Heats Cp/Cv (Real)</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> <tr> <td>Estimate Sound Press Level</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>dBA</td> </tr> </tbody> </table>	Process Data Case	General		General							Sizing Case	Open		Closed							Calculated C _v	Inlet	Outlet	Inlet	Outlet	Inlet	Outlet	Inlet	Outlet	Units	Liquid Flow at Flowing Cond									m3/h	Vapor Flow at Base Cond									normal m3/h	Steam Flow									kg/h	Temperature	0		0						°C	Liquid Specific Gravity									----	Vapor Spec. Gravity (Air=1)									----	Pressure	6.75	6.75	6.75	2.50					kg/cm2(g)	Liq Critical Press (Pseudo)									kg/cm2(a)	Liquid Vapor Pressure									kg/cm2(a)	Liquid Viscosity									cP	Compressibility Z										Specific Heats Cp/Cv (Real)										Estimate Sound Press Level									dBA
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DATA PAGE 1 REMARKS		§ The information is based on Masoneilan 35002 series. ¶¶ Line size valve. † Provide an adjustable limit stop with locking feature, per vendor design practice, to limit travel in the closed position. Field set limit stop to maintain the minimum flow requirement for associated equipment or service.																																																																																																																																																															
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ALL ITEMS SHALL COMPLY WITH THE GENERAL REQUIREMENTS SHOWN ON THE GENERAL REQUIREMENTS SECTION OF THIS SPECIFICATION.



CONTROL VALVES

PROJECT SPECIFICATION

963766 — **616**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Item Number: 31 Quantity: 1		Valve Tag Number: HV-23-1001		Data Page: 2 of 2	
TRANS-DUCER	Transducer Tag Number:		HANDWHEEL	Handwheel Position:	
	Transducer Input Signal Range:			Usage:	
	Transducer Output Signal Range:				
	Manufacturer:				
SOLENOID VALVE	Solenoid Valve Tag Number:		POSITION SWITCH	Open Position Switch Tag Number:	
	Body Type: Body Material:			Close Position Switch Tag Number:	
	Seat Material: Disc Material:			Sensor Type:	
	Connection Size/Type:			Contact Form:	
	Packing Style:			Open Position Switch Contact Action:	
	Fluid:			Close Position Switch Contact Action:	
	Pressure:			Contact Load: Contact Rating:	
	Minimum Pressure Port Orifice C_v : Size:			Mounting Location:	
	Minimum Exhaust Port Orifice C_v : Size:			Manufacturer:	
	Operation Style:			POSITION TRANSMITTER	Position Transmitter Tag Number:
	Coil Configuration:		Sensor Type:		
	Operating Voltage: Frequency:		Output Signal Range:		
	Action Coil 1 Deenergized:		Travel Measurement:		
	Action Coil 1 Energized:		Mounting Location:		
	Action Coil 2 Deenergized:		LOCK-UP VALVE	Manufacturer:	
	Action Coil 2 Energized:			Action Style:	
	Manual Reset:			Mounting Location:	
	Manual Operator:		QUICK EXHAUST	Model Number:	
	Mounting Location:			Manufacturer:	
	Manufacturer:			Connection Size/Type:	
POSITIONER	Positioner Tag Number:		VOLUME BOOSTER	Body Material:	
	Positioner Type: Digital			Seat Ring Material:	
	Positioner Input Signal Range: 4 - 20 mA			Diaphragm Material:	
	Positioner Output Signal Range: 0 - 100 %			Mounting Location:	
AIR FILTER	Quantity: one		TUBING	Manufacturer:	
	Type: Filter/Regulator			Connection Size/Type:	
TUBING	Manufacturer:		TUBING	Body Material:	
	Tubing Size:			Seat Ring Material:	
Tubing Material:		Diaphragm Material:		Mounting Location:	
		Manufacturer:			
DATA PAGE 2 REMARKS					

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CONTROL VALVES

PROJECT SPECIFICATION

963766 — **616**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				
1	28AUG12	MVD	MW				

GENERAL	Item Number: 32 Quantity: 1 Valve Tag Number: PCV-23-1003 Data Page: 1 of 2																																																																																																																																						
	Service: Nitrogen Purge to Concentration Vacuum Producing Equipment Process Material: Nitrogen P&ID Drawing: 10																																																																																																																																						
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VALVE BODY & BONNET	Body Type: Regulator § Body Size: Port Type: Single 1 Inlet Conn Size: 1/2" Type: Flanged Class: 300 1 Outlet Conn Size: 1/2" Type: Flanged Class: 300 Flange Facing: Flange Face Finish: Body & Bonnet Material: 316 Stainless Steel ‡ Bonnet Style: Packing Style: Packing Material: Body/Bonnet Gasket Material: Lub/Iso Valve Material: Lub Material: Manufacturer: Masoneilan (or equal) Model Number:	TRIM	Approximate Valve C _v : 1.0 F _L : 0.9 x _T : 0.72 Trim Style: Trim Size: 1/2" orifice Inherent Characteristic: Manufacturer's Standard Balance Style: Full Open Position: Guiding Style: Closure Member Material: Stainless Steel Seat Ring Material: Stainless Steel Seat Ring Retainer Material: Bearing Material: Bushing/Cage Material: Stem/Shaft Material: Stainless Steel Flow Direction: Seat Leakage Class:																																																																																																																																				
	ACTUATOR		INSTALLATION	Type: Action Style: Service Type: Max Shutoff Press: In: Out: kg/cm2(g) Maximum Stroke Time to Open: Maximum Stroke Time to Close: Stroke Cycle Service:	Inlet Pipe Nominal Size: Est Wall Thick: Outlet Pipe Nominal Size: Est Wall Thick: Est Insulation Type: Est Insulation Thickness: Installation: Design Pressure : 10 kg/cm2g Design Temperature : 120 C																																																																																																																																		
				§ The information is based on Masoneilan 17 series pressure regulator. Provide a Masoneilan type 17-22 (or equal) pressure reducing regulator with an 8 to 40 psig spring set at 12 psig (0.84kg/cm2 (g)).																																																																																																																																			
				‡ Preferred material is L grade stainless steel. Standard grade stainless steel that is not welded may be used for process wetted parts including the body. Stainless steel with weld repair is acceptable if low carbon (<= 0.03 %) filler material is used.																																																																																																																																			

ALL ITEMS SHALL COMPLY WITH THE GENERAL REQUIREMENTS SHOWN ON THE GENERAL REQUIREMENTS SECTION OF THIS SPECIFICATION.

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CONTROL VALVES

PROJECT SPECIFICATION

963766 — **616**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Item Number: 32 Quantity: 1		Valve Tag Number: PCV-23-1003		Data Page: 2 of 2	
TRANSducer	Transducer Tag Number:		HANDWHEEL	Handwheel Position:	
	Transducer Input Signal Range:			Usage:	
	Transducer Output Signal Range:				
	Manufacturer:				
SOLENOID VALVE	Solenoid Valve Tag Number:		POSITION SWITCH	Open Position Switch Tag Number:	
	Body Type: Body Material:			Close Position Switch Tag Number:	
	Seat Material: Disc Material:			Sensor Type:	
	Connection Size/Type:			Contact Form:	
	Packing Style:			Open Position Switch Contact Action:	
	Fluid:			Close Position Switch Contact Action:	
	Pressure:			Contact Load: Contact Rating:	
	Minimum Pressure Port Orifice C_v : Size:			Mounting Location:	
	Minimum Exhaust Port Orifice C_v : Size:			Manufacturer:	
	Operation Style:			POSITION TRANSMITTER	Position Transmitter Tag Number:
	Coil Configuration:		Sensor Type:		
	Operating Voltage: Frequency:		Output Signal Range:		
	Action Coil 1 Deenergized:		Travel Measurement:		
	Action Coil 1 Energized:		Mounting Location:		
	Action Coil 2 Deenergized:		LOCK-UP VALVE	Manufacturer:	
	Action Coil 2 Energized:			Action Style:	
	Manual Reset:			Mounting Location:	
	Manual Operator:		QUICK EXHAUST	Model Number:	
	Mounting Location:			Manufacturer:	
	Manufacturer:			Connection Size/Type:	
POSITIONER	Positioner Tag Number:		VOLUME BOOSTER	Body Material:	
	Positioner Type:			Diaphragm Material:	
	Positioner Input Signal Range:			Exhaust Action:	
AIR FILTER	Positioner Output Signal Range:		TUBING	Mounting Location:	
	Manufacturer:			Manufacturer:	
	Quantity:			Connection Size/Type:	
TUBING	Type:		TUBING	Body Material:	
	Manufacturer:			Seat Ring Material:	
	Tubing Size:			Diaphragm Material:	
DATA PAGE 2 REMARKS	Tubing Material:		TUBING	Mounting Location:	
				Manufacturer:	

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CONTROL VALVES

PROJECT SPECIFICATION

963766 — **616**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				
1	28AUG12	MVD	MW				

GENERAL	Item Number: 33 Quantity: 1 Valve Tag Number: FV-23-1001 Data Page: 1 of 2																																																																																																																																																																										
	Service: Nitrogen to Concentration Vacuum Producing Equipment Process Material: Nitrogen P&ID Drawing: 10																																																																																																																																																																										
SIZING CONDITIONS	<table border="1"> <tr> <th>Process Data Case</th> <th colspan="2">Design</th> <th colspan="2">Design</th> <th colspan="2"></th> <th colspan="2"></th> <th></th> </tr> <tr> <td>Sizing Case</td> <td colspan="2">Normal</td> <td colspan="2">Maximum</td> <td colspan="2"></td> <td colspan="2"></td> <td></td> </tr> <tr> <td>Calculated C_v</td> <td colspan="2">0.286</td> <td colspan="2">0.364</td> <td colspan="2"></td> <td colspan="2"></td> <td></td> </tr> <tr> <td></td> <td>Inlet</td> <td>Outlet</td> <td>Inlet</td> <td>Outlet</td> <td>Inlet</td> <td>Outlet</td> <td>Inlet</td> <td>Outlet</td> <td>Units</td> </tr> <tr> <td>Liquid Flow at Flowing Cond</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>m3/h</td> </tr> <tr> <td>Vapor Flow at Base Cond</td> <td>29</td> <td></td> <td>37</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>normal m3/h</td> </tr> <tr> <td>Steam Flow</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>kg/h</td> </tr> <tr> <td>Temperature</td> <td>38</td> <td></td> <td>38</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>°C</td> </tr> <tr> <td>Liquid Specific Gravity</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>----</td> </tr> <tr> <td>Vapor Spec. Gravity (Air=1)</td> <td>0.967</td> <td></td> <td>0.967</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>----</td> </tr> <tr> <td>Pressure</td> <td>7.00</td> <td>0.07</td> <td>7.00</td> <td>0.07</td> <td></td> <td></td> <td></td> <td></td> <td>kg/cm2(g)</td> </tr> <tr> <td>Liq Critical Press (Pseudo)</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>kg/cm2(a)</td> </tr> <tr> <td>Liquid Vapor Pressure</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>kg/cm2(a)</td> </tr> <tr> <td>Liquid Viscosity</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>cP</td> </tr> <tr> <td>Compressibility Z</td> <td>1.000</td> <td></td> <td>1.000</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> <tr> <td>Specific Heats Cp/Cv (Real)</td> <td>1.400</td> <td></td> <td>1.400</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> <tr> <td>Estimate Sound Press Level</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>dBA</td> </tr> </table>	Process Data Case	Design		Design							Sizing Case	Normal		Maximum							Calculated C _v	0.286		0.364								Inlet	Outlet	Inlet	Outlet	Inlet	Outlet	Inlet	Outlet	Units	Liquid Flow at Flowing Cond									m3/h	Vapor Flow at Base Cond	29		37						normal m3/h	Steam Flow									kg/h	Temperature	38		38						°C	Liquid Specific Gravity									----	Vapor Spec. Gravity (Air=1)	0.967		0.967						----	Pressure	7.00	0.07	7.00	0.07					kg/cm2(g)	Liq Critical Press (Pseudo)									kg/cm2(a)	Liquid Vapor Pressure									kg/cm2(a)	Liquid Viscosity									cP	Compressibility Z	1.000		1.000							Specific Heats Cp/Cv (Real)	1.400		1.400							Estimate Sound Press Level									dBA
	Process Data Case	Design		Design																																																																																																																																																																							
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VALVE BODY & BONNET	Body Type: Globe § Body Size: 1" Port Type: Single Inlet Conn Size: 1" Type: Flanged Class: 300 Outlet Conn Size: 1" Type: Flanged Class: 300 Flange Facing: Raised Face Flange Face Finish: 1 Body & Bonnet Material: 316L Stainless Steel ‡ Bonnet Style: Integral or Bolted Packing Style: Manufacturer's Recommendation Packing Material: Manufacturer's Recommendation Body/Bonnet Gasket Material: Lub/Iso Valve Material: Lub Material: Manufacturer: Masoneilan (or equal) Model Number:	TRIM	1 Approximate Valve C _v : 0.60 F _L : 0.9 x _T : 0.72 1 Trim Style: Man. Standard Trim Size: No. 3 Inherent Characteristic: Linear Balance Style: Unbalanced Full Open Position: 100 % Guiding Style: Top 1 Closure Member Material: Stellite Seat Ring Material: 316 Stainless Steel 1 Seat Ring Retainer Material: 316 Stainless Steel Bearing Material: 1 Bushing/Cage Material: 316 Stainless Steel Stem/Shaft Material: 316 Stainless Steel Flow Direction: Seat Leakage Class: ANSI Class IV																																																																																																																																																																								
	Type: Spring Diaphragm, Pneumatic Action Style: Fail Open Service Type: Modulating Max Shutoff Press: In: 9.00 Out: 0 kg/cm2(g) Maximum Stroke Time to Open: Maximum Stroke Time to Close: Stroke Cycle Service:		INSTALLATION	Inlet Pipe Nominal Size: Est Wall Thick: Outlet Pipe Nominal Size: Est Wall Thick: Est Insulation Type: Est Insulation Thickness: Installation: Design Pressure : 10 kg/cm2g Design Temperature : 120 C																																																																																																																																																																							
	1 § The information is based on Masoneilan 28000 series.																																																																																																																																																																										
	‡ Preferred material is L grade stainless steel. Standard grade stainless steel that is not welded may be used for process wetted parts including the body. Stainless steel with weld repair is acceptable if low carbon (<= 0.03 %) filler material is used.																																																																																																																																																																										

ALL ITEMS SHALL COMPLY WITH THE GENERAL REQUIREMENTS SHOWN ON THE GENERAL REQUIREMENTS SECTION OF THIS SPECIFICATION.

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CONTROL VALVES

PROJECT SPECIFICATION

963766 — **616**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Item Number: 33 Quantity: 1		Valve Tag Number: FV-23-1001		Data Page: 2 of 2	
TRANS-DUCER	Transducer Tag Number:		HANDWHEEL	Handwheel Position:	
	Transducer Input Signal Range:			Usage:	
	Transducer Output Signal Range:				
	Manufacturer:				
SOLENOID VALVE	Solenoid Valve Tag Number:		POSITION SWITCH	Open Position Switch Tag Number:	
	Body Type: Body Material:			Close Position Switch Tag Number:	
	Seat Material: Disc Material:			Sensor Type:	
	Connection Size/Type:			Contact Form:	
	Packing Style:			Open Position Switch Contact Action:	
	Fluid:			Close Position Switch Contact Action:	
	Pressure:			Contact Load: Contact Rating:	
	Minimum Pressure Port Orifice C_v : Size:			Mounting Location:	
	Minimum Exhaust Port Orifice C_v : Size:			Manufacturer:	
	Operation Style:			POSITION TRANSMITTER	Position Transmitter Tag Number:
	Coil Configuration:		Sensor Type:		
	Operating Voltage: Frequency:		Output Signal Range:		
	Action Coil 1 Deenergized:		Travel Measurement:		
	Action Coil 1 Energized:		Mounting Location:		
	Action Coil 2 Deenergized:		LOCK-UP VALVE	Manufacturer:	
	Action Coil 2 Energized:			Action Style:	
	Manual Reset:			Mounting Location:	
	Manual Operator:		QUICK EXHAUST	Model Number:	
	Mounting Location:			Manufacturer:	
	Manufacturer:			Connection Size/Type:	
POSITIONER	Positioner Tag Number:		VOLUME BOOSTER	Body Material:	
	Positioner Type: Digital			Diaphragm Material:	
	Positioner Input Signal Range: 4 - 20 mA			Exhaust Action:	
	Positioner Output Signal Range: 0 - 100 %			Mounting Location:	
AIR FILTER	Manufacturer:		Manufacturer:		
	Quantity: one		Connection Size/Type:		
	Type: Filter/Regulator		Body Material:		
TUBING	Manufacturer:		Seat Ring Material:		
	Tubing Size:		Diaphragm Material:		
	Tubing Material:		Mounting Location:		
Manufacturer:		Manufacturer:			
DATA PAGE 2 REMARKS					

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CONTROL VALVES

PROJECT SPECIFICATION

963766 — **616**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Note: The information in this document is confidential and the property of UOP LLC and must not be disclosed to others or reproduced in any manner or used for any purpose whatsoever without its written permission.

GENERAL	Item Number: 34 Quantity: 1 Valve Tag Number: UXV-23-1001 Data Page: 1 of 2 Service: Concentration Vacuum Producing Equipment Vent Process Material: Air with trace Aromatic Hydrocarbon P&ID Drawing: 10																																																																																																																																																																																	
SIZING CONDITIONS	<table border="1" style="width: 100%; border-collapse: collapse;"> <tr> <th rowspan="2">Process Data Case</th> <th colspan="2">General</th> <th colspan="2">General</th> <th colspan="2"></th> <th colspan="2"></th> <th rowspan="2">Units</th> </tr> <tr> <th colspan="2">Open</th> <th colspan="2">Closed</th> <th colspan="2"></th> <th colspan="2"></th> </tr> <tr> <td>Sizing Case</td> <td colspan="2"></td> <td colspan="2"></td> <td colspan="2"></td> <td colspan="2"></td> <td></td> </tr> <tr> <td>Calculated C_v</td> <td colspan="2"></td> <td colspan="2"></td> <td colspan="2"></td> <td colspan="2"></td> <td></td> </tr> <tr> <td>Liquid Flow at Flowing Cond</td> <td>Inlet</td> <td>Outlet</td> <td>Inlet</td> <td>Outlet</td> <td>Inlet</td> <td>Outlet</td> <td>Inlet</td> <td>Outlet</td> <td>m³/h</td> </tr> <tr> <td>Vapor Flow at Base Cond</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>normal m³/h</td> </tr> <tr> <td>Steam Flow</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>kg/h</td> </tr> <tr> <td>Temperature</td> <td colspan="2">20</td> <td colspan="2">20</td> <td colspan="2"></td> <td colspan="2"></td> <td>°C</td> </tr> <tr> <td>Liquid Specific Gravity</td> <td colspan="2"></td> <td colspan="2"></td> <td colspan="2"></td> <td colspan="2"></td> <td>----</td> </tr> <tr> <td>Vapor Spec. Gravity (Air=1)</td> <td colspan="2"></td> <td colspan="2"></td> <td colspan="2"></td> <td colspan="2"></td> <td>----</td> </tr> <tr> <td>Pressure</td> <td>0.07</td> <td>0.07</td> <td>0.07</td> <td>0.06</td> <td colspan="2"></td> <td colspan="2"></td> <td>kg/cm²(g)</td> </tr> <tr> <td>Liq Critical Press (Pseudo)</td> <td colspan="2"></td> <td colspan="2"></td> <td colspan="2"></td> <td colspan="2"></td> <td>kg/cm²(a)</td> </tr> <tr> <td>Liquid Vapor Pressure</td> <td colspan="2"></td> <td colspan="2"></td> <td colspan="2"></td> <td colspan="2"></td> <td>kg/cm²(a)</td> </tr> <tr> <td>Liquid Viscosity</td> <td colspan="2"></td> <td colspan="2"></td> <td colspan="2"></td> <td colspan="2"></td> <td>cP</td> </tr> <tr> <td>Compressibility Z</td> <td colspan="2"></td> <td colspan="2"></td> <td colspan="2"></td> <td colspan="2"></td> <td></td> </tr> <tr> <td>Specific Heats Cp/Cv (Real)</td> <td colspan="2"></td> <td colspan="2"></td> <td colspan="2"></td> <td colspan="2"></td> <td></td> </tr> <tr> <td>Estimate Sound Press Level</td> <td colspan="2"></td> <td colspan="2"></td> <td colspan="2"></td> <td colspan="2"></td> <td>dBA</td> </tr> </table>										Process Data Case	General		General						Units	Open		Closed						Sizing Case										Calculated C _v										Liquid Flow at Flowing Cond	Inlet	Outlet	Inlet	Outlet	Inlet	Outlet	Inlet	Outlet	m ³ /h	Vapor Flow at Base Cond									normal m ³ /h	Steam Flow									kg/h	Temperature	20		20						°C	Liquid Specific Gravity									----	Vapor Spec. Gravity (Air=1)									----	Pressure	0.07	0.07	0.07	0.06					kg/cm ² (g)	Liq Critical Press (Pseudo)									kg/cm ² (a)	Liquid Vapor Pressure									kg/cm ² (a)	Liquid Viscosity									cP	Compressibility Z										Specific Heats Cp/Cv (Real)										Estimate Sound Press Level									dBA
Process Data Case	General		General						Units																																																																																																																																																																									
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Liquid Flow at Flowing Cond	Inlet	Outlet	Inlet	Outlet	Inlet	Outlet	Inlet	Outlet	m ³ /h																																																																																																																																																																									
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VALVE BODY & BONNET	Body Type: Globe \$ Body Size: 2" #1 Port Type: Single Inlet Conn Size: 2" #1 Type: Flanged Class: 300 Outlet Conn Size: 2" #1 Type: Flanged Class: 300 Flange Facing: Raised Face Flange Face Finish: Body & Bonnet Material: 316 Stainless Steel # Bonnet Style: Integral or Bolted Packing Style: Manufacturer's Recommendation Packing Material: Manufacturer's Recommendation Body/Bonnet Gasket Material: Manufacturer's Recommend Lub/Iso Valve Material: Lub Material: Manufacturer: Masoneilan (or equal) Model Number:					TRIM	Approximate Valve C _v : 46 F _L : x _T : Trim Style: Man. Standard Trim Size: Inherent Characteristic: Equal Percentage Balance Style: Unbalanced Full Open Position: 100 % Guiding Style: Top Closure Member Material: 316 Stainless Steel + soft seat Seat Ring Material: 316 Stainless Steel Seat Ring Retainer Material: 304 Stainless Steel Bearing Material: Bushing/Cage Material: Stellite Stem/Shaft Material: 316 Stainless Steel Flow Direction: Flow to close Seat Leakage Class: ANSI Class VI																																																																																																																																																																											
ACTUATOR	Type: Spring Diaphragm, Pneumatic Action Style: Fail Service Type: On-Off Max Shutoff Press: In: 3.85 Out: 0 kg/cm ² (g) Maximum Stroke Time to Open: Maximum Stroke Time to Close: 5 Seconds Stroke Cycle Service:					INSTALLATION	Inlet Pipe Nominal Size: Est Wall Thick: Outlet Pipe Nominal Size: Est Wall Thick: Est Insulation Type: Est Insulation Thickness: Installation: Design Pressure : 4.5 kg/cm ² g (F.V) Design Temperature : 120 C																																																																																																																																																																											
DATA PAGE 1 REMARKS	§ The information is based on Masoneilan 21000 series. # Line size valve. # Preferred material is L grade stainless steel. Standard grade stainless steel that is not welded may be used for process wetted parts including the body. Stainless steel with weld repair is acceptable if low carbon (<= 0.03 %) filler material is used.																																																																																																																																																																																	

ALL ITEMS SHALL COMPLY WITH THE GENERAL REQUIREMENTS SHOWN ON THE GENERAL REQUIREMENTS SECTION OF THIS SPECIFICATION.



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CONTROL VALVES

PROJECT SPECIFICATION

963766 — **616**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Item Number: 34 Quantity: 1 Valve Tag Number: UXV-23-1001 Data Page: 2 of 2	
TRANSUDUCER Transducer Tag Number: Transducer Input Signal Range: Transducer Output Signal Range: Manufacturer:	HANDWHEEL Handwheel Position: Usage:
SOLENOID VALVE Solenoid Valve Tag Number: UXY-23-1001 Body Type: 3-Way Body Material: Brass Seat Material: Stainless Steel Disc Material: Stainless Steel Connection Size/Type: † Packing Style: Packless Fluid: Instrument Air Pressure: Contractor to determine Minimum Pressure Port Orifice C_v: MAX Size: MAX Minimum Exhaust Port Orifice C_v: MAX Size: MAX Operation Style: † Coil Configuration: Single Operating Voltage: \$\$ Frequency: Action Coil 1 Deenergized: Vent air from valve Action Coil 1 Energized: Supply air to valve Action Coil 2 Deenergized: Action Coil 2 Energized: Manual Reset: Manual Operator: Mounting Location: Nearest item to actuator in air line. † Manufacturer: ASCO (or equal)	POSITION SWITCH Open Position Switch Tag Number: Close Position Switch Tag Number: Sensor Type: Contact Form: Open Position Switch Contact Action: Close Position Switch Contact Action: Contact Load: Contact Rating: Mounting Location: Manufacturer: POSITION TRANSMITTER Position Transmitter Tag Number: Sensor Type: Output Signal Range: Travel Measurement: Mounting Location: Manufacturer: LOCK-UP VALVE Action Style: Mounting Location: Model Number: Manufacturer: QUICK EXHAUST Connection Size/Type: Body Material: Diaphragm Material: Exhaust Action: Mounting Location: Manufacturer: VOLUME BOOSTER Connection Size/Type: Body Material: Seat Ring Material: Diaphragm Material: Mounting Location: Manufacturer:
POSITIONER Positioner Tag Number: Positioner Type: Positioner Input Signal Range: Positioner Output Signal Range: Manufacturer:	
AIR FILTER Quantity: one Type: Filter/Regulator Manufacturer:	
TUBING Tubing Size: Tubing Material:	
DATA PAGE 2 REMARKS † The solenoid valve shall be mounted and tubed by the control valve vendor. The solenoid valve size and type, the tubing size, the actuator air connection size, and the air/filter regulator capacity shall be determined by the control valve vendor to meet the maximum stroking time requirements specified for the control valve. \$\$ The Contractor shall coordinate the solenoid valve operating voltage and coil watt rating with the supplier of the Concentration Vacuum Producing Equipment Shutdown System (UC-2303). See UOP Project Specification 660, this project.	

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CONTROL VALVES

PROJECT SPECIFICATION

963766 — **616**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Note: The information in this document is confidential and the property of UOP LLC and must not be disclosed to others or reproduced in any manner or used for any purpose whatsoever without its written permission.

GENERAL	Item Number: 35 Quantity: 1 Valve Tag Number: LV-23-1101 Data Page: 1 of 2																																																																																																																																																																										
	Service: Cumene to Cumene Flush Drum Process Material: Hydrocarbon with aromatics P&ID Drawing: 11																																																																																																																																																																										
SIZING CONDITIONS	<table border="1"> <tr> <th>Process Data Case</th> <th colspan="2">Single Case</th> <th colspan="2"></th> <th colspan="2"></th> <th colspan="2"></th> <th></th> </tr> <tr> <th>Sizing Case</th> <th colspan="2">Maximum</th> <th colspan="2"></th> <th colspan="2"></th> <th colspan="2"></th> <th></th> </tr> <tr> <td>Calculated C_v</td> <td colspan="2">30.0</td> <td colspan="2"></td> <td colspan="2"></td> <td colspan="2"></td> <td></td> </tr> <tr> <td></td> <td>Inlet</td> <td>Outlet</td> <td>Inlet</td> <td>Outlet</td> <td>Inlet</td> <td>Outlet</td> <td>Inlet</td> <td>Outlet</td> <td>Units</td> </tr> <tr> <td>Liquid Flow at Flowing Cond</td> <td>33.7</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>m3/h</td> </tr> <tr> <td>Vapor Flow at Base Cond</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>normal m3/h</td> </tr> <tr> <td>Steam Flow</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>kg/h</td> </tr> <tr> <td>Temperature</td> <td>46</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>°C</td> </tr> <tr> <td>Liquid Specific Gravity</td> <td>0.841</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>----</td> </tr> <tr> <td>Vapor Spec. Gravity (Air=1)</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>----</td> </tr> <tr> <td>Pressure</td> <td>4.00</td> <td>2.55</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>kg/cm2(g)</td> </tr> <tr> <td>Liq Critical Press (Pseudo)</td> <td>32.72</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>kg/cm2(a)</td> </tr> <tr> <td>Liquid Vapor Pressure</td> <td>0.90</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>kg/cm2(a)</td> </tr> <tr> <td>Liquid Viscosity</td> <td>0.425</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>cP</td> </tr> <tr> <td>Compressibility Z</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> <tr> <td>Specific Heats Cp/Cv (Real)</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> <tr> <td>Estimate Sound Press Level</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>dBA</td> </tr> </table>	Process Data Case	Single Case									Sizing Case	Maximum									Calculated C _v	30.0										Inlet	Outlet	Inlet	Outlet	Inlet	Outlet	Inlet	Outlet	Units	Liquid Flow at Flowing Cond	33.7								m3/h	Vapor Flow at Base Cond									normal m3/h	Steam Flow									kg/h	Temperature	46								°C	Liquid Specific Gravity	0.841								----	Vapor Spec. Gravity (Air=1)									----	Pressure	4.00	2.55							kg/cm2(g)	Liq Critical Press (Pseudo)	32.72								kg/cm2(a)	Liquid Vapor Pressure	0.90								kg/cm2(a)	Liquid Viscosity	0.425								cP	Compressibility Z										Specific Heats Cp/Cv (Real)										Estimate Sound Press Level									dBA
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VALVE BODY & BONNET	Body Type: Globe § Body Size: 2" Port Type: Single Inlet Conn Size: 2" Type: Flanged Class: 300 Outlet Conn Size: 2" Type: Flanged Class: 300 Flange Facing: Raised Face Flange Face Finish: Body & Bonnet Material: Carbon Steel Bonnet Style: Integral or Bolted Packing Style: Manufacturer's Recommendation Packing Material: Manufacturer's Recommendation Body/Bonnet Gasket Material: Manufacturer's Recommend Lub/Iso Valve Material: Lub Material: Manufacturer: Masoneilan (or equal) Model Number:	TRIM	Approximate Valve C _v : 46 F _L : 0.9 x _T : 0.72 Trim Style: Man. Standard Trim Size: Inherent Characteristic: Equal Percentage Balance Style: Unbalanced Full Open Position: 100 % Guiding Style: Top Closure Member Material: 416 Stainless Steel Seat Ring Material: 416 Stainless Steel Seat Ring Retainer Material: 304 Stainless Steel Bearing Material: Bushing/Cage Material: 440C Stainless Steel Stem/Shaft Material: 316 Stainless Steel Flow Direction: Seat Leakage Class: ANSI Class IV																																																																																																																																																																								
	ACTUATOR		Type: Spring Diaphragm, Pneumatic Action Style: Fail Close Service Type: Modulating Max Shutoff Press: In: 4.22 Out: 0.70 kg/cm2(g) Maximum Stroke Time to Open: Maximum Stroke Time to Close: Stroke Cycle Service:	INSTALLATION	Inlet Pipe Nominal Size: Est Wall Thick: Outlet Pipe Nominal Size: Est Wall Thick: Est Insulation Type: Est Insulation Thickness: Installation: Design Pressure : 10.6 kg/cm2g (F.V) Design Temperature : 120 C																																																																																																																																																																						
			DATA PAGE 1 REMARKS		§ The information is based on Masoneilan 21000 series. Valve will normally be in the closed position. Valve opening will be intermittent and generally short-term.																																																																																																																																																																						

ALL ITEMS SHALL COMPLY WITH THE GENERAL REQUIREMENTS SHOWN ON THE GENERAL REQUIREMENTS SECTION OF THIS SPECIFICATION.



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CONTROL VALVES

PROJECT SPECIFICATION

963766 — **616**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Item Number: 35 Quantity: 1		Valve Tag Number: LV-23-1101		Data Page: 2 of 2	
TRANS-DUCER	Transducer Tag Number:		HANDWHEEL	Handwheel Position:	
	Transducer Input Signal Range:			Usage:	
	Transducer Output Signal Range:				
	Manufacturer:				
SOLENOID VALVE	Solenoid Valve Tag Number:		POSITION SWITCH	Open Position Switch Tag Number:	
	Body Type: Body Material:			Close Position Switch Tag Number:	
	Seat Material: Disc Material:			Sensor Type:	
	Connection Size/Type:			Contact Form:	
	Packing Style:			Open Position Switch Contact Action:	
	Fluid:			Close Position Switch Contact Action:	
	Pressure:			Contact Load: Contact Rating:	
	Minimum Pressure Port Orifice C _v : Size:			Mounting Location:	
	Minimum Exhaust Port Orifice C _v : Size:			Manufacturer:	
	Operation Style:			POSITION TRANSMITTER	Position Transmitter Tag Number:
	Coil Configuration:		Sensor Type:		
	Operating Voltage: Frequency:		Output Signal Range:		
	Action Coil 1 Deenergized:		Travel Measurement:		
	Action Coil 1 Energized:		Mounting Location:		
	Action Coil 2 Deenergized:		LOCK-UP VALVE	Manufacturer:	
	Action Coil 2 Energized:			Action Style:	
	Manual Reset:			Mounting Location:	
	Manual Operator:		QUICK EXHAUST	Model Number:	
	Mounting Location:			Manufacturer:	
	Manufacturer:			Connection Size/Type:	
POSITIONER	Positioner Tag Number:		VOLUME BOOSTER	Body Material:	
	Positioner Type: Digital			Seat Ring Material:	
	Positioner Input Signal Range: 4 - 20 mA			Diaphragm Material:	
	Positioner Output Signal Range: 0 - 100 %			Mounting Location:	
AIR FILTER	Quantity: one		TUBING	Manufacturer:	
	Type: Filter/Regulator			Connection Size/Type:	
TUBING	Manufacturer:		TUBING	Body Material:	
	Tubing Size:			Seat Ring Material:	
Tubing Material:		Diaphragm Material:		Mounting Location:	
		Manufacturer:			
DATA PAGE 2 REMARKS					

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CONTROL VALVES

PROJECT SPECIFICATION

963766 — **616**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Note: The information in this document is confidential and the property of UOP LLC and must not be disclosed to others or reproduced in any manner or used for any purpose whatsoever without its written permission.

GENERAL	Item Number: 36 Quantity: 1 Valve Tag Number: PV-23-1101A Data Page: 1 of 2																																																																																																																																																																										
	Service: Nitrogen to Cumene Flush Drum Process Material: Nitrogen P&ID Drawing: 11																																																																																																																																																																										
SIZING CONDITIONS	<table border="1"> <tr> <th>Process Data Case</th> <th colspan="2">General</th> <th colspan="2"></th> <th colspan="2"></th> <th colspan="2"></th> <th></th> </tr> <tr> <th>Sizing Case</th> <th colspan="2">Maximum</th> <th colspan="2"></th> <th colspan="2"></th> <th colspan="2"></th> <th></th> </tr> <tr> <td>Calculated C_v</td> <td colspan="2">0.184</td> <td colspan="2"></td> <td colspan="2"></td> <td colspan="2"></td> <td></td> </tr> <tr> <td></td> <td>Inlet</td> <td>Outlet</td> <td>Inlet</td> <td>Outlet</td> <td>Inlet</td> <td>Outlet</td> <td>Inlet</td> <td>Outlet</td> <td>Units</td> </tr> <tr> <td>Liquid Flow at Flowing Cond</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>m3/h</td> </tr> <tr> <td>Vapor Flow at Base Cond</td> <td>19</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>normal m3/h</td> </tr> <tr> <td>Steam Flow</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>kg/h</td> </tr> <tr> <td>Temperature</td> <td>38</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>°C</td> </tr> <tr> <td>Liquid Specific Gravity</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>----</td> </tr> <tr> <td>Vapor Spec. Gravity (Air=1)</td> <td>0.967</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>----</td> </tr> <tr> <td>Pressure</td> <td>7.00</td> <td>0.70</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>kg/cm2(g)</td> </tr> <tr> <td>Liq Critical Press (Pseudo)</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>kg/cm2(a)</td> </tr> <tr> <td>Liquid Vapor Pressure</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>kg/cm2(a)</td> </tr> <tr> <td>Liquid Viscosity</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>cP</td> </tr> <tr> <td>Compressibility Z</td> <td>1.000</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> <tr> <td>Specific Heats Cp/Cv (Real)</td> <td>1.400</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> <tr> <td>Estimate Sound Press Level</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>dBA</td> </tr> </table>	Process Data Case	General									Sizing Case	Maximum									Calculated C _v	0.184										Inlet	Outlet	Inlet	Outlet	Inlet	Outlet	Inlet	Outlet	Units	Liquid Flow at Flowing Cond									m3/h	Vapor Flow at Base Cond	19								normal m3/h	Steam Flow									kg/h	Temperature	38								°C	Liquid Specific Gravity									----	Vapor Spec. Gravity (Air=1)	0.967								----	Pressure	7.00	0.70							kg/cm2(g)	Liq Critical Press (Pseudo)									kg/cm2(a)	Liquid Vapor Pressure									kg/cm2(a)	Liquid Viscosity									cP	Compressibility Z	1.000									Specific Heats Cp/Cv (Real)	1.400									Estimate Sound Press Level									dBA
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VALVE BODY & BONNET	Body Type: Globe \$ Body Size: 1" Port Type: Single Inlet Conn Size: 1" Type: Flanged Class: 300 Outlet Conn Size: 1" Type: Flanged Class: 300 Flange Facing: Raised Face Flange Face Finish: Body & Bonnet Material: 316L Stainless Steel \$\$ Bonnet Style: Integral or Bolted Packing Style: Manufacturer's Recommendation Packing Material: Manufacturer's Recommendation Body/Bonnet Gasket Material: Lub/Iso Valve Material: Lub Material: Manufacturer: Masoneilan (or equal) Model Number:	TRIM	Approximate Valve C _v : 0.25 F _L : 0.9 x _T : 0.72 Trim Style: Man. Standard Trim Size: No. 4 Inherent Characteristic: Linear Balance Style: Unbalanced Full Open Position: 100 % Guiding Style: Top Closure Member Material: Stellite Seat Ring Material: 17-4PH Stainless Steel Seat Ring Retainer Material: 17-4PH Stainless Steel Bearing Material: Bushing/Cage Material: 17-4PH Stainless Steel Stem/Shaft Material: 316 Stainless Steel Flow Direction: Seat Leakage Class: ANSI Class IV																																																																																																																																																																								
	ACTUATOR		Type: Spring Diaphragm, Pneumatic Action Style: Fail Close Service Type: Modulating Max Shutoff Press: In: 9.00 Out: 0.70 kg/cm2(g) Maximum Stroke Time to Open: Maximum Stroke Time to Close: Stroke Cycle Service:	INSTALLATION	Inlet Pipe Nominal Size: Est Wall Thick: Outlet Pipe Nominal Size: Est Wall Thick: Est Insulation Type: Est Insulation Thickness: Installation: Design Pressure : 10 kg/cm2g Design Temperature : 120 C																																																																																																																																																																						
			DATA PAGE 1 REMARKS		§ The information is based on Masoneilan 28000 series. §§ Carbon steel is an acceptable alternate.																																																																																																																																																																						

ALL ITEMS SHALL COMPLY WITH THE GENERAL REQUIREMENTS SHOWN ON THE GENERAL REQUIREMENTS SECTION OF THIS SPECIFICATION.



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CONTROL VALVES

PROJECT SPECIFICATION

963766 — **616**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Item Number: 36 Quantity: 1		Valve Tag Number: PV-23-1101A		Data Page: 2 of 2	
TRANSducer	Transducer Tag Number:		HANDWHEEL	Handwheel Position:	
	Transducer Input Signal Range:			Usage:	
	Transducer Output Signal Range:				
	Manufacturer:				
SOLENOID VALVE	Solenoid Valve Tag Number:		POSITION SWITCH	Open Position Switch Tag Number:	
	Body Type: Body Material:			Close Position Switch Tag Number:	
	Seat Material: Disc Material:			Sensor Type:	
	Connection Size/Type:			Contact Form:	
	Packing Style:			Open Position Switch Contact Action:	
	Fluid:			Close Position Switch Contact Action:	
	Pressure:			Contact Load: Contact Rating:	
	Minimum Pressure Port Orifice C _v : Size:			Mounting Location:	
	Minimum Exhaust Port Orifice C _v : Size:			Manufacturer:	
	Operation Style:			POSITION TRANSMITTER	Position Transmitter Tag Number:
	Coil Configuration:		Sensor Type:		
	Operating Voltage: Frequency:		Output Signal Range:		
	Action Coil 1 Deenergized:		Travel Measurement:		
	Action Coil 1 Energized:		Mounting Location:		
	Action Coil 2 Deenergized:		LOCK-UP VALVE	Manufacturer:	
	Action Coil 2 Energized:			Action Style:	
	Manual Reset:			Mounting Location:	
	Manual Operator:		QUICK EXHAUST	Model Number:	
	Mounting Location:			Manufacturer:	
	Manufacturer:			Connection Size/Type:	
POSITIONER	Positioner Tag Number:		VOLUME BOOSTER	Body Material:	
	Positioner Type: Digital			Diaphragm Material:	
	Positioner Input Signal Range: 4 - 20 mA			Exhaust Action:	
	Positioner Output Signal Range: 0 - 100 %			Mounting Location:	
AIR FILTER	Manufacturer:		Manufacturer:		
	Quantity: one		Connection Size/Type:		
	Type: Filter/Regulator		Body Material:		
TUBING	Manufacturer:		Seat Ring Material:		
	Tubing Size:		Diaphragm Material:		
	Tubing Material:		Mounting Location:		
Manufacturer:		Manufacturer:			
DATA PAGE 2 REMARKS					

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PROJECT SPECIFICATION

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REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Note: The information in this document is confidential and the property of UOP LLC and must not be disclosed to others or reproduced in any manner or used for any purpose whatsoever without its written permission.

GENERAL	Item Number: 37 Quantity: 1 Valve Tag Number: PV-23-1101B Data Page: 1 of 2																																																																																																																																																																										
	Service: Vent Gas from Cumene Flush Drum Process Material: Nitrogen P&ID Drawing: 11																																																																																																																																																																										
SIZING CONDITIONS	<table border="1"> <thead> <tr> <th>Process Data Case</th> <th colspan="2">General</th> <th colspan="2"></th> <th colspan="2"></th> <th colspan="2"></th> <th></th> </tr> <tr> <th>Sizing Case</th> <th colspan="2">Maximum</th> <th colspan="2"></th> <th colspan="2"></th> <th colspan="2"></th> <th></th> </tr> <tr> <td>Calculated C_v</td> <td colspan="2">2.51</td> <td colspan="2"></td> <td colspan="2"></td> <td colspan="2"></td> <td></td> </tr> </thead> <tbody> <tr> <td></td> <td>Inlet</td> <td>Outlet</td> <td>Inlet</td> <td>Outlet</td> <td>Inlet</td> <td>Outlet</td> <td>Inlet</td> <td>Outlet</td> <td>Units</td> </tr> <tr> <td>Liquid Flow at Flowing Cond</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>m3/h</td> </tr> <tr> <td>Vapor Flow at Base Cond</td> <td>48</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>normal m3/h</td> </tr> <tr> <td>Steam Flow</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>kg/h</td> </tr> <tr> <td>Temperature</td> <td>46</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>°C</td> </tr> <tr> <td>Liquid Specific Gravity</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>----</td> </tr> <tr> <td>Vapor Spec. Gravity (Air=1)</td> <td>0.967</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>----</td> </tr> <tr> <td>Pressure</td> <td>0.70</td> <td>0.05</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>kg/cm2(g)</td> </tr> <tr> <td>Liq Critical Press (Pseudo)</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>kg/cm2(a)</td> </tr> <tr> <td>Liquid Vapor Pressure</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>kg/cm2(a)</td> </tr> <tr> <td>Liquid Viscosity</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>cP</td> </tr> <tr> <td>Compressibility Z</td> <td>1.000</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> <tr> <td>Specific Heats Cp/Cv (Real)</td> <td>1.400</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> <tr> <td>Estimate Sound Press Level</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>dBA</td> </tr> </tbody> </table>	Process Data Case	General									Sizing Case	Maximum									Calculated C _v	2.51										Inlet	Outlet	Inlet	Outlet	Inlet	Outlet	Inlet	Outlet	Units	Liquid Flow at Flowing Cond									m3/h	Vapor Flow at Base Cond	48								normal m3/h	Steam Flow									kg/h	Temperature	46								°C	Liquid Specific Gravity									----	Vapor Spec. Gravity (Air=1)	0.967								----	Pressure	0.70	0.05							kg/cm2(g)	Liq Critical Press (Pseudo)									kg/cm2(a)	Liquid Vapor Pressure									kg/cm2(a)	Liquid Viscosity									cP	Compressibility Z	1.000									Specific Heats Cp/Cv (Real)	1.400									Estimate Sound Press Level									dBA
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VALVE BODY & BONNET	Body Type: Globe § Body Size: 1" Port Type: Single Inlet Conn Size: 1" Type: Flanged Class: 300 Outlet Conn Size: 1" Type: Flanged Class: 300 Flange Facing: Raised Face Flange Face Finish: Body & Bonnet Material: 316 Stainless Steel ‡ Bonnet Style: Integral or Bolted Packing Style: Manufacturer's Recommendation Packing Material: Manufacturer's Recommendation Body/Bonnet Gasket Material: Manufacturer's Recommend Lub/Iso Valve Material: Lub Material: Manufacturer: Masoneilan (or equal) Model Number:	TRIM	Approximate Valve C _v : 3.8 F _L : 0.9 x _T : 0.72 Trim Style: Man. Standard Trim Size: Inherent Characteristic: Equal Percentage Balance Style: Unbalanced Full Open Position: 100 % Guiding Style: Top Closure Member Material: 316 Stainless Steel Seat Ring Material: 316 Stainless Steel Seat Ring Retainer Material: 304 Stainless Steel Bearing Material: Bushing/Cage Material: Stellite Stem/Shaft Material: 316 Stainless Steel Flow Direction: Seat Leakage Class: ANSI Class IV																																																																																																																																																																								
	ACTUATOR		INSTALLATION	Type: Spring Diaphragm, Pneumatic Action Style: Fail Close Service Type: Modulating Max Shutoff Press: In: 11.00 Out: 0 kg/cm2(g) Maximum Stroke Time to Open: Maximum Stroke Time to Close: Stroke Cycle Service:	Inlet Pipe Nominal Size: Est Wall Thick: Outlet Pipe Nominal Size: Est Wall Thick: Est Insulation Type: Est Insulation Thickness: Installation: Design Pressure : 10.0 kg/cm2g Design Temperature : 205 C																																																																																																																																																																						
				§ The information is based on Masoneilan 21000 series.																																																																																																																																																																							
				‡ Preferred material is L grade stainless steel. Standard grade stainless steel that is not welded may be used for process wetted parts including the body. Stainless steel with weld repair is acceptable if low carbon (<= 0.03 %) filler material is used.																																																																																																																																																																							

ALL ITEMS SHALL COMPLY WITH THE GENERAL REQUIREMENTS SHOWN ON THE GENERAL REQUIREMENTS SECTION OF THIS SPECIFICATION.



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CONTROL VALVES

PROJECT SPECIFICATION

963766 — **616**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Item Number: 37 Quantity: 1		Valve Tag Number: PV-23-1101B		Data Page: 2 of 2	
TRANS-DUCER	Transducer Tag Number:		HANDWHEEL	Handwheel Position:	
	Transducer Input Signal Range:			Usage:	
	Transducer Output Signal Range:				
	Manufacturer:				
SOLENOID VALVE	Solenoid Valve Tag Number:		POSITION SWITCH	Open Position Switch Tag Number:	
	Body Type: Body Material:			Close Position Switch Tag Number:	
	Seat Material: Disc Material:			Sensor Type:	
	Connection Size/Type:			Contact Form:	
	Packing Style:			Open Position Switch Contact Action:	
	Fluid:			Close Position Switch Contact Action:	
	Pressure:			Contact Load: Contact Rating:	
	Minimum Pressure Port Orifice C_v : Size:			Mounting Location:	
	Minimum Exhaust Port Orifice C_v : Size:			Manufacturer:	
	Operation Style:			POSITION TRANSMITTER	Position Transmitter Tag Number:
	Coil Configuration:		Sensor Type:		
	Operating Voltage: Frequency:		Output Signal Range:		
	Action Coil 1 Deenergized:		Travel Measurement:		
	Action Coil 1 Energized:		Mounting Location:		
	Action Coil 2 Deenergized:		LOCK-UP VALVE	Manufacturer:	
	Action Coil 2 Energized:			Action Style:	
	Manual Reset:			Mounting Location:	
	Manual Operator:		QUICK EXHAUST	Model Number:	
	Mounting Location:			Manufacturer:	
	Manufacturer:			Connection Size/Type:	
POSITIONER	Positioner Tag Number:		VOLUME BOOSTER	Body Material:	
	Positioner Type: Digital			Seat Ring Material:	
	Positioner Input Signal Range: 4 - 20 mA			Diaphragm Material:	
AIR FILTER	Positioner Output Signal Range: 0 - 100 %		TUBING	Mounting Location:	
	Manufacturer:			Manufacturer:	
	Quantity: one				
TUBING	Type: Filter/Regulator				
	Manufacturer:				
	Tubing Size:				
DATA PAGE 2 REMARKS	Tubing Material:				

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CONTROL VALVES

PROJECT SPECIFICATION

963766 — **616**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

GENERAL	Item Number: 38 Quantity: 1 Valve Tag Number: FV-23-1202 Data Page: 1 of 2																																																																																																																																																							
	Service: Decomposer Feed Startup bypass Process Material: Hydrocarbon with aromatics P&ID Drawing: 12																																																																																																																																																							
SIZING CONDITIONS	<table border="1"> <tr> <td>Process Data Case</td> <td colspan="2">Single Case</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> <tr> <td>Sizing Case</td> <td colspan="2">Maximum</td> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> <tr> <td>Calculated C_v</td> <td colspan="2">133</td> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> <tr> <td></td> <td>Inlet</td> <td>Outlet</td> <td>Inlet</td> <td>Outlet</td> <td>Inlet</td> <td>Outlet</td> <td>Inlet</td> <td>Outlet</td> </tr> <tr> <td>Liquid Flow at Flowing Cond</td> <td>62.7</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>m3/h</td> </tr> <tr> <td>Vapor Flow at Base Cond</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>normal m3/h</td> </tr> <tr> <td>Steam Flow</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>kg/h</td> </tr> <tr> <td>Temperature</td> <td>60</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>°C</td> </tr> <tr> <td>Liquid Specific Gravity</td> <td>0.972</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>----</td> </tr> <tr> <td>Vapor Spec. Gravity (Air=1)</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>----</td> </tr> <tr> <td>Pressure</td> <td>3.29</td> <td>2.99</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>kg/cm2(g)</td> </tr> <tr> <td>Liq Critical Press (Pseudo)</td> <td>33.89</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>kg/cm2(a)</td> </tr> <tr> <td>Liquid Vapor Pressure</td> <td>0.01</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>kg/cm2(a)</td> </tr> <tr> <td>Liquid Viscosity</td> <td>3.432</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>cP</td> </tr> <tr> <td>Compressibility Z</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> <tr> <td>Specific Heats Cp/Cv (Real)</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> <tr> <td>Estimate Sound Press Level</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>dBA</td> </tr> </table>	Process Data Case	Single Case								Sizing Case	Maximum							Calculated C _v	133								Inlet	Outlet	Inlet	Outlet	Inlet	Outlet	Inlet	Outlet	Liquid Flow at Flowing Cond	62.7							m3/h	Vapor Flow at Base Cond								normal m3/h	Steam Flow								kg/h	Temperature	60							°C	Liquid Specific Gravity	0.972							----	Vapor Spec. Gravity (Air=1)								----	Pressure	3.29	2.99						kg/cm2(g)	Liq Critical Press (Pseudo)	33.89							kg/cm2(a)	Liquid Vapor Pressure	0.01							kg/cm2(a)	Liquid Viscosity	3.432							cP	Compressibility Z									Specific Heats Cp/Cv (Real)									Estimate Sound Press Level								dBA
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Estimate Sound Press Level								dBA																																																																																																																																																
VALVE BODY & BONNET	Body Type: Eccentric Rotary Plug § Body Size: 3" Port Type: Single Inlet Conn Size: 3" Type: Flanged Class: 300 Outlet Conn Size: 3" Type: Flanged Class: 300 Flange Facing: Raised Face Flange Face Finish: Body & Bonnet Material: 316 Stainless Steel ‡ Bonnet Style: Integral or Bolted Packing Style: Manufacturer's Recommendation Packing Material: Manufacturer's Recommendation Body/Bonnet Gasket Material: Lub/Iso Valve Material: Lub Material: Manufacturer: Masoneilan (or equal) Model Number:	TRIM	Approximate Valve C _v : 135 F _L : 0.68 x _T : 0.4 Trim Style: Man. Standard Trim Size: Inherent Characteristic: Linear Balance Style: Full Open Position: 100 % Guiding Style: Shaft Closure Member Material: 316 SS, Stellite sheath Seat Ring Material: 316 Stainless Steel Seat Ring Retainer Material: 316 Stainless Steel Bearing Material: Stellite Bushing/Cage Material: Stem/Shaft Material: 316 Stainless Steel Flow Direction: Seat Leakage Class: ANSI Class IV																																																																																																																																																					
	ACTUATOR		INSTALLATION	Type: Spring Diaphragm, Pneumatic Action Style: Fail Open Service Type: Modulating Max Shutoff Press: In: 8.86 Out: 0 kg/cm2(g) Maximum Stroke Time to Open: 5 Seconds Maximum Stroke Time to Close: Stroke Cycle Service:	Inlet Pipe Nominal Size: Est Wall Thick: Outlet Pipe Nominal Size: Est Wall Thick: Est Insulation Type: Est Insulation Thickness: Installation: Design Pressure : 13.55 kg/cm2g Design Temperature : 250 C																																																																																																																																																			
				§ The information is based on Masoneilan 35002 series.																																																																																																																																																				
				‡ Preferred material is L grade stainless steel. Standard grade stainless steel that is not welded may be used for process wetted parts including the body. Stainless steel with weld repair is acceptable if low carbon (<= 0.03 %) filler material is used.																																																																																																																																																				
				Valve will normally be in the closed position. Valve opening will be intermittent and generally short-term.																																																																																																																																																				
	ALL ITEMS SHALL COMPLY WITH THE GENERAL REQUIREMENTS SHOWN ON THE GENERAL REQUIREMENTS SECTION OF THIS SPECIFICATION.																																																																																																																																																							

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CONTROL VALVES

PROJECT SPECIFICATION

963766 — **616**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Item Number: 38 Quantity: 1		Valve Tag Number: FV-23-1202		Data Page: 2 of 2		
TRANS-DUCER	Transducer Tag Number:		HANDWHEEL	Handwheel Position:		
	Transducer Input Signal Range:			Usage:		
TRANS-DUCER	Transducer Output Signal Range:		HANDWHEEL			
	Manufacturer:					
SOLENOID VALVE	Solenoid Valve Tag Number: FXY-23-1202		POSITION SWITCH	Open Position Switch Tag Number:		
	Body Type: 3-Way Body Material: Brass			Close Position Switch Tag Number:		
	Seat Material: Stainless Steel Disc Material: Stainless Steel			Sensor Type:		
	Connection Size/Type: †			Contact Form:		
	Packing Style: Packless			Open Position Switch Contact Action:		
	Fluid: Control Signal Air			Close Position Switch Contact Action:		
	Pressure: Contractor to determine			Contact Load: Contact Rating:		
	Minimum Pressure Port Orifice C_v : MAX Size: MAX			Mounting Location:		
	Minimum Exhaust Port Orifice C_v : MAX Size: MAX			Manufacturer:		
	Operation Style: †					
	Coil Configuration: Single					
	Operating Voltage: \$\$ Frequency:					
	Action Coil 1 Deenergized: Vent air from valve					
	Action Coil 1 Energized: Supply air to valve					
Action Coil 2 Deenergized:						
Action Coil 2 Energized:						
Manual Reset:						
Manual Operator:						
Mounting Location: Nearest item to actuator in air line. †						
Manufacturer: ASCO (or equal)						
POSITIONER	Positioner Tag Number:		QUICK EXHAUST	Connection Size/Type:		
	Positioner Type: Digital			Body Material:		
	Positioner Input Signal Range: 4 - 20 mA			Diaphragm Material:		
	Positioner Output Signal Range: 0 - 100 %			Exhaust Action:		
Manufacturer:		Mounting Location:		Manufacturer:		
AIR FILTER	Quantity: one		VOLUME BOOSTER	Connection Size/Type:		
	Type: Filter/Regulator			Body Material:		
	Manufacturer:			Seat Ring Material:		
				Diaphragm Material:		
TUBING	Tubing Size:		Mounting Location:		Manufacturer:	
	Tubing Material:					
DATA PAGE 2 REMARKS	† The solenoid valve shall be mounted and tubed by the control valve vendor. The solenoid valve size and type, the tubing size, the actuator air connection size, and the air/filter regulator capacity shall be determined by the control valve vendor to meet the maximum stroking time requirements specified for the control valve. Pilot operated solenoid valves shall be suitable for operation in low pressure control signal air.					
	\$\$ The Contractor shall coordinate the solenoid valve operating voltage and coil watt rating with the supplier of the Decomposition Section Shutdown System (UC-2302). See UOP Project Specification 660, this project.					

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PROJECT SPECIFICATION

963766 — **616**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

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GENERAL	Item Number: 39 Quantity: 1 Valve Tag Number: UXV-23-1203 Data Page: 1 of 2																																																																																																																																																																								
	Service: Decomposer Feed Startup Bypass to Oxidate Circulation Line Process Material: Hydrocarbon with aromatics P&ID Drawing: 12																																																																																																																																																																								
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CONTROL VALVES

PROJECT SPECIFICATION

963766 — **616**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Item Number: 39 Quantity: 1		Valve Tag Number: UXV-23-1203		Data Page: 2 of 2	
TRANS-DUCER	Transducer Tag Number:		HANDWHEEL	Handwheel Position:	
	Transducer Input Signal Range:			Usage:	
Transducer Output Signal Range:					
Manufacturer:					
SOLENOID VALVE	Solenoid Valve Tag Number: UXY-23-1203A, B		POSITION SWITCH	Open Position Switch Tag Number:	
	Body Type: 3-Way Body Material: Brass			Close Position Switch Tag Number:	
	Seat Material: Stainless Steel Disc Material: Stainless Steel			Sensor Type:	
	Connection Size/Type: †			Contact Form:	
	Packing Style: Packless			Open Position Switch Contact Action:	
	Fluid: Instrument Air			Close Position Switch Contact Action:	
	Pressure: Contractor to determine			Contact Load: Contact Rating:	
	Minimum Pressure Port Orifice C _v : MAX Size: MAX			Mounting Location:	
	Minimum Exhaust Port Orifice C _v : MAX Size: MAX			Manufacturer:	
	Operation Style: †				
	Coil Configuration: Single		POSITION TRANSMITTER	Position Transmitter Tag Number:	
	Operating Voltage: \$\$ Frequency:			Sensor Type:	
	Action Coil 1 Deenergized: Supply air to valve			Output Signal Range:	
	Action Coil 1 Energized: Vent air from valve			Travel Measurement:	
	Action Coil 2 Deenergized:		Mounting Location:		
	Action Coil 2 Energized:		Manufacturer:		
	Manual Reset:		LOCK-UP VALVE	Action Style:	
	Manual Operator:			Mounting Location:	
Mounting Location: Nearest item to actuator in air line. †		Model Number:			
Manufacturer: ASCO (or equal)		Manufacturer:			
POSITIONER	Positioner Tag Number:		QUICK EXHAUST	Connection Size/Type:	
	Positioner Type:			Body Material:	
	Positioner Input Signal Range:			Diaphragm Material:	
	Positioner Output Signal Range:			Exhaust Action:	
Manufacturer:		Mounting Location:			
AIR FILTER	Quantity: one		VOLUME BOOSTER	Connection Size/Type:	
	Type: Filter/Regulator			Body Material:	
	Manufacturer:			Seat Ring Material:	
				Diaphragm Material:	
TUBING	Tubing Size:		Mounting Location:		
	Tubing Material:		Manufacturer:		
DATA PAGE 2 REMARKS	† Provide two solenoid valves. The solenoid valves shall be mounted and tubed by the control valve vendor. See Sketch "A". The solenoid valves size and type, the tubing size, the actuator air connection size, and the air/filter regulator capacity shall be determined by the control valve vendor to meet the maximum stroking time requirements specified for the control valve.				
	\$\$ The Contractor shall coordinate the solenoid valve operating voltage and coil watt rating with the supplier of the Decomposition Section Shutdown System (UC-2302). See UOP Project Specification 660, this project.				

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CONTROL VALVES

PROJECT SPECIFICATION

963766 — **616**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				
1	28AUG12	MVD	MW				

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GENERAL	Item Number: 40 Quantity: 1 Valve Tag Number: UXV-23-1202 Data Page: 1 of 2																																																																																																																																																																						
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CONTROL VALVES

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963766 — **616**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Item Number: 40 Quantity: 1		Valve Tag Number: UXV-23-1202		Data Page: 2 of 2	
TRANSDUCER	Transducer Tag Number:	HANDWHEEL	Handwheel Position:		
	Transducer Input Signal Range:		Usage:		
	Transducer Output Signal Range:				
	Manufacturer:				
SOLENOID VALVE	Solenoid Valve Tag Number: UXY-23-1202	POSITION SWITCH	Open Position Switch Tag Number:		
	Body Type: 3-Way Body Material: Brass		Close Position Switch Tag Number:		
	Seat Material: Stainless Steel Disc Material: Stainless Steel		Sensor Type:		
	Connection Size/Type: †		Contact Form:		
	Packing Style: Packless		Open Position Switch Contact Action:		
	Fluid: Instrument Air		Close Position Switch Contact Action:		
	Pressure: Contractor to determine		Contact Load: Contact Rating:		
	Minimum Pressure Port Orifice C _v : MAX Size: MAX		Mounting Location:		
	Minimum Exhaust Port Orifice C _v : MAX Size: MAX		Manufacturer:		
	Operation Style: †				
	Coil Configuration: Single	POSITION TRANSMITTER	Position Transmitter Tag Number:		
	Operating Voltage: \$\$ Frequency:		Sensor Type:		
	Action Coil 1 Deenergized: Vent air from valve		Output Signal Range:		
	Action Coil 1 Energized: Supply air to valve		Travel Measurement:		
	Action Coil 2 Deenergized:		Mounting Location:		
	Action Coil 2 Energized:		Manufacturer:		
	Manual Reset:	LOCK-UP VALVE	Action Style:		
	Manual Operator:		Mounting Location:		
Mounting Location: Nearest item to actuator in air line. †	Model Number:				
Manufacturer: ASCO (or equal)		Manufacturer:			
POSITIONER	Positioner Tag Number:	QUICK EXHAUST	Connection Size/Type:		
	Positioner Type:		Body Material:		
	Positioner Input Signal Range:		Diaphragm Material:		
	Positioner Output Signal Range:		Exhaust Action:		
	Manufacturer:		Mounting Location:		
AIR FILTER	Quantity: one	VOLUME BOOSTER	Connection Size/Type:		
	Type: Filter/Regulator		Body Material:		
	Manufacturer:		Seat Ring Material:		
			Diaphragm Material:		
TUBING	Tubing Size:		Mounting Location:		
	Tubing Material:		Manufacturer:		
DATA PAGE 2 REMARKS	† The solenoid valve shall be mounted and tubed by the control valve vendor. The solenoid valve size and type, the tubing size, the actuator air connection size, and the air/filter regulator capacity shall be determined by the control valve vendor to meet the maximum stroking time requirements specified for the control valve.				
	\$\$ The Contractor shall coordinate the solenoid valve operating voltage and coil watt rating with the supplier of the Decomposition Section Shutdown System (UC-2302). See UOP Project Specification 660, this project.				

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CONTROL VALVES

PROJECT SPECIFICATION

963766 — **616**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

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GENERAL	Item Number: 41 Quantity: 1 Valve Tag Number: UXV-23-1204 Data Page: 1 of 2																																																																																																																																																																								
	Service: Decomposer Feed Shutoff Process Material: Hydrocarbon with aromatics P&ID Drawing: 12																																																																																																																																																																								
SIZING CONDITIONS	<table border="1"> <thead> <tr> <th rowspan="2">Process Data Case</th> <th colspan="2">General</th> <th colspan="2">General</th> <th colspan="2"></th> <th colspan="2"></th> <th rowspan="2">Units</th> </tr> <tr> <th colspan="2">Open</th> <th colspan="2">Closed</th> <th colspan="2"></th> <th colspan="2"></th> </tr> </thead> <tbody> <tr> <td>Sizing Case</td> <td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td> </tr> <tr> <td>Calculated C_v</td> <td>Inlet</td><td>Outlet</td><td>Inlet</td><td>Outlet</td><td>Inlet</td><td>Outlet</td><td>Inlet</td><td>Outlet</td><td></td> </tr> <tr> <td>Liquid Flow at Flowing Cond</td> <td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>m3/h</td> </tr> <tr> <td>Vapor Flow at Base Cond</td> <td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>normal m3/h</td> </tr> <tr> <td>Steam Flow</td> <td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>kg/h</td> </tr> <tr> <td>Temperature</td> <td>60</td><td></td><td>60</td><td></td><td></td><td></td><td></td><td></td><td>°C</td> </tr> <tr> <td>Liquid Specific Gravity</td> <td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>----</td> </tr> <tr> <td>Vapor Spec. Gravity (Air=1)</td> <td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>----</td> </tr> <tr> <td>Pressure</td> <td>3.47</td><td>3.47</td><td>3.47</td><td>3.40</td><td></td><td></td><td></td><td></td><td>kg/cm2(g)</td> </tr> <tr> <td>Liq Critical Press (Pseudo)</td> <td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>kg/cm2(a)</td> </tr> <tr> <td>Liquid Vapor Pressure</td> <td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>kg/cm2(a)</td> </tr> <tr> <td>Liquid Viscosity</td> <td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>cP</td> </tr> <tr> <td>Compressibility Z</td> <td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td> </tr> <tr> <td>Specific Heats Cp/Cv (Real)</td> <td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td> </tr> <tr> <td>Estimate Sound Press Level</td> <td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>dBA</td> </tr> </tbody> </table>	Process Data Case	General		General						Units	Open		Closed						Sizing Case										Calculated C _v	Inlet	Outlet	Inlet	Outlet	Inlet	Outlet	Inlet	Outlet		Liquid Flow at Flowing Cond									m3/h	Vapor Flow at Base Cond									normal m3/h	Steam Flow									kg/h	Temperature	60		60						°C	Liquid Specific Gravity									----	Vapor Spec. Gravity (Air=1)									----	Pressure	3.47	3.47	3.47	3.40					kg/cm2(g)	Liq Critical Press (Pseudo)									kg/cm2(a)	Liquid Vapor Pressure									kg/cm2(a)	Liquid Viscosity									cP	Compressibility Z										Specific Heats Cp/Cv (Real)										Estimate Sound Press Level									dBA
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Estimate Sound Press Level									dBA																																																																																																																																																																
VALVE BODY & BONNET	Body Type: Eccentric Disc § Body Size: 6" #1 Port Type: Single Inlet Conn Size: 6" #1 Type: Flanged Class: 300 Outlet Conn Size: 6" #1 Type: Flanged Class: 300 Flange Facing: Flange Face Finish: Body & Bonnet Material: 316 Stainless Steel ‡ Bonnet Style: Packing Style: Manufacturer's Recommendation Packing Material: Manufacturer's Recommendation Body/Bonnet Gasket Material: Lub/Iso Valve Material: Lub Material: Manufacturer: Metso Automation (or equal) Model Number:	TRIM	Approximate Valve C _v : 510 F _L : x _T : Trim Style: Man. Standard Trim Size: Inherent Characteristic: Equal Percentage Balance Style: Full Open Position: 90° Guiding Style: Shaft Closure Member Material: 316 Stainless Steel Seat Ring Material: Incoloy 825 with Hard Chrome Seat Ring Retainer Material: 316 Stainless Steel Bearing Material: Stellite Bushing/Cage Material: Stem/Shaft Material: 17-4PH Flow Direction: Seat Leakage Class: ANSI Class V																																																																																																																																																																						
	ACTUATOR		INSTALLATION	Type: Spring Return Piston, Pneumatic Action Style: Fail Close Service Type: On-Off Max Shutoff Press: In: 8.86 Out: 0 kg/cm2(g) Maximum Stroke Time to Open: 5 Seconds Maximum Stroke Time to Close: 5 Seconds Stroke Cycle Service:	Inlet Pipe Nominal Size: Est Wall Thick: Outlet Pipe Nominal Size: Est Wall Thick: Est Insulation Type: Est Insulation Thickness: Installation: Design Pressure : 10 kg/cm2g (F.V) Design Temperature : 250 C																																																																																																																																																																				
				§ The information is based on Neldisc L6 series. Valve selected shall be of double flanged construction with minimal flange to flange dimension and no pockets within the valve body.																																																																																																																																																																					
				‡ Preferred material is L grade stainless steel. Standard grade Stainless steel that is not welded may be used for process-wetted parts including the body. Stainless steel with weld repair is acceptable if low carbon (<0.03 %) filler material is used.																																																																																																																																																																					
				#1 Line size valve.																																																																																																																																																																					
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CONTROL VALVES

PROJECT SPECIFICATION

963766 — **616**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Item Number: 41 Quantity: 1		Valve Tag Number: UXV-23-1204		Data Page: 2 of 2	
TRANSDUCER	Transducer Tag Number:	HANDWHEEL	Handwheel Position:		
	Transducer Input Signal Range:		Usage:		
	Transducer Output Signal Range:				
	Manufacturer:				
SOLENOID VALVE	Solenoid Valve Tag Number: UXY-23-1204	POSITION SWITCH	Open Position Switch Tag Number:		
	Body Type: 3-Way Body Material: Brass		Close Position Switch Tag Number:		
	Seat Material: Stainless Steel Disc Material: Stainless Steel		Sensor Type:		
	Connection Size/Type: †		Contact Form:		
	Packing Style: Packless		Open Position Switch Contact Action:		
	Fluid: Instrument Air		Close Position Switch Contact Action:		
	Pressure: Contractor to determine		Contact Load:	Contact Rating:	
	Minimum Pressure Port Orifice C _v : MAX Size: MAX		Mounting Location:		
	Minimum Exhaust Port Orifice C _v : MAX Size: MAX		Manufacturer:		
	Operation Style: †	POSITION TRANSMITTER	Position Transmitter Tag Number:		
	Coil Configuration: Single		Sensor Type:		
	Operating Voltage: \$\$ Frequency:		Output Signal Range:		
	Action Coil 1 Deenergized: Vent air from valve		Travel Measurement:		
	Action Coil 1 Energized: Supply air to valve		Mounting Location:		
	Action Coil 2 Deenergized:		Manufacturer:		
	Action Coil 2 Energized:	LOCK-UP VALVE	Action Style:		
	Manual Reset:		Mounting Location:		
	Manual Operator:		Model Number:		
Mounting Location: Nearest item to actuator in air line. †		Manufacturer:			
Manufacturer: ASCO (or equal)	QUICK EXHAUST	Connection Size/Type:			
Positioner Tag Number:		Body Material:			
Positioner Type:		Diaphragm Material:			
Positioner Input Signal Range:		Exhaust Action:			
Positioner Output Signal Range:		Mounting Location:			
Manufacturer:		Manufacturer:			
AIR FILTER	Quantity: one	VOLUME BOOSTER	Connection Size/Type:		
	Type: Filter/Regulator		Body Material:		
	Manufacturer:		Seat Ring Material:		
	Diaphragm Material:				
TUBING	Tubing Size:		Mounting Location:		
	Tubing Material:		Manufacturer:		
DATA PAGE 2 REMARKS	† The solenoid valve shall be mounted and tubed by the control valve vendor. The solenoid valve size and type, the tubing size, the actuator air connection size, and the air/filter regulator capacity shall be determined by the control valve vendor to meet the maximum stroking time requirements specified for the control valve.				
	\$\$ The Contractor shall coordinate the solenoid valve operating voltage and coil watt rating with the supplier of the Decomposition Section Shutdown System (UC-2302). See UOP Project Specification 660, this project.				

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CONTROL VALVES

PROJECT SPECIFICATION

963766 — **616**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

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GENERAL	Item Number: 42 Quantity: 1 Valve Tag Number: UXV-23-1201A Data Page: 1 of 2																																																																																																																																																									
	Service: Cumene Flush from Decomposer Feed Line Process Material: Hydrocarbon with aromatics P&ID Drawing: 12																																																																																																																																																									
SIZING CONDITIONS	<table border="1"> <thead> <tr> <th rowspan="2">Process Data Case</th> <th colspan="2">General</th> <th colspan="2">General</th> <th colspan="2"></th> <th colspan="2"></th> <th rowspan="2">Units</th> </tr> <tr> <th colspan="2">Open</th> <th colspan="2">Closed</th> <th colspan="2"></th> <th colspan="2"></th> </tr> </thead> <tbody> <tr> <td>Sizing Case</td> <td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr> <tr> <td>Calculated C_v</td> <td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr> <tr> <td>Liquid Flow at Flowing Cond</td> <td>Inlet</td><td>Outlet</td><td>Inlet</td><td>Outlet</td><td>Inlet</td><td>Outlet</td><td>Inlet</td><td>Outlet</td></tr> <tr> <td>Vapor Flow at Base Cond</td> <td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>m3/h</td></tr> <tr> <td>Steam Flow</td> <td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>normal m3/h</td></tr> <tr> <td>Temperature</td> <td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>kg/h</td></tr> <tr> <td>Liquid Specific Gravity</td> <td>40</td><td></td><td>40</td><td></td><td></td><td></td><td></td><td>°C</td></tr> <tr> <td>Vapor Spec. Gravity (Air=1)</td> <td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>----</td></tr> <tr> <td>Pressure</td> <td>3.40</td><td>3.40</td><td>3.47</td><td>2.96</td><td></td><td></td><td></td><td>kg/cm2(g)</td></tr> <tr> <td>Liq Critical Press (Pseudo)</td> <td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>kg/cm2(a)</td></tr> <tr> <td>Liquid Vapor Pressure</td> <td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>kg/cm2(a)</td></tr> <tr> <td>Liquid Viscosity</td> <td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>cP</td></tr> <tr> <td>Compressibility Z</td> <td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr> <tr> <td>Specific Heats Cp/Cv (Real)</td> <td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr> <tr> <td>Estimate Sound Press Level</td> <td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>dBA</td></tr> </tbody> </table>	Process Data Case	General		General						Units	Open		Closed						Sizing Case									Calculated C _v									Liquid Flow at Flowing Cond	Inlet	Outlet	Inlet	Outlet	Inlet	Outlet	Inlet	Outlet	Vapor Flow at Base Cond								m3/h	Steam Flow								normal m3/h	Temperature								kg/h	Liquid Specific Gravity	40		40					°C	Vapor Spec. Gravity (Air=1)								----	Pressure	3.40	3.40	3.47	2.96				kg/cm2(g)	Liq Critical Press (Pseudo)								kg/cm2(a)	Liquid Vapor Pressure								kg/cm2(a)	Liquid Viscosity								cP	Compressibility Z									Specific Heats Cp/Cv (Real)									Estimate Sound Press Level								dBA
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VALVE BODY & BONNET	Body Type: Globe § Body Size: 1-1/2" ¶¶ Port Type: Single Inlet Conn Size: 1-1/2" ¶¶ Type: Flanged Class: 300 Outlet Conn Size: 1-1/2" ¶¶ Type: Flanged Class: 300 Flange Facing: Raised Face Flange Face Finish: Body & Bonnet Material: 316 Stainless Steel ‡ Bonnet Style: Integral or Bolted Packing Style: Manufacturer's Recommendation Packing Material: Manufacturer's Recommendation Body/Bonnet Gasket Material: Manufacturer's Recommend Lub/Iso Valve Material: Lub Material: Manufacturer: Masoneilan (or equal) Model Number:																																																																																																																																																									
	TRIM	Approximate Valve C _v : 25 F _L : x _T : Trim Style: Man. Standard Trim Size: Inherent Characteristic: Equal Percentage Balance Style: Unbalanced Full Open Position: 100 % Guiding Style: Top Closure Member Material: 316 Stainless Steel Seat Ring Material: 316 Stainless Steel Seat Ring Retainer Material: 304 Stainless Steel Bearing Material: Bushing/Cage Material: Stellite Stem/Shaft Material: 316 Stainless Steel Flow Direction: Flow to close Seat Leakage Class: ANSI Class V																																																																																																																																																								
		ACTUATOR	Type: Spring Diaphragm, Pneumatic Action Style: Fail Open Service Type: On-Off Max Shutoff Press: In: 8.86 Out: 0 kg/cm2(g) Maximum Stroke Time to Open: 5 Seconds Maximum Stroke Time to Close: 5 Seconds Stroke Cycle Service:																																																																																																																																																							
			INSTALLATION	Inlet Pipe Nominal Size: Est Wall Thick: Outlet Pipe Nominal Size: Est Wall Thick: Est Insulation Type: Est Insulation Thickness: Installation: Design Pressure : 13.55 kg/cm2g (F.V) Design Temperature : 250 C																																																																																																																																																						
				DATA PAGE 1 REMARKS	§ The information is based on Masoneilan 21000 series. ‡ Preferred material is L grade stainless steel. Standard grade stainless steel that is not welded may be used for process wetted parts including the body. Stainless steel with weld repair is acceptable if low carbon (<= 0.03%) filler material is used. ¶¶ Line size valve. Valve will normally be in the closed position. Valve opening will be intermittent and generally short-term.																																																																																																																																																					
					ALL ITEMS SHALL COMPLY WITH THE GENERAL REQUIREMENTS SHOWN ON THE GENERAL REQUIREMENTS SECTION OF THIS SPECIFICATION.																																																																																																																																																					



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CONTROL VALVES

PROJECT SPECIFICATION

963766 — **616**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Item Number: 42 Quantity: 1		Valve Tag Number: UXV-23-1201A		Data Page: 2 of 2	
TRANSDUCER	Transducer Tag Number:		HANDWHEEL	Handwheel Position:	
	Transducer Input Signal Range:			Usage:	
	Transducer Output Signal Range:				
	Manufacturer:				
SOLENOID VALVE	Solenoid Valve Tag Number: UXY-23-1201 ††		POSITION SWITCH	Open Position Switch Tag Number:	
	Body Type: 3-Way Body Material: Brass			Close Position Switch Tag Number:	
	Seat Material: Stainless Steel Disc Material: Stainless Steel			Sensor Type:	
	Connection Size/Type: †			Contact Form:	
	Packing Style: Packless			Open Position Switch Contact Action:	
	Fluid: Instrument Air			Close Position Switch Contact Action:	
	Pressure: Contractor to determine			Contact Load: Contact Rating:	
	Minimum Pressure Port Orifice C_v : MAX Size: MAX			Mounting Location:	
	Minimum Exhaust Port Orifice C_v : MAX Size: MAX			Manufacturer:	
	Operation Style: †		POSITION TRANSMITTER	Position Transmitter Tag Number:	
	Coil Configuration: Single			Sensor Type:	
	Operating Voltage: \$\$ Frequency:			Output Signal Range:	
	Action Coil 1 Deenergized: Vent air from valve			Travel Measurement:	
	Action Coil 1 Energized: Supply air to valve			Mounting Location:	
	Action Coil 2 Deenergized:			Manufacturer:	
	Action Coil 2 Energized:		LOCK-UP VALVE	Action Style:	
	Manual Reset:			Mounting Location:	
	Manual Operator:			Model Number:	
Mounting Location: Nearest item to actuator in air line. †			Manufacturer:		
Manufacturer: ASCO (or equal)		QUICK EXHAUST	Connection Size/Type:		
Positioner Tag Number:			Body Material:		
Positioner Type:			Diaphragm Material:		
Positioner Input Signal Range:			Exhaust Action:		
Positioner Output Signal Range:			Mounting Location:		
Manufacturer:			Manufacturer:		
AIR FILTER	Quantity: one		VOLUME BOOSTER	Connection Size/Type:	
	Type: Filter/Regulator			Body Material:	
	Manufacturer:			Seat Ring Material:	
Tubing Size:		Diaphragm Material:			
Tubing Material:			Mounting Location:		
			Manufacturer:		
DATA PAGE 2 REMARKS	† The solenoid valve shall be mounted and tubed by the control valve vendor. The solenoid valve size and type, the tubing size, the actuator air connection size, and the air/filter regulator capacity shall be determined by the control valve vendor to meet the maximum stroking time requirements specified for the control valve.				
	\$\$ The Contractor shall coordinate the solenoid valve operating voltage and coil watt rating with the supplier of the Decomposition Section Shutdown System (UC-2302). See UOP Project Specification 660, this project.				
	†† The solenoid valve is connected to both UV-23-1201A and UV-23-1201B shutoff valves.				

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CONTROL VALVES

PROJECT SPECIFICATION

963766 — **616**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Note: The information in this document is confidential and the property of UOP LLC and must not be disclosed to others or reproduced in any manner or used for any purpose whatsoever without its written permission.

GENERAL	Item Number: 43 Quantity: 1		Valve Tag Number: FV-23-1205		Data Page: 1 of 2		
	Service: Decomposer Feed		P&ID Drawing: 12				
Process Material: Hydrocarbon with aromatics							
SIZING CONDITIONS	Process Data Case	Single Case		Single Case		Single Case	
	Sizing Case	Normal		Maximum		Minimum	
	Calculated C _v	54		64		23.5	
		Inlet	Outlet	Inlet	Outlet	Inlet	Outlet
	Liquid Flow at Flowing Cond	62.7		69.0		31.4	
	Vapor Flow at Base Cond						
	Steam Flow						
	Temperature	60		60		60	
	Liquid Specific Gravity	0.972		0.972		0.972	
	Vapor Spec. Gravity (Air=1)						
	Pressure	3.30	1.55	3.11	1.56	3.85	1.49
	Liq Critical Press (Pseudo)	33.89		33.89		33.89	
	Liquid Vapor Pressure	0.01		0.01		0.01	
	Liquid Viscosity	3.432		3.432		3.432	
	Compressibility Z						
	Specific Heats Cp/Cv (Real)						
	Estimate Sound Press Level						dBA
VALVE BODY & BONNET	Body Type: Globe §	Approximate Valve C _v : 75 F _L : 0.9 x _T : 0.72					
	Body Size: 3" Port Type: Single	Trim Style: Man. Standard Trim Size:					
	Inlet Conn Size: 3" Type: Flanged Class: 300	Inherent Characteristic: Linear					
	Outlet Conn Size: 3" Type: Flanged Class: 300	Balance Style: Balanced Full Open Position: 100 %					
	Flange Facing: Raised Face Flange Face Finish:	Guiding Style: Cage					
	Body & Bonnet Material: 316 Stainless Steel ‡	Closure Member Material: 316 Stainless Steel					
	Bonnet Style: Integral or Bolted	Seat Ring Material: 316 SS with Stellite sheath					
	Packing Style: Manufacturer's Recommendation	Seat Ring Retainer Material:					
	Packing Material: Manufacturer's Recommendation	Bearing Material:					
	Body/Bonnet Gasket Material: Manufacturer's Recommend	Bushing/Cage Material: 316 SS hard chrome plated					
	Lub/Iso Valve Material: Lub Material:	Stem/Shaft Material: 316 Stainless Steel					
	Manufacturer: Masoneilan (or equal)	Flow Direction:					
	Model Number:	Seat Leakage Class: ANSI Class IV					
	ACTUATOR	Type: Spring Diaphragm, Pneumatic †	Inlet Pipe Nominal Size: Est Wall Thick:				
Action Style: Fail Open		Outlet Pipe Nominal Size: Est Wall Thick:					
Service Type: Modulating		Est Insulation Type:					
Max Shutoff Press: In: 8.86 Out: 0 kg/cm2(g)		Est Insulation Thickness:					
Maximum Stroke Time to Open:		Installation:					
Maximum Stroke Time to Close:		Design Pressure : 8.66 kg/cm2g (F.V)					
Stroke Cycle Service:		Design Temperature : 250 C					
DATA PAGE 1 REMARKS		§ The information is based on Masoneilan 41000 series.					
	‡ Preferred material is L grade stainless steel. Standard grade stainless steel that is not welded may be used for process wetted parts including the body. Stainless steel with weld repair is acceptable if low carbon (<= 0.03 %) filler material is used.						
	† Provide an adjustable limit stop with locking feature, per vendor design practice, to limit travel in the closed position. Field set limit stop to maintain the minimum flow requirement for associated equipment or service.						

ALL ITEMS SHALL COMPLY WITH THE GENERAL REQUIREMENTS SHOWN ON THE GENERAL REQUIREMENTS SECTION OF THIS SPECIFICATION.



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CONTROL VALVES

PROJECT SPECIFICATION

963766 — **616**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Item Number: 43 Quantity: 1		Valve Tag Number: FV-23-1205		Data Page: 2 of 2	
TRANS-DUCER	Transducer Tag Number:		HANDWHEEL	Handwheel Position: Manufacturer's Standard	
	Transducer Input Signal Range:			Usage: Manual Operation	
TRANS-DUCER	Transducer Output Signal Range:		HANDWHEEL		
	Manufacturer:				
SOLENOID VALVE	Solenoid Valve Tag Number:		POSITION SWITCH	Open Position Switch Tag Number:	
	Body Type: Body Material:			Close Position Switch Tag Number:	
SOLENOID VALVE	Seat Material: Disc Material:		POSITION SWITCH	Sensor Type:	
	Connection Size/Type:			Contact Form:	
SOLENOID VALVE	Packing Style:		POSITION SWITCH	Open Position Switch Contact Action:	
	Fluid:			Close Position Switch Contact Action:	
SOLENOID VALVE	Pressure:		POSITION SWITCH	Contact Load: Contact Rating:	
	Minimum Pressure Port Orifice C _v : Size:			Mounting Location:	
SOLENOID VALVE	Minimum Exhaust Port Orifice C _v : Size:		POSITION SWITCH	Manufacturer:	
	Operation Style:				
SOLENOID VALVE	Coil Configuration:		POSITION TRANSMITTER	Position Transmitter Tag Number:	
	Operating Voltage: Frequency:			Sensor Type:	
SOLENOID VALVE	Action Coil 1 Deenergized:		POSITION TRANSMITTER	Output Signal Range:	
	Action Coil 1 Energized:			Travel Measurement:	
SOLENOID VALVE	Action Coil 2 Deenergized:		POSITION TRANSMITTER	Mounting Location:	
	Action Coil 2 Energized:			Manufacturer:	
SOLENOID VALVE	Manual Reset:		LOCK-UP VALVE	Action Style:	
	Manual Operator:			Mounting Location:	
SOLENOID VALVE	Mounting Location:		LOCK-UP VALVE	Model Number:	
	Manufacturer:			Manufacturer:	
POSITIONER	Positioner Tag Number:		QUICK EXHAUST	Connection Size/Type:	
	Positioner Type: Digital			Body Material:	
POSITIONER	Positioner Input Signal Range: 4 - 20 mA		QUICK EXHAUST	Diaphragm Material:	
	Positioner Output Signal Range: 0 - 100 %			Exhaust Action:	
AIR FILTER	Quantity: one		QUICK EXHAUST	Mounting Location:	
	Type: Filter/Regulator			Manufacturer:	
TUBING	Manufacturer:		VOLUME BOOSTER	Connection Size/Type:	
	Tubing Size:			Body Material:	
TUBING	Tubing Material:		VOLUME BOOSTER	Seat Ring Material:	
				Diaphragm Material:	
DATA PAGE 2 REMARKS			VOLUME BOOSTER	Mounting Location:	
				Manufacturer:	

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CONTROL VALVES

PROJECT SPECIFICATION

963766 — **616**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				
1	28AUG12	MVD	MW				

GENERAL	Item Number: 44 Quantity: 1 Valve Tag Number: UXV-23-1201B Data Page: 1 of 2																																																																																																																																																																								
	Service: Cumene Flush to Decomposer Feed Line Process Material: Hydrocarbon with aromatics P&ID Drawing: 12																																																																																																																																																																								
SIZING CONDITIONS	<table border="1"> <thead> <tr> <th rowspan="2">Process Data Case</th> <th colspan="2">General</th> <th colspan="2">General</th> <th colspan="2"></th> <th colspan="2"></th> <th rowspan="2">Units</th> </tr> <tr> <th colspan="2">Open</th> <th colspan="2">Closed</th> <th colspan="2"></th> <th colspan="2"></th> </tr> </thead> <tbody> <tr> <td>Sizing Case</td> <td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td> </tr> <tr> <td>Calculated C_v</td> <td>Inlet</td><td>Outlet</td><td>Inlet</td><td>Outlet</td><td>Inlet</td><td>Outlet</td><td>Inlet</td><td>Outlet</td><td></td> </tr> <tr> <td>Liquid Flow at Flowing Cond</td> <td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>m³/h</td> </tr> <tr> <td>Vapor Flow at Base Cond</td> <td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>normal m³/h</td> </tr> <tr> <td>Steam Flow</td> <td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>kg/h</td> </tr> <tr> <td>Temperature</td> <td>46</td><td></td><td>46</td><td></td><td></td><td></td><td></td><td></td><td>°C</td> </tr> <tr> <td>Liquid Specific Gravity</td> <td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>----</td> </tr> <tr> <td>Vapor Spec. Gravity (Air=1)</td> <td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>----</td> </tr> <tr> <td>Pressure</td> <td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>kg/cm²(g)</td> </tr> <tr> <td>1 Liq Critical Press (Pseudo)</td> <td>1.55</td><td>1.55</td><td>1.60</td><td>1.55</td><td></td><td></td><td></td><td></td><td>kg/cm²(a)</td> </tr> <tr> <td>Liquid Vapor Pressure</td> <td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>kg/cm²(a)</td> </tr> <tr> <td>Liquid Viscosity</td> <td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>cP</td> </tr> <tr> <td>Compressibility Z</td> <td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td> </tr> <tr> <td>Specific Heats Cp/Cv (Real)</td> <td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td> </tr> <tr> <td>Estimate Sound Press Level</td> <td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>dBA</td> </tr> </tbody> </table>	Process Data Case	General		General						Units	Open		Closed						Sizing Case										Calculated C _v	Inlet	Outlet	Inlet	Outlet	Inlet	Outlet	Inlet	Outlet		Liquid Flow at Flowing Cond									m ³ /h	Vapor Flow at Base Cond									normal m ³ /h	Steam Flow									kg/h	Temperature	46		46						°C	Liquid Specific Gravity									----	Vapor Spec. Gravity (Air=1)									----	Pressure									kg/cm ² (g)	1 Liq Critical Press (Pseudo)	1.55	1.55	1.60	1.55					kg/cm ² (a)	Liquid Vapor Pressure									kg/cm ² (a)	Liquid Viscosity									cP	Compressibility Z										Specific Heats Cp/Cv (Real)										Estimate Sound Press Level									dBA
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VALVE BODY & BONNET	Body Type: Globe § Body Size: 1-1/2" ¶¶ Port Type: Single Inlet Conn Size: 1-1/2" ¶¶ Type: Flanged Class: 300 Outlet Conn Size: 1-1/2" ¶¶ Type: Flanged Class: 300 Flange Facing: Raised Face Flange Face Finish: Body & Bonnet Material: 316 Stainless Steel ‡ Bonnet Style: Integral or Bolted Packing Style: Manufacturer's Recommendation Packing Material: Manufacturer's Recommendation Body/Bonnet Gasket Material: Manufacturer's Recommend Lub/Iso Valve Material: Lub Material: Manufacturer: Masoneilan (or equal) Model Number:	TRIM	Approximate Valve C _v : 25 F _L : x _T : Trim Style: Man. Standard Trim Size: Inherent Characteristic: Equal Percentage Balance Style: Unbalanced Full Open Position: 100 % Guiding Style: Top Closure Member Material: 316 Stainless Steel Seat Ring Material: 316 Stainless Steel Seat Ring Retainer Material: 304 Stainless Steel Bearing Material: Bushing/Cage Material: Stellite Stem/Shaft Material: 316 Stainless Steel Flow Direction: Flow to open Seat Leakage Class: ANSI Class V																																																																																																																																																																						
	ACTUATOR		INSTALLATION	Type: Spring Diaphragm, Pneumatic Action Style: Fail Open Service Type: On-Off Max Shutoff Press: In: 0.70 Out: 8.86 kg/cm ² (g) Maximum Stroke Time to Open: 5 Seconds Maximum Stroke Time to Close: 5 Seconds Stroke Cycle Service:	Inlet Pipe Nominal Size: Est Wall Thick: Outlet Pipe Nominal Size: Est Wall Thick: Est Insulation Type: Est Insulation Thickness: Installation:																																																																																																																																																																				
				Design Pressure : 10 kg/cm ² g (F.V) Design Temperature : 250 C																																																																																																																																																																					
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				¶¶ Line size valve.																																																																																																																																																																					
				Valve will normally be in the closed position. Valve opening will be intermittent and generally short-term.																																																																																																																																																																					
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ALL ITEMS SHALL COMPLY WITH THE GENERAL REQUIREMENTS SHOWN ON THE GENERAL REQUIREMENTS SECTION OF THIS SPECIFICATION.

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**CONTROL VALVES****PROJECT SPECIFICATION**963766 — **616**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Item Number: 44 Quantity: 1		Valve Tag Number: UXV-23-1201B		Data Page: 2 of 2	
TRANS-DUCER	Transducer Tag Number:		HANDWHEEL	Handwheel Position:	
	Transducer Input Signal Range:			Usage:	
	Transducer Output Signal Range:				
	Manufacturer:				
SOLENOID VALVE	Solenoid Valve Tag Number:		POSITION SWITCH	Open Position Switch Tag Number:	
	Body Type: Body Material:			Close Position Switch Tag Number:	
	Seat Material: Disc Material:			Sensor Type:	
	Connection Size/Type:			Contact Form:	
	Packing Style:			Open Position Switch Contact Action:	
	Fluid:			Close Position Switch Contact Action:	
	Pressure:			Contact Load: Contact Rating:	
	Minimum Pressure Port Orifice C_v : Size:			Mounting Location:	
	Minimum Exhaust Port Orifice C_v : Size:			Manufacturer:	
	Operation Style:			POSITION TRANSMITTER	Position Transmitter Tag Number:
	Coil Configuration:		Sensor Type:		
	Operating Voltage: Frequency:		Output Signal Range:		
	Action Coil 1 Deenergized:		Travel Measurement:		
	Action Coil 1 Energized:		Mounting Location:		
	Action Coil 2 Deenergized:		LOCK-UP VALVE	Manufacturer:	
	Action Coil 2 Energized:			Action Style:	
	Manual Reset:			Mounting Location:	
	Manual Operator:		QUICK EXHAUST	Model Number:	
	Mounting Location:			Manufacturer:	
	Manufacturer:			Connection Size/Type:	
POSITIONER	Positioner Tag Number:		VOLUME BOOSTER	Body Material:	
	Positioner Type:			Diaphragm Material:	
	Positioner Input Signal Range:			Exhaust Action:	
AIR FILTER	Positioner Output Signal Range:		TUBING	Mounting Location:	
	Manufacturer:			Manufacturer:	
	Quantity:			Connection Size/Type:	
TUBING	Type:		TUBING	Body Material:	
	Manufacturer:			Seat Ring Material:	
	Tubing Size:			Diaphragm Material:	
DATA PAGE 2 REMARKS	Tubing Material:		TUBING	Mounting Location:	
				Manufacturer:	

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CONTROL VALVES

PROJECT SPECIFICATION

963766 — **616**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

GENERAL	Item Number: 45 Quantity: 1 Valve Tag Number: UXV-23-1205 Data Page: 1 of 2																																																																																																																																																																								
	Service: Decomposer Feed Shutoff Process Material: Hydrocarbon with aromatics P&ID Drawing: 12																																																																																																																																																																								
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Estimate Sound Press Level									dBA																																																																																																																																																																
VALVE BODY & BONNET	Body Type: Eccentric Disc § Body Size: 6" #1 Port Type: Single Inlet Conn Size: 6" #1 Type: Flanged Class: 300 Outlet Conn Size: 6" #1 Type: Flanged Class: 300 Flange Facing: Flange Face Finish: Body & Bonnet Material: 316 Stainless Steel ‡ Bonnet Style: Packing Style: Manufacturer's Recommendation Packing Material: Manufacturer's Recommendation Body/Bonnet Gasket Material: Lub/Iso Valve Material: Lub Material: Manufacturer: Metso Automation (or equal) Model Number:	TRIM	Approximate Valve C _v : 510 F _L : x _T : Trim Style: Man. Standard Trim Size: Inherent Characteristic: Equal Percentage Balance Style: Full Open Position: 90° Guiding Style: Shaft Closure Member Material: 316 Stainless Steel Seat Ring Material: Incoloy 825 with Hard Chrome Seat Ring Retainer Material: 316 Stainless Steel Bearing Material: Stellite Bushing/Cage Material: Stem/Shaft Material: 17-4PH Flow Direction: Seat Leakage Class: ANSI Class V																																																																																																																																																																						
	ACTUATOR		INSTALLATION	Type: Spring Return Piston, Pneumatic Action Style: Fail Close Service Type: On-Off Max Shutoff Press: In: 8.86 Out: 0 kg/cm2(g) Maximum Stroke Time to Open: 5 Seconds Maximum Stroke Time to Close: 5 Seconds Stroke Cycle Service:	Inlet Pipe Nominal Size: Est Wall Thick: Outlet Pipe Nominal Size: Est Wall Thick: Est Insulation Type: Est Insulation Thickness: Installation: Design Pressure : 10 kg/cm2g (F.V) Design Temperature : 250 C																																																																																																																																																																				
				§ The information is based on Neldisc L6 series. Valve selected shall be of double flanged construction with minimal flange to flange dimension and no pockets within the valve body.																																																																																																																																																																					
				‡ Preferred material is L grade stainless steel. Standard grade Stainless steel that is not welded may be used for process-wetted parts including the body. Stainless steel with weld repair is acceptable if low carbon (<0.03 %) filler material is used.																																																																																																																																																																					
				#1 Line size valve.																																																																																																																																																																					

ALL ITEMS SHALL COMPLY WITH THE GENERAL REQUIREMENTS SHOWN ON THE GENERAL REQUIREMENTS SECTION OF THIS SPECIFICATION.

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CONTROL VALVES

PROJECT SPECIFICATION

963766 — **616**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Item Number: 45 Quantity: 1		Valve Tag Number: UXV-23-1205		Data Page: 2 of 2		
TRANSDUCER	Transducer Tag Number:		HANDWHEEL	Handwheel Position:		
	Transducer Input Signal Range:			Usage:		
Transducer Output Signal Range:						
Manufacturer:						
SOLENOID VALVE	Solenoid Valve Tag Number: UXY-23-1205		POSITION SWITCH	Open Position Switch Tag Number:		
	Body Type: 3-Way Body Material: Brass			Close Position Switch Tag Number:		
	Seat Material: Stainless Steel Disc Material: Stainless Steel			Sensor Type:		
	Connection Size/Type: †			Contact Form:		
	Packing Style: Packless			Open Position Switch Contact Action:		
	Fluid: Instrument Air			Close Position Switch Contact Action:		
	Pressure: Contractor to determine			Contact Load: Contact Rating:		
	Minimum Pressure Port Orifice C _v : MAX Size: MAX			Mounting Location:		
	Minimum Exhaust Port Orifice C _v : MAX Size: MAX			Manufacturer:		
	Operation Style: †					
	Coil Configuration: Single			POSITION TRANSMITTER	Position Transmitter Tag Number:	
	Operating Voltage: \$\$ Frequency:				Sensor Type:	
	Action Coil 1 Deenergized: Vent air from valve				Output Signal Range:	
	Action Coil 1 Energized: Supply air to valve				Travel Measurement:	
	Action Coil 2 Deenergized:			LOCK-UP VALVE	Mounting Location:	
	Action Coil 2 Energized:				Manufacturer:	
	Manual Reset:				Action Style:	
	Manual Operator:			QUICK EXHAUST	Mounting Location:	
Mounting Location: Nearest item to actuator in air line. †		Model Number:				
Manufacturer: ASCO (or equal)		Manufacturer:				
POSITIONER	Positioner Tag Number:		VOLUME BOOSTER	Connection Size/Type:		
	Positioner Type:			Body Material:		
	Positioner Input Signal Range:			Seat Ring Material:		
	Positioner Output Signal Range:			Diaphragm Material:		
Manufacturer:		Mounting Location:				
AIR FILTER	Quantity: one					
	Type: Filter/Regulator					
Manufacturer:						
TUBING	Tubing Size:					
	Tubing Material:					
† The solenoid valve shall be mounted and tubed by the control valve vendor. The solenoid valve size and type, the tubing size, the actuator air connection size, and the air/filter regulator capacity shall be determined by the control valve vendor to meet the maximum stroking time requirements specified for the control valve. \$\$ The Contractor shall coordinate the solenoid valve operating voltage and coil watt rating with the supplier of the Decomposition Section Shutdown System (UC-2302). See UOP Project Specification 660, this project.						

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DATA PAGE 2 REMARKS



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CONTROL VALVES

PROJECT SPECIFICATION

963766 — **616**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

GENERAL	Item Number: 46 Quantity: 1 Valve Tag Number: UXV-23-1206 Data Page: 1 of 2																																																																																																																																																																
	Service: Decomposer Feed Flush Drum to Decomposer Feed Line Shuoff Process Material: Hydrocarbon with aromatics P&ID Drawing: 12A																																																																																																																																																																
SIZING CONDITIONS	<table border="1"> <thead> <tr> <th>Process Data Case</th> <th colspan="2">General</th> <th colspan="2">General</th> <th colspan="2"></th> <th colspan="2"></th> <th></th> </tr> <tr> <th>Sizing Case</th> <th colspan="2">Open</th> <th colspan="2">Closed</th> <th colspan="2"></th> <th colspan="2"></th> <th></th> </tr> <tr> <th>Calculated C_v</th> <th>Inlet</th> <th>Outlet</th> <th>Inlet</th> <th>Outlet</th> <th>Inlet</th> <th>Outlet</th> <th>Inlet</th> <th>Outlet</th> <th>Units</th> </tr> </thead> <tbody> <tr> <td>Liquid Flow at Flowing Cond</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>m3/h</td> </tr> <tr> <td>Vapor Flow at Base Cond</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>normal m3/h</td> </tr> <tr> <td>Steam Flow</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>kg/h</td> </tr> <tr> <td>Temperature</td> <td>46</td> <td></td> <td>46</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>°C</td> </tr> <tr> <td>Liquid Specific Gravity</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>----</td> </tr> <tr> <td>Vapor Spec. Gravity (Air=1)</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>----</td> </tr> <tr> <td>Pressure</td> <td>1.65</td> <td>1.65</td> <td>1.65</td> <td>1.51</td> <td></td> <td></td> <td></td> <td></td> <td>kg/cm2(g)</td> </tr> <tr> <td>Liq Critical Press (Pseudo)</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>kg/cm2(a)</td> </tr> <tr> <td>Liquid Vapor Pressure</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>kg/cm2(a)</td> </tr> <tr> <td>Liquid Viscosity</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>cP</td> </tr> <tr> <td>Compressibility Z</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> <tr> <td>Specific Heats Cp/Cv (Real)</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> <tr> <td>Estimate Sound Press Level</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>dBA</td> </tr> </tbody> </table>	Process Data Case	General		General							Sizing Case	Open		Closed							Calculated C _v	Inlet	Outlet	Inlet	Outlet	Inlet	Outlet	Inlet	Outlet	Units	Liquid Flow at Flowing Cond									m3/h	Vapor Flow at Base Cond									normal m3/h	Steam Flow									kg/h	Temperature	46		46						°C	Liquid Specific Gravity									----	Vapor Spec. Gravity (Air=1)									----	Pressure	1.65	1.65	1.65	1.51					kg/cm2(g)	Liq Critical Press (Pseudo)									kg/cm2(a)	Liquid Vapor Pressure									kg/cm2(a)	Liquid Viscosity									cP	Compressibility Z										Specific Heats Cp/Cv (Real)										Estimate Sound Press Level									dBA
	Process Data Case	General		General																																																																																																																																																													
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Estimate Sound Press Level									dBA																																																																																																																																																								
VALVE BODY & BONNET	Body Type: Eccentric Disc § Body Size: 4" #2 Port Type: Single Inlet Conn Size: 4" #2 Type: Flanged Class: 300 Outlet Conn Size: 4" #2 Type: Flanged Class: 300 Flange Facing: Flange Face Finish: Body & Bonnet Material: 316 Stainless Steel ‡ Bonnet Style: Packing Style: Manufacturer's Recommendation Packing Material: Manufacturer's Recommendation Body/Bonnet Gasket Material: Lub/Iso Valve Material: Lub Material: Manufacturer: Metso Automation (or equal) Model Number:																																																																																																																																																																
	TRIM	Approximate Valve C_v: 180 F_L: x_T: Trim Style: Man. Standard Trim Size: Inherent Characteristic: Equal Percentage Balance Style: Full Open Position: 90° Guiding Style: Shaft Closure Member Material: 316 Stainless Steel Seat Ring Material: Incoloy 825 with Hard Chrome Seat Ring Retainer Material: 316 Stainless Steel Bearing Material: Stellite Bushing/Cage Material: Stem/Shaft Material: 17-4PH Flow Direction: Seat Leakage Class: ANSI Class V																																																																																																																																																															
		INSTALLATION	Type: Spring Return Piston, Pneumatic Action Style: Fail Open Service Type: On-Off Max Shutoff Press: In: 0 Out: 8.86 kg/cm2(g) Maximum Stroke Time to Open: 5 Seconds Maximum Stroke Time to Close: Stroke Cycle Service:																																																																																																																																																														
			ACTUATOR	Inlet Pipe Nominal Size: Est Wall Thick: Outlet Pipe Nominal Size: Est Wall Thick: Est Insulation Type: Est Insulation Thickness: Installation: Design Pressure : 10 kg/cm2g (F.V) Design Temperature : 250 C																																																																																																																																																													
				DATA PAGE 1 REMARKS	§ The information is based on Neldisc L6 series. Valve selected shall be of double flanged construction with minimal flange-to-flange dimension and no pockets with the valve body. ‡ Preferred material is L grade stainless steel. Standard grade stainless steel that is not welded may be used for process wetted parts including the body. Stainless steel with weld repair is acceptable if low carbon (<= 0.03 %) filler material is used. #2 Line size valve. Valve will normally be in the closed position. Valve opening will be intermittent and generally short-term.																																																																																																																																																												

ALL ITEMS SHALL COMPLY WITH THE GENERAL REQUIREMENTS SHOWN ON THE GENERAL REQUIREMENTS SECTION OF THIS SPECIFICATION.

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CONTROL VALVES

PROJECT SPECIFICATION

963766 — **616**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Item Number: 46 Quantity: 1		Valve Tag Number: UXV-23-1206		Data Page: 2 of 2		
TRANS-DUCER	Transducer Tag Number:		HANDWHEEL	Handwheel Position:		
	Transducer Input Signal Range:			Usage:		
Transducer Output Signal Range:						
Manufacturer:						
SOLENOID VALVE	Solenoid Valve Tag Number: UXY-23-1206		POSITION SWITCH	Open Position Switch Tag Number:		
	Body Type: 3-Way Body Material: Brass			Close Position Switch Tag Number:		
	Seat Material: Stainless Steel Disc Material: Stainless Steel			Sensor Type:		
	Connection Size/Type: †			Contact Form:		
	Packing Style: Packless			Open Position Switch Contact Action:		
	Fluid: Instrument Air			Close Position Switch Contact Action:		
	Pressure: Contractor to determine			Contact Load: Contact Rating:		
	Minimum Pressure Port Orifice C _v : MAX Size: MAX			Mounting Location:		
	Minimum Exhaust Port Orifice C _v : MAX Size: MAX			Manufacturer:		
	Operation Style: †					
	Coil Configuration: Single			POSITION TRANSMITTER	Position Transmitter Tag Number:	
	Operating Voltage: \$\$ Frequency:				Sensor Type:	
	Action Coil 1 Deenergized: Vent air from valve				Output Signal Range:	
	Action Coil 1 Energized: Supply air to valve				Travel Measurement:	
	Action Coil 2 Deenergized:			LOCK-UP VALVE	Mounting Location:	
	Action Coil 2 Energized:				Manufacturer:	
	Manual Reset:				Action Style:	
	Manual Operator:			QUICK EXHAUST	Mounting Location:	
Mounting Location: Nearest item to actuator in air line. †		Model Number:				
Manufacturer: ASCO (or equal)		Manufacturer:				
POSITIONER	Positioner Tag Number:		VOLUME BOOSTER	Connection Size/Type:		
	Positioner Type:			Body Material:		
	Positioner Input Signal Range:			Seat Ring Material:		
	Positioner Output Signal Range:			Diaphragm Material:		
Manufacturer:		Mounting Location:				
AIR FILTER	Quantity: one					
	Type: Filter/Regulator					
Manufacturer:						
TUBING	Tubing Size:					
	Tubing Material:					
† The solenoid valve shall be mounted and tubed by the control valve vendor. The solenoid valve size and type, the tubing size, the actuator air connection size, and the air/filter regulator capacity shall be determined by the control valve vendor to meet the maximum stroking time requirements specified for the control valve. \$\$ The Contractor shall coordinate the solenoid valve operating voltage and coil watt rating with the supplier of the Decomposition Section Shutdown System (UC-2302). See UOP Project Specification 660, this project.						

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DATA PAGE 2 REMARKS



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CONTROL VALVES

PROJECT SPECIFICATION

963766 — **616**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

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GENERAL	Item Number: 47 Quantity: 1		Valve Tag Number: PV-23-1202A				Data Page: 1 of 2			
	Service: Nitrogen to Decomposer Feed Flush Drum		P&ID Drawing: 12A							
SIZING CONDITIONS	Process Data Case		General							
	Sizing Case		Maximum							
	Calculated C _v		2.09							
		Inlet	Outlet	Inlet	Outlet	Inlet	Outlet	Inlet	Outlet	Units
	Liquid Flow at Flowing Cond									m3/h
	Vapor Flow at Base Cond	212								normal m3/h
	Steam Flow									kg/h
	Temperature	38								°C
	Liquid Specific Gravity									----
	Vapor Spec. Gravity (Air=1)	0.967								----
	Pressure	7.00	0.70							kg/cm2(g)
	Liq Critical Press (Pseudo)									kg/cm2(a)
	Liquid Vapor Pressure									kg/cm2(a)
	Liquid Viscosity									cP
	Compressibility Z	1.000								
	Specific Heats Cp/Cv (Real)	1.400								
	Estimate Sound Press Level									dBA
	VALVE BODY & BONNET	Body Type: Globe §				Approximate Valve C _v : 3.8		F _L : 0.9		x _T : 0.72
Body Size: 1" Port Type: Single				Trim Style: Man. Standard		Trim Size:				
Inlet Conn Size: 1" Type: Flanged Class: 300				Inherent Characteristic: Equal Percentage						
Outlet Conn Size: 1" Type: Flanged Class: 300				Balance Style: Unbalanced Full Open Position: 100 %						
Flange Facing: Raised Face Flange Face Finish:				Guiding Style: Top						
Body & Bonnet Material: Carbon Steel				Closure Member Material: 416 Stainless Steel						
Bonnet Style: Integral or Bolted				Seat Ring Material: 416 Stainless Steel						
Packing Style: Manufacturer's Recommendation				Seat Ring Retainer Material: 304 Stainless Steel						
Packing Material: Manufacturer's Recommendation				Bearing Material:						
Body/Bonnet Gasket Material: Manufacturer's Recommend				Bushing/Cage Material: 440C Stainless Steel						
Lub/Iso Valve Material: Lub Material:				Stem/Shaft Material: 316 Stainless Steel						
Manufacturer: Masoneilan (or equal)				Flow Direction:						
Model Number:				Seat Leakage Class: ANSI Class IV						
ACTUATOR		Type: Spring Diaphragm, Pneumatic				Inlet Pipe Nominal Size:		Est Wall Thick:		
	Action Style: Fail Close				Outlet Pipe Nominal Size:		Est Wall Thick:			
	Service Type: Modulating				Est Insulation Type:					
	Max Shutoff Press: In: 9.00 Out: 0.70 kg/cm2(g)				Est Insulation Thickness:					
	Maximum Stroke Time to Open:				Installation:					
	Maximum Stroke Time to Close:				Design Pressure : 10 kg/cm2g					
	Stroke Cycle Service:				Design Temperature : 120 C					
DATA PAGE 1 REMARKS	§ The information is based on Masoneilan 21000 series.									
	UOP calculations indicate that the noise level of this valve may exceed 85 dBA. The estimated noise level shall be verified based on final design criteria and Vendor/Contractor's detailed calculations. It is the responsibility of the detailed design Contractor to select a valve size, style and/or trim style that complies with all the specified process requirements, including noise limitation.									

ALL ITEMS SHALL COMPLY WITH THE GENERAL REQUIREMENTS SHOWN ON THE GENERAL REQUIREMENTS SECTION OF THIS SPECIFICATION.



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CONTROL VALVES

PROJECT SPECIFICATION

963766 — **616**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Item Number: 47 Quantity: 1		Valve Tag Number: PV-23-1202A		Data Page: 2 of 2	
TRANSducer	Transducer Tag Number:		HANDWHEEL	Handwheel Position:	
	Transducer Input Signal Range:			Usage:	
	Transducer Output Signal Range:				
	Manufacturer:				
SOLENOID VALVE	Solenoid Valve Tag Number:		POSITION SWITCH	Open Position Switch Tag Number:	
	Body Type: Body Material:			Close Position Switch Tag Number:	
	Seat Material: Disc Material:			Sensor Type:	
	Connection Size/Type:			Contact Form:	
	Packing Style:			Open Position Switch Contact Action:	
	Fluid:			Close Position Switch Contact Action:	
	Pressure:			Contact Load: Contact Rating:	
	Minimum Pressure Port Orifice C _v : Size:			Mounting Location:	
	Minimum Exhaust Port Orifice C _v : Size:			Manufacturer:	
	Operation Style:			POSITION TRANSMITTER	Position Transmitter Tag Number:
	Coil Configuration:		Sensor Type:		
	Operating Voltage: Frequency:		Output Signal Range:		
	Action Coil 1 Deenergized:		Travel Measurement:		
	Action Coil 1 Energized:		Mounting Location:		
	Action Coil 2 Deenergized:		LOCK-UP VALVE	Manufacturer:	
	Action Coil 2 Energized:			Action Style:	
	Manual Reset:			Mounting Location:	
	Manual Operator:		QUICK EXHAUST	Model Number:	
	Mounting Location:			Manufacturer:	
	Manufacturer:			Connection Size/Type:	
POSITIONER	Positioner Tag Number:		VOLUME BOOSTER	Body Material:	
	Positioner Type: Digital			Diaphragm Material:	
	Positioner Input Signal Range: 4 - 20 mA			Exhaust Action:	
AIR FILTER	Positioner Output Signal Range: 0 - 100 %		TUBING	Mounting Location:	
	Manufacturer:			Manufacturer:	
	Quantity: one			Connection Size/Type:	
TUBING	Type: Filter/Regulator		TUBING	Body Material:	
	Manufacturer:			Seat Ring Material:	
	Tubing Size:			Diaphragm Material:	
DATA PAGE 2 REMARKS	Tubing Material:		TUBING	Mounting Location:	
				Manufacturer:	

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CONTROL VALVES

PROJECT SPECIFICATION

963766 — **616**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Note: The information in this document is confidential and the property of UOP LLC and must not be disclosed to others or reproduced in any manner or used for any purpose whatsoever without its written permission.

GENERAL	Item Number: 48 Quantity: 1		Valve Tag Number: PV-23-1202B				Data Page: 1 of 2					
	Service: Vent Gas from Decomposer Feed Flush Drum						P&ID Drawing: 12A					
SIZING CONDITIONS	Process Data Case		General									
	Sizing Case		Maximum									
	Calculated C _v		0.77									
		Inlet	Outlet	Inlet	Outlet	Inlet	Outlet	Inlet	Outlet	Units		
	Liquid Flow at Flowing Cond									m3/h		
	Vapor Flow at Base Cond	14								normal m3/h		
	Steam Flow									kg/h		
	Temperature	46								°C		
	Liquid Specific Gravity									----		
	Vapor Spec. Gravity (Air=1)	0.967								----		
	Pressure	0.70	0.20							kg/cm2(g)		
	Liq Critical Press (Pseudo)									kg/cm2(a)		
	Liquid Vapor Pressure									kg/cm2(a)		
	Liquid Viscosity									cP		
	Compressibility Z	1.000										
	Specific Heats Cp/Cv (Real)	1.400										
	Estimate Sound Press Level									dBA		
	VALVE BODY & BONNET	Body Type: Globe \$				Approximate Valve C _v : 1.2				F _L : 0.9		x _T : 0.72
Body Size: 1" Port Type: Single				Trim Style: Man. Standard				Trim Size: No. 2				
Inlet Conn Size: 1" Type: Flanged Class: 300				Inherent Characteristic: Linear								
Outlet Conn Size: 1" Type: Flanged Class: 300				Balance Style: Unbalanced				Full Open Position: 100 %				
Flange Facing: Raised Face Flange Face Finish:				Guiding Style: Top								
Body & Bonnet Material: 316L Stainless Steel \$\$				Closure Member Material: Stellite								
Bonnet Style: Integral or Bolted				Seat Ring Material: 17-4PH Stainless Steel								
Packing Style: Manufacturer's Recommendation				Seat Ring Retainer Material: 17-4PH Stainless Steel								
Packing Material: Manufacturer's Recommendation				Bearing Material:								
Body/Bonnet Gasket Material:				Bushing/Cage Material: 17-4PH Stainless Steel								
Lub/Iso Valve Material: Lub Material:				Stem/Shaft Material: 316 Stainless Steel								
Manufacturer: Masoneilan (or equal)				Flow Direction:								
Model Number:				Seat Leakage Class: ANSI Class IV								
ACTUATOR		Type: Spring Diaphragm, Pneumatic				Inlet Pipe Nominal Size:				Est Wall Thick:		
	Action Style: Fail Close				Outlet Pipe Nominal Size:				Est Wall Thick:			
	Service Type: Modulating				Est Insulation Type:							
	Max Shutoff Press: In: 11.00 Out: 0 kg/cm2(g)				Est Insulation Thickness:							
	Maximum Stroke Time to Open:				Installation:							
	Maximum Stroke Time to Close:				Design Pressure : 10.0 kg/cm2g							
	Stroke Cycle Service:				Design Temperature : 120 C							
DATA PAGE 1 REMARKS	§ The information is based on Masoneilan 28000 series.											
	§§ Carbon steel is an acceptable alternate.											

ALL ITEMS SHALL COMPLY WITH THE GENERAL REQUIREMENTS SHOWN ON THE GENERAL REQUIREMENTS SECTION OF THIS SPECIFICATION.



CONTROL VALVES

PROJECT SPECIFICATION

963766 — **616**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Item Number: 48 Quantity: 1		Valve Tag Number: PV-23-1202B		Data Page: 2 of 2	
TRANS-DUCER	Transducer Tag Number:		HANDWHEEL	Handwheel Position:	
	Transducer Input Signal Range:			Usage:	
	Transducer Output Signal Range:				
	Manufacturer:				
SOLENOID VALVE	Solenoid Valve Tag Number:		POSITION SWITCH	Open Position Switch Tag Number:	
	Body Type: Body Material:			Close Position Switch Tag Number:	
	Seat Material: Disc Material:			Sensor Type:	
	Connection Size/Type:			Contact Form:	
	Packing Style:			Open Position Switch Contact Action:	
	Fluid:			Close Position Switch Contact Action:	
	Pressure:			Contact Load: Contact Rating:	
	Minimum Pressure Port Orifice C _v : Size:			Mounting Location:	
	Minimum Exhaust Port Orifice C _v : Size:			Manufacturer:	
	Operation Style:			POSITION TRANSMITTER	Position Transmitter Tag Number:
	Coil Configuration:		Sensor Type:		
	Operating Voltage: Frequency:		Output Signal Range:		
	Action Coil 1 Deenergized:		Travel Measurement:		
	Action Coil 1 Energized:		Mounting Location:		
	Action Coil 2 Deenergized:		LOCK-UP VALVE	Manufacturer:	
	Action Coil 2 Energized:			Action Style:	
	Manual Reset:			Mounting Location:	
	Manual Operator:		QUICK EXHAUST	Model Number:	
	Mounting Location:			Manufacturer:	
	Manufacturer:			Connection Size/Type:	
POSITIONER	Positioner Tag Number:		VOLUME BOOSTER	Body Material:	
	Positioner Type: Digital			Diaphragm Material:	
	Positioner Input Signal Range: 4 - 20 mA			Exhaust Action:	
AIR FILTER	Positioner Output Signal Range: 0 - 100 %		TUBING	Mounting Location:	
	Manufacturer:			Manufacturer:	
	Quantity: one			Connection Size/Type:	
TUBING	Type: Filter/Regulator		TUBING	Body Material:	
	Manufacturer:			Seat Ring Material:	
	Tubing Size:			Diaphragm Material:	
DATA PAGE 2 REMARKS	Tubing Material:		TUBING	Mounting Location:	
				Manufacturer:	

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CONTROL VALVES

PROJECT SPECIFICATION

963766 — **616**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

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GENERAL	Item Number: 49 Quantity: 1		Valve Tag Number: HV-23-1201		Data Page: 1 of 2	
	Service: Makeup Cumene to Decoposer Feed Flush Drum		P&ID Drawing: 12A			
SIZING CONDITIONS	Process Data Case		Design			
	Sizing Case		Maximum			
	Calculated C _v		7.7			
		Inlet	Outlet	Inlet	Outlet	Units
	Liquid Flow at Flowing Cond	9.5				m ³ /h
	Vapor Flow at Base Cond					normal m ³ /h
	Steam Flow					kg/h
	Temperature	46				°C
	Liquid Specific Gravity	0.841				----
	Vapor Spec. Gravity (Air=1)					----
	Pressure	4.13	2.37			kg/cm ² (g)
	Liq Critical Press (Pseudo)	32.72				kg/cm ² (a)
	Liquid Vapor Pressure	0.90				kg/cm ² (a)
	Liquid Viscosity	0.425				cP
	Compressibility Z					
	Specific Heats Cp/Cv (Real)					
	Estimate Sound Press Level					dBA
	VALVE BODY & BONNET	Body Type: Globe §		Approximate Valve C _v : 12		F _L : 0.9 x _T : 0.72
Body Size: 1" Port Type: Single		Trim Style: Man. Standard		Trim Size:		
Inlet Conn Size: 1" Type: Flanged Class: 300		Inherent Characteristic: Linear				
Outlet Conn Size: 1" Type: Flanged Class: 300		Balance Style: Unbalanced		Full Open Position: 100 %		
Flange Facing: Raised Face Flange Face Finish:		Guiding Style: Top				
Body & Bonnet Material: Carbon Steel		Closure Member Material: 416 Stainless Steel				
Bonnet Style: Integral or Bolted		Seat Ring Material: 416 Stainless Steel				
Packing Style: Manufacturer's Recommendation		Seat Ring Retainer Material: 304 Stainless Steel				
Packing Material: Manufacturer's Recommendation		Bearing Material:				
Body/Bonnet Gasket Material: Manufacturer's Recommend		Bushing/Cage Material: 440C Stainless Steel				
Lub/Iso Valve Material: Lub Material:		Stem/Shaft Material: 316 Stainless Steel				
ACTUATOR		Manufacturer: Masoneilan (or equal)		Flow Direction:		
	Model Number:		Seat Leakage Class: ANSI Class V			
	Type: Spring Diaphragm, Pneumatic		Inlet Pipe Nominal Size:		Est Wall Thick:	
	Action Style: Fail Close		Outlet Pipe Nominal Size:		Est Wall Thick:	
	Service Type: Modulating		Est Insulation Type:			
	Max Shutoff Press: In: 6.42 Out: 0 kg/cm ² (g)		Est Insulation Thickness:			
	Maximum Stroke Time to Open:		Installation:			
	Maximum Stroke Time to Close:		Design Pressure : 10.59 kg/cm ² g			
DATA PAGE 1 REMARKS	Stroke Cycle Service:		Design Temperature : 120 C			
	§ The information is based on Masoneilan 21000 series. Valve will normally be in the closed position. Valve opening will be intermittent and generally short-term.					

ALL ITEMS SHALL COMPLY WITH THE GENERAL REQUIREMENTS SHOWN ON THE GENERAL REQUIREMENTS SECTION OF THIS SPECIFICATION.



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CONTROL VALVES

PROJECT SPECIFICATION

963766 — **616**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Item Number: 49 Quantity: 1		Valve Tag Number: HV-23-1201		Data Page: 2 of 2	
TRANS-DUCER	Transducer Tag Number:		HANDWHEEL	Handwheel Position:	
	Transducer Input Signal Range:			Usage:	
	Transducer Output Signal Range:				
	Manufacturer:				
SOLENOID VALVE	Solenoid Valve Tag Number:		POSITION SWITCH	Open Position Switch Tag Number:	
	Body Type: Body Material:			Close Position Switch Tag Number:	
	Seat Material: Disc Material:			Sensor Type:	
	Connection Size/Type:			Contact Form:	
	Packing Style:			Open Position Switch Contact Action:	
	Fluid:			Close Position Switch Contact Action:	
	Pressure:			Contact Load: Contact Rating:	
	Minimum Pressure Port Orifice C _v : Size:			Mounting Location:	
	Minimum Exhaust Port Orifice C _v : Size:			Manufacturer:	
	Operation Style:			POSITION TRANSMITTER	Position Transmitter Tag Number:
	Coil Configuration:		Sensor Type:		
	Operating Voltage: Frequency:		Output Signal Range:		
	Action Coil 1 Deenergized:		Travel Measurement:		
	Action Coil 1 Energized:		Mounting Location:		
	Action Coil 2 Deenergized:		LOCK-UP VALVE	Manufacturer:	
	Action Coil 2 Energized:			Action Style:	
	Manual Reset:			Mounting Location:	
	Manual Operator:		QUICK EXHAUST	Model Number:	
	Mounting Location:			Manufacturer:	
	Manufacturer:			Connection Size/Type:	
POSITIONER	Positioner Tag Number:		VOLUME BOOSTER	Body Material:	
	Positioner Type: Digital			Seat Ring Material:	
	Positioner Input Signal Range: 4 - 20 mA			Diaphragm Material:	
AIR FILTER	Positioner Output Signal Range: 0 - 100 %		TUBING	Mounting Location:	
	Manufacturer:			Manufacturer:	
	Quantity: one				
TUBING	Type: Filter/Regulator				
	Manufacturer:				
	Tubing Size:				
DATA PAGE 2 REMARKS	Tubing Material:				

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CONTROL VALVES

PROJECT SPECIFICATION

963766 — **616**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

GENERAL	Item Number: 50 Quantity: 1		Valve Tag Number: FV-23-1201		Data Page: 1 of 2		
	Service: Process Water to Decomposer Feed		P&ID Drawing: 12A				
SIZING CONDITIONS	Process Data Case	Single Case		Single Case		Single Case	
	Sizing Case	Normal		Maximum		Minimum	
	Calculated C _v	0.233		0.56		0.112	
		Inlet	Outlet	Inlet	Outlet	Inlet	
	Liquid Flow at Flowing Cond	0.4		0.8		0.2	
	Vapor Flow at Base Cond						
	Steam Flow						
	Temperature	50		50		50	
	Liquid Specific Gravity	0.989		0.989		0.989	
	Vapor Spec. Gravity (Air=1)						
	Pressure	5.10	1.52	4.05	1.56	5.36	
	Liq Critical Press (Pseudo)	224.90		224.90		224.90	
	Liquid Vapor Pressure	0.13		0.13		0.13	
	Liquid Viscosity	0.547		0.547		0.547	
	Compressibility Z						
Specific Heats Cp/Cv (Real)							
Estimate Sound Press Level							
VALVE BODY & BONNET	Body Type: Globe §					Approximate Valve C _v : 1.7 F _L : 0.9 x _T : 0.72	
	Body Size: 1" Port Type: Single					Trim Style: Man. Standard Trim Size:	
	Inlet Conn Size: 1" Type: Flanged Class: 300					Inherent Characteristic: Equal Percentage	
	Outlet Conn Size: 1" Type: Flanged Class: 300					Balance Style: Unbalanced Full Open Position: 100 %	
	Flange Facing: Raised Face Flange Face Finish:					Guiding Style: Top	
	Body & Bonnet Material: 316 Stainless Steel ‡					Closure Member Material: 316 Stainless Steel	
	Bonnet Style: Integral or Bolted					Seat Ring Material: 316 Stainless Steel	
	Packing Style: Manufacturer's Recommendation					Seat Ring Retainer Material: 304 Stainless Steel	
	Packing Material: Manufacturer's Recommendation					Bearing Material:	
	Body/Bonnet Gasket Material: Manufacturer's Recommend					Bushing/Cage Material: Stellite	
	Lub/Iso Valve Material: Lub Material:					Stem/Shaft Material: 316 Stainless Steel	
	Manufacturer: Masoneilan (or equal)					Flow Direction:	
	Model Number:					Seat Leakage Class: ANSI Class IV	
	ACTUATOR	Type: Spring Diaphragm, Pneumatic					Inlet Pipe Nominal Size: Est Wall Thick:
		Action Style: Fail Close					Outlet Pipe Nominal Size: Est Wall Thick:
Service Type: Modulating						Est Insulation Type:	
Max Shutoff Press: In: 6.44 Out: 0.70 kg/cm2(g)						Est Insulation Thickness:	
Maximum Stroke Time to Open:						Installation:	
Maximum Stroke Time to Close:						Design Pressure : 9.72 kg/cm2g	
Stroke Cycle Service:						Design Temperature : 120 C	
DATA PAGE 1 REMARKS		§ The information is based on Masoneilan 21000 series.					
	‡ Preferred material is L grade stainless steel. Standard grade stainless steel that is not welded may be used for process wetted parts including the body. Stainless steel with weld repair is acceptable if low carbon (<= 0.03 %) filler material is used.						

ALL ITEMS SHALL COMPLY WITH THE GENERAL REQUIREMENTS SHOWN ON THE GENERAL REQUIREMENTS SECTION OF THIS SPECIFICATION.

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CONTROL VALVES

PROJECT SPECIFICATION

963766 — 616

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Item Number: 50 Quantity: 1 Valve Tag Number: FV-23-1201 Data Page: 2 of 2																																																																																																					
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See UOP Project Specification 660, this project. </td> </tr> </table> </td></tr></table>	SOLENOID VALVE	Solenoid Valve Tag Number: FXY-23-1201	POSITION SWITCH	Open Position Switch Tag Number:	Body Type: 3-Way Body Material: Brass	Close Position Switch Tag Number:	Seat Material: Stainless Steel Disc Material: Stainless Steel	Sensor Type:	Connection Size/Type: †	Contact Form:	Packing Style: Packless	Open Position Switch Contact Action:	Fluid: Control Signal Air	Close Position Switch Contact Action:	Pressure: Contractor to determine	Contact Load: Contact Rating:	Minimum Pressure Port Orifice C_v : MAX Size: MAX	Mounting Location:	Minimum Exhaust Port Orifice C_v : MAX Size: MAX	Manufacturer:	Operation Style: †	POSITION TRANSMITTER	Position Transmitter Tag Number:	Coil Configuration: Single	Sensor Type:	Operating Voltage: \$\$ Frequency:	Output Signal Range:	Action Coil 1 Deenergized: Vent air from valve	Travel Measurement:	Action Coil 1 Energized: Supply air to valve	LOCK-UP VALVE	Mounting Location:	Action Coil 2 Deenergized:	Manufacturer:	Action Coil 2 Energized:	Action Style:	Manual Reset:	Mounting Location:	Manual Operator:	QUICK EXHAUST	Model Number:	Mounting Location: Nearest item to actuator in air line. †	Manufacturer:	Manufacturer: ASCO (or equal)	Connection Size/Type:		Body Material:	<table border="1"> <tr> <td rowspan="3">POSITIONER</td> <td>Positioner Tag Number:</td> <td rowspan="3">QUICK EXHAUST</td> <td>Diaphragm Material:</td> </tr> <tr> <td>Positioner Type: Digital</td> <td>Exhaust Action:</td> </tr> <tr> <td>Positioner Input Signal Range: 4 - 20 mA</td> <td>Mounting Location:</td> </tr> <tr> <td>Positioner Output Signal Range: 0 - 100 %</td> <td>Manufacturer:</td> <td rowspan="4">VOLUME BOOSTER</td> <td>Connection Size/Type:</td> </tr> <tr> <td>Manufacturer:</td> <td>Body Material:</td> </tr> <tr> <td>Quantity: one</td> <td>Seat Ring Material:</td> </tr> <tr> <td>Type: Filter/Regulator</td> <td>Diaphragm Material:</td> </tr> <tr> <td> <table border="1"> <tr> <td rowspan="2">AIR FILTER</td> <td>Manufacturer:</td> <td rowspan="2">VOLUME BOOSTER</td> <td>Mounting Location:</td> </tr> <tr> <td></td> <td>Manufacturer:</td> </tr> <tr> <td rowspan="2">TUBING</td> <td>Tubing Size:</td> <td rowspan="2">VOLUME BOOSTER</td> <td></td> </tr> <tr> <td>Tubing Material:</td> <td></td> </tr> </table> </td> <td> <table border="1"> <tr> <td rowspan="2">TUBING</td> <td>Tubing Size:</td> <td rowspan="2">VOLUME BOOSTER</td> <td></td> </tr> <tr> <td>Tubing Material:</td> <td></td> </tr> </table> </td> </tr> <tr> <td colspan="4"> † The solenoid valve shall be mounted and tubed by the control valve vendor. 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CONTROL VALVES

PROJECT SPECIFICATION

963766 — 616

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				
1	28AUG12	MVD	MW				

GENERAL	Item Number: 51 Quantity: 1 Valve Tag Number: PV-23-1301A Data Page: 1 of 2																																																																																																																																																																										
	Service: Nitrogen to Decomposer Drum Process Material: Nitrogen P&ID Drawing: 13																																																																																																																																																																										
SIZING CONDITIONS	<table border="1"> <tr> <th>Process Data Case</th> <th colspan="2">General</th> <th colspan="2"></th> <th colspan="2"></th> <th colspan="2"></th> <th></th> </tr> <tr> <th>Sizing Case</th> <th colspan="2">Maximum</th> <th colspan="2"></th> <th colspan="2"></th> <th colspan="2"></th> <th></th> </tr> <tr> <th>Calculated C_v</th> <th colspan="2">1.02</th> <th colspan="2"></th> <th colspan="2"></th> <th colspan="2"></th> <th></th> </tr> <tr> <th></th> <th>Inlet</th> <th>Outlet</th> <th>Inlet</th> <th>Outlet</th> <th>Inlet</th> <th>Outlet</th> <th>Inlet</th> <th>Outlet</th> <th>Units</th> </tr> <tr> <td>Liquid Flow at Flowing Cond</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>m³/h</td> </tr> <tr> <td>Vapor Flow at Base Cond</td> <td>103</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>normal m³/h</td> </tr> <tr> <td>Steam Flow</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>kg/h</td> </tr> <tr> <td>Temperature</td> <td>38</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>°C</td> </tr> <tr> <td>Liquid Specific Gravity</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>----</td> </tr> <tr> <td>Vapor Spec. Gravity (Air=1)</td> <td>0.967</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>----</td> </tr> <tr> <td>Pressure</td> <td>7.00</td> <td>0.70</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>kg/cm²(g)</td> </tr> <tr> <td>Liq Critical Press (Pseudo)</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>kg/cm²(a)</td> </tr> <tr> <td>Liquid Vapor Pressure</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>kg/cm²(a)</td> </tr> <tr> <td>Liquid Viscosity</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>cP</td> </tr> <tr> <td>Compressibility Z</td> <td>1.000</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> <tr> <td>Specific Heats Cp/Cv (Real)</td> <td>1.400</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> <tr> <td>Estimate Sound Press Level</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>dBA</td> </tr> </table>	Process Data Case	General									Sizing Case	Maximum									Calculated C _v	1.02										Inlet	Outlet	Inlet	Outlet	Inlet	Outlet	Inlet	Outlet	Units	Liquid Flow at Flowing Cond									m ³ /h	Vapor Flow at Base Cond	103								normal m ³ /h	Steam Flow									kg/h	Temperature	38								°C	Liquid Specific Gravity									----	Vapor Spec. Gravity (Air=1)	0.967								----	Pressure	7.00	0.70							kg/cm ² (g)	Liq Critical Press (Pseudo)									kg/cm ² (a)	Liquid Vapor Pressure									kg/cm ² (a)	Liquid Viscosity									cP	Compressibility Z	1.000									Specific Heats Cp/Cv (Real)	1.400									Estimate Sound Press Level									dBA
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VALVE BODY & BONNET	Body Type: Globe § Body Size: 1" Port Type: Single Inlet Conn Size: 1" Type: Flanged Class: 300 Outlet Conn Size: 1" Type: Flanged Class: 300 Flange Facing: Raised Face Flange Face Finish: Body & Bonnet Material: 316L Stainless Steel §§ Bonnet Style: Integral or Bolted Packing Style: Manufacturer's Recommendation Packing Material: Manufacturer's Recommendation Body/Bonnet Gasket Material: Lub/Iso Valve Material: Lub Material: Manufacturer: Masoneilan (or equal) Model Number:	TRIM	Approximate Valve C _v : 1.2 F _L : 0.9 x _T : 0.72 Trim Style: Man. Standard Trim Size: No. 2 Inherent Characteristic: Linear Balance Style: Unbalanced Full Open Position: 100 % Guiding Style: Top Closure Member Material: Stellite 1 Seat Ring Material: 316 Stainless Steel 1 Seat Ring Retainer Material: 316 Stainless Steel Bearing Material: 1 Bushing/Cage Material: 316 Stainless Steel Stem/Shaft Material: 316 Stainless Steel Flow Direction: Seat Leakage Class: ANSI Class IV																																																																																																																																																																								
	ACTUATOR		INSTALLATION	Type: Spring Diaphragm, Pneumatic Action Style: Fail Close Service Type: Modulating Max Shutoff Press: In: 9.00 Out: 0.70 kg/cm²(g) Maximum Stroke Time to Open: Maximum Stroke Time to Close: Stroke Cycle Service:	Inlet Pipe Nominal Size: Est Wall Thick: Outlet Pipe Nominal Size: Est Wall Thick: Est Insulation Type: Est Insulation Thickness: Installation: Design Pressure : 10 kg/cm ² g Design Temperature : 250 C																																																																																																																																																																						
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ALL ITEMS SHALL COMPLY WITH THE GENERAL REQUIREMENTS SHOWN ON THE GENERAL REQUIREMENTS SECTION OF THIS SPECIFICATION.

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CONTROL VALVES

PROJECT SPECIFICATION

963766 — 616

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Item Number: 51 Quantity: 1		Valve Tag Number: PV-23-1301A		Data Page: 2 of 2	
TRANS-DUCER	Transducer Tag Number:		HANDWHEEL	Handwheel Position:	
	Transducer Input Signal Range:			Usage:	
	Transducer Output Signal Range:				
	Manufacturer:				
SOLENOID VALVE	Solenoid Valve Tag Number:		POSITION SWITCH	Open Position Switch Tag Number:	
	Body Type: Body Material:			Close Position Switch Tag Number:	
	Seat Material: Disc Material:			Sensor Type:	
	Connection Size/Type:			Contact Form:	
	Packing Style:			Open Position Switch Contact Action:	
	Fluid:			Close Position Switch Contact Action:	
	Pressure:			Contact Load: Contact Rating:	
	Minimum Pressure Port Orifice C_v : Size:			Mounting Location:	
	Minimum Exhaust Port Orifice C_v : Size:			Manufacturer:	
	Operation Style:			POSITION TRANSMITTER	Position Transmitter Tag Number:
	Coil Configuration:		Sensor Type:		
	Operating Voltage: Frequency:		Output Signal Range:		
	Action Coil 1 Deenergized:		Travel Measurement:		
	Action Coil 1 Energized:		Mounting Location:		
	Action Coil 2 Deenergized:		Manufacturer:		
	Action Coil 2 Energized:		LOCK-UP VALVE	Action Style:	
	Manual Reset:			Mounting Location:	
	Manual Operator:			Model Number:	
Mounting Location:		QUICK EXHAUST	Manufacturer:		
Manufacturer:			Connection Size/Type:		
Positioner Tag Number:			Body Material:		
Positioner Type: Digital			Diaphragm Material:		
Positioner Input Signal Range: 4 - 20 mA		VOLUME BOOSTER	Exhaust Action:		
Positioner Output Signal Range: 0 - 100 %			Mounting Location:		
Manufacturer:			Manufacturer:		
AIR FILTER	Quantity: one		Connection Size/Type:		
	Type: Filter/Regulator		Body Material:		
TUBING	Manufacturer:		Seat Ring Material:		
	Tubing Size:		Diaphragm Material:		
Tubing Material:		Mounting Location:			
		Manufacturer:			
DATA PAGE 2 REMARKS					

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CONTROL VALVES

PROJECT SPECIFICATION

963766 — 616

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

GENERAL	Item Number: 52 Quantity: 1		Valve Tag Number: PV-23-1301B				Data Page: 1 of 2			
	Service: Vent Gas from Decomposer Drum		P&ID Drawing: 13							
SIZING CONDITIONS	Process Data Case		General							
	Sizing Case		Maximum							
	Calculated C _v		5.4							
		Inlet	Outlet	Inlet	Outlet	Inlet	Outlet	Inlet	Outlet	Units
	Liquid Flow at Flowing Cond									m3/h
	Vapor Flow at Base Cond	103								normal m3/h
	Steam Flow									kg/h
	Temperature	60								°C
	Liquid Specific Gravity									----
	Vapor Spec. Gravity (Air=1)	0.967								----
	Pressure	0.70	0.01							kg/cm2(g)
	Liq Critical Press (Pseudo)									kg/cm2(a)
	Liquid Vapor Pressure									kg/cm2(a)
	Liquid Viscosity									cP
	Compressibility Z	1.000								
	Specific Heats Cp/Cv (Real)	1.400								
	Estimate Sound Press Level									dBA
VALVE BODY & BONNET	Body Type: Globe §								Approximate Valve C _v : 6 F _L : 0.9 x _T : 0.72	
	Body Size: 1-1/2" Port Type: Single								Trim Style: Man. Standard Trim Size:	
	Inlet Conn Size: 1-1/2" Type: Flanged Class: 300								Inherent Characteristic: Equal Percentage	
	Outlet Conn Size: 1-1/2" Type: Flanged Class: 300								Balance Style: Unbalanced Full Open Position: 100 %	
	Flange Facing: Raised Face Flange Face Finish:								Guiding Style: Top	
	Body & Bonnet Material: 316 Stainless Steel ‡								Closure Member Material: 316 Stainless Steel	
	Bonnet Style: Integral or Bolted								Seat Ring Material: 316 Stainless Steel	
	Packing Style: Manufacturer's Recommendation								Seat Ring Retainer Material: 304 Stainless Steel	
	Packing Material: Manufacturer's Recommendation								Bearing Material:	
	Body/Bonnet Gasket Material: Manufacturer's Recommend								Bushing/Cage Material: Stellite	
	Lub/Iso Valve Material: Lub Material:								Stem/Shaft Material: 316 Stainless Steel	
	Manufacturer: Masoneilan (or equal)								Flow Direction:	
	Model Number:								Seat Leakage Class: ANSI Class IV	
	ACTUATOR	Type: Spring Diaphragm, Pneumatic								Inlet Pipe Nominal Size: Est Wall Thick:
Action Style: Fail Close								Outlet Pipe Nominal Size: Est Wall Thick:		
Service Type: Modulating								Est Insulation Type:		
Max Shutoff Press: In: 12.10 Out: 0 kg/cm2(g)								Est Insulation Thickness:		
Maximum Stroke Time to Open:								Installation:		
Maximum Stroke Time to Close:								Design Pressure : 11.5 kg/cm2g (F.V)		
Stroke Cycle Service:								Design Temperature : 250 C		
DATA PAGE 1	§ The information is based on Masoneilan 21000 series. ‡ Preferred material is L grade stainless steel. Standard grade stainless steel that is not welded may be used for process wetted parts including the body. Stainless steel with weld repair is acceptable if low carbon (<= 0.03 %) filler material is used.									

ALL ITEMS SHALL COMPLY WITH THE GENERAL REQUIREMENTS SHOWN ON THE GENERAL REQUIREMENTS SECTION OF THIS SPECIFICATION.

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CONTROL VALVES

PROJECT SPECIFICATION

963766 — 616

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Item Number: 52 Quantity: 1		Valve Tag Number: PV-23-1301B		Data Page: 2 of 2		
TRANS-DUCER	Transducer Tag Number:		HANDWHEEL	Handwheel Position:		
	Transducer Input Signal Range:			Usage:		
	Transducer Output Signal Range:					
	Manufacturer:					
SOLENOID VALVE	Solenoid Valve Tag Number:		POSITION SWITCH	Open Position Switch Tag Number:		
	Body Type: Body Material:			Close Position Switch Tag Number:		
	Seat Material: Disc Material:			Sensor Type:		
	Connection Size/Type:			Contact Form:		
	Packing Style:			Open Position Switch Contact Action:		
	Fluid:			Close Position Switch Contact Action:		
	Pressure:			Contact Load: Contact Rating:		
	Minimum Pressure Port Orifice C _v : Size:			Mounting Location:		
	Minimum Exhaust Port Orifice C _v : Size:			Manufacturer:		
	Operation Style:					
	POSITIONER	Coil Configuration:		POSITION TRANSMITTER	Position Transmitter Tag Number:	
		Operating Voltage: Frequency:			Sensor Type:	
		Action Coil 1 Deenergized:			Output Signal Range:	
		Action Coil 1 Energized:			Travel Measurement:	
		Action Coil 2 Deenergized:		Mounting Location:		
		Action Coil 2 Energized:		Manufacturer:		
		Manual Reset:		LOCK-UP VALVE	Action Style:	
		Manual Operator:			Mounting Location:	
Mounting Location:		Model Number:				
Manufacturer:		Manufacturer:				
AIR FILTER	Positioner Tag Number:		QUICK EXHAUST	Connection Size/Type:		
	Positioner Type: Digital			Body Material:		
	Positioner Input Signal Range: 4 - 20 mA			Diaphragm Material:		
	Positioner Output Signal Range: 0 - 100 %			Exhaust Action:		
TUBING	Quantity: one		VOLUME BOOSTER	Mounting Location:		
	Type: Filter/Regulator			Manufacturer:		
	Manufacturer:			Connection Size/Type:		
	Tubing Size:			Body Material:		
Tubing Material:		Seat Ring Material:		Diaphragm Material:		
		Mounting Location:		Manufacturer:		
DATA PAGE 2 REMARKS						

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CONTROL VALVES

PROJECT SPECIFICATION

963766 — 616

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				
1	18SEP12	RED	MW				

GENERAL	Item Number: 53 Quantity: 1 Valve Tag Number: FV-23-1301 Data Page: 1 of 2																																																																																																																																						
	Service: Decomposer Product to Dehydrators Process Material: Hydrocarbon with aromatics P&ID Drawing: 13																																																																																																																																						
SIZING CONDITIONS	<table border="1"> <thead> <tr> <th rowspan="2">Process Data Case</th> <th colspan="2">Single Case</th> <th colspan="2">Single Case</th> <th colspan="2">Single Case</th> <th rowspan="2">Units</th> </tr> <tr> <th colspan="2">Normal</th> <th colspan="2">Maximum</th> <th colspan="2">Minimum</th> </tr> </thead> <tbody> <tr> <td>1 Calculated C_v</td> <td colspan="2">46.4</td> <td colspan="2">58</td> <td colspan="2">17.5</td> <td></td> </tr> <tr> <td></td> <td>Inlet</td> <td>Outlet</td> <td>Inlet</td> <td>Outlet</td> <td>Inlet</td> <td>Outlet</td> <td></td> </tr> <tr> <td>Liquid Flow at Flowing Cond</td> <td>68.7</td> <td></td> <td>75.6</td> <td></td> <td>34.4</td> <td></td> <td>m3/h</td> </tr> <tr> <td>Vapor Flow at Base Cond</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>normal m3/h</td> </tr> <tr> <td>Steam Flow</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>kg/h</td> </tr> <tr> <td>Temperature</td> <td>83</td> <td></td> <td>83</td> <td></td> <td>83</td> <td></td> <td>°C</td> </tr> <tr> <td>Liquid Specific Gravity</td> <td>0.892</td> <td></td> <td>0.892</td> <td></td> <td>0.892</td> <td></td> <td>----</td> </tr> <tr> <td>Vapor Spec. Gravity (Air=1)</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>----</td> </tr> <tr> <td>1 Pressure</td> <td>9.17</td> <td>6.50</td> <td>8.83</td> <td>6.79</td> <td>10.12</td> <td>5.45</td> <td>kg/cm2(g)</td> </tr> <tr> <td>Liq Critical Press (Pseudo)</td> <td>59.70</td> <td></td> <td>59.70</td> <td></td> <td>59.70</td> <td></td> <td>kg/cm2(a)</td> </tr> <tr> <td>Liquid Vapor Pressure</td> <td>1.50</td> <td></td> <td>1.50</td> <td></td> <td>1.50</td> <td></td> <td>kg/cm2(a)</td> </tr> <tr> <td>Liquid Viscosity</td> <td>0.532</td> <td></td> <td>0.532</td> <td></td> <td>0.532</td> <td></td> <td>cP</td> </tr> <tr> <td>Compressibility Z</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> <tr> <td>Specific Heats Cp/Cv (Real)</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> <tr> <td>Estimate Sound Press Level</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>dBA</td> </tr> </tbody> </table>	Process Data Case	Single Case		Single Case		Single Case		Units	Normal		Maximum		Minimum		1 Calculated C _v	46.4		58		17.5				Inlet	Outlet	Inlet	Outlet	Inlet	Outlet		Liquid Flow at Flowing Cond	68.7		75.6		34.4		m3/h	Vapor Flow at Base Cond							normal m3/h	Steam Flow							kg/h	Temperature	83		83		83		°C	Liquid Specific Gravity	0.892		0.892		0.892		----	Vapor Spec. Gravity (Air=1)							----	1 Pressure	9.17	6.50	8.83	6.79	10.12	5.45	kg/cm2(g)	Liq Critical Press (Pseudo)	59.70		59.70		59.70		kg/cm2(a)	Liquid Vapor Pressure	1.50		1.50		1.50		kg/cm2(a)	Liquid Viscosity	0.532		0.532		0.532		cP	Compressibility Z								Specific Heats Cp/Cv (Real)								Estimate Sound Press Level							dBA
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VALVE BODY & BONNET	Body Type: Globe § Body Size: 3" Port Type: Single Inlet Conn Size: 3" Type: Flanged Class: 300 Outlet Conn Size: 3" Type: Flanged Class: 300 Flange Facing: Raised Face Flange Face Finish: Body & Bonnet Material: 316 Stainless Steel ‡ Bonnet Style: Integral or Bolted Packing Style: Manufacturer's Recommendation Packing Material: Manufacturer's Recommendation Body/Bonnet Gasket Material: Manufacturer's Recommend Lub/Iso Valve Material: Lub Material: Manufacturer: Masoneilan (or equal) Model Number:	TRIM	Approximate Valve C _v : 75 F _L : 0.9 x _T : 0.72 Trim Style: Man. Standard Trim Size: Inherent Characteristic: Linear Balance Style: Balanced Full Open Position: 100 % Guiding Style: Cage Closure Member Material: 316 Stainless Steel Seat Ring Material: 316 SS with Stellite sheath Seat Ring Retainer Material: Bearing Material: Bushing/Cage Material: 316 SS hard chrome plated Stem/Shaft Material: 316 Stainless Steel Flow Direction: Seat Leakage Class: ANSI Class IV																																																																																																																																				
	ACTUATOR		INSTALLATION	Type: Spring Diaphragm, Pneumatic Action Style: Fail Open Service Type: Modulating Max Shutoff Press: In: 21.00 Out: 5.10 kg/cm2(g) Maximum Stroke Time to Open: Maximum Stroke Time to Close: Stroke Cycle Service:	Inlet Pipe Nominal Size: Est Wall Thick: Outlet Pipe Nominal Size: Est Wall Thick: Est Insulation Type: Est Insulation Thickness: Installation: Design Pressure : 21 kg/cm2g (F.V) Design Temperature : 250 C																																																																																																																																		
				§ The information is based on Masoneilan 41000 series.																																																																																																																																			
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ALL ITEMS SHALL COMPLY WITH THE GENERAL REQUIREMENTS SHOWN ON THE GENERAL REQUIREMENTS SECTION OF THIS SPECIFICATION.

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CONTROL VALVES

PROJECT SPECIFICATION

963766 — 616

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Item Number: 53 Quantity: 1 Valve Tag Number: FV-23-1301 Data Page: 2 of 2																																																																																														
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PROJECT SPECIFICATION

963766 — **616**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

GENERAL	Item Number: 54 Quantity: 1 Valve Tag Number: UXV-23-1401 Data Page: 1 of 2																																																																																																																																																																
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VALVE BODY & BONNET	Body Type: Eccentric Disc § Body Size: 8" ± Port Type: Single Inlet Conn Size: 8" ± Type: Flanged Class: 300 Outlet Conn Size: 8" ± Type: Flanged Class: 300 Flange Facing: Flange Face Finish: Body & Bonnet Material: Carbon Steel Bonnet Style: Packing Style: Manufacturer's Recommendation Packing Material: Manufacturer's Recommendation Body/Bonnet Gasket Material: Lub/Iso Valve Material: Lub Material: Manufacturer: Masoneilan (or equal) Model Number:	TRIM	Approximate Valve C _v : 1911 F _L : x _T : Trim Style: Man. Standard Trim Size: Inherent Characteristic: Equal Percentage Balance Style: Full Open Position: 90° Guiding Style: Shaft Closure Member Material: 316 Stainless Steel Seat Ring Material: RTFE Seat Ring Retainer Material: Carbon Steel Bearing Material: Nitrided Stainless Steel Bushing/Cage Material: Stem/Shaft Material: 17-4PH Stainless Steel Flow Direction: Flow to close Seat Leakage Class: ANSI Class VI																																																																																																																																																														
	ACTUATOR		Type: Spring Diaphragm, Pneumatic Action Style: Fail Close Service Type: On-Off Max Shutoff Press: In: 4.50 Out: 0 kg/cm2(g) Maximum Stroke Time to Open: Maximum Stroke Time to Close: 5 Seconds Stroke Cycle Service:	INSTALLATION	Inlet Pipe Nominal Size: Est Wall Thick: Outlet Pipe Nominal Size: Est Wall Thick: Est Insulation Type: Est Insulation Thickness: Installation: Design Pressure : 16.5 kg/cm2g (F.V) Design Temperature : 195 C																																																																																																																																																												
			DATA PAGE 1 REMARKS		§ The information is based on Masoneilan 39003 series. ± Line size valve.																																																																																																																																																												

ALL ITEMS SHALL COMPLY WITH THE GENERAL REQUIREMENTS SHOWN ON THE GENERAL REQUIREMENTS SECTION OF THIS SPECIFICATION.

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CONTROL VALVES

PROJECT SPECIFICATION

963766 — 616

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Item Number: 54 Quantity: 1		Valve Tag Number: UXV-23-1401		Data Page: 2 of 2			
TRANS-DUCER	Transducer Tag Number:		HANDWHEEL	Handwheel Position:			
	Transducer Input Signal Range:			Usage:			
	Transducer Output Signal Range:						
	Manufacturer:						
SOLENOID VALVE	Solenoid Valve Tag Number: UXY-23-1401		POSITION SWITCH	Open Position Switch Tag Number:			
	Body Type: 3-Way Body Material: Brass			Close Position Switch Tag Number:			
	Seat Material: Stainless Steel Disc Material: Stainless Steel			Sensor Type:			
	Connection Size/Type: †			Contact Form:			
	Packing Style: Packless			Open Position Switch Contact Action:			
	Fluid: Instrument Air			Close Position Switch Contact Action:			
	Pressure: Contractor to determine			Contact Load: Contact Rating:			
	Minimum Pressure Port Orifice C _v : MAX Size: MAX			Mounting Location:			
	Minimum Exhaust Port Orifice C _v : MAX Size: MAX			Manufacturer:			
	Operation Style: †						
	Coil Configuration: Single						
	Operating Voltage: \$\$ Frequency:						
	Action Coil 1 Deenergized: Vent air from valve						
	Action Coil 1 Energized: Supply air to valve						
	Action Coil 2 Deenergized:						
	Action Coil 2 Energized:						
	POSITIONER	Manual Reset:		POSITION TRANSMITTER	Position Transmitter Tag Number:		
		Manual Operator:			Sensor Type:		
Mounting Location: Nearest item to actuator in air line. †		Output Signal Range:					
Manufacturer: ASCO (or equal)		Travel Measurement:					
		Mounting Location:					
		Manufacturer:					
AIR FILTER		Positioner Tag Number:			LOCK-UP VALVE	Action Style:	
		Positioner Type:				Mounting Location:	
		Positioner Input Signal Range:				Model Number:	
	Positioner Output Signal Range:		Manufacturer:				
	Manufacturer:						
	TUBING	Quantity: one		QUICK EXHAUST		Connection Size/Type:	
		Type: Filter/Regulator				Body Material:	
		Manufacturer:				Diaphragm Material:	
		Exhaust Action:					
		Mounting Location:					
		Manufacturer:					
VOLUME BOOSTER		Tubing Size:			VOLUME BOOSTER	Connection Size/Type:	
		Tubing Material:				Body Material:	
						Seat Ring Material:	
						Diaphragm Material:	
			Mounting Location:				
			Manufacturer:				
	REMARKS	† The solenoid valve shall be mounted and tubed by the control valve vendor. The solenoid valve size and type, the tubing size, the actuator air connection size, and the air/filter regulator capacity shall be determined by the control valve vendor to meet the maximum stroking time requirements specified for the control valve. \$\$ The Contractor shall coordinate the solenoid valve operating voltage and coil watt rating with the supplier of the Decomposition Section Shutdown System (UC-2302). See UOP Project Specification 660, this project.					

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CONTROL VALVES

PROJECT SPECIFICATION

963766 — 616

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

GENERAL	Item Number: 55 Quantity: 1	Valve Tag Number: TV-23-1401		Data Page: 1 of 2					
	Service: S4 Steam to Dehydrators		P&ID Drawing: 14						
SIZING CONDITIONS	Process Data Case	Design		Design		Design			
	Sizing Case	Normal		Maximum		Minimum			
	Calculated C _v	95		166		37.3			
		Inlet	Outlet	Inlet	Outlet	Inlet	Outlet	Inlet	Outlet
	Liquid Flow at Flowing Cond								Units
	Vapor Flow at Base Cond								m3/h
	Steam Flow	3421		3763		1711			normal m3/h
	Temperature	160		147		165			kg/h
	Liquid Specific Gravity								°C
	Vapor Spec. Gravity (Air=1)	0.622		0.622		0.622			----
	Pressure	4.00	3.19	3.50	3.19	4.50	3.19		----
	Liq Critical Press (Pseudo)								kg/cm2(g)
	Liquid Vapor Pressure								kg/cm2(a)
	Liquid Viscosity								kg/cm2(a)
	Compressibility Z	0.960		0.958		0.958			cP
	Specific Heats Cp/Cv (Real)	1.317		1.318		1.316			
	Estimate Sound Press Level								dBA
VALVE BODY & BONNET	Body Type: Globe §		Approximate Valve C _v : 225		F _L : 0.9		x _T : 0.72		
	Body Size: 4" Port Type: Single		Trim Style: Man. Standard		Trim Size:				
	Inlet Conn Size: 4" Type: Flanged Class: 300		Inherent Characteristic: Equal Percentage						
	Outlet Conn Size: 4" Type: Flanged Class: 300		Balance Style: Balanced		Full Open Position: 100 %				
	Flange Facing: Raised Face Flange Face Finish:		Guiding Style: Cage						
	Body & Bonnet Material: Carbon Steel		Closure Member Material: 17-4 PH Stainless Steel						
	Bonnet Style: Integral or Bolted		Seat Ring Material: 316 SS with Stellite sheath						
	Packing Style: Manufacturer's Recommendation		Seat Ring Retainer Material:						
	Packing Material: Manufacturer's Recommendation		Bearing Material:						
	Body/Bonnet Gasket Material: Manufacturer's Recommendation		Bushing/Cage Material: CA6NM hard chrome plated						
	Lub/Iso Valve Material: Lub Material:		Stem/Shaft Material: 17-4 PH Stainless Steel						
	Manufacturer: Masoneilan (or equal)		Flow Direction:						
	Model Number:		Seat Leakage Class: ANSI Class IV						
	ACTUATOR	Type: Spring Diaphragm, Pneumatic		Inlet Pipe Nominal Size:		Est Wall Thick:			
Action Style: Fail Close		Outlet Pipe Nominal Size:		Est Wall Thick:					
Service Type: Modulating		Est Insulation Type:							
Max Shutoff Press: In: 4.50 Out: 0 kg/cm2(g)		Est Insulation Thickness:							
Maximum Stroke Time to Open:		Installation:							
Maximum Stroke Time to Close:		Design Pressure :		16.5 kg/cm2g (F.V)					
Stroke Cycle Service:		Design Temperature :		195 C					
DATA PAGE 1 REMARKS	§ The information is based on Masoneilan 41000 series.								

ALL ITEMS SHALL COMPLY WITH THE GENERAL REQUIREMENTS SHOWN ON THE GENERAL REQUIREMENTS SECTION OF THIS SPECIFICATION.

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CONTROL VALVES

PROJECT SPECIFICATION

963766 — 616

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Item Number: 55 Quantity: 1		Valve Tag Number: TV-23-1401		Data Page: 2 of 2	
TRANS-DUCER	Transducer Tag Number:		HANDWHEEL	Handwheel Position:	
	Transducer Input Signal Range:			Usage:	
	Transducer Output Signal Range:				
	Manufacturer:				
SOLENOID VALVE	Solenoid Valve Tag Number:		POSITION SWITCH	Open Position Switch Tag Number:	
	Body Type: Body Material:			Close Position Switch Tag Number:	
	Seat Material: Disc Material:			Sensor Type:	
	Connection Size/Type:			Contact Form:	
	Packing Style:			Open Position Switch Contact Action:	
	Fluid:			Close Position Switch Contact Action:	
	Pressure:			Contact Load: Contact Rating:	
	Minimum Pressure Port Orifice C _v : Size:			Mounting Location:	
	Minimum Exhaust Port Orifice C _v : Size:			Manufacturer:	
	Operation Style:			POSITION TRANSMITTER	Position Transmitter Tag Number:
	Coil Configuration:		Sensor Type:		
	Operating Voltage: Frequency:		Output Signal Range:		
	Action Coil 1 Deenergized:		Travel Measurement:		
	Action Coil 1 Energized:		Mounting Location:		
	Action Coil 2 Deenergized:		Manufacturer:		
	Action Coil 2 Energized:		LOCK-UP VALVE	Action Style:	
	Manual Reset:			Mounting Location:	
	Manual Operator:			Model Number:	
	Mounting Location:		QUICK EXHAUST	Manufacturer:	
	Manufacturer:			Connection Size/Type:	
		Body Material:			
		Diaphragm Material:			
POSITIONER	Positioner Tag Number:		VOLUME BOOSTER	Exhaust Action:	
	Positioner Type: Digital			Mounting Location:	
	Positioner Input Signal Range: 4 - 20 mA			Manufacturer:	
	Positioner Output Signal Range: 0 - 100 %			Connection Size/Type:	
AIR FILTER	Quantity: one		Body Material:		
	Type: Filter/Regulator		Seat Ring Material:		
	Manufacturer:		Diaphragm Material:		
TUBING	Tubing Size:		Mounting Location:		
	Tubing Material:		Manufacturer:		
DATA PAGE 2 REMARKS					

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CONTROL VALVES

PROJECT SPECIFICATION

963766 — **616**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

GENERAL	Item Number: 56 Quantity: 1 Valve Tag Number: UXV-23-1403 Data Page: 1 of 2																																																																																																																																																																
	Service: Decomposer Product Return to Decomposer Process Material: Hydrocarbon with aromatics P&ID Drawing: 14A																																																																																																																																																																
SIZING CONDITIONS	<table border="1"> <thead> <tr> <th>Process Data Case</th> <th colspan="2">General</th> <th colspan="2">General</th> <th colspan="2"></th> <th colspan="2"></th> <th></th> </tr> <tr> <th>Sizing Case</th> <th colspan="2">Open</th> <th colspan="2">Closed</th> <th colspan="2"></th> <th colspan="2"></th> <th></th> </tr> <tr> <th>Calculated C_v</th> <th>Inlet</th> <th>Outlet</th> <th>Inlet</th> <th>Outlet</th> <th>Inlet</th> <th>Outlet</th> <th>Inlet</th> <th>Outlet</th> <th>Units</th> </tr> </thead> <tbody> <tr> <td>Liquid Flow at Flowing Cond</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>m3/h</td> </tr> <tr> <td>Vapor Flow at Base Cond</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>normal m3/h</td> </tr> <tr> <td>Steam Flow</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>kg/h</td> </tr> <tr> <td>Temperature</td> <td>140</td> <td></td> <td>140</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>°C</td> </tr> <tr> <td>Liquid Specific Gravity</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>----</td> </tr> <tr> <td>Vapor Spec. Gravity (Air=1)</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>----</td> </tr> <tr> <td>Pressure</td> <td>2.16</td> <td>2.16</td> <td>5.11</td> <td>2.16</td> <td></td> <td></td> <td></td> <td></td> <td>kg/cm2(g)</td> </tr> <tr> <td>Liq Critical Press (Pseudo)</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>kg/cm2(a)</td> </tr> <tr> <td>Liquid Vapor Pressure</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>kg/cm2(a)</td> </tr> <tr> <td>Liquid Viscosity</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>cP</td> </tr> <tr> <td>Compressibility Z</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> <tr> <td>Specific Heats Cp/Cv (Real)</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> <tr> <td>Estimate Sound Press Level</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>dBa</td> </tr> </tbody> </table>	Process Data Case	General		General							Sizing Case	Open		Closed							Calculated C _v	Inlet	Outlet	Inlet	Outlet	Inlet	Outlet	Inlet	Outlet	Units	Liquid Flow at Flowing Cond									m3/h	Vapor Flow at Base Cond									normal m3/h	Steam Flow									kg/h	Temperature	140		140						°C	Liquid Specific Gravity									----	Vapor Spec. Gravity (Air=1)									----	Pressure	2.16	2.16	5.11	2.16					kg/cm2(g)	Liq Critical Press (Pseudo)									kg/cm2(a)	Liquid Vapor Pressure									kg/cm2(a)	Liquid Viscosity									cP	Compressibility Z										Specific Heats Cp/Cv (Real)										Estimate Sound Press Level									dBa
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VALVE BODY & BONNET	Body Type: Eccentric Disc § Body Size: 4"## Port Type: Single Inlet Conn Size: 4"## Type: Flanged Class: 300 Outlet Conn Size: 4"## Type: Flanged Class: 300 Flange Facing: Flange Face Finish: Body & Bonnet Material: 316 Stainless Steel ‡ Bonnet Style: Packing Style: Manufacturer's Recommendation Packing Material: Manufacturer's Recommendation Body/Bonnet Gasket Material: Lub/Iso Valve Material: Lub Material: Manufacturer: Metso Automation (or equal) Model Number:	TRIM	Approximate Valve C _v : 510 F _L : x _T : Trim Style: Man. Standard Trim Size: Inherent Characteristic: Equal Percentage Balance Style: Full Open Position: 90° Guiding Style: Shaft Closure Member Material: 316 Stainless Steel Seat Ring Material: Incoloy 825 with Hard Chrome Seat Ring Retainer Material: 316 Stainless Steel Bearing Material: Stellite Bushing/Cage Material: Stem/Shaft Material: 17-4PH Flow Direction: Seat Leakage Class: ANSI Class V																																																																																																																																																														
	ACTUATOR		INSTALLATION	Type: Spring Return Piston, Pneumatic Action Style: Fail Open Service Type: On-Off Max Shutoff Press: In: 21.00 Out: 0.70 kg/cm2(g) Maximum Stroke Time to Open: 5 Seconds Maximum Stroke Time to Close: 5 Seconds Stroke Cycle Service:	Inlet Pipe Nominal Size: Est Wall Thick: Outlet Pipe Nominal Size: Est Wall Thick: Est Insulation Type: Est Insulation Thickness: Installation: Design Pressure : 21 kg/cm2g (F.V) Design Temperature : 250 C																																																																																																																																																												
				§ The information is based on Neldisc L6 series. Valve selected shall be of double flanged construction with minimal flange-to-flange dimension and no pockets with the valve body.																																																																																																																																																													
				## Line size valve.																																																																																																																																																													
				‡ Preferred material is L grade stainless steel. Standard grade stainless steel that is not welded may be used for process wetted parts including the body. Stainless steel with weld repair is acceptable if low carbon (<= 0.03 %) filler material is used.																																																																																																																																																													
				Valve will normally be in the closed position. Valve opening will be intermittent and generally short-term.																																																																																																																																																													

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CONTROL VALVES

PROJECT SPECIFICATION

963766 — 616

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Item Number: 56 Quantity: 1 Valve Tag Number: UXV-23-1403 Data Page: 2 of 2																																																																																	
<table border="1"> <tr> <td rowspan="4">TRANSDUCER</td> <td>Transducer Tag Number:</td> <td rowspan="4">HANDWHEEL</td> <td>Handwheel Position:</td> </tr> <tr> <td>Transducer Input Signal Range:</td> <td>Usage:</td> </tr> <tr> <td>Transducer Output Signal Range:</td> <td></td> </tr> <tr> <td>Manufacturer:</td> <td></td> </tr> </table>	TRANSDUCER	Transducer Tag Number:	HANDWHEEL	Handwheel Position:	Transducer Input Signal Range:	Usage:	Transducer Output Signal Range:		Manufacturer:		<table border="1"> <tr> <td rowspan="14">SOLENOID VALVE</td> <td>Solenoid Valve Tag Number: UXY-23-1403</td> <td rowspan="14">POSITION SWITCH</td> <td>Open Position Switch Tag Number:</td> </tr> <tr> <td>Body Type: 3-Way Body Material: Brass</td> <td>Close Position Switch Tag Number:</td> </tr> <tr> <td>Seat Material: Stainless Steel Disc Material: Stainless Steel</td> <td>Sensor Type:</td> </tr> <tr> <td>Connection Size/Type: †</td> <td>Contact Form:</td> </tr> <tr> <td>Packing Style: Packless</td> <td>Open Position Switch Contact Action:</td> </tr> <tr> <td>Fluid: Instrument Air</td> <td>Close Position Switch Contact Action:</td> </tr> <tr> <td>Pressure: Contractor to determine</td> <td>Contact Load: Contact Rating:</td> </tr> <tr> <td>Minimum Pressure Port Orifice C_v: MAX Size: MAX</td> <td>Mounting Location:</td> </tr> <tr> <td>Minimum Exhaust Port Orifice C_v: MAX Size: MAX</td> <td>Manufacturer:</td> </tr> <tr> <td>Operation Style: †</td> <td rowspan="4">POSITION TRANSMITTER</td> <td>Position Transmitter Tag Number:</td> </tr> <tr> <td>Coil Configuration: Single</td> <td>Sensor Type:</td> </tr> <tr> <td>Operating Voltage: \$\$ Frequency:</td> <td>Output Signal Range:</td> </tr> <tr> <td>Action Coil 1 Deenergized: Vent air from valve</td> <td>Travel Measurement:</td> </tr> <tr> <td>Action Coil 1 Energized: Supply air to valve</td> <td rowspan="2">LOCK-UP VALVE</td> <td>Mounting Location:</td> </tr> <tr> <td>Action Coil 2 Deenergized:</td> <td>Manufacturer:</td> </tr> <tr> <td>Action Coil 2 Energized:</td> <td rowspan="3">QUICK EXHAUST</td> <td>Action Style:</td> </tr> <tr> <td>Manual Reset:</td> <td>Mounting Location:</td> </tr> <tr> <td>Manual Operator:</td> <td>Model Number:</td> </tr> <tr> <td>Mounting Location: Nearest item to actuator in air line. †</td> <td rowspan="3">VOLUME BOOSTER</td> <td>Manufacturer:</td> </tr> <tr> <td>Manufacturer: ASCO (or equal)</td> <td>Connection Size/Type:</td> </tr> <tr> <td>Positioner Tag Number:</td> <td>Body Material:</td> </tr> <tr> <td>Positioner Type:</td> <td>Diaphragm Material:</td> </tr> <tr> <td>Positioner Input Signal Range:</td> <td>Exhaust Action:</td> </tr> <tr> <td>Positioner Output Signal Range:</td> <td>Mounting Location:</td> </tr> <tr> <td>Manufacturer:</td> <td>Manufacturer:</td> </tr> <tr> <td>Quantity: one</td> <td rowspan="4">TUBING</td> <td>Connection Size/Type:</td> </tr> <tr> <td>Type: Filter/Regulator</td> <td>Body Material:</td> </tr> <tr> <td>Manufacturer:</td> <td>Seat Ring Material:</td> </tr> <tr> <td></td> <td>Diaphragm Material:</td> </tr> <tr> <td>Tubing Size:</td> <td rowspan="2">DATA PAGE 2 REMARKS</td> <td>Mounting Location:</td> </tr> <tr> <td>Tubing Material:</td> <td>Manufacturer:</td> </tr> </table>	SOLENOID VALVE	Solenoid Valve Tag Number: UXY-23-1403	POSITION SWITCH	Open Position Switch Tag Number:	Body Type: 3-Way Body Material: Brass	Close Position Switch Tag Number:	Seat Material: Stainless Steel Disc Material: Stainless Steel	Sensor Type:	Connection Size/Type: †	Contact Form:	Packing Style: Packless	Open Position Switch Contact Action:	Fluid: Instrument Air	Close Position Switch Contact Action:	Pressure: Contractor to determine	Contact Load: Contact Rating:	Minimum Pressure Port Orifice C _v : MAX Size: MAX	Mounting Location:	Minimum Exhaust Port Orifice C _v : MAX Size: MAX	Manufacturer:	Operation Style: †	POSITION TRANSMITTER	Position Transmitter Tag Number:	Coil Configuration: Single	Sensor Type:	Operating Voltage: \$\$ Frequency:	Output Signal Range:	Action Coil 1 Deenergized: Vent air from valve	Travel Measurement:	Action Coil 1 Energized: Supply air to valve	LOCK-UP VALVE	Mounting Location:	Action Coil 2 Deenergized:	Manufacturer:	Action Coil 2 Energized:	QUICK EXHAUST	Action Style:	Manual Reset:	Mounting Location:	Manual Operator:	Model Number:	Mounting Location: Nearest item to actuator in air line. †	VOLUME BOOSTER	Manufacturer:	Manufacturer: ASCO (or equal)	Connection Size/Type:	Positioner Tag Number:	Body Material:	Positioner Type:	Diaphragm Material:	Positioner Input Signal Range:	Exhaust Action:	Positioner Output Signal Range:	Mounting Location:	Manufacturer:	Manufacturer:	Quantity: one	TUBING	Connection Size/Type:	Type: Filter/Regulator	Body Material:	Manufacturer:	Seat Ring Material:		Diaphragm Material:	Tubing Size:	DATA PAGE 2 REMARKS	Mounting Location:	Tubing Material:	Manufacturer:
TRANSDUCER		Transducer Tag Number:		HANDWHEEL	Handwheel Position:																																																																												
		Transducer Input Signal Range:			Usage:																																																																												
		Transducer Output Signal Range:																																																																															
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SOLENOID VALVE	Solenoid Valve Tag Number: UXY-23-1403	POSITION SWITCH	Open Position Switch Tag Number:																																																																														
	Body Type: 3-Way Body Material: Brass		Close Position Switch Tag Number:																																																																														
	Seat Material: Stainless Steel Disc Material: Stainless Steel		Sensor Type:																																																																														
	Connection Size/Type: †		Contact Form:																																																																														
	Packing Style: Packless		Open Position Switch Contact Action:																																																																														
	Fluid: Instrument Air		Close Position Switch Contact Action:																																																																														
	Pressure: Contractor to determine		Contact Load: Contact Rating:																																																																														
	Minimum Pressure Port Orifice C _v : MAX Size: MAX		Mounting Location:																																																																														
	Minimum Exhaust Port Orifice C _v : MAX Size: MAX		Manufacturer:																																																																														
	Operation Style: †		POSITION TRANSMITTER	Position Transmitter Tag Number:																																																																													
	Coil Configuration: Single			Sensor Type:																																																																													
	Operating Voltage: \$\$ Frequency:			Output Signal Range:																																																																													
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	Action Coil 1 Energized: Supply air to valve		LOCK-UP VALVE	Mounting Location:																																																																													
Action Coil 2 Deenergized:	Manufacturer:																																																																																
Action Coil 2 Energized:	QUICK EXHAUST	Action Style:																																																																															
Manual Reset:		Mounting Location:																																																																															
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Positioner Type:	Diaphragm Material:																																																																																
Positioner Input Signal Range:	Exhaust Action:																																																																																
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Tubing Size:	DATA PAGE 2 REMARKS	Mounting Location:																																																																															
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† The solenoid valve shall be mounted and tubed by the control valve vendor. The solenoid valve size and type, the tubing size, the actuator air connection size, and the air/filter regulator capacity shall be determined by the control valve vendor to meet the maximum stroking time requirements specified for the control valve. \$\$ The Contractor shall coordinate the solenoid valve operating voltage and coil watt rating with the supplier of the Decomposition Section Shutdown System (UC-2302). See UOP Project Specification 660, this project.																																																																																	

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CONTROL VALVES

PROJECT SPECIFICATION

963766 — 616

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

GENERAL	Item Number: 57 Quantity: 1 Valve Tag Number: UXV-23-1402 Data Page: 1 of 2																																																																																																																																																																
	Service: Decomposer Product to Crude Product Cooler Shutoff Process Material: Hydrocarbon with aromatics P&ID Drawing: 14A																																																																																																																																																																
SIZING CONDITIONS	<table border="1"> <thead> <tr> <th>Process Data Case</th> <th colspan="2">General</th> <th colspan="2">General</th> <th colspan="2"></th> <th colspan="2"></th> <th></th> </tr> <tr> <th>Sizing Case</th> <th colspan="2">Open</th> <th colspan="2">Closed</th> <th colspan="2"></th> <th colspan="2"></th> <th></th> </tr> <tr> <th>Calculated C_v</th> <th>Inlet</th> <th>Outlet</th> <th>Inlet</th> <th>Outlet</th> <th>Inlet</th> <th>Outlet</th> <th>Inlet</th> <th>Outlet</th> <th>Units</th> </tr> </thead> <tbody> <tr> <td>Liquid Flow at Flowing Cond</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>m3/h</td> </tr> <tr> <td>Vapor Flow at Base Cond</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>normal m3/h</td> </tr> <tr> <td>Steam Flow</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>kg/h</td> </tr> <tr> <td>Temperature</td> <td>140</td> <td></td> <td>140</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>°C</td> </tr> <tr> <td>Liquid Specific Gravity</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>----</td> </tr> <tr> <td>Vapor Spec. Gravity (Air=1)</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>----</td> </tr> <tr> <td>Pressure</td> <td>5.11</td> <td>5.11</td> <td>2.16</td> <td>5.10</td> <td></td> <td></td> <td></td> <td></td> <td>kg/cm2(g)</td> </tr> <tr> <td>Liq Critical Press (Pseudo)</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>kg/cm2(a)</td> </tr> <tr> <td>Liquid Vapor Pressure</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>kg/cm2(a)</td> </tr> <tr> <td>Liquid Viscosity</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>cP</td> </tr> <tr> <td>Compressibility Z</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> <tr> <td>Specific Heats Cp/Cv (Real)</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> <tr> <td>Estimate Sound Press Level</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>dBa</td> </tr> </tbody> </table>	Process Data Case	General		General							Sizing Case	Open		Closed							Calculated C _v	Inlet	Outlet	Inlet	Outlet	Inlet	Outlet	Inlet	Outlet	Units	Liquid Flow at Flowing Cond									m3/h	Vapor Flow at Base Cond									normal m3/h	Steam Flow									kg/h	Temperature	140		140						°C	Liquid Specific Gravity									----	Vapor Spec. Gravity (Air=1)									----	Pressure	5.11	5.11	2.16	5.10					kg/cm2(g)	Liq Critical Press (Pseudo)									kg/cm2(a)	Liquid Vapor Pressure									kg/cm2(a)	Liquid Viscosity									cP	Compressibility Z										Specific Heats Cp/Cv (Real)										Estimate Sound Press Level									dBa
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VALVE BODY & BONNET	Body Type: Eccentric Disc § Body Size: 4" #1 Port Type: Single Inlet Conn Size: 4" #1 Type: Flanged Class: 300 Outlet Conn Size: 4" #1 Type: Flanged Class: 300 Flange Facing: Flange Face Finish: Body & Bonnet Material: 316 Stainless Steel ‡ Bonnet Style: Packing Style: Manufacturer's Recommendation Packing Material: Manufacturer's Recommendation Body/Bonnet Gasket Material: Lub/Iso Valve Material: Lub Material: Manufacturer: Metso Automation (or equal) Model Number:	TRIM	Approximate Valve C _v : 510 F _L : x _T : Trim Style: Man. Standard Trim Size: Inherent Characteristic: Equal Percentage Balance Style: Full Open Position: 90° Guiding Style: Shaft Closure Member Material: 316 Stainless Steel Seat Ring Material: Incoloy 825 with Hard Chrome Seat Ring Retainer Material: 316 Stainless Steel Bearing Material: Stellite Bushing/Cage Material: Stem/Shaft Material: 17-4PH Flow Direction: Seat Leakage Class: ANSI Class V																																																																																																																																																														
	ACTUATOR		INSTALLATION	Type: Spring Return Piston, Pneumatic Action Style: Fail Close Service Type: On-Off Max Shutoff Press: In: 21.00 Out: 5.10 kg/cm2(g) Maximum Stroke Time to Open: 5 Seconds Maximum Stroke Time to Close: 5 Seconds Stroke Cycle Service:	Inlet Pipe Nominal Size: Est Wall Thick: Outlet Pipe Nominal Size: Est Wall Thick: Est Insulation Type: Est Insulation Thickness: Installation: Design Pressure : 21 kg/cm2g (F.V) Design Temperature : 250 C																																																																																																																																																												
				§ The information is based on Neldisc L6 series. Valve selected shall be of double flanged construction with minimal flange-to-flange dimension and no pockets with the valve body.																																																																																																																																																													
				#1 Line size valve.																																																																																																																																																													
				‡ Preferred material is L grade stainless steel. Standard grade stainless steel that is not welded may be used for process wetted parts including the body. Stainless steel with weld repair is acceptable if low carbon (<= 0.03 %) filler material is used.																																																																																																																																																													

ALL ITEMS SHALL COMPLY WITH THE GENERAL REQUIREMENTS SHOWN ON THE GENERAL REQUIREMENTS SECTION OF THIS SPECIFICATION.

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CONTROL VALVES

PROJECT SPECIFICATION

963766 — 616

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Item Number: 57 Quantity: 1 Valve Tag Number: UXV-23-1402 Data Page: 2 of 2																																																																																												
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CONTROL VALVES

PROJECT SPECIFICATION

963766 — 616

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

GENERAL	Item Number: 58 Quantity: 1 Valve Tag Number: TV-23-1302B Data Page: 1 of 2								
	Service: Reliable Cooling Water bypassing Decomposer Cooler Process Material: Water P&ID Drawing: 17								
SIZING CONDITIONS	Process Data Case	Single Case							
	Sizing Case	Maximum							
	Calculated C _v	844							
		Inlet	Outlet	Inlet	Outlet	Inlet	Outlet	Inlet	Outlet
	Liquid Flow at Flowing Cond	1057.6							Units
	Vapor Flow at Base Cond								m3/h
	Steam Flow								normal m3/h
	Temperature	33							kg/h
	Liquid Specific Gravity	0.996							°C
	Vapor Spec. Gravity (Air=1)								----
	Pressure	4.81	2.68						----
	Liq Critical Press (Pseudo)	224.90							kg/cm2(g)
	Liquid Vapor Pressure	0.05							kg/cm2(a)
	Liquid Viscosity	0.749							kg/cm2(a)
	Compressibility Z								cP
Specific Heats Cp/Cv (Real)									
Estimate Sound Press Level								dBA	
VALVE BODY & BONNET	Body Type: Eccentric Rotary Plug §	Approximate Valve C _v : 1050 F _L : 0.68 x _T : 0.4							
	Body Size: 12" Port Type: Single	Trim Style: Man. Standard Trim Size:							
	Inlet Conn Size: 12" Type: Flanged Class: 150	Inherent Characteristic: Linear							
	Outlet Conn Size: 12" Type: Flanged Class: 150	Balance Style: Full Open Position: 100 %							
	Flange Facing: Raised Face Flange Face Finish:	Guiding Style: Shaft							
	Body & Bonnet Material: Carbon Steel	Closure Member Material: 316 SS, Stellite sheath							
	Bonnet Style: Integral or Bolted	Seat Ring Material: 316 SS with Stellite sheath							
	Packing Style: Manufacturer's Recommendation	Seat Ring Retainer Material: Stellite							
	Packing Material: Manufacturer's Recommendation	Bearing Material: Stellite							
	Body/Bonnet Gasket Material:	Bushing/Cage Material:							
	Lub/Iso Valve Material: Lub Material:	Stem/Shaft Material: 17-4 PH Stainless Steel							
	Manufacturer: Masoneilan (or equal)	Flow Direction: Flow to close							
	Model Number:	Seat Leakage Class: ANSI Class IV							
	ACTUATOR	Type: Spring Return Piston, Pneumatic	Inlet Pipe Nominal Size: Est Wall Thick:						
Action Style: Fail Close		Outlet Pipe Nominal Size: Est Wall Thick:							
Service Type: Modulating		Est Insulation Type:							
Max Shutoff Press: In: 8.16 Out: 0.74 kg/cm2(g)		Est Insulation Thickness:							
Maximum Stroke Time to Open:		Installation:							
Maximum Stroke Time to Close:		Design Pressure : 13 kg/cm2g							
Stroke Cycle Service:		Design Temperature : 120 C							
DATA PAGE 1 REMARKS	§ The information is based on Masoneilan 35002 series.								
	Hardened trim is recommended for this service based upon the possibility of cavitation occurring.								

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CONTROL VALVES

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REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Item Number: 58 Quantity: 1		Valve Tag Number: TV-23-1302B		Data Page: 2 of 2	
TRANS-DUCER	Transducer Tag Number:		HANDWHEEL	Handwheel Position: Manufacturer's Standard	
	Transducer Input Signal Range:			Usage: Manual Operation	
	Transducer Output Signal Range:				
	Manufacturer:				
SOLENOID VALVE	Solenoid Valve Tag Number:		POSITION SWITCH	Open Position Switch Tag Number:	
	Body Type: Body Material:			Close Position Switch Tag Number:	
	Seat Material: Disc Material:			Sensor Type:	
	Connection Size/Type:			Contact Form:	
	Packing Style:			Open Position Switch Contact Action:	
	Fluid:			Close Position Switch Contact Action:	
	Pressure:			Contact Load: Contact Rating:	
	Minimum Pressure Port Orifice C _v : Size:			Mounting Location:	
	Minimum Exhaust Port Orifice C _v : Size:			Manufacturer:	
	Operation Style:			POSITION TRANSMITTER	Position Transmitter Tag Number:
	Coil Configuration:		Sensor Type:		
	Operating Voltage: Frequency:		Output Signal Range:		
	Action Coil 1 Deenergized:		Travel Measurement:		
	Action Coil 1 Energized:		Mounting Location:		
	Action Coil 2 Deenergized:		Manufacturer:		
	Action Coil 2 Energized:		LOCK-UP VALVE	Action Style:	
	Manual Reset:			Mounting Location:	
	Manual Operator:			Model Number:	
	Mounting Location:		QUICK EXHAUST	Manufacturer:	
	Manufacturer:			Connection Size/Type:	
Positioner Tag Number:		Body Material:			
Positioner Type: Digital		Diaphragm Material:			
Positioner Input Signal Range: 4 - 20 mA		VOLUME BOOSTER	Exhaust Action:		
Positioner Output Signal Range: 0 - 100 %			Mounting Location:		
Manufacturer:			Manufacturer:		
AIR FILTER	Quantity: one		Connection Size/Type:		
	Type: Filter/Regulator		Body Material:		
TUBING	Manufacturer:		Seat Ring Material:		
	Tubing Size:		Diaphragm Material:		
Tubing Material:		Mounting Location:			
		Manufacturer:			
DATA PAGE 2 REMARKS					

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CONTROL VALVES

PROJECT SPECIFICATION

963766 — **616**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

GENERAL	Item Number: 59 Quantity: 1		Valve Tag Number: TV-23-1302A				Data Page: 1 of 2				
	Service: Reliable Cooling Water to Decomposer Cooler						P&ID Drawing: 17				
SIZING CONDITIONS	Process Data Case	Single Case									
	Sizing Case	Maximum									
	Calculated C _v	4532									
		Inlet	Outlet	Inlet	Outlet	Inlet	Outlet	Inlet	Outlet	Units	
	Liquid Flow at Flowing Cond	4230.3								m3/h	
	Vapor Flow at Base Cond									normal m3/h	
	Steam Flow									kg/h	
	Temperature	33								°C	
	Liquid Specific Gravity	0.996								----	
	Vapor Spec. Gravity (Air=1)									----	
	Pressure	4.79	3.61							kg/cm2(g)	
	Liq Critical Press (Pseudo)	224.90								kg/cm2(a)	
	Liquid Vapor Pressure	0.05								kg/cm2(a)	
	Liquid Viscosity	0.749								cP	
	Compressibility Z										
	Specific Heats Cp/Cv (Real)										
	Estimate Sound Press Level									dBa	
VALVE BODY & BONNET	Body Type: Eccentric Disc §				Approximate Valve C _v : 13139 ‡		F _L : 0.6		x _T : 0.35		
	Body Size: 20" Port Type: Single				Trim Style: Man. Standard		Trim Size:				
	Inlet Conn Size: 20" Type: Flanged Class: 150				Inherent Characteristic: Equal Percentage						
	Outlet Conn Size: 20" Type: Flanged Class: 150				Balance Style:		Full Open Position: 90°				
	Flange Facing:		Flange Face Finish:		Guiding Style: Shaft						
	Body & Bonnet Material: Carbon Steel				Closure Member Material: 316 Stainless Steel						
	Bonnet Style:				Seat Ring Material: Inconel						
	Packing Style: Manufacturer's Recommendation				Seat Ring Retainer Material: Carbon Steel						
	Packing Material: Manufacturer's Recommendation				Bearing Material: Manufacturer's Recommendation						
	Body/Bonnet Gasket Material:				Bushing/Cage Material:						
	Lub/Iso Valve Material:		Lub Material:		Stem/Shaft Material: 17-4PH Stainless Steel						
	Manufacturer: Masoneilan (or equal)				Flow Direction:						
	Model Number:				Seat Leakage Class: ANSI Class IV						
	ACTUATOR	Type: Spring Diaph, Piston/Double Act Piston, Pneu ‡‡				Inlet Pipe Nominal Size:		Est Wall Thick:			
Action Style: Fail Open				Outlet Pipe Nominal Size:		Est Wall Thick:					
Service Type: Modulating				Est Insulation Type:							
Max Shutoff Press: In: 8.16 Out: 0.74 kg/cm2(g)				Est Insulation Thickness:							
Maximum Stroke Time to Open:				Installation:							
Maximum Stroke Time to Close:				Design Pressure :		13 kg/cm2g					
Stroke Cycle Service:				Design Temperature :		120 C					
DATA PAGE 1 REMARKS	§ The information is based on Masoneilan 39003 series.										
	‡ Incorporates a 0.9112 installation factor.										
	Hardened trim is recommended for this service based upon the possibility of cavitation occurring.										
	‡‡ If a double acting piston actuator is provided for this valve, then a volume tank with the required accessories shall be supplied to move the control valve to the minimum signal position as specified, upon loss of instrument air pressure.										

ALL ITEMS SHALL COMPLY WITH THE GENERAL REQUIREMENTS SHOWN ON THE GENERAL REQUIREMENTS SECTION OF THIS SPECIFICATION.

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CONTROL VALVES

PROJECT SPECIFICATION

963766 — 616

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Item Number: 59 Quantity: 1		Valve Tag Number: TV-23-1302A		Data Page: 2 of 2	
TRANS-DUCER	Transducer Tag Number:		HANDWHEEL	Handwheel Position: Manufacturer's Standard	
	Transducer Input Signal Range:			Usage: Manual Operation	
	Transducer Output Signal Range:				
	Manufacturer:				
SOLENOID VALVE	Solenoid Valve Tag Number:		POSITION SWITCH	Open Position Switch Tag Number:	
	Body Type: Body Material:			Close Position Switch Tag Number:	
	Seat Material: Disc Material:			Sensor Type:	
	Connection Size/Type:			Contact Form:	
	Packing Style:			Open Position Switch Contact Action:	
	Fluid:			Close Position Switch Contact Action:	
	Pressure:			Contact Load: Contact Rating:	
	Minimum Pressure Port Orifice C _v : Size:			Mounting Location:	
	Minimum Exhaust Port Orifice C _v : Size:			Manufacturer:	
	Operation Style:			POSITION TRANSMITTER	Position Transmitter Tag Number:
	Coil Configuration:		Sensor Type:		
	Operating Voltage: Frequency:		Output Signal Range:		
	Action Coil 1 Deenergized:		Travel Measurement:		
	Action Coil 1 Energized:		Mounting Location:		
	Action Coil 2 Deenergized:		LOCK-UP VALVE	Manufacturer:	
	Action Coil 2 Energized:			Action Style:	
	Manual Reset:			Mounting Location:	
	Manual Operator:		QUICK EXHAUST	Model Number:	
	Mounting Location:			Manufacturer:	
	Manufacturer:			Connection Size/Type:	
AIR FILTER	Positioner Tag Number:		VOLUME BOOSTER	Body Material:	
	Positioner Type: Digital			Diaphragm Material:	
	Positioner Input Signal Range: 4 - 20 mA			Exhaust Action:	
TUBING	Positioner Output Signal Range: 0 - 100 %			Mounting Location:	
	Manufacturer:			Manufacturer:	
	Quantity: one			Connection Size/Type:	
DATA PAGE 2 REMARKS	Type: Filter/Regulator			Body Material:	
	Manufacturer:			Seat Ring Material:	
	Tubing Size:			Diaphragm Material:	
	Tubing Material:			Mounting Location:	
				Manufacturer:	

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CONTROL VALVES

PROJECT SPECIFICATION

963766 — **616**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				
1	28AUG12	MVD	MW				

GENERAL	Item Number: 60 Quantity: 2 Valve Tag Number: FV-23-1801, 1802 Data Page: 1 of 2																																																																																																																																																																										
	Service: Circulating Decomposer Liquid to Calorimeter Process Material: Hydrocarbon with aromatics P&ID Drawing: 18																																																																																																																																																																										
SIZING CONDITIONS	<table border="1"> <tr> <th>Process Data Case</th> <th colspan="2">Single Case</th> <th colspan="2"></th> <th colspan="2"></th> <th colspan="2"></th> <th></th> </tr> <tr> <td>Sizing Case</td> <td colspan="2">Normal</td> <td colspan="2"></td> <td colspan="2"></td> <td colspan="2"></td> <td></td> </tr> <tr> <td>Calculated C_v</td> <td colspan="2">0.132</td> <td colspan="2"></td> <td colspan="2"></td> <td colspan="2"></td> <td></td> </tr> <tr> <td></td> <td>Inlet</td> <td>Outlet</td> <td>Inlet</td> <td>Outlet</td> <td>Inlet</td> <td>Outlet</td> <td>Inlet</td> <td>Outlet</td> <td>Units</td> </tr> <tr> <td>Liquid Flow at Flowing Cond</td> <td>0.2</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>m³/h</td> </tr> <tr> <td>Vapor Flow at Base Cond</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>normal m³/h</td> </tr> <tr> <td>Steam Flow</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>kg/h</td> </tr> <tr> <td>Temperature</td> <td>79</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>°C</td> </tr> <tr> <td>Liquid Specific Gravity</td> <td>0.898</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>----</td> </tr> <tr> <td>Vapor Spec. Gravity (Air=1)</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>----</td> </tr> <tr> <td>Pressure</td> <td>3.72</td> <td>2.10</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>kg/cm²(g)</td> </tr> <tr> <td>Liq Critical Press (Pseudo)</td> <td>59.64</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>kg/cm²(a)</td> </tr> <tr> <td>Liquid Vapor Pressure</td> <td>1.41</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>kg/cm²(a)</td> </tr> <tr> <td>Liquid Viscosity</td> <td>0.571</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>cP</td> </tr> <tr> <td>Compressibility Z</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> <tr> <td>Specific Heats Cp/Cv (Real)</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> <tr> <td>Estimate Sound Press Level</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>dBA</td> </tr> </table>	Process Data Case	Single Case									Sizing Case	Normal									Calculated C _v	0.132										Inlet	Outlet	Inlet	Outlet	Inlet	Outlet	Inlet	Outlet	Units	Liquid Flow at Flowing Cond	0.2								m ³ /h	Vapor Flow at Base Cond									normal m ³ /h	Steam Flow									kg/h	Temperature	79								°C	Liquid Specific Gravity	0.898								----	Vapor Spec. Gravity (Air=1)									----	Pressure	3.72	2.10							kg/cm ² (g)	Liq Critical Press (Pseudo)	59.64								kg/cm ² (a)	Liquid Vapor Pressure	1.41								kg/cm ² (a)	Liquid Viscosity	0.571								cP	Compressibility Z										Specific Heats Cp/Cv (Real)										Estimate Sound Press Level									dBA
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Estimate Sound Press Level									dBA																																																																																																																																																																		
VALVE BODY & BONNET	Body Type: Globe § Body Size: 1/2" Port Type: Single Inlet Conn Size: 1/2" Type: Flanged Class: 300 Outlet Conn Size: 1/2" Type: Flanged Class: 300 Flange Facing: Raised Face Flange Face Finish: Body & Bonnet Material: 316L Stainless Steel §§ Bonnet Style: Integral or Bolted Packing Style: Manufacturer's Recommendation Packing Material: Manufacturer's Recommendation Body/Bonnet Gasket Material: Lub/Iso Valve Material: Lub Material: Manufacturer: Masoneilan (or equal) Model Number:	TRIM	Approximate Valve C _v : 0.25 F _L : 0.9 x _T : 0.72 Trim Style: Man. Standard Trim Size: No. 4 Inherent Characteristic: Linear Balance Style: Unbalanced Full Open Position: 100 % Guiding Style: Top Closure Member Material: Stellite 1 Seat Ring Material: 316 Stainless Steel 1 Seat Ring Retainer Material: 316 Stainless Steel Bearing Material: 1 Bushing/Cage Material: 316 Stainless Steel Stem/Shaft Material: 316 Stainless Steel Flow Direction: Seat Leakage Class: ANSI Class IV																																																																																																																																																																								
	ACTUATOR		INSTALLATION	Type: Spring Diaphragm, Pneumatic Action Style: Fail Open Service Type: Modulating Max Shutoff Press: In: 16.50 Out: 1.38 kg/cm²(g) Maximum Stroke Time to Open: Maximum Stroke Time to Close: Stroke Cycle Service:	Inlet Pipe Nominal Size: Est Wall Thick: Outlet Pipe Nominal Size: Est Wall Thick: Est Insulation Type: Est Insulation Thickness: Installation: Design Pressure : 16.5 kg/cm²(g) (F.V) Design Temperature : 250 C																																																																																																																																																																						
				§ The information is based on Masoneilan 28000 series.																																																																																																																																																																							
				§§ Preferred material is L grade stainless steel. Standard grade stainless steel that is not welded may be used for process wetted parts including the body. Stainless steel with weld repair is acceptable if low carbon (<= 0.03 %) filler material is used.																																																																																																																																																																							

ALL ITEMS SHALL COMPLY WITH THE GENERAL REQUIREMENTS SHOWN ON THE GENERAL REQUIREMENTS SECTION OF THIS SPECIFICATION.

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CONTROL VALVES

PROJECT SPECIFICATION

963766 — 616

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Item Number: 60 Quantity: 2		Valve Tag Number: FV-23-1801, 1802		Data Page: 2 of 2		
TRANS-DUCER	Transducer Tag Number:		HANDWHEEL	Handwheel Position:		
	Transducer Input Signal Range:			Usage:		
	Transducer Output Signal Range:					
	Manufacturer:					
SOLENOID VALVE	Solenoid Valve Tag Number:		POSITION SWITCH	Open Position Switch Tag Number:		
	Body Type: Body Material:			Close Position Switch Tag Number:		
	Seat Material: Disc Material:			Sensor Type:		
	Connection Size/Type:			Contact Form:		
	Packing Style:			Open Position Switch Contact Action:		
	Fluid:			Close Position Switch Contact Action:		
	Pressure:			Contact Load: Contact Rating:		
	Minimum Pressure Port Orifice C _v : Size:			Mounting Location:		
	Minimum Exhaust Port Orifice C _v : Size:			Manufacturer:		
	Operation Style:					
	Coil Configuration:		POSITION TRANSMITTER	Position Transmitter Tag Number:		
	Operating Voltage: Frequency:			Sensor Type:		
	Action Coil 1 Deenergized:			Output Signal Range:		
	Action Coil 1 Energized:			Travel Measurement:		
	Action Coil 2 Deenergized:			Mounting Location:		
	Action Coil 2 Energized:			Manufacturer:		
	Manual Reset:		LOCK-UP VALVE	Action Style:		
	Manual Operator:			Mounting Location:		
Mounting Location:		Model Number:				
Manufacturer:		Manufacturer:				
POSITIONER	Positioner Tag Number:		QUICK EXHAUST	Connection Size/Type:		
	Positioner Type: Digital			Body Material:		
	Positioner Input Signal Range: 4 - 20 mA			Diaphragm Material:		
	Positioner Output Signal Range: 0 - 100 %			Exhaust Action:		
Manufacturer:		Mounting Location:				
AIR FILTER	Quantity: one		VOLUME BOOSTER	Connection Size/Type:		
	Type: Filter/Regulator			Body Material:		
	Manufacturer:			Seat Ring Material:		
TUBING	Tubing Size:			Diaphragm Material:		
	Tubing Material:		Mounting Location:			
Manufacturer:						
DATA PAGE 2 REMARKS						

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PROJECT SPECIFICATION

963766 — **616**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

GENERAL	Item Number: 61 Quantity: 1		Valve Tag Number: PV-23-1401		Data Page: 1 of 2	
	Service: Crude Product to Fractionation Feed Tanks at Fractionation Section					
	Process Material: Hydrocarbon with aromatics			P&ID Drawing: 19		
SIZING CONDITIONS	Process Data Case	Single Case		Single Case		Single Case
	Sizing Case	Normal		Maximum †		Minimum
	Calculated C _v	64		92		26.3
		Inlet	Outlet	Inlet	Outlet	Inlet
	Liquid Flow at Flowing Cond	68.1		89.9		34.0
	Vapor Flow at Base Cond					
	Steam Flow					
	Temperature	43		43		43
	Liquid Specific Gravity	0.930		0.930		0.930
	Vapor Spec. Gravity (Air=1)					
	Pressure	3.06	1.41	2.63	1.42	4.59
	Liq Critical Press (Pseudo)	61.12		61.12		61.12
	Liquid Vapor Pressure	1.13		1.13		1.13
	Liquid Viscosity	1.005		1.005		1.005
	Compressibility Z					
	Specific Heats Cp/Cv (Real)					
	Estimate Sound Press Level					dBA
VALVE BODY & BONNET	Body Type: Eccentric Rotary Plug §					Approximate Valve C _v : 135
	Body Size: 3" Port Type: Single					F _L : 0.68 x _T : 0.4
	Inlet Conn Size: 3" Type: Flanged Class: 300					Trim Style: Man. Standard Trim Size:
	Outlet Conn Size: 3" Type: Flanged Class: 300					Inherent Characteristic: Linear
	Flange Facing: Raised Face Flange Face Finish:					Balance Style: Full Open Position: 100 %
	Body & Bonnet Material: 316 Stainless Steel ‡					Guiding Style: Shaft
	Bonnet Style: Integral or Bolted					Closure Member Material: 316 SS, Stellite sheath
	Packing Style: Manufacturer's Recommendation					Seat Ring Material: 316 SS with Stellite sheath
	Packing Material: Manufacturer's Recommendation					Seat Ring Retainer Material: Stellite
	Body/Bonnet Gasket Material:					Bearing Material: Stellite
	Lub/Iso Valve Material: Lub Material:					Bushing/Cage Material:
	Manufacturer: Masoneilan (or equal)					Stem/Shaft Material: 316 Stainless Steel
	Model Number:					Flow Direction: Flow to close
						Seat Leakage Class: ANSI Class IV
ACTUATOR	Type: Spring Diaphragm, Pneumatic					Inlet Pipe Nominal Size: Est Wall Thick:
	Action Style: Fail Close					Outlet Pipe Nominal Size: Est Wall Thick:
	Service Type: Modulating					Est Insulation Type:
	Max Shutoff Press: In: 21.00 Out: 0 kg/cm2(g)					Est Insulation Thickness:
	Maximum Stroke Time to Open:					Installation:
	Maximum Stroke Time to Close:					Design Pressure : 21 kg/cm2g (F.V)
	Stroke Cycle Service:					Design Temperature : 250 C
DATA PAGE 1 REMARKS	§ The information is based on Masoneilan 35002 series.					
	Hardened trim is recommended for this service based upon the possibility of cavitation occurring.					
	‡ Preferred material is L grade stainless steel. Standard grade stainless steel that is not welded may be used for process wetted parts including the body. Stainless steel with weld repair is acceptable if low carbon (<= 0.03 %) filler material is used.					
	† Maximum sizing case includes 60% of sump pump rated capacity for neutralizing/reprocessing sump material.					

ALL ITEMS SHALL COMPLY WITH THE GENERAL REQUIREMENTS SHOWN ON THE GENERAL REQUIREMENTS SECTION OF THIS SPECIFICATION.

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CONTROL VALVES

PROJECT SPECIFICATION

963766 — **616**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Item Number: 61 Quantity: 1		Valve Tag Number: PV-23-1401		Data Page: 2 of 2	
TRANS-DUCER	Transducer Tag Number:		HANDWHEEL	Handwheel Position:	
	Transducer Input Signal Range:			Usage:	
	Transducer Output Signal Range:				
	Manufacturer:				
SOLENOID VALVE	Solenoid Valve Tag Number:		POSITION SWITCH	Open Position Switch Tag Number:	
	Body Type: Body Material:			Close Position Switch Tag Number:	
	Seat Material: Disc Material:			Sensor Type:	
	Connection Size/Type:			Contact Form:	
	Packing Style:			Open Position Switch Contact Action:	
	Fluid:			Close Position Switch Contact Action:	
	Pressure:			Contact Load: Contact Rating:	
	Minimum Pressure Port Orifice C_v : Size:			Mounting Location:	
	Minimum Exhaust Port Orifice C_v : Size:			Manufacturer:	
	Operation Style:			POSITION TRANSMITTER	Position Transmitter Tag Number:
	Coil Configuration:		Sensor Type:		
	Operating Voltage: Frequency:		Output Signal Range:		
	Action Coil 1 Deenergized:		Travel Measurement:		
	Action Coil 1 Energized:		Mounting Location:		
	Action Coil 2 Deenergized:		Manufacturer:		
	Action Coil 2 Energized:		LOCK-UP VALVE	Action Style:	
	Manual Reset:			Mounting Location:	
	Manual Operator:			Model Number:	
	Mounting Location:		QUICK EXHAUST	Manufacturer:	
	Manufacturer:			Connection Size/Type:	
Positioner Tag Number:		Body Material:			
Positioner Type: Digital		Diaphragm Material:			
Positioner Input Signal Range: 4 - 20 mA		VOLUME BOOSTER	Exhaust Action:		
Positioner Output Signal Range: 0 - 100 %			Mounting Location:		
Manufacturer:			Manufacturer:		
AIR FILTER	Quantity: one		Connection Size/Type:		
	Type: Filter/Regulator		Body Material:		
TUBING	Manufacturer:		Seat Ring Material:		
	Tubing Size:		Diaphragm Material:		
Tubing Material:		Mounting Location:			
		Manufacturer:			
DATA PAGE 2 REMARKS					

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CONTROL VALVES

PROJECT SPECIFICATION

963766 — 616

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

GENERAL	Item Number: 62 Quantity: 1 Valve Tag Number: FV-23-2001 Data Page: 1 of 2									
	Service: Acid Aromatics to Direct Neutralization Static Mixers Process Material: Hydrocarbon with aromatics P&ID Drawing: 20A									
SIZING CONDITIONS	Process Data Case	Single Case		Single Case		Single Case				
	Sizing Case	Normal		Maximum		Minimum				
	Calculated C _v	19.5		22.5		8.4				
		Inlet	Outlet	Inlet	Outlet	Inlet	Outlet	Inlet	Outlet	Units
	Liquid Flow at Flowing Cond	22.8		25.1		11.4				m ³ /h
	Vapor Flow at Base Cond									normal m ³ /h
	Steam Flow									kg/h
	Temperature	38		38		38				°C
	Liquid Specific Gravity	0.939		0.939		0.939				----
	Vapor Spec. Gravity (Air=1)									----
	Pressure	6.16	4.41	5.86	4.26	7.28	4.92			kg/cm ² (g)
	Liq Critical Press (Pseudo)	59.64		59.64		59.64				kg/cm ² (a)
	Liquid Vapor Pressure	1.18		1.18		1.18				kg/cm ² (a)
	Liquid Viscosity	1.160		1.160		1.160				cP
	Compressibility Z									
Specific Heats Cp/Cv (Real)										
Estimate Sound Press Level									dBA	
VALVE BODY & BONNET	Body Type: Globe §					Approximate Valve C _v : 46 F _L : 0.9 x _T : 0.72				
	Body Size: 2" Port Type: Single					Trim Style: Man. Standard Trim Size:				
	Inlet Conn Size: 2" Type: Flanged Class: 300					Inherent Characteristic: Linear				
	Outlet Conn Size: 2" Type: Flanged Class: 300					Balance Style: Unbalanced Full Open Position: 100 %				
	Flange Facing: Raised Face Flange Face Finish:					Guiding Style: Top				
	Body & Bonnet Material: 316 Stainless Steel ‡					Closure Member Material: 316 Stainless Steel				
	Bonnet Style: Integral or Bolted					Seat Ring Material: 316 Stainless Steel				
	Packing Style: Manufacturer's Recommendation					Seat Ring Retainer Material: 304 Stainless Steel				
	Packing Material: Manufacturer's Recommendation					Bearing Material:				
	Body/Bonnet Gasket Material: Manufacturer's Recommend					Bushing/Cage Material: Stellite				
	Lub/Iso Valve Material: Lub Material:					Stem/Shaft Material: 316 Stainless Steel				
	Manufacturer: Masoneilan (or equal)					Flow Direction:				
	Model Number:					Seat Leakage Class: ANSI Class IV				
	ACTUATOR	Type: Spring Diaphragm, Pneumatic					Inlet Pipe Nominal Size: Est Wall Thick:			
Action Style: Fail Close					Outlet Pipe Nominal Size: Est Wall Thick:					
Service Type: Modulating					Est Insulation Type:					
Max Shutoff Press: In: 13.06 Out: 4.34 kg/cm ² (g)					Est Insulation Thickness:					
Maximum Stroke Time to Open:					Installation:					
Maximum Stroke Time to Close:					Design Pressure : 21 kg/cm ² g					
Stroke Cycle Service:					Design Temperature : 250 C					
DATA PAGE 1	§ The information is based on Masoneilan 21000 series. ‡ Preferred material is L grade stainless steel. Standard grade stainless steel that is not welded may be used for process wetted parts including the body. Stainless steel with weld repair is acceptable if low carbon (<= 0.03 %) filler material is used.									

ALL ITEMS SHALL COMPLY WITH THE GENERAL REQUIREMENTS SHOWN ON THE GENERAL REQUIREMENTS SECTION OF THIS SPECIFICATION.

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CONTROL VALVES

PROJECT SPECIFICATION

963766 — 616

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Item Number: 62 Quantity: 1		Valve Tag Number: FV-23-2001		Data Page: 2 of 2	
TRANS-DUCER	Transducer Tag Number:		HANDWHEEL	Handwheel Position:	
	Transducer Input Signal Range:			Usage:	
	Transducer Output Signal Range:				
	Manufacturer:				
SOLENOID VALVE	Solenoid Valve Tag Number:		POSITION SWITCH	Open Position Switch Tag Number:	
	Body Type: Body Material:			Close Position Switch Tag Number:	
	Seat Material: Disc Material:			Sensor Type:	
	Connection Size/Type:			Contact Form:	
	Packing Style:			Open Position Switch Contact Action:	
	Fluid:			Close Position Switch Contact Action:	
	Pressure:			Contact Load: Contact Rating:	
	Minimum Pressure Port Orifice C _v : Size:			Mounting Location:	
	Minimum Exhaust Port Orifice C _v : Size:			Manufacturer:	
	Operation Style:				
	Coil Configuration:		POSITION TRANSMITTER	Position Transmitter Tag Number:	
	Operating Voltage: Frequency:			Sensor Type:	
	Action Coil 1 Deenergized:			Output Signal Range:	
	Action Coil 1 Energized:			Travel Measurement:	
	Action Coil 2 Deenergized:			Mounting Location:	
	Action Coil 2 Energized:			Manufacturer:	
	Manual Reset:		LOCK-UP VALVE	Action Style:	
	Manual Operator:			Mounting Location:	
Mounting Location:		Model Number:			
Manufacturer:		Manufacturer:			
POSITIONER	Positioner Tag Number:		QUICK EXHAUST	Connection Size/Type:	
	Positioner Type: Digital			Body Material:	
	Positioner Input Signal Range: 4 - 20 mA			Diaphragm Material:	
	Positioner Output Signal Range: 0 - 100 %			Exhaust Action:	
	Manufacturer:			Mounting Location:	
AIR FILTER	Quantity: one		VOLUME BOOSTER	Connection Size/Type:	
	Type: Filter/Regulator			Body Material:	
	Manufacturer:			Seat Ring Material:	
TUBING	Tubing Size:			Diaphragm Material:	
	Tubing Material:			Mounting Location:	
			Manufacturer:		
DATA PAGE 2 REMARKS					

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CONTROL VALVES

PROJECT SPECIFICATION

963766 — 616

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				
1	28AUG12	MVD	MW				

GENERAL		Item Number: 63 Quantity: 1		Valve Tag Number: PCV-13-2001				Data Page: 1 of 2				
		Service: Nitrogen Purge to Acid Aromatics Sump										
		Process Material: Nitrogen				P&ID Drawing: 20A						
SIZING CONDITIONS	Process Data Case		General									
	Sizing Case		Normal									
	Calculated C _v		0.058									
			Inlet	Outlet	Inlet	Outlet	Inlet	Outlet	Inlet	Outlet	Units	
	Liquid Flow at Flowing Cond										m3/h	
	Vapor Flow at Base Cond		6								normal m3/h	
	Steam Flow										kg/h	
	Temperature		38								°C	
	Liquid Specific Gravity										----	
	Vapor Spec. Gravity (Air=1)		0.967								----	
	Pressure		7.00	0.84							kg/cm2(g)	
	Liq Critical Press (Pseudo)										kg/cm2(a)	
	Liquid Vapor Pressure										kg/cm2(a)	
	Liquid Viscosity										cP	
	Compressibility Z		1.000									
	Specific Heats Cp/Cv (Real)		1.400									
	Estimate Sound Press Level										dBA	
	VALVE BODY & BONNET	Body Type: Regulator §				TRIM	Approximate Valve C _v : 1.0 F _L : 0.9 x _T : 0.72					
		Body Size: Port Type: Single					Trim Style: Trim Size: 1/2" orifice					
		1	Inlet Conn Size: 1/2" Type: Flanged Class: 300				Inherent Characteristic: Manufacturer's Standard					
1		Outlet Conn Size: 1/2" Type: Flanged Class: 300			Balance Style: Full Open Position:							
Flange Facing: Flange Face Finish:				Guiding Style:								
Body & Bonnet Material: Carbon Steel				Closure Member Material: Stainless Steel								
Bonnet Style:				Seat Ring Material: Stainless Steel								
Packing Style:				Seat Ring Retainer Material:								
Packing Material:				Bearing Material:								
Body/Bonnet Gasket Material:				Bushing/Cage Material:								
Lub/Iso Valve Material: Lub Material:				Stem/Shaft Material: Stainless Steel								
Manufacturer: Masoneilan (or equal)				Flow Direction:								
Model Number:				Seat Leakage Class:								
ACTUATOR		Type:					INSTALLATION	Inlet Pipe Nominal Size:		Est Wall Thick:		
		Action Style:						Outlet Pipe Nominal Size:		Est Wall Thick:		
	Service Type:				Est Insulation Type:							
	Max Shutoff Press: In:		Out:		Est Insulation Thickness:							
	Maximum Stroke Time to Open:				Installation:							
	Maximum Stroke Time to Close:											
	Stroke Cycle Service:				Design Pressure :			10 kg/cm2g				
					Design Temperature :			325 C				
DATA PAGE 1 REMARKS	§ The information is based on Masoneilan 17 series pressure regulator. Provide a Masoneilan type 17-22 (or equal) pressure reducing regulator with an 8 to 40 psig spring set at 12 psig (0.84 kg/cm2 (g)).											
ALL ITEMS SHALL COMPLY WITH THE GENERAL REQUIREMENTS SHOWN ON THE GENERAL REQUIREMENTS SECTION OF THIS SPECIFICATION.												



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CONTROL VALVES

PROJECT SPECIFICATION

963766 — 616

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Item Number: 63 Quantity: 1		Valve Tag Number: PCV-13-2001		Data Page: 2 of 2	
TRANS-DUCER	Transducer Tag Number:		HANDWHEEL	Handwheel Position:	
	Transducer Input Signal Range:			Usage:	
	Transducer Output Signal Range:				
	Manufacturer:				
SOLENOID VALVE	Solenoid Valve Tag Number:		POSITION SWITCH	Open Position Switch Tag Number:	
	Body Type: Body Material:			Close Position Switch Tag Number:	
	Seat Material: Disc Material:			Sensor Type:	
	Connection Size/Type:			Contact Form:	
	Packing Style:			Open Position Switch Contact Action:	
	Fluid:			Close Position Switch Contact Action:	
	Pressure:			Contact Load: Contact Rating:	
	Minimum Pressure Port Orifice C _v : Size:			Mounting Location:	
	Minimum Exhaust Port Orifice C _v : Size:			Manufacturer:	
	Operation Style:				
	Coil Configuration:		POSITION TRANSMITTER	Position Transmitter Tag Number:	
	Operating Voltage: Frequency:			Sensor Type:	
	Action Coil 1 Deenergized:			Output Signal Range:	
	Action Coil 1 Energized:			Travel Measurement:	
	Action Coil 2 Deenergized:			Mounting Location:	
	Action Coil 2 Energized:			Manufacturer:	
	Manual Reset:		LOCK-UP VALVE	Action Style:	
	Manual Operator:			Mounting Location:	
Mounting Location:		Model Number:			
Manufacturer:		Manufacturer:			
POSITIONER	Positioner Tag Number:		QUICK EXHAUST	Connection Size/Type:	
	Positioner Type:			Body Material:	
	Positioner Input Signal Range:			Diaphragm Material:	
	Positioner Output Signal Range:			Exhaust Action:	
	Manufacturer:			Mounting Location:	
AIR FILTER	Quantity:		VOLUME BOOSTER	Connection Size/Type:	
	Type:			Body Material:	
	Manufacturer:			Seat Ring Material:	
TUBING	Tubing Size:			Diaphragm Material:	
	Tubing Material:			Mounting Location:	
			Manufacturer:		
DATA PAGE 2 REMARKS					

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CONTROL VALVE PROCESS SPECIFICATION

CUSTOMERS NAME: PTT PHENOL COMPANY LIMITED					PROJECT No: 120117				
LOCATION: MAP TA PHUT, THAILAND					UNIT No: CDN UNIT				
REV	F1	F2	F3	Z1	DOCUMENT No. 14780-8120-PS-0010				
DATE	JUN 10, 2014	FEB 06, 2015	MAR 04, 2015	MAY 10, 2016	SHEET 134 OF 136				
PREP. BY	S.W.Park	S.W.Park	S.W.Park	S.W.Park					
REVIEW BY	J.S.Chung	J.S.Chung	J.S.Chung	J.S.Chung					
APP. BY	S.W.Lee	S.W.Lee	S.W.Lee	S.W.Lee					
1	General	Item No.	UXV-23-0803		PCV-23-0901		PCV-23-0902		PCV-23-1702
		P&ID No.	14780-8120-25-23-0008A		14780-8120-25-23-0001J		14780-8120-25-23-0001J		14780-8120-25-23-0001K
		Service	RCS TO E-2301		N ₂ CONTROL VALVE		N ₂ CONTROL VALVE		N ₂ CONTROL VALVE
3	Line Size	inch		28	3/4"		3/4"		3/4"
4	Fluid Properties	Type	Water		NITROGEN		NITROGEN		NITROGEN
		State	LIQ		VAP		VAP		VAP
		Description	RELIABLE COOLING WATER		NITROGEN		NITROGEN		NITROGEN
		Corrosion/ Erosion due to							
5									
7	Normal Flow kg/h	Vap. in/out			7.5		7.5		7.5
		Liq. in/out	3,759,096	3,756,096					
8	Max. Flow %		100%		120		120		120
9	Min. Flow %		N/A		50		50		50
	LIQUID PROPERTIES				-		-		-
10	S.G. @ flow conditions		0.996		-		-		-
11	Viscosity @ flow conditions cP		0.749		-		-		-
13	Liquid Vapour Press kg/cm ² (a)		0.05		-		-		-
14	Liquid Critical Press kg/cm ² (a)		224.90		-		-		-
	VAPOUR PROPERTIES				-		-		-
16	Mol. Weight				28		28		28
17	Vapour Compressibility (PV/RT)				0.996		0.996		0.996
18	Vapour Density kg/m ³				17.1		17.1		17.1
19	Viscosity @ flow conditions cP				0.019		0.019		0.019
20	Vapour Cp/Cv @ flow conditions				1.425		1.425		1.425
21	Operating temp °C	Minimum			0		0		0
		Normal	33		38		38		38
		Maximum			50		50		50
22	Pressures kg/cm ² g	Normal Flow	7.2		7	5.42	7	5.42	15 14.5
23		Upstream / Downstream	Maximum Flow						
24		@	Minimum Flow						
26	Max. Shutoff Pressure kg/cm ²		10.50		10		10		17
27	System DP at normal flow kg/cm ²				N/A		N/A		N/A
28	Downstream flashing (Yes/ No) Wt% Flash		NO		NO		NO		NO
29	Tight shut off class		CLASS 6		CLASS 4		CLASS 4		CLASS 4
30	Air failure		O		O		O		O
31	Time to close s		5 seconds		N /A		N /A		N /A
32	Valve type		Ecentric Disc		GLOBE		GLOBE		GLOBE
33	Design pressure kg/cm ² (g)		10.50		10		10		17
34	Design Temperature °C		120		120		120		120
35	Remarks		Notes		P-2301A		P-2301B		P-2302A
36	Revision								

NOTES

Provide two solenoid valves. The solenoid valves shall be mounted and tubed by the control valve vendor.

The solenoid valves size and type, the tubing size, the actuator air connection size, and the air/filter regulator capacity shall be determined by the control valve vendor to meet the maximum stroking time requirements specified for the control valve.

The contractor shall coordinate the solenoid valve operating voltage and coil watt rating with the supplier of the Concentration Section Shutdown System(UC-2301). See UOP Project Specification 660, this project



CONTROL VALVE PROCESS SPECIFICATION

CUSTOMERS NAME: PTT PHENOL COMPANY LIMITED					PROJECT No: 120117				
LOCATION: MAP TA PHUT, THAILAND					UNIT No: CDN UNIT				
REV	F1	F2	F3	Z1	DOCUMENT No. 14780-8120-PS-0010				
DATE	JUN 10, 2014	FEB 06, 2015	MAR 04, 2015	MAY 10, 2016	SHEET	135	OF	136	
PREP. BY	S.W.Park	S.W.Park	S.W.Park	S.W.Park					
REVIEW BY	J.S.Chung	J.S.Chung	J.S.Chung	J.S.Chung					
APP. BY	S.W.Lee	S.W.Lee	S.W.Lee	S.W.Lee					
1	General	Item No.	PCV-23-1704		PCV-23-1302		PCV-23-1303		PCV-23-0801
		P&ID No.	14780-8120-25-23-0001K		14780-8120-25-23-0001J		14780-8120-25-23-0001J		14780-8120-25-23-0001J
		Service	N ₂ CONTROL VALVE		N ₂ CONTROL VALVE		N ₂ CONTROL VALVE		N ₂ CONTROL VALVE
3	Line Size	inch	3/4"		3/4"		3/4"		3/4"
4	Fluid Properties	Type	NITROGEN		NITROGEN		NITROGEN		NITROGEN
		State	VAP		VAP		VAP		VAP
		Description	NITROGEN		NITROGEN		NITROGEN		NITROGEN
		Corrosion/ Erosion due to							
5									
7	Normal Flow kg/h	Vap. in/out	7.5		7.5		7.5		7.5
		Liq. in/out							
8	Max. Flow %		120		120		120		120
9	Min. Flow %		50		50		50		50
	LIQUID PROPERTIES		-		-		-		-
10	S.G. @ flow conditions		-		-		-		-
11	Viscosity @ flow conditions cP		-		-		-		-
13	Liquid Vapour Press kg/cm ² (a)		-		-		-		-
14	Liquid Critical Press kg/cm ² (a)		-		-		-		-
	VAPOUR PROPERTIES		-		-		-		-
16	Mol. Weight		28		28		28		28
17	Vapour Compressibility (PV/RT)		0.996		0.996		0.996		0.998
18	Vapour Density kg/m ³		17.1		17.1		17.1		8.511
19	Viscosity @ flow conditions cP		0.019		0.019		0.019		0.019
20	Vapour Cp/Cv @ flow conditions		1.425		1.425		1.425		1.411
21	Operating temp °C	Minimum	0		0		0		0
		Normal	38		38		38		38
		Maximum	50		50		50		50
22	Pressures kg/cm ² g	Normal Flow	15	14.5	15	14	15	14	7 6.54
23		Upstream / Downstream	Maximum Flow						
24		@	Minimum Flow						
26	Max. Shutoff Pressure kg/cm ²		17		17		17		10
27	System DP at normal flow kg/cm ²		N/A		N/A		N/A		N/A
28	Downstream flashing (Yes/ No) Wt% Flash		NO		NO		NO		NO
29	Tight shut off class		CLASS 4		CLASS 4		CLASS 4		CLASS 4
30	Air failure		O		O		O		O
31	Time to close s		N /A		N /A		N /A		N /A
32	Valve type		GLOBE		GLOBE		GLOBE		GLOBE
33	Design pressure kg/cm ² (g)		17		17		17		10
34	Design Temperature °C		120		120		120		120
35	Remarks		P-2302B		P-2303A		P-2303B		P-2307A
36	Revision								

NOTES



CONTROL VALVE PROCESS SPECIFICATION

CUSTOMERS NAME: PTT PHENOL COMPANY LIMITED					PROJECT No: 120117				
LOCATION: MAP TA PHUT, THAILAND					UNIT No: CDN UNIT				
REV	F1	F2	F3	Z1	DOCUMENT No. 14780-8120-PS-0010				
DATE	JUN 10, 2014	FEB 06, 2015	MAR 04, 2015	MAY 10, 2016	SHEET	136	OF	136	
PREP. BY	S.W.Park	S.W.Park	S.W.Park	S.W.Park					
REVIEW BY	J.S.Chung	J.S.Chung	J.S.Chung	J.S.Chung					
APP. BY	S.W.Lee	S.W.Lee	S.W.Lee	S.W.Lee					
1	General	Item No.	PCV-23-0802						
		P&ID No.	14780-8120-25-23-0001J						
		Service	N ₂ CONTROL VALVE						
3	Line Size	inch		3/4"					
4	Fluid Properties	Type	NITROGEN						
		State	VAP						
		Description	NITROGEN						
		Corrosion/ Erosion due to							
5									
7	Normal Flow kg/h	Vap. in/out	7.5						
		Liq. in/out							
8	Max. Flow %		120						
9	Min. Flow %		50						
	LIQUID PROPERTIES		-						
10	S.G. @ flow conditions		-						
11	Viscosity @ flow conditions cP		-						
13	Liquid Vapour Press kg/cm ² (a)		-						
14	Liquid Critical Press kg/cm ² (a)		-						
	VAPOUR PROPERTIES		-						
16	Mol. Weight		28						
17	Vapour Compressibility (PV/RT)		0.998						
18	Vapour Density kg/m ³		8.511						
19	Viscosity @ flow conditions cP		0.019						
20	Vapour Cp/Cv @ flow conditions		1.411						
21	Operating temp °C	Minimum	0						
		Normal	38						
		Maximum	50						
22	Pressures kg/cm ² g	Normal Flow	7	6.54					
23		Maximum Flow							
24		Minimum Flow							
26	Max. Shutoff Pressure kg/cm ²		10						
27	System DP at normal flow kg/cm ²		N/A						
28	Downstream flashing (Yes/ No) Wt% Flash		NO						
29	Tight shut off class		CLASS 4						
30	Air failure		O						
31	Time to close s		N /A						
32	Valve type		GLOBE						
33	Design pressure kg/cm ² (g)		10						
34	Design Temperature °C		120						
35	Remarks		P-2307B						
36	Revision								

NOTES